

SRI CHAITANYA IIT ACADEMY, INDIA

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MATHEMATICS
QUADRATIC EQUATIONS | ASSIGNMENT

1. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is **[AIEEE-2009]**

(1) Less than $4ab$ (2) Greater than $-4ab$
(3) Less than $-4ab$ (4) Greater than $4ab$

2. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$ **[AIEEE-2010]**

(1) -2 (2) -1
(3) 1 (4) 2

3. Let for $a \neq a_1 \neq 0$,
 $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and
 $p(x) = f(x) - g(x)$.

If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is **[AIEEE-2011]**

(1) 6 (2) 18
(3) 3 (4) 9

4. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots $(4, 3)$. Rahul made a mistake in writing down coefficient of x to get roots $(3, 2)$. The correct roots of equation are: **[AIEEE-2011]**

(1) $-6, -1$ (2) $-4, -3$
(3) $6, 1$ (4) $4, 3$

5. The real number k for which the equation $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$ **[JEE (Main)-2013]**

(1) Lies between 1 and 2
(2) Lies between 2 and 3
(3) Lies between -1 and 0
(4) Does not exist

6. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in R$, have a common root, then $a : b : c$ is **[JEE (Main)-2013]**

(1) $1 : 2 : 3$ (2) $3 : 2 : 1$
(3) $1 : 3 : 2$ (4) $3 : 1 : 2$

7. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$.

If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$

is equal to

[JEE (Main)-2015]

- (1) 6 (2) -6
(3) 3 (4) -3

8. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is

[JEE (Main)-2016]

- (1) -4 (2) 6
(3) 5 (4) 3

9. If, for a positive integer n , the quadratic equation, $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ has two consecutive integral solutions, then n is equal to

[JEE (Main)-2017]

- (1) 9 (2) 10
(3) 11 (4) 12

10. Let $S = \{x \in R : x \geq 0 \text{ and}$

$$2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}.$$

[JEE (Main)-2018]

- (1) Is an empty set
(2) Contains exactly one element
(3) Contains exactly two elements
(4) Contains exactly four elements

11. If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval $[1, 5]$ then m lies in the interval

[JEE (Main)-2019]

- (1) $(-5, -4)$ (2) $(3, 4)$
(3) $(4, 5)$ (4) $(5, 6)$

12. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is

[JEE (Main)-2019]

- (1) 4 (2) 5
(3) 2 (4) 3

13. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval $(0, 2)$ and its other root lies in the interval $(2, 3)$. Then the number of elements in S is

[JEE (Main)-2019]

- (1) 11 (2) 18
(3) 12 (4) 10

14. The value of λ such that sum of the squares of the roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is **[JEE (Main)-2019]**

(1) 2 (2) 1
(3) $\frac{15}{8}$ (4) $\frac{4}{9}$

15. If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube of the other root, then a value of k is **[JEE (Main)-2019]**

(1) -300 (2) 144
(3) -81 (4) 100

16. Let α and β the roots of the quadratic equation $x^2 \sin \theta - x (\sin \theta \cos \theta + 1) + \cos \theta = 0$ ($0 < \theta < 45^\circ$), and $\alpha < \beta$. Then

$\sum_{n=0}^{\infty} \left(\alpha^n + \frac{(-1)^n}{\beta^n} \right)$ is equal to **[JEE (Main)-2019]**

(1) $\frac{1}{1+\cos \theta} - \frac{1}{1-\sin \theta}$ (2) $\frac{1}{1-\cos \theta} + \frac{1}{1+\sin \theta}$
(3) $\frac{1}{1-\cos \theta} - \frac{1}{1+\sin \theta}$ (4) $\frac{1}{1+\cos \theta} + \frac{1}{1-\sin \theta}$

17. If λ be the ratio of the roots of the quadratic equation in x , $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is

[JEE (Main)-2019]

(1) $4 - 2\sqrt{3}$ (2) $4 - 3\sqrt{2}$
(3) $2 - \sqrt{3}$ (4) $-2 + \sqrt{2}$

18. The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, $x \in R$, is always positive, is

[JEE (Main)-2019]

(1) 8 (2) 3
(3) 6 (4) 7

19. If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$

is **[JEE (Main)-2019]**

(1) 4 (2) 5
(3) 3 (4) 2

20. The sum of the solutions of the equation

$$|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0, (x > 0) \text{ is equal to}$$

[JEE (Main)-2019]

- (1) 4 (2) 10
(3) 9 (4) 12

21. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

[JEE (Main)-2019]

- (1) d, e, f are in A.P.
(2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
(3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.
(4) d, e, f are in G.P.

22. The number of integral values of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :

[JEE (Main)-2019]

- (1) Infinitely many (2) 3
(3) 2 (4) 1

23. Let $p, q \in \mathbb{R}$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then

[JEE (Main)-2019]

- (1) $q^2 - 4p - 16 = 0$ (2) $p^2 - 4q + 12 = 0$
(3) $p^2 - 4q - 12 = 0$ (4) $q^2 + 4p + 14 = 0$

24. If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is

[JEE (Main)-2019]

- (1) $8\sqrt{3}$ (2) $10\sqrt{5}$
(3) $4\sqrt{3}$ (4) $8\sqrt{5}$

25. If α and β are the roots of the equation

$$375x^2 - 25x - 2 = 0, \text{ then } \lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$$

is equal to

[JEE (Main)-2019]

- (1) $\frac{21}{346}$ (2) $\frac{7}{116}$
(3) $\frac{29}{358}$ (4) $\frac{1}{12}$

26. If α , β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to

[JEE (Main)-2019]

- (1) 0 (2) $\alpha\gamma$
(3) $\beta\gamma$ (4) $\alpha\beta$

27. Let α and β be two real roots of the equation $(k+1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, where $k(\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is

[JEE (Main)-2020]

- (1) 10 (2) $10\sqrt{2}$
(3) 5 (4) $5\sqrt{2}$

28. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

[JEE (Main)-2020]

- (1) $p_3 = p_5 - p_4$
(2) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
(3) $p_5 = 11$
(4) $p_5 = p_2 \cdot p_3$

29. Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S

[JEE (Main)-2020]

- (1) Contains at least four elements
(2) Is a singleton
(3) Contains exactly two elements
(4) Is an empty set

30. The number of real roots of the equation, $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is [JEE (Main)-2020]

- (1) 4 (2) 2
(3) 3 (4) 1

31. Let $a, b \in \mathbb{R}$, $a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to

[JEE (Main)-2020]

- (1) 25 (2) 24
(3) 26 (4) 28

32. Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \dots$, then [JEE (Main)-2020]

(1) $5S_6 + 6S_5 = 2S_4$ (2) $6S_6 + 5S_5 + 2S_4 = 0$
(3) $6S_6 + 5S_5 = 2S_4$ (4) $5S_6 + 6S_5 + 2S_4 = 0$

33. Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in [JEE (Main)-2020]

(1) $(-1, 0)$ (2) $(-3, -1)$
(3) $(0, 1)$ (4) $(1, 3)$

34. If α and β are the roots of the equation

$x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then

$\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to

[JEE (Main)-2020]

(1) $\frac{9}{4}(9 - q^2)$ (2) $\frac{9}{4}(9 + p^2)$
(3) $\frac{9}{4}(9 + q^2)$ (4) $\frac{9}{4}(9 - p^2)$

35. The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is [JEE (Main)-2020]

(1) $(-3, -1)$ (2) $(2, 4]$
(3) $(0, 2)$ (4) $(1, 3]$

36. Let $\lambda \neq 0$ be in R . If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then

$\frac{\beta\gamma}{\lambda}$ is equal to [JEE (Main)-2020]

(1) 18 (2) 9
(3) 27 (4) 36

37. The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is [JEE (Main)-2020]

(1) $\frac{25}{9}$ (2) $\frac{25}{81}$
(3) $\frac{5}{9}$ (4) $\frac{5}{27}$

38. If α and β are the roots of the equation,

$7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to [JEE (Main)-2020]

- (1) $\frac{1}{24}$ (2) $\frac{27}{32}$
(3) $\frac{3}{8}$ (4) $\frac{27}{16}$

39. If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of

$\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is [JEE (Main)-2020]

- (1) 3 (2) 2
(3) 4 (4) 1

40. If α and β are the roots of the equation $2x(2x + 1) = 1$, then β is equal to

[JEE (Main)-2020]

- (1) $2\alpha^2$
(2) $-2\alpha(\alpha + 1)$
(3) $2\alpha(\alpha - 1)$
(4) $2\alpha(\alpha + 1)$

41. The least positive value of 'a' for which the equation, $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is _____ [JEE (Main)-2020]

42. The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R, is : [JEE (Main)-2021]

- (1) 2 (2) 3
(3) 4 (4) 0

43. Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is : [JEE (Main)-2021]

- (1) 2 (2) 4
(3) 3 (4) 1

44. Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

[JEE (Main)-2021]

45. The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is :

[JEE (Main)-2021]

- (1) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$ (2) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$
 (3) $2 + \frac{2}{5}\sqrt{30}$ (4) $5 + \frac{2}{5}\sqrt{30}$

46. The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to

- (1) $2 + \sqrt{3}$
 (2) $3 + 2\sqrt{3}$
 (3) $4 + \sqrt{3}$
 (4) $1.5 + \sqrt{3}$

[JEE (Main)-2021]

47. The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is

[JEE (Main)-2021]

- (1) 1 (2) 2
 (3) 6 (4) 4

48. If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n , then the value of

$\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2} \right)$ is equal to _____.

[JEE (Main)-2021]

49. The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is

[JEE (Main)-2021]

- (1) 4 (2) 2
 (3) 1 (4) 3

50. If $a + b + c = 1$, $ab + bc + ca = 2$ and $abc = 3$, then the value of $a^4 + b^4 + c^4$ is equal to _____.

51. [JEE (Main)-2021]

Let α, β be two roots of the equation

$$x^2 + (20)^{\frac{1}{4}}x + (5)^{\frac{1}{2}} = 0. \text{ Then } \alpha^8 + \beta^8 \text{ is equal to}$$

[JEE (Main)-2021]

- (1) 160 (2) 10
(3) 50 (4) 100

52. Let $\alpha = \max_{x \in \mathbb{R}} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}$ and

$$\beta = \min_{x \in \mathbb{R}} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}.$$

If $8x^2 + bx + c = 0$ is a quadratic equation whose

roots are $\alpha^{\frac{1}{5}}$ and $\beta^{\frac{1}{5}}$, then the value of $c - b$ is equal to:

[JEE (Main)-2021]

- (1) 43 (2) 42
(3) 50 (4) 47

53. The number of real roots of the equation

$$e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0 \text{ is equal to } \underline{\hspace{2cm}}.$$

54. [JEE (Main)-2021]

The sum of all integral values of $k (k \neq 0)$ for which

the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real

55. roots, is _____. [JEE (Main)-2021]

Let $\lambda \neq 0$ be in $\mathbb{R}$. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$, and α and γ are the roots

of the equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to _____.

[JEE (Main)-2021]

56. The set of all values of $k > -1$, for which the equation $(3x^2 + 4x + 3)^2 - (k + 1)(3x^2 + 4x + 3)(3x^2 + 4x + 2) + k(3x^2 + 4x + 2)^2 = 0$ has real roots, is

[JEE (Main)-2021]

- (1) $\left(\frac{1}{2}, \frac{3}{1}\right) - \{1\}$ (2) $\left[-\frac{1}{2}, 1\right)$
(3) $[2, 3)$ (4) $\left(1, \frac{5}{2}\right]$

57. $\operatorname{cosec} 18^\circ$ is a root of the equation

[JEE (Main)-2021]

- (1) $x^2 - 2x + 4 = 0$ (2) $x^2 + 2x - 4 = 0$
(3) $x^2 - 2x - 4 = 0$ (4) $4x^2 + 2x - 1 = 0$

58. The number of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is :

[JEE (Main)-2021]

- (1) 8 (2) 4
(3) 6 (4) 2

59. Let $f(x)$ be a polynomial of degree 3 such that

$$f(k) = -\frac{2}{k} \text{ for } k = 2, 3, 4, 5. \text{ Then the value of } 52 - 10 f(10) \text{ is equal to } \underline{\hspace{2cm}}.$$

60. [JEE (Main)-2021]

If α and β are the roots of the quadratic equation,

$$x^2 + x \sin \theta - 2 \sin \theta = 0, \quad \theta \in \left(0, \frac{\pi}{2}\right), \text{ then}$$

$$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}) \cdot (\alpha - \beta)^{24}} \text{ is equal to}$$

[JEE (Main)-2021]

$$(1) \frac{2^{12}}{(\sin \theta - 8)^6} \quad (2) \frac{2^{12}}{(\sin \theta - 4)^{12}}$$

$$(3) \frac{2^6}{(\sin \theta + 8)^{12}} \quad (4) \frac{2^{12}}{(\sin \theta + 8)^{12}}$$

61. If for some $p, q, r \in \mathbf{R}$, not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x$

$$- 8 = 0, \text{ then } \frac{q^2 + r^2}{p^2} \text{ is equal to } \underline{\hspace{2cm}}.$$

[JEE (Main)-2022]

62. If α, β are the roots of the equation

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_3 3}}\right) + 3 \left(3^{\frac{(\log_3 5)^2}{2}} - 5^{\frac{(\log_3 3)^2}{3}} - 1\right) = 0,$$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and

$$\beta + \frac{1}{\alpha}, \text{ is} \quad \text{[JEE (Main)-2022]}$$

$$(1) 3x^2 - 20x - 12 = 0 \quad (2) 3x^2 - 10x - 4 = 0$$

$$(3) 3x^2 - 10x + 2 = 0 \quad (4) 3x^2 - 20x + 16 = 0$$

63. Let a, b be the roots of the equation

$$x^2 - \sqrt{2}x + \sqrt{6} = 0 \text{ and } \frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1, \frac{1}{\beta^2} + 1 \text{ be}$$

the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are

[JEE (Main)-2022]

(1) non-real complex number

(2) real and both negative

(3) real and both positive

(4) real and exactly one of them is positive

64. Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3$, $f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to [JEE (Main)-2022]

- (1) -4 (2) $\frac{13}{2}$
(3) $\frac{23}{2}$ (4) 4

65. Let $a, b (a > b)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $n \in \mathbb{N}$,

then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____ [JEE (Main)-2022]

66. If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

- (1) 18 (2) 24
(3) 36 (4) 96

67. [JEE (Main)-2022]

The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

- (1) $\log_e 3$ (2) $-\log_e 3$
(3) $\log_e 6$ (4) $-\log_e 6$

68. [JEE (Main)-2022]

Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to

- (1) 37 (2) 58
(3) 68 (4) 92

[JEE (Main)-2022]

69. The sum of the cubes of all the roots of the equation $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is _____.

70. [JEE (Main)-2022]

Let p and q be two real numbers such that $p + q = 3$

and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

71. [JEE (Main)-2022]

If the sum of all the roots of the equation $e^{2x} - 11e^x -$

$45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.

[JEE (Main)-2022]



72. Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation

$$x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0, \quad \lambda > 0. \text{ If}$$

$\beta + \lambda = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

73. **[JEE (Main)-2022]**

The number of real solutions of the equation

$$e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0 \text{ is } \underline{\hspace{2cm}}.$$

74. Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1 , then the sum of the roots of $f(x) = 0$ is equal to:

(1) $\frac{11}{3}$

(2) $\frac{7}{3}$

(3) $\frac{13}{3}$

(4) $\frac{14}{3}$

75. **[JEE (Main)-2022]**

Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$ and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

[JEE (Main)-2022]

76. Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p$, $p \neq 0$, and

$$f(1) = \frac{1}{3}. \text{ If the equations } f(x) = 0 \text{ and } f(f(x)) = 0$$

have a common real root, then $f(-3)$ is equal to _____.

[JEE (Main)-2022]

77. The sum of all real value of x for which

$$\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12} \text{ is equal to}$$

_____.

[JEE (Main)-2022]

78. The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is **[JEE (Main)-2022]**

(1) 4

(2) 5

(3) 6

(4) 8

1 ☐

79. Find integral values of k for which the equation $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ possess no real roots ?

80. For what values of 'a' does the equation $x^2 + 2a\sqrt{a^2 - 3}x + 4 = 0$ possess equal roots ?
-
81. Form a quadratic equation whose roots are the numbers $\frac{1}{10 - \sqrt{72}}$ and $\frac{1}{10 + 6\sqrt{2}}$.
-
82. For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a + 2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$?
-
83. Find a such that one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other.
-
84. The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p.
-
85. to 3 ?
Find b in the equation $5x^2 + bx - 28 = 0$ if the roots x_1 and x_2 of the equation are related as $5x_1 + 2x_2 = 1$ and b is an integer.
-
86. product ?
Find all the values of a for which the sum of the roots of the equation $x^2 - 2a(x - 1) - 1 = 0$ is equal to the sum of the squares of its roots.
-
87. Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0$, $m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.
-
88. Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.
-
89. Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x_1^3 + x_2^3$.
-
90. Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.
-
91. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.
-
92. Find all values of p for which the roots of the equation $(p - 3)x^2 - 2px + 5p = 0$ are real and positive.
-
93. If the equation $x^2 - 15 - m(2x - 8) = 0$ has equal roots, find the values of m.
-
94. For what values of m will the equation $x^2 - 2x(1 + 3m) + 7(3 + 2m) = 0$ have equal roots ?
-
95. For what value of m will the equation $\frac{x^2 - bx}{ax - c} = \frac{m - 1}{m + 1}$ have roots equal in magnitude but opposite in sign ?
-
96. Prove that the roots of the following equations are rational, where a, b, c $\in \mathbb{Q}$:
(1) $(a + c - b)x^2 + 2cx + (b + c - a) = 0$,
(2) $abc^2x^2 + 3a^2cx + b^2cx - 6a^2 - ab + 2b^2 = 0$.
-
97. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the values of
(i) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$
(ii) $\alpha^4\beta^7 + \alpha^7\beta^4$
-
98. Find the value of $x^3 + x^2 - x + 22$ when $x = 1 + 2i$.
-

99. If α and β are the roots of $x^2 + px + q = 0$, form the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$.

100 Prove that the roots of $(x-a)(x-b) = h^2$ are always real.

101 If α, β are the roots of $ax^2 + bx + c = 0$, form the equation whose roots are $\alpha^2 + \beta^2$ and $\alpha^{-2} + \beta^{-2}$.

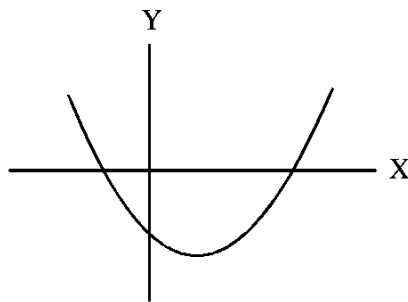
102 Form the equation whose roots are the squares of the sum and of the difference of the roots of $2x^2 + 2(m+n)x + m^2 + n^2 = 0$.

103 Solve the following for x :

(i) $12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$ (ii) $2\sqrt{x} + 2x^{\frac{1}{2}} = 5$

(iii) $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = 2\frac{1}{6}$ (iv) $2^{2x+3} - 57 = 65(2^x - 1)$ (v) $x(2x+1)(x-2)(2x-3) = 63$

104 Given the graph of quadratic equation $f(x) = ax^2 + bx + c = 0$. Which of the following is correct?



- (A) $a < 0$ (B) $D < 0$ (C) $b < 0$ (D) $c > 0$

105 Let $p, q \in \{1, 2, 3, 4\}$. The number of equation of the form $px^2 + qx + 1 = 0$ having real roots is

- (A) 15 (B) 9 (C) 7 (D) 8

106 If $b \in \mathbb{R}^+$ then roots of the equation $(2+b)x^2 + (3+b)x + (4+b) = 0$ is

- (A) Real and distinct (B) Real and equal
(C) Imaginary (D) Cannot be predicted

107 Suppose $a, b, c > 0$, then the number of real roots of the equation $ax^2 + b|x| + c = 0$ is

- (A) 1 (B) 4 (C) 2 (D) None of these

108 If $x = \sqrt{3+2\sqrt{2}}$, then $x^2 + \frac{1}{x^2}$ is equal to

- (A) $2\sqrt{2}$ (B) 8 (C) 6 (D) 1

109 If $x = 3 - \sqrt{8}$, then $x^3 + \frac{1}{x^3}$ is equal to

- (A) 6 (B) 198 (C) $6\sqrt{2}$ (D) 102

$3 + 2\sqrt{5}$

110 If $\frac{3\pi + 4\sqrt{2}}{3 - \sqrt{2}} = a + b\sqrt{2}$, then a and b ($a, b \in \mathbb{Q}$) are respectively equal to

- (A) $\frac{13}{7}, \frac{9}{7}$ (B) $\frac{9}{7}, \frac{13}{7}$ (C) $\frac{13}{7}, \frac{7}{9}$ (D) $\frac{7}{9}, \frac{7}{13}$

111 The number of solution of the equation, $\log(-2x) = 2 \log(x + 1)$ is

- (A) zero (B) 1 (C) 2 (D) none

112 The solution set of the equation $e^{4x} - 5e^{2x} + 4 = 0$ over $\mathbb{R}$ is

- (A) $\{1, 4\}$ (B) $\{-4, -1\}$ (C) $\{-\log_e 2, 0, \log_e 2\}$ (D) $\{0, \log_e 2\}$

113 The sum of the solutions of the equation $9^x - 6 \cdot 3^x + 8 = 0$ is

- (A) $\log_3 2$ (B) $\log_3 6$ (C) $\log_3 8$ (D) $\log_3 4$

114 If $\sin \theta$ and $\cos \theta$ are the roots of the equation $ax^2 - bx + c = 0$, then

- (A) $a^2 - b^2 = 2ac$ (B) $a^2 + b^2 = 2ac$ (C) $a^2 + b^2 + 2ac = 0$ (D) $b^2 - a^2 = 2ac$

115 The roots of the equation $x^2 - 2\sqrt{2}x + 1 = 0$ are

- (A) real and different (B) imaginary and different
(C) real and equal (D) rational and different

116 The roots of the equation $(b + c)x^2 - (a + b + c)x + a = 0$ ($a, b, c \in \mathbb{Q}$, $b + c \neq a$) are

- (A) irrational and different (B) rational and different
(C) imaginary and different (D) real and equal

117 If the roots of the equation $ax^2 + x + b = 0$ be real and different, then the roots of the equation

$x^2 - 4\sqrt{ab}x + 1 = 0$ will be

- (A) rational (B) irrational (C) real (D) imaginary

118 If $a < c < b$ then the roots of the equation $(a - b)^2 x^2 + 2(a + b - 2c)x + 1 = 0$ are

- (A) imaginary (B) real
(C) one real & imaginary (D) equal & imaginary

119 The number of real solutions of $x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$ is

- (A) 0 (B) 1 (C) 2 (D) infinite

120 Sum of roots of the equation $(x + 3)^2 - 4|x + 3| + 3 = 0$ is

- (A) 4 (B) 12 (C) -12 (D) -4

121 If α, β are roots of the equation $x^2 + px - q = 0$ & γ, δ are roots of $x^2 + px + r = 0$, then the value of $(\alpha - \gamma)(\alpha - \delta)$ is

- (A) $p + r$ (B) $p - r$ (C) $q - r$ (D) $q + r$

122 If α, β are roots of $Ax^2 + Bx + C = 0$ and α^2, β^2 are roots of $x^2 + px + q = 0$ then p is equal to

- (A) $\frac{B^2 - 4AC}{A^2}$ (B) $\frac{2AC - B^2}{A^2}$ (C) $\frac{4AC - B^2}{A^2}$ (D) None of these

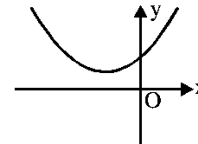
- 123 If α, β are roots of the equation $x^2 - 5x + 6 = 0$ then the equation whose roots are $\alpha + 3$ and $\beta + 3$ is
- (A) $x^2 - 11x + 30 = 0$ (B) $(x - 3)^2 - 5(x - 3) + 6 = 0$
(C) Both (A) and (B) (D) None of these
-
- 124 If α, β are the root of a quadratic equation $x^2 - 3x + 5 = 0$ then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is
- (A) $x^2 + 4x + 1 = 0$ (B) $x^2 - 4x + 4 = 0$ (C) $x^2 - 4x - 1 = 0$ (D) $x^2 + 2x + 3 = 0$
-
- 125 The number of values of a for which $(a^2 - 3a + 2)x^2 + (a^2 - 5a + 6)x + a^2 - 4 = 0$ is an identity in x is
- (A) 0 (B) 2 (C) 1 (D) 3
-
- 126 If a, b , be the roots of the equation $ax^2 - 35x + 2 = 0$ then the value of $(2a-35)^3(2b-35)^3$ is equal to
- (A) 8 (B) 1 (C) 64 (D) None
-
- 127 If $(2x + 1)^2 - |2x + 1| - 6 < 0$, then
- (A) $-1 < x < 2$ (B) $-2 < x < 1$ (C) $-2 < x < -1$ (D) $1 < x < 2$
-
- 128 If exactly one root of the quadratic equation $f(x) = ax^2 + bx + c = 0$ is at infinity then
- (A) a tends to zero (B) b tends to zero (C) b must not be zero (D) Both (A) and (C)
-
- 129 If $(3 - 4\sin^2 1)(3 - 4\sin^2 3)(3 - 4\sin^2 5) \dots (3 - 4\sin^2(3^{n-1})) = \sin a / \sin b$, where $n \in \mathbb{N}$ and $n > 1$ & a, b are integers in radian, then the digit at the unit place of $(a + b)$ cannot be-
- (A) 4 (B) 1 (C) 8 (D) 2
-
- 130 The values of the parameter ' a ' for which the quadratic equations $(1 - 2a)x^2 - 6ax - 1 = 0$ and $ax^2 - x + 1 = 0$ have at least one root common, are
- (A) 0, $1/2$ (B) $1/2, 2/9$ (C) $2/9$ (D) $1/3, 1/2, 2/9$
-
- 131 If $\alpha + \beta = 3$ and $\alpha^3 + \beta^3 = 7$, then α and β are the roots of the equation
- (A) $3x^2 + 9x + 7 = 0$ (B) $9x^2 - 27x + 20 = 0$ (C) $2x^2 - 6x + 15 = 0$ (D) none of these
-
- 132 If α, β are roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $2\alpha + 3\beta$ and $3\alpha + 2\beta$ is
- (A) $abx^2 - (a+b)cx + (a+b)^2 = 0$ (B) $acx^2 - (a+c)bx + (a+c)^2 = 0$
(C) $acx^2 + (a+c)bx - (a+c)^2 = 0$ (D) None of these
-
- 133 The equations $ax^2 + bx + a = 0 (a, b \in \mathbb{R})$ and $x^3 - 2x^2 + 2x - 1 = 0$ have 2 roots common. Then $a + b$ must be equal to
- (A) 1 (B) -1 (C) 0 (D) None of these
-
- 134 The value of m for which the equation $\frac{a}{x+a+m} + \frac{b}{x+b+m} = 1$ has roots equal in magnitude and opposite in signs is
- (A) $\frac{a-b}{a+b}$ (B) -1 (C) 0 (D) $\frac{a+b}{a-b}$
-
- 135 If the product of 2 positive numbers is 9, then the possible value of the sum of their reciprocals lies in the interval
- (A) $\left[\frac{1}{3}, \infty\right)$ (B) $[1, \infty)$ (C) $\left[\frac{4}{9}, \infty\right)$ (D) $\left[\frac{2}{3}, \infty\right)$

136 If $(49)^{3\log_{\sqrt{343}}\sqrt{x}} - 2x - 3 = 0$, then x is equal to

- (A) -1 (B) 3 (C) -1, 3 (D) 2, 3

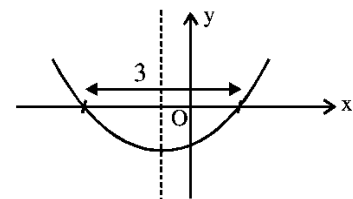
137 The curve of the quadratic expression $y = ax^2 + bx + c$ is shown in the figure and α, β be the roots of the equation $ax^2 + bx + c = 0$ then correct option is [D is the discriminant]

- (A) $a > 0, b > 0, c > 0, D > 0, \alpha + \beta > 0, \alpha\beta > 0$
 (B) $a > 0, b > 0, c > 0, D < 0, \alpha + \beta < 0, \alpha\beta < 0$
 (C) $a > 0, b > 0, c > 0, D < 0, \alpha + \beta < 0, \alpha\beta > 0$
 (D) $a > 0, b < 0, c > 0, D < 0, \alpha + \beta > 0, \alpha\beta > 0$



138 If graph of $f(x) = x^2 + bx + c$ is drawn in adjacent diagram, where $b, c \in I$, then number of such quadratic equation $f(x) = 0$ is

- (A) 1
 (B) 2
 (C) 3
 (D) 4



139 The value of 'a' for which the equation $x^7 + ax^2 + 3 = 0$ and $x^8 + ax^3 + 3 = 0$ have a common root, can be

- (A) 1 (B) -2 (C) -3 (D) -4

140 If $x^2 + 3x + 3 = 0$ and $ax^2 + bx + 1 = 0$, $a, b \in Q$ have a common root, then value of $(3a + b)$ is equal to

- (A) $1/3$ (B) 1 (C) 2 (D) 4

141 The number of integral values of k for which the curve $y = x^2 + kx + 4$ touches the x axis is

- (A) 0 (B) 1 (C) 2 (D) 4

142 If α, β are the roots of $x^2 - px + r = 0$ and $\alpha + 1, \beta - 1$ are the roots of $x^2 - qx + r = 0$, then r is

- (A) $\frac{p-1}{4}$ (B) $\frac{q+1}{4}$ (C) $\frac{p^2-1}{4}$ (D) $\frac{q^2+1}{4}$

143 If the roots of the equation $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ equals.

- (A) -2 (B) 3 (C) 2 (D) 1

144 The sum of all values of p for which the vertex of the parabola $y = x^2 + 2px + 13$ lie at a distance of 5 from the origin, is

- (A) 0 (B) 6 (C) 7 (D) 8

[MULTIPLE CORRECT CHOICE TYPE]

145 to Q.17 has four choices (A), (B), (C), (D) out of which ONE OR MORE may be correct

16. If equations $ax^2 - (a+b)x + b = 0$ & $bx^2 + (b-c)x - c = 0$ has exactly one root in common $\{a, b, c \neq 0\}$, then which of the following can be correct

- (A) $b^2 = ac$ (B) $-a = b \neq c$ (C) $b = a \neq c$ (D) $-a \neq b = c$

146 If one of the roots of $x^2 - bx + c = 0$, $b, c \in Q$ is $\sqrt{7-4\sqrt{3}}$ then

- (A) $\log_b c = 0$ (B) $b + c = 5$ (C) $\log_c b = 0$ (D) $bc = -4$

[COMPREHENSION TYPE]

147 Consider the quadratic expression $y = x^2 - px + q$ where $p, q \in \mathbb{R}$.

- On the basis of above information, answer the following questions :

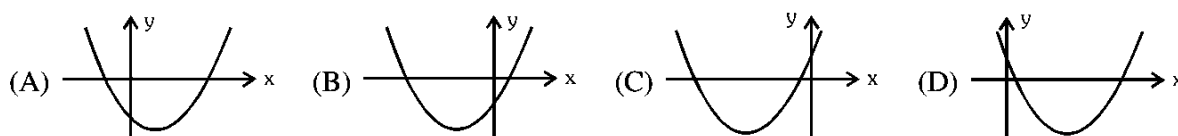
If $p = 4$ and $q = 9$, then minimum value of the expression is

- (A) 3 (B) 4 (C) 5 (D) 6

148 Consider the quadratic expression $y = x^2 - px + q$ where $p, q \in \mathbb{R}$.

- On the basis of above information, answer the following questions :

If $p < 0$ & $q < 0$, then the possible graph of $y = x^2 - px + q$ is



149 Consider the quadratic expression $y = x^2 - px + q$ where $p, q \in \mathbb{R}$.

- On the basis of above information, answer the following questions :

If p is positive odd integer and roots of equation $y = 0$ are prime numbers and $p + q = 23$, then absolute value of difference of roots is

- (A) 1 (B) 2 (C) 3 (D) 5

150 Let $x^2 - (m-3)x + m = 0$ ($m \in \mathbb{R}$) be a quadratic equation. Find the value of m for which the roots of the equation are

- (a) Positive
- (b) Negative
- (c) Such that at least one is positive
- (d) One root is smaller than 2 and other root is greater than 2
- (e) Both the roots are greater than 2
- (f) Both the roots are smaller than 2
- (g) Exactly one root lies in the interval (1, 2)
- (h) Both the roots lie in the interval (1, 2)
- (i) Such that at least one root lie in the interval (1, 2)
- (j) One root is greater than 2 and the other root is smaller than 1

151 If α, β are the roots of $x^2 - p(x+1) - c = 0$, $c \neq 1$ then show that

- (i) $(\alpha + 1)(\beta + 1) = 1 - c$ (ii) $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + c} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + c} = 1$

152 Find the value of k for which $\left| \frac{x^2 + kx + 1}{x^2 + x + 1} \right| < 2 \quad \forall x \in \mathbb{R}$

153 If x and y are two real numbers connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then find range of x and y .

154 If x is real, then find the range of the expression $\frac{x^2 + 14x + 9}{x^2 + 2x + 3}$.

155 Find the smallest and greatest value of $\frac{x^2 + x + 1}{x^2 + 1} \quad \forall x \in \mathbb{R}$.

156 If α and β are the roots of $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{\alpha-1}{\alpha-2}$ and $\frac{\beta-1}{\beta-2}$ is

- (A) $a(x+1)^2 + b(x+1)(x-2) + c(x-2)^2 = 0$ (B) $a(x-2)^2 + b(x+1)(x-2) + c(x+1)^2 = 0$
 (C) $a(2x+3)^2 + b(x+1)(x+2) + c(x+2)^2 = 0$ (D) $a(2x+1)^2 + b(2x+1)(x-1) + c(x-1)^2 = 0$

157 If α and β are roots of the equation $2x^2 + 4x - 5 = 0$, then the equation whose roots are $2\alpha - 3$ and $2\beta - 3$ is

- (A) $x^2 + 10x - 11 = 0$ (B) $11x^2 + 10x - 1 = 0$ (C) $x^2 + 10x + 11 = 0$ (D) $11x^2 - 10x + 1 = 0$

158 If α and β are roots of $ax^2 + bx + c = 0$, then the equation whose roots are $\alpha + (c/a)$ and $\beta + (c/a)$ is

- (A) $a^2x^2 - 2(ac+b)x + c(a+b) = 0$ (B) $a^2x^2 - (ca+b)x + c(a+b) = 0$
 (C) $a^2x^2 + 2(ac+b)x - c(a+b+c) = 0$ (D) $a^2x^2 - a(-b+2c)x + c(a-b+c) = 0$

159 If the equation $(a^2+1)x^2 - (a+1)x + (a^2-a-2) = 0$ has one positive and one negative root, then which one of the following is possible?

- (A) $a \leq -1$ (B) $-1 < a < 2$ (C) $2 \leq a \leq 5$ (D) $a > 5$

160 If $mx^2 - (m+1)x + 3 = 0$ has roots belonging to $(1, 2)$, then

- (A) $0 < m < 1$ (B) $1 \leq m \leq 2$ (C) $m < 0$ (D) no real value for m exists

161 If $x^2 - (m+1)x + m^2 + m - 8 = 0$ has one root in $(-\infty, 1)$ and the other root in $(1, \infty)$, then

- (A) $m < -2\sqrt{2}$ (B) $m > 2\sqrt{2}$ (C) $-2\sqrt{2} < m < 2\sqrt{2}$ (D) no real value for m exists

162 If the function $y = \frac{x - x}{1 - mx}$ takes all real values for real values of x , then

- (A) $m < 0$ (B) $0 < m < 1$ (C) $m > 0$ (D) $m > 1$

163 If $y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ takes all real values, then

- (A) $0 < c < 1$ (B) $c < -1$ (C) $c > 1$ (D) $c > 0$

164 If $\frac{x^2 + ax + 1}{x^2 + x + 1} < 3$ for all real x , then

- (A) $a < 0$ (B) $a < -1$ (C) $-1 < a < 7$ (D) $a > 7$

165

ALL

- **Assertion (A)** : Line $5x + 7y + 9 = 0$ cannot intersect the parabola $y = x^2$ in a rational point (whose abscissa & ordinate both are rational)

Reason (R) : For odd integer a,b and c equation $ax^2 + bx + c = 0$ cannot have rational roots.

- (A) Both **A** and **R** are correct and **R** is the correct explanation of **A**
- (B) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
- (C) **A** is correct but **R** is not correct
- (D) **A** is not correct but **R** is correct

166 If the quadratic equation

- $\alpha^2(x+1)^2 + \beta^2(2x^2 - x + 1) - 5x^2 - 3 = 0$ is satisfied for all $x \in \mathbb{R}$, then number of ordered pair (α, β) which is possible is/are

167

If α and β are the roots of the equation

- $x^2 - 6x + 12 = 0$ and the value of $(\alpha - 2)^{24} - \frac{(\beta - 6)^8}{\alpha^8} + 1$ is a^b where a and b are natural numbers then least value of $(a + b)$ is

168

ALL

- If $a > 2$, roots of the equation $(2 - a)x^2 + 3ax - 1 = 0$ are

- (A) One positive and one negative
- (B) Both negative
- (C) Both positive
- (D) Both imaginary

169 Let equation $x^3 - 9x^2 + 24x + \lambda = 0$ has three real and distinct roots a, b, c. If $\lambda \in (-20, -18)$, then the value of

$([a] + [b] + [c])$ will be
(Where $[.]$ represents greatest integer function.)

- (A) 8 (B) 6 (C) 7 (D) 0

170	Smallest possible natural number 'n' for which the equation $x^2 - nx + 2014 = 0$ has integer roots, is
171	If 1 lies between the roots of equation $x^2 - px + 3 = 0$ and $t \in \mathbb{R}$ then the value of $\left[\left(\frac{2 t }{ t ^2 + 4} \right)^p \right]$ is (where $[.]$ denotes greatest integer function)
172	The sum of real roots of the equation $(2x - 1)^2 - 3 2x - 1 + 2 = 0$ is - (A) 3 (B) 2 (C) 1 (D) 0
173	Three pairs of real numbers (x_1, y_1) (x_2, y_2) and (x_3, y_3) satisfy both equations $x^3 - 3xy^2 = 2005$ and $y^3 - 3x^2y = 2004$ find value of $\frac{y_1 y_2 y_3}{2(y_1 - x_1)(y_2 - x_2)(y_3 - x_3)}$
174	Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation (A) $7x^2 - 245x + 250 = 0$ (B) $49x^2 - 245x + 250 = 0$ (C) $49x^2 + 245x + 250 = 0$ (D) $7x^2 + 245x - 250 = 0$
175	Consider the quadratic polynomial $f(x) = x^2 - px + q$ where $f(x) = 0$ has prime roots. If $p + q = 11$ and $a = p^2 + q^2$, then find the value of $f(a)$ (where a is an odd positive integer).

- 176 . If $x^2 - 4x + \log_{\frac{1}{4}}(a^2) = 0$ does not have two distinct real roots, then minimum value of a is
 (A) $1/16$ (B) $-1/4$
 (C) $-1/16$ (D) $1/4$
-
- 177 . The smallest possible natural number 'p' for which the equation $x^2 - px + 1258 = 0$ has integral roots ?
-
- 178 . Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$, α and $1/\beta$ are the roots of the equation $ax^2 + 2bx + c = 0$ where $\beta^2 \notin \{-1, 0, 1\}$.
Statement-I : $(p^2 - q)(b^2 - ac) \geq 0$
Statement-II : $b \neq pa$ or $c \neq qa$
 (A) Both Statement-I and Statement-II are false.
 (B) Both Statement-I and Statement-II are true.
 (C) Statement-I is true but Statement-II is false.
 (D) Statement-I is false but Statement-II is true.
-
- 179 . Consider the equation $\frac{1}{x+p} + \frac{1}{x+q} + \frac{1}{x+r} = \frac{3}{x}$
 (Where p, q, r are distinct real numbers)
 The number of real roots of the given equation is(are)
-
- 180 . Consider the equation $\frac{1}{x+p} + \frac{1}{x+q} + \frac{1}{x+r} = \frac{3}{x}$
 If $p, q, r \in \mathbb{R}^+$ the number of positive real roots of given equation is(are)
-
- 181 . If α, β are roots of equation $x^2 - ax + b = 0$, $b > 1$ and roots of equation $bx^2 - 4x + 4 = 0$ are $\alpha + \frac{\beta^2}{\alpha}, \beta + \frac{\alpha^2}{\beta}$, then $|a| + b$ is
-
- 182 . If the quadratic equation $x^2 - px + 1 = 0$ has roots a, b^2 and $x^2 - qx + 8 = 0$ has roots b, a^2 ($a, b \in \mathbb{R}$). If $S = \{x \mid x \in \mathbb{R}, x^2 - px + q = 0\}$ then $n(S)$ is
-

183

Let α, β be roots of equation $x^2 - x - 3 = 0$.

If $S_n = \frac{\alpha^{n+1}}{\alpha^{n+2} - \alpha^{n+1} - \alpha^n + \alpha^{n-1}} + \frac{\beta^{n+1}}{\beta^{n+2} - \beta^{n+1} - \beta^n + \beta^{n-1}}$, then $|S_3|$ equals

- (A) $\frac{2}{9}$ (B) $\frac{4}{9}$
(C) $\frac{1}{9}$ (D) $\frac{8}{9}$

184

ALL

If α, β are the roots of the equation $ax^2 + bx + c = 0$ and $A_n = \alpha^n + \beta^n$. then $aA_{n+2} + bA_{n+1} + cA_n$ is equal to

- (A) 0 (B) 1
(C) $a + b + c$ (D) abc

185 $f(x) = x^2 + \lambda x + \mu \cos x$, $\lambda \in \mathbb{Z}$, $\mu \in \mathbb{R}$. The

number of ordered pairs (λ, μ) for which $f(x) = 0$ and $f(f(x)) = 0$ have same set of real roots.

- (A) 4 (B) 6 (C) 8 (D) 10

186 Consider the quadratic trinomial $f(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter. Match the range of p in

List-II with the conditions given in List-I.

List-I		List-II	
(I)	If both roots of $f(x) = 0$ are confined in $(-1, 1)$ then	(P)	$\left(\frac{2}{5}, \infty\right)$
(II)	Exactly one root of $f(x) = 0$ lies in $(-1, 1)$	(Q)	ϕ
(III)	Both roots of $f(x) = 0$ are greater than 1	(R)	$\left(-\frac{1}{17}, \frac{1}{3}\right)$
(IV)	One root of $f(x) = 0$ is greater than 1 and other root of $f(x) = 0$ is less than -1	(S)	$\left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$
		(T)	$\left(-\infty, -\frac{1}{17}\right) \cup \left(\frac{1}{3}, \infty\right)$

(A) $I \rightarrow R; II \rightarrow Q; III \rightarrow S; IV \rightarrow T$ (B) $I \rightarrow R; II \rightarrow S; III \rightarrow Q; IV \rightarrow Q$ (C) $I \rightarrow R; II \rightarrow T; III \rightarrow S; IV \rightarrow Q$ (D) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow T$

- 187 If every solution of the equation $x([x] - 5) + 2\{x\} + 6 = 0$, satisfies the inequality $(a - 3)x^2 + 2(a + 3)x - 8a \leq 0$, where $[x]$ and $\{x\}$ denote greatest integer and fractional part functions respectively then value(s) of a can be
- (A) $(-2, 1)$ (B) $(-9, 0]$
 (C) $\{0, 1\}$ (D) $(1, 2)$

$$\left(1 - \cos\left\{1 - \tan\left(\frac{\pi}{4} - x\right)\right\}\right)(x + 1)^n + \lambda \sin(n - \sqrt{n^2 - 8n})x$$

- 188 **Assertion (A)** : Line $7x + 5y + 1 = 0$ cannot intersect the parabola $y = x^2$ in a rational point (whose abscissa & ordinate both are rational)
Reason (R) : For odd integer a, b and c equation $ax^2 + bx + c = 0$ cannot have rational roots.
- (A) Both **A** and **R** are correct and **R** is the correct explanation of **A**
 (B) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
 (C) **A** is correct but **R** is not correct
 (D) **A** is not correct but **R** is correct

- 189 The smallest possible natural number 'p' for which the equation $x^2 - px + 1258 = 0$ has integral roots ?

- 190 If p, q, r and s are non-zero numbers such that r, s are roots of $x^2 + px + q = 0$ and p, q are the roots of $x^2 + rx + s = 0$, then the value of $(4p + q + 2r + s)$ is
- (A) 0
 (B) 1
 (C) 2
 (D) 4

- 191 Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -7$ suppose that $\sqrt{2}i$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If $\alpha_1, \alpha_2, \alpha_3$ and α_4 are all the roots of the equation $f(x) = 0$, then $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ is equal to

- 192 The range of 'a' for which roots of $x^2 - 2x - a^2 + 1 = 0$ lie between the roots (exclusive) of the equation $x^2 - 2(a+1)x + a(a-1) = 0$, is

- (A) $\left(-\frac{1}{3}, 1\right)$ (B) $\left(-\frac{1}{3}, -\frac{1}{4}\right)$
 (C) $\left(-\frac{1}{4}, 1\right)$ (D) $\left(-\frac{1}{3}, \infty\right)$

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Space for Rough Work

- 193 If $x^2 - (2a+1)x + (a^2+a) < 0$ for all $x \in (-1, 2)$, then number of possible integral value(s) of a is
 (A) 0 (B) 2 (C) 3 (D) 4

- 194 Find smallest prime value of a, if roots of the equation $x^2 + ax + 1 = 0$ are distinct and both lie between 0 and 1.
 (A) 2
 (B) 3
 (C) 5
 (D) Does not exist any such value of a.

- 195 If roots of the equation $2x^3 + 4x^2 - 5x + 8 = 0$ are α, β, γ then
 (A) $\sqrt{(2-\alpha)(2-\beta)(\gamma-2)(1-\alpha^2)(1-\beta^2)(1-\gamma^2)} = 22.5$
 (B) $1+\alpha, 1+\beta, 1+\gamma$ are roots of $2x^3 - 2x^2 - 7x + 15 = 0$
 (C) $\sqrt{(2-\alpha)(2-\beta)(2-\gamma)(1-\alpha^2)(1-\beta^2)(1-\gamma^2)} = 22.5$
 (D) $1+\alpha, 1+\beta, 1+\gamma$ are roots of $2x^3 - 6x^2 - 7x - 15 = 0$

- 196 Consider $f(x) = ax^2 + bx + c \forall x \in \mathbb{R}$, a quadratic expression such that for $x \in [0, 2]$ range of $f(x)$ is $[0, 1]$ and $-\frac{b}{2a} \notin (0, 2)$

Then which of following are correct

- (A) The value of C may be 0 or 1
 (B) The sum of roots of $ax^2 + bx + c$ is $P^2 + P + 3$ for $P \in \mathbb{R}$, then $|a|$ can be 2
 (C) The value of $|2a + b|$ is $\frac{1}{2}$
 (D) If sum of roots of $ax^2 + bx + c$ is $P^2 + P + 3$ for $P \in \mathbb{R}$ then $|a| \leq \frac{2}{3}$

- 197 Given that, $A = 2^{\log_3 e^{\log \sqrt{e^3}}}$

$$B = \sqrt{17 + 12\sqrt{2}} - \sqrt{17 - 12\sqrt{2}};$$

$$\text{and } (\sqrt{2} - 1)^{-C} = \frac{4}{A + B}$$

On the basis of above information, answer the following questions :

If 2 lies between the roots of quadratic equation $Ax^2 + \sqrt{2}Bx + t = 0$, then $|t|$ can be

- (A) 32
 (B) 31
 (C) 33
 (D) 30
- 198 If the equation $x^4 + ax^3 + 2x^2 + bx + 1 = 0$; ($a, b \in \mathbb{R}$) has a real solution, then minimum value of $a^2 + b^2$ is

- 199 α, β are real and distinct roots of $x^2 + 4bx + c = 0$ where $b, c \in \mathbb{R}$.

List-I		List-II	
(P)	$f(x) = x^3 + 6bx^2 + 3cx + 2$; $b < 0, c > 0, f(\alpha)f(\beta) > 0$	(1)	$f(x) = 0$ has three real roots out of which two are positive real roots and one is negative real root
(Q)	$f(x) = cx^3 + 6bx^2 + 3x - 2$; $c < 0, f(1/\alpha)f(1/\beta) > 0$	(2)	$f(x) = 0$ has exactly one real root which is positive
(R)	$f(x) = x^3 + 6bx^2 + 3cx - 2$; $c < 0, f(\alpha)f(\beta) > 0$	(3)	$f(x) = 0$ has three real roots out of which two are negative real roots and one is positive real root
(S)	$f(x) = cx^3 + 6bx^2 + 3x - 2$; $c < 0, f(1/\alpha)f(1/\beta) < 0$	(4)	$f(x) = 0$ has exactly one real root which is negative
		(5)	$f(x) = 0$ has three positive real roots

- (A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 3$
 (B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
 (C) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
 (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

200 Let $P(x)$ be a quadratic polynomial with real coefficients satisfying $x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3$ for all real numbers x , and suppose $P(11) = 181$. Find $P(16)$.

201 Positive integers a , b and c are all integral power of positive integer k . If the equation $ax^2 - bx + c = 0$ has exactly one real solution r and this value r is less than 100, then maximum possible value of r is

- (A) 16 (B) 64
(C) 74 (D) 99

202 The curve $y = \ell x^2 + \ell x + \frac{1}{48}$ and $x = \ell y^2 + \ell y + \frac{1}{48}$ ($\ell \neq 0$) touch each other for

- (A) two rational values of ℓ
(B) two irrational values of ℓ
(C) four rational values of ℓ
(D) four irrational values of ℓ

203 If p and q are prime numbers and the equation $x^2 - px + q = 0$ has distinct integral roots then which of the following statements are true

- I : The difference of the roots is odd
II : Atleast one root is prime
III : $p^2 - q$ is prime
IV : $p + q$ is prime

- (A) I only (B) II only
(C) II and III only (D) All are true

204 remainder when $|w|$ divided by 7 is

Let $x^2 + bx + 6b = 0$ has only integer roots. If number of real values of 'b', is n then value of $\frac{n}{2}$ equals

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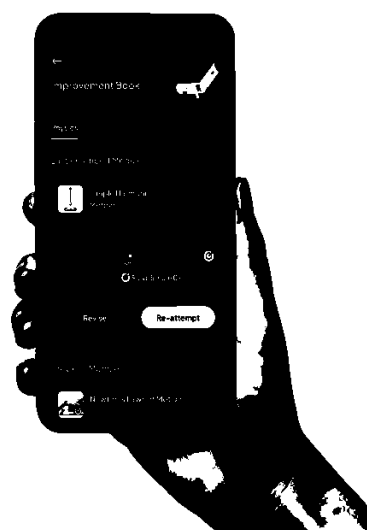
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205 Let $f(x) = x^2 - 3x - 7$. Let $g(x)$ and $h(x)$ be two quadratic polynomials also, both with the coefficient of x^2 equal to 1. Rohan computes each of the three sums $f(x) + g(x)$, $g(x) + h(x)$, $h(x) + f(x)$, and is surprised to find that each pair of these sums has a common root, and these three common roots are distinct. If $g(0) = 2$, then $h(0)$ is

- (A) $\frac{33}{52}$ (B) $\frac{19}{33}$ (C) $\frac{52}{19}$ (D) $\frac{71}{19}$

206 Let $f(x) = x^2 + (a+2)x + a^2 - a + 2$. Given a, α, β ($\alpha < \beta$) be real numbers and α, β are the roots of the equation $f(x) = 0$.

If both roots of the equation $f(x) = 0$ lies in interval $(0, \infty)$, then range of a , is :

- (A) $\left(\frac{2}{3}, 2\right)$ (B) $(-\infty, -2)$
(C) $(0, \infty)$ (D) ϕ

207 If α and β are roots of the equation $x^2 - 10x - 6 = 0$ and $P_n = \alpha^n - \beta^n$, then the value of $\frac{P_{16} - 5P_{14}}{2P_{15}}$ is equal to

208 If the equation $x^2 + 2|a|x + 4 = 0$ has integral roots, then minimum value of a is

- (A) 4 (B) $-5/2$
(C) 0 (D) -4

209 The sum of integral values of α for which the equation $(x + \alpha)(x + 1991) + 1 = 0$ has integral roots is/are

- (A) 1989
(B) 1997
(C) 3982
(D) none of these

210 If α and β are the roots of equation $x^2 - a(x + 1) - b = 0$, then $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + b} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + b}$ is/are

- (A) 1
(B) $\frac{a+1}{b} + \frac{b+1}{a}$
(C) $a + \frac{1}{a} + b + \frac{1}{b}$
(D) none of these

211 The integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots are

- (A) (1, 6)
(B) (8, 12)
(C) (4, 6)
(D) (3, 15)

212 If $a^2 + b^2 + c^2 + 6 = 2a + 2b + 4c$ and quadratic equations $ax^2 + bx + c = 0$ and $2px^2 + qx + 4r = 0$ have at least one root in common (where $p, q, r \in \mathbb{N}$), then the minimum value of $(2p + q + 3r)$ is

- (A) 10
(B) 7
(C) 8
(D) 6

213 If α, β are the roots of the equation $x^2 - 2x + 3 = 0$, then find the constant term of the equation whose roots are $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ & $\beta^3 - \beta^2 + \beta + 5$.

214 Find the integral value of a , for which the quadratic expression $ax^2 + (a - 2)x - 2$ is negative for exactly two integral values of x .

215 The set of values of ' a ' for which $ax^2 - (4 - 2a)x - 8 < 0$ for exactly three integral value of x is

- (A) $2 \leq a < 4$
- (B) $1 \leq a < 2$
- (C) $2 < a < 4$
- (D) $1 < a < 2$

216 The sum of all the real roots of the equation $(x^2 - 8x + 6)^{x^2 - 9} = 1$ is/are

217 Let equations $x^2 + 2x + a = 0$ and $ax^2 + x + 2 = 0$, $a \in \mathbb{R}$ have atleast one common root, then product of all the possible values of a is

- (A) 1
- (B) 3
- (C) 0
- (D) -3

218 If both roots of the equation $x^2 - (k + 1)x + k = 0$ lies between the roots of the equation $x^2 - 2kx + k^2 - 1 = 0$, then number of integral value(s) of k is/are

- (A) 0
- (B) 1
- (C) 2
- (D) infinite

219 Number of value(s) of ' a ' for which the real roots of the equation $x^2 + (a^2 - 4)x + a^2 - 3a + 1 = 0$ are equal in magnitude and opposite in sign is/are

- (A) 0
- (B) 1
- (C) 2
- (D) 4

220 Let S be the set of all non zero real number ' a ' such that the quadratic equation $ax^2 + (a - 1)x + (a - 2) = 0$ has two distinct real roots α and β satisfying the inequality $|\alpha - \beta| < 2$. Then which of the following interval(s) is/are a subset of S ?

- (A) $\left(1 + \frac{2}{\sqrt{3}}, \infty\right)$
- (B) $\left(1 - \frac{2}{\sqrt{3}}, 1\right)$
- (C) $\left(1, 1 + \frac{2}{\sqrt{3}}\right)$
- (D) $\left(1 - \frac{2}{\sqrt{3}}, -\frac{1}{7}\right)$

221 Let a, b, c, d are non zero real number such that a and $2b$ are roots of $x^2 - 9cx - 4d = 0$ and c and $2d$ are roots of $x^2 - 9ax - 4b = 0$. Choose the option(s) which can be true?

- (A) a and c are the roots of the equation $x^2 + 4x + 4 = 0$
- (B) a and c are the roots of the equation $x^2 - 20x + 4 = 0$
- (C) If a and c are distinct then $a^2 + c^2 = 392$
- (D) If a and c are equal then $b + d = -16$

- 222 If roots of quadratic polynomial $f(x) = a^2x^2 - 8ax + 7, a \in R - \{0\}$ lies on either side of unity, then the possible value(s) of a is/are:
- (A) 1
(B) 3
(C) 5
(D) 7

- 223 Let a, b, c be real numbers with $a > 0$ such that the quadratic polynomial $p(x) = ax^2 + bcx + b^3 + c^3 - 4abc$ has non-real zeroes. Then which of the following statement is/are correct?
- (A) Exactly one of the polynomial $T_1(x) = ax^2 + bx + c = 0$ and $T_2(x) = ax^2 + cx + b$ takes only positive values
(B) Both the polynomial $T_1(x) = ax^2 + bx + c$ and $T_2(x) = ax^2 + cx + b$ takes only positive values
(C) $(ax^2 + bx + c)(ax^2 + cx + b) = 0$ has four real solution
(D) Only one of the equation $ax^2 + bx + c = 0$ and $ax^2 + cx + b = 0$ has real solution

- 224 If the roots of $x^2 + x + a = 0$ exceed a , then
- (A) $2 < a < 3$
(B) $a > 3$
(C) $-3 < a < 3$
(D) $a < -2$

- 225 One of the roots of the given equation $\begin{vmatrix} b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$ is
- (A) $-(a+b)$
(B) $-(b+c)$
(C) $-a$
(D) $-(a+b+c)$

- 226 The value of ' a ' for which the equation $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0$ has more than two roots is $-k$, then the value of $k + 2020$ is

- 227 The sum of real roots of the equation of the equation $z^5 - 5z^4 + 9z^3 - 9z^2 + 5z - 1 = 0$, is
- (A) 5
(B) 4
(C) 3
(D) 1

- 228 Given that x_1, x_3 are roots of the equation $ax^2 - 4x + 1 = 0$ and x_2, x_4 are roots of the equation $bx^2 - 6x + 1 = 0$. If x_1, x_2, x_3, x_4 are in harmonic progression, then
- (A) $3a - b = 1$
(B) $a^2 + b^2 = 73$
(C) $2a < 3b$
(D) $\frac{1}{a} > \frac{1}{b}$

- 229 If a, b, c be the sides of $\triangle ABC$ and equations $ax^2 + bx + c = 0$ and $5x^2 + 12x + 13 = 0$ have a common root, then $\angle C$ is

- 230 If a, b, c are real numbers such that $\frac{2a+3c}{b+c} + 3 = 0$, then the quadratic equation $ax^2 + bx + c = 0$ has
- (A) atleast one root in $(-1, 0)$
 (B) atleast one root in $(0, 1)$
 (C) no root in $(-1, 1)$
 (D) none of these
-
- 231 The complete set of real values of ' a ' $\in \mathbb{R}^+$ for which both roots of the equation $x^2 + 2ax + 9 = 0$ lies in $(-6, 1)$ is
- (A) $\left[3, \frac{15}{4}\right)$ (B) $\left(\frac{39}{4}, 12\right)$
 (C) $[9, 12)$ (D) $(-2, 0) \cup (2, 9)$
-
- 232 Let α, β be roots of the equation $x^2 - 3x - \sqrt{7} = 0$ and $\alpha^2 + \frac{1}{\beta}, \beta^2 + \frac{1}{\alpha}$ be the roots of the equation $x^2 + px + q = 0$, then for the equation $x^2 + p^2q^2x - (11q + p) = 0$, roots are
- (A) positive (B) negative
 (C) one positive and other negative (D) none of these
-
- 233 Let a, b be the roots of the quadratic equation $x^2 - (k-2)cx - (k-1)d = 0$ and c, d be the roots of the quadratic equation $x^2 - (k-2)ax - (k-1)b = 0$. If a, b, c and d are distinct non-zero real numbers such that $a + b + c + d = 100$, then the integral value of k is _____.
-
- 234 A monic quadratic trinomial $P(x)$ is such that $P(x) = 0$ and $P(P(P(x))) = 0$ have a common root, then
- (A) $P(0) \cdot P(1) > 0$ (B) $P(0)P(1) < 0$
 (C) $P(0)P(1) = 0$ (D) Data insufficient
-
- 235 Let $a, b \in \mathbb{N}, a \neq b$ and the quadratic equations $(a-1)x^2 - (a^2+2)x + a^2+2a = 0$, $(b-1)x^2 - (b^2+2)x + b^2+2b = 0$ have a common root, then
- (A) $a = 3$ (B) $a = 2$
 (C) $b = 2$ (D) $a = 4$
-
- 236 Let $P(x) = ax^2 + bx + c$ where a, b, c are real numbers and $a \neq 0$, the polynomial $P(x^2 + 4x - 7)$ has $x = 1$ as a root and at least one repeated roots, then the roots of polynomial $P(x^2 + 4x - 7)$ can be
- (A) $x = 1$ (B) $x = -5$
 (C) $x = -2$ (D) $x = 2$
-
- 237 If lines $\frac{x}{2} = \frac{y}{x_1} = \frac{z}{x_2}$ and $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$ lies in the same plane then for equation $x_1t^2 + (x_2+2)t + a = 0$
- (A) $2x_1 - x_2 = 1$ (B) sum of roots of above equation $= -2$
 (C) $2x_1 + x_2 = -4$ (D) sum of root is 0
-
- 238 A quadratic equation with roots α and β where $\alpha\beta = 4$, satisfies the equation $\frac{\alpha}{\alpha-1} + \frac{\beta}{\beta-1} = \frac{a^2-7}{a^2-4}$. Number of integral values of a for which $\alpha, \beta \in (1, 4)$ is equal to

- 239 Let $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$ be two quadratic equations having roots α, β and γ, δ respectively. If all the coefficients are real and $\frac{\alpha^{2024} + \beta^{2024}}{\alpha^{-2024} + \beta^{-2024}} + \frac{\gamma^{2024} + \delta^{2024}}{\gamma^{-2024} + \delta^{-2024}} = 0$, then which of the following must be true ?
- (A) $a + a' = 0$ (B) $b + b' = 0$
(C) $c + c' = 0$ (D) none of these

- 240 Let α, β roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, ($n \in \mathbb{N}$) then the value of $(aS_{2023} + bS_{2022} + c \cdot S_{2021})$ is equal to
- (A) abc (B) $a + b + c$
(C) c (D) 0

- 241 If α, β be two roots of the equation $x^2 + (24)^{1/4}x + 6^{1/2} = 0$, then $\alpha^8 + \beta^8$ is equal to
- (A) 12 (B) 144
(C) 72 (D) 228

2022

- 242 If α and β be the roots of the equation $x^2 + kx - \frac{1}{2k^2} = 0$, where $k \in \mathbb{R}$. Then the maximum value of $\alpha^4 + \beta^4$ is
- (A) $4\sqrt{2}$ (B) $2 - 2\sqrt{2}$
(C) $2 + \sqrt{2}$ (D) 4

- 243 If α, β be the roots of $x^2 - 2x - a + 1 = 0$ and γ, δ the roots of $x^2 - 2(a + 1)x + a(a - 1) = 0$ such that α and β lie in the (γ, δ) , then
- (A) $a \in \left(-\infty, \frac{1}{4}\right) \cup (1, \infty)$
(B) $a \in \left(-\frac{1}{4}, 1\right)$
(C) a lies in $(-1, 1)$
(D) $a \in (1, \infty)$

- 244 If α, β, γ are roots of equation $x^3 + 2x^2 + 3x + 3 = 0$. Then the value of

$\left(\frac{\alpha}{\alpha+1}\right)^2 + \left(\frac{\beta}{\beta+1}\right)^2 + \left(\frac{\gamma}{\gamma+1}\right)^2$ value of on 1(D) 0

$\left(\frac{\alpha}{\alpha+1}\right)^3 + \left(\frac{\beta}{\beta+1}\right)^3 + \left(\frac{\gamma}{\gamma+1}\right)^3$ value of on 1(D) 0 to is equal to

- (A) 0
(B) 3
(C) 11
(D) 44

- 245 If $\alpha, \beta, \gamma, \sigma$ are the roots of the equation $x^4 + 4x^3 - 6x^2 + 7x - 9 = 0$, then the value of $(1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)(1 + \sigma^2)$ is
- (A) 9
(B) 11
(C) 13
(D) 5

- 246 If α and β are the roots of the equation $x^2 + px + q = 0$ and also $x^{3900} + p^{1950}x^{1950} + q^{1950} = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $x^n + 1 + (x+1)^n = 0$, then the value of n must be equal to:

- 247 The sum of all integral values of a for which one root of equation $(a-5)x^2 - 2ax + a - 4 = 0$ is smaller than 1 and the other greater than 2, must be equal to_____.

- 248 If α, β are the roots of the equation $2x^2 + 4x - 5 = 0$, the equation whose roots are the reciprocals of $2\alpha - 3$, and $2\beta - 3$ is

(A) $11x^2 + 10x + 1 = 0$
 (B) $x^2 + 10x + 11 = 0$
 (C) $x^2 + 10x - 11 = 0$
 (D) $11x^2 - 10x + 1 = 0$

- 249 If the equation $ax^2 + 2bx + 4c = 16$ has no real roots and $a + c > b + 4$, then integral value of c can be equal to

(A) 2
 (B) 6
 (C) 12
 (D) 20

- 250 If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the value(s) of n for which

$\left(\frac{\alpha}{\beta}\right)^n = 1$ is:
 (A) 4
 (B) 6
 (C) 8
 (D) 10

- 251 If atleast
 c are sid

$(k > 12),$

(A) 1
 (C) $\sqrt{3}$

- 252 If the c
 values c

(A) $(0, 1)$
 (C) $(0, c)$

- 253 If p_1, p_2

quadrat
 terms, t

(A) -1
 (C) $\frac{1}{2}$

- 254 If a, b, c

(A) ax^2
 (B) $(a +$
 (C) abc^2
 (D) $(a +$

- 255 If both r

(A) $3a >$
 (C) $c <$

256 The value of the common

- (A) $\frac{17}{8}$
(C) $\frac{15}{8}$

257 If n is even

- $x^{2n} + p^n x$
(A) -1
(C) 1

258 Sum of all the roots of the equation $x^2 - 2x + |x - 1| - 5 = 0$ is

- (A) 0 (B) 2
(C) 1 (D) 5

259 If the equations $px^2 + qx + r = 0$ and $rx^2 + qx + p = 0$ ($p \neq r \neq 0$) have a negative common root, then the value of $(p - q + r)$ is equal to

- (A) -1 (B) 1
(C) 2 (D) 0

260 If $\frac{a^{\frac{n-1}{2}} + b^{\frac{n-1}{2}}}{a^{\frac{n+1}{2}} + b^{\frac{n+1}{2}}}$ is geometric mean of $\frac{1}{a}, \frac{1}{b}$ ($a \neq b$), then the quadratic equation $bx^2 + nx + b = 0$ has

• (A) rational and equal roots
(B) rational and distinct roots
(C) irrational roots
(D) imaginary roots

261 If $\alpha + \frac{1}{\alpha}$ & $2 - \beta - \frac{1}{\beta}$ ($\alpha, \beta > 0$) are the roots of the quadratic equation $x^2 - 2(a+1)x + a - 3 = 0$, then the sum of integral value of 'a'.

262 The number of irrational roots of the equation $(x^2 - 3x + 1)(x^2 + 3x + 2)(x^2 - 9x + 20) = -30$ is

• (A) 6
(B) 4
(C) 2
(D) 0

263 If all the equations $x^2 + (2a + 3b)x + 60 = 0$, $x^2 + ax + 10 = 0$ and $x^2 + bx + 8 = 0$ where $a, b \in \mathbb{R}$, have a common root, then value of $|a - b|$ is

• (A) 0
(B) 1
(C) 2
(D) none of these

264 If $ax^2 + bx + c = 0$ has no real roots and $a, b, c \in \mathbb{R}$ and $a \neq 0$ and also $a + c > 0$, then

• (A) $a - b + c < 0$
(B) $a - b + c > 0$
(C) $a + c = b$
(D) $b > 0$

- 265 If α and β are the roots of the equation $ax^2 + bx + c = 0$ such that $\beta < \alpha < 0$, then the quadratic equation whose roots are $|\alpha|, |\beta|$ is given by :

(A) $|a|x^2 + |b|x + |c| = 0$ (B) $ax^2 - b|x| + c = 0$
 (C) $|a|x^2 - |b|x + |c| = 0$ (D) $a|x|^2 + b|x| + c = 0$

- 266 If $bx^2 + ax + c = 0$, $ax^2 + bx + c = 0$ and $ax^2 + cx + b = 0$, each equation has equal roots,

then $\begin{vmatrix} 3 & a & bc \\ 3 & b & ac \\ 3 & c & ab \end{vmatrix}$ is equal to

(A) $3(a-b)(b-c)(c-a)$ (B) $\frac{3}{4}(a-b)(b-c)(c-a)$
 (C) $3abc$ (D) abc

- 267 Given that the quadratic equation $ax^2 + bx + c = 0$ has no real roots, but Adam got two roots 2 and 4 since he wrote down a wrong value of a . Ben also got two roots -1 and 4 because he wrote down the sign of a term wrongly, then the value of $\frac{2b+3c}{a}$

(A) 3 (B) 4
 (C) 5 (D) 6

- 268 If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to sum of the square of their reciprocal, then, $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in

(A) AP (B) GP
 (C) HP (D) none of these

- 269 If a, b, c ,

(A) 6 real roots
 (C) 4 real roots

- 270 If the equation $x^2 + \lambda x + \mu = 0$ has equal roots and one root of the equation $x^2 + \lambda x - 12 = 0$ is 2,

then $(\lambda, \mu) =$
 (A) $(4, 4)$ (B) $(-4, 4)$
 (C) $(4, -4)$ (D) $(-4, -4)$

- 271 If the quadratic equation has integer roots,

(A) greatest value of a
 (C) least value of a

- 272 If each equation has equal roots,

(A) a, b, c are in AP
 (C) a, b, c are in GP

- 273 If $f(x) =$

(A) odd function
 (B) At least one root
 (C) Greater than 1
 (D) All roots are real

- 274 Consider the quadratic equation $(a + c - b)x^2 + (a + c - b)x + (a + c - b) = 0$. If both the roots are real numbers and $a + c - b \neq 0$. If both the roots must be rational are

(A) a, b, c

(C) $\frac{b+c-a}{a+c-b}$

- 275 If the roots of the equation $x^2 + mx + n = 0$ are cubes of the roots of the equation $x^2 + px + q = 0$, then

(A) $m = p^3 + 3pq$

(B) $m = p^3 - 3pq$

(C) $m + n = p^3$

(D) $\frac{m}{n} = \left(\frac{p}{q}\right)^3$

- 276 If α, β are the roots of the equation $x^2 + px + q = 0$ and also of the equation $x^{2n} + p^n x^n + q^n = 0$ and

- if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of the equation $x^n + 1 + (x + 1)^n = 0$ then n must be an

(A) even integer

(B) odd integer

(C) irrational number

(D) none of these

277 If the qu

- (A) both
- (B) both
- (C) both

(D) exac

of a variable triangle are $(3, 4)$, $(5 \cos \theta, 5 \sin \theta)$ and $(5 \sin \theta, 5 \cos \theta)$ then locus of its
itre is

- $(y - 1)^2 + (x - y - 7)^2 = 100$ (B) $(x + y - 7)^2 + (x - y - 1)^2 = 100$
- $y - 7)^2 + (x + y - 1)^2 = 100$ (D) $(x + y - 7)^2 + (x - y + 1)^2 = 100$

ability that at least one of the events A and B occur simultaneously with probability $3/5$.

B occur simultaneously with probability $1/5$ then $P(\bar{A}) + P(\bar{B})$ is

- (B) $\frac{4}{5}$
- (D) $\frac{7}{5}$

are three complex numbers selected randomly (without repetition) from the set

...2013) then the probability that $\arg\left(\frac{a-b}{a-c}\right) = 0$ is

- (B) $\frac{2}{3}$

$\frac{1}{3}$

(D) none of these

the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$ then c is equal to

- (B) -1
- (D) 3

adrate equation $4x^2 - 2(a + c - 1)x + ac - b = 0$ ($a > b > c$)

roots are greater than a

roots are less than c

roots be between $\left(\frac{c}{2}\right)$ and $\left(\frac{a}{2}\right)$

atly one of the roots has between $\left(\frac{c}{2}\right)$ and $\left(\frac{a}{2}\right)$

278 The value of a for which $2x^2 - 2(2a + 1)x + a(a + 1) = 0$ may have one root less than a and other
root greater than 'a' are given by

- (A) $1 > a > 0$ (B) $-1 < a < 0$
- (C) $a \geq 0$ (D) $a > 0$ or $a < -1$

279

If the equation $ax^2 + 25x - 3c = 0$ has non real roots and $\frac{3c}{4} < (a + b)$, then c is always

- (A) negative (B) 0
- (C) non-negative (D) zero

Space for rough work

280 If α and β are the distinct real roots of $x^2 + ax + b = 0$ and α^4 and β^4 are the roots of
 $x^2 - px + q = 0$ then the roots of $x^2 - 4bx + 2b^2 - p = 0$ are always

- (A) both non-real (B) both positive
- (C) both negative (D) one positive and other negative

- 281 Let a, b,
 . equation
 (A) 2
 (C) -1

- 282 The value of m for which one of the roots of $x^2 - 3x + 2m = 0$ is double of one of the roots of
 . $x^2 - x + m = 0$ is
 (A) -2 (B) 1
 (C) 2 (D) none of these

- 283 If the roc
 . then the
 (A) 1
 (C) 3

- 284 If α and
 . is
 (A) 2
 (C) 0

- 285 A quadr
 . the coeff

- 286 If $ax^2 + b$
 . (A) $4a -$
 (C) $13a$

- 287 If both roots of the quadratic $ax^2 + bx + c = 0$ lie in $(0, 2)$ then $25ac + 20bc + 16c^2$ is always
 . (A) positive (B) negative
 (C) equal to zero (D) cannot say with given information

- 288 If α and β are the roots of the quadratic equation, $x^2 + x \sin \theta - 2 \sin \theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$ then

. $\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12})(\alpha - \beta)^{24}}$ is equal to:

- (A) $\frac{2^{12}}{(\sin \theta + 8)^{12}}$ (B) $\frac{2^{12}}{(\sin \theta - 4)^{12}}$
 (C) $\frac{2^{12}}{(\sin \theta - 8)^6}$ (D) $\frac{2^6}{(\sin \theta + 8)^{12}}$

- 289 The value of 'a' for which the equation $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0$ has more than
 . two roots is $-k$, then the value of $k+2020$? is

Ans. 02021.00

Sol. $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0 \Rightarrow (a + 1)(a + 3)x^2 + (a + 1)(a - 2)x + (a + 1)a = 0$
 $(a + 1)[(a + 3)x^2 + (a - 2)x + a] = 0$
 if $a = -1$, then quadratic have more than two roots.

- 290 If the quadratic equation $(2m + 1)x^2 + (2n + 1)x + (2p + 1) = 0$, where $m, n, p \in \mathbb{I}$, has real roots then
 . both roots are
 (A) Rational (B) Irrational
 (C) Positive (D) Can't say

- 291 Let α, β be roots of the equation $x^2 - 3x - \sqrt{7} = 0$ and $\alpha^2 + \frac{1}{\beta}, \beta^2 + \frac{1}{\alpha}$ be the roots of the equation $x^2 + px + q = 0$, then for the equation $x^2 + p^2q^2x - (11q + p) = 0$, roots are
 (A) positive (B) negative
 (C) one positive and other negative (D) none of these
- 292 If the roots of $\ell x^2 + 2mx + n = 0$ are real & distinct, then the roots of $(\ell + n)(\ell x^2 + 2mx + n) = 2(\ell n - m^2)(x^2 + 1)$ will be -
 (A) real & equal (B) real & different
 (C) imaginary (D) may be real or imaginary
- 293 If α and β (where $\alpha, \beta \in \mathbb{R}$) are the roots of equation $x^2 + 4x + b = 0$ and $|\alpha - \beta| \leq 2$, then the complete range of b is-
 (A) (3,4) (B) (4,5) (C) [3,4] (D) (3,5)
- 294 Let $f(x) = \frac{x^2 - 5x + 6}{x^2 - 3x + 2}$, then range of $f(x)$ is -
 (A) $(-\infty, 1) \cup (1, \infty)$ (B) $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$
 (C) $(-\infty, \infty)$ (D) $(-\infty, -1) \cup (-1, \infty)$
- 295 The smallest positive number 'k' such that there exists $\alpha, \beta \in \mathbb{R}$ satisfying $\alpha + \beta = k$ and $\alpha\beta = k$, is
 (A) 4 (B) 1 (C) 2 (D) 8
- 296 Quadratic equation in x , $(k^2 - 2k + 2)x^2 - 2(\sqrt{k^4 + 4})x + (k^2 + 2k + 2) = 0 \forall k \in \mathbb{R}$ has
 (A) real & distinct roots (B) equal roots
 (C) imaginary roots (D) roots of opposite sign
- 297 If equation $x^2 - px + p = 0$ has positive integral roots α, β , then value of $\alpha^{2016} - \beta^{2016}$ is (where $p \in \mathbb{R}$)
 (A) 4 (B) 2016 (C) $p^2 - 4$ (D) 0
- 298 If $a + b + c > \frac{9c}{4}$ and quadratic equation $ax^2 + 2bx - 5c = 0$ has non-real roots, then -
 (A) $a > 0, c > 0$ (B) $a > 0, c < 0$ (C) $a < 0, c < 0$ (D) $a < 0, c > 0$
- 299 Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then value of $P(9)$ is -
 (A) 98 (B) 899 (C) 80 (D) 898
- 300 If α and β are roots of the equation $ax^2 + bx + c = 0$ then roots of the equation $a(2x + 1)^2 - b(2x + 1)(3 - x) + c(3 - x)^2 = 0$ are -
 (A) $\frac{2\alpha + 1}{\alpha - 3}, \frac{2\beta + 1}{\beta - 3}$ (B) $\frac{3\alpha + 1}{\alpha - 2}, \frac{3\beta + 1}{\beta - 2}$ (C) $\frac{2\alpha - 1}{\alpha - 2}, \frac{2\beta - 1}{\beta - 2}$ (D) none of these
- 301 The set of values of 'a' for which the inequality $x^2 - (a + 2)x - (a + 3) < 0$ is satisfied by atleast one positive real x , is -
 (A) $[-3, \infty)$ (B) $(-3, \infty)$ (C) $(-\infty, -3)$ (D) $(-\infty, 3]$

- 302 If one root of the equations $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ is common, then the value of $\left(\frac{a^3 + b^3 + c^3}{abc}\right)^2$ is -
 (A) 1 (B) 3 (C) 9 (D) 16
- 303 Set of the values of parameter 'm' for which every solution of the inequality $x^2 - 3x + 2 \leq 0$ is also a solution of the inequality $2x^2 - (m + 1)x - (2m + 3) < 0$, is -
 (A) $\left(\frac{3}{4}, \infty\right)$ (B) $\left(-\frac{2}{3}, \infty\right)$ (C) $\left(-\frac{2}{3}, \frac{3}{4}\right)$ (D) $(-\infty, \infty)$
- 304 Number of rational roots of the equation $x^3 - 3x + 1 = 0$
 (A) 0 (B) 1 (C) 2 (D) 3
- 305 The integral values of k for which the roots of $kx(x+2) + k - (x+2) = 0$ are rational is given by -
 (A) 5 (B) -3 (C) -6 (D) 12
- 306 Number of values of $x \in \mathbb{R}$ satisfying $\left|x - \frac{1}{x}\right| + \left|x + \frac{1}{x}\right| = 2$ are -
 (A) 1 (B) 2 (C) 3 (D) 4
- 307 If one of the roots of $ax^2 + bx + c = 0$ is greater than 1 and the other is less than -3 and if the roots of $cx^2 + bx + a = 0$ are α & β , then -
 (A) $\alpha < 1$ & $\beta > -\frac{1}{3}$ (B) $\alpha < 1$ & $\beta < -\frac{1}{3}$ (C) $\alpha > 1$ & $\beta > 1$ (D) $\alpha > 1$ & $\beta < -\frac{1}{3}$
- 308 If $px^2 - qx + r = 0$ has imaginary roots and $q < \frac{4p+r}{2}$ then set of real values of x satisfying the equation $\left|p\left(x^2 + \frac{6}{p}\right) - x(q+5) + r\left(1 + \frac{x^2}{r}\right)\right| = \|px^2 - qx + r\| - \|x^2 - 5x + 6\|$, is -
 (A) $(-\infty, 2] \cup [3, \infty)$ (B) $[2, 3]$ (C) $\mathbb{R}$ (D) no value of x
- 309 The set of values of 'a' for which $f(x) = ax^2 + 2x(1-a) - 4$ is negative for exactly three integral values of x, is -
 (A) $(0, 2)$ (B) $(0, 1]$ (C) $[1, 2)$ (D) $[2, \infty)$
- 310 The number of possible value of 'α' for which the expression $y = \frac{\alpha x^2 + 7x - 2}{\alpha + 7x - 2x^2}$ has atleast one common linear factor in numerator and denominator, is -
 (A) 0 (B) 1 (C) 2 (D) 3
- 311 If α, β are real numbers such that $\beta > \alpha$ & $|\alpha - \beta| < 2$, then the equation $(x - \alpha)(x - \beta) + 1 = 0$ has -
 (A) both roots in (α, β) (B) both roots in $(-\infty, \alpha)$ (C) both roots in (β, ∞) (D) no real roots
- 312 If α, β, γ are roots of equation $111x^3 - 11x - 1 = 0$, then $(\alpha\beta)^{-2} + (\beta\gamma)^{-2} + (\gamma\alpha)^{-2}$ equals
 (A) 2332 (B) 1331 (C) 1210 (D) 2442
- 313 If $4(a-1)x^2 + 4(a-1)x + (119 + 5a - 4a^2) > 0 \forall x \in \mathbb{R}$, then number of integral values of 'a' is -
 (A) 4 (B) 5 (C) 9 (D) 10

- 314 Let $f(x)$ is polynomial of two degree such that $f(1) = 6$, $f(-1) = 30$ and $f(3) > 0$. If one root is twice of the other root of the equation $f(x) = 0$, then the value of $f(3)$ is -
 (A) 54 (B) 78 (C) 96 (D) 110
- 315 If $\frac{9 - |2x + 3|}{2 - |x|} < 0$, then number of integral values of x is -
 (A) 3 (B) 4 (C) 5 (D) more than 5
- 316 If $a_n = \sin^n \theta + \cos^n \theta \forall n \in \mathbb{N}$ and $4a_{n+2} - 2\sqrt{6}a_{n+1} + a_n = 0 \forall n \in \mathbb{N}$, then value of $(\sin \theta + \cos \theta)$ is -
 (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{\sqrt{2}}$
- 317 The number of integers 'x' satisfying the equation $(x^2 - x - 1)^{x+2} = 1$ is -
 (A) 2 (B) 3 (C) 4 (D) 5
- 318 Complete set of values of 'm' for which $\operatorname{cosec}^4 x + 2m \operatorname{cosec}^2 x + m^2 - 3 > 0 \forall x \in \mathbb{R} - \{n\pi\}$, is -
 (A) $\mathbb{R}$ (B) $(\sqrt{3} - 1, \infty)$ (C) $(-\infty, -\sqrt{3} - 1)$ (D) $(-1, \infty)$
- 319 If sum of the roots of quadratic equation $ax^2 + bx + c = 0$ is equal to sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to -
 (A) 2 (B) -2 (C) 1 (D) -1
- 320 If $\alpha = \log_3 4$ is one of the roots of equation $x^3 + ax^2 + bx + 1 = 0$, $a, b \in \mathbb{R}$, then equation which always have a root $\beta = \log_2 48$, is -
 (A) $x^3 + 2(b - 6)x^2 + 4(a - 4b + 12)x + 8(4b - 2a - 7) = 0$
 (B) $x^3 + bx^2 + ax + 1 = 0$
 (C) $(65 + 16a + 4b)x^3 + 2(48 + 8a + b)x^2 + 4(12 + a)x + 8 = 0$
 (D) $(65 + 16a + 4b)x^3 + (48 + 8a + b)x^2 + (12 + a)x + 8 = 0$
- 321 If $3a + 4b = 5$, where $a, b \in \mathbb{R}$, then minimum value of $a^2 + b^2$ is -
 (A) 1 (B) 4 (C) 5 (D) 10
- 322 If the equation $x^3 + 2x^2 - 4x + 5 = 0$ has roots α, β, γ , then the value of $\frac{(\alpha^3 + 5)(\beta^3 + 5)(\gamma^3 + 5)}{8\alpha\beta\gamma}$ is
 (A) 104 (B) 8 (C) 13 (D) -104
- 323 If $p, q, r, s \in \mathbb{R}$, then equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ must has -
 (A) 6 real roots (B) atleast two real roots
 (C) 2 real & 4 imaginary roots (D) 4 real & 2 imaginary roots
- 324 If $a^2 \sin^2 x - 2 \sin x - (2a + 1) = 0$ ($a > 0$) has exactly one root in $\left[0, \frac{\pi}{2}\right]$, then range of a is
 (A) $[2, \infty)$ (B) $[1, \infty)$ (C) $[3, \infty)$ (D) none of these
- 325 The number of solutions of equation $|2ax - 3| + |ax + 1| + |5 - ax| = \frac{1}{2}$ is/are -
 (A) 1 (B) 0 (C) 2 (D) 3

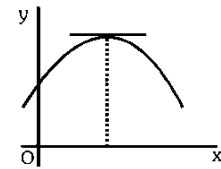
- 326 If α, β are roots of $x^2 + px + q = 0$, $p \neq 0$ and also of $x^{2n} + p^n x^n + q^n = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are roots of $x^n + 1 + (x + 1)^n = 0$, then $n \in \mathbb{Z}$ is
 (A) an odd prime (B) odd integer (C) even integer (D) none of these
- 327 Roots of the equation $2(x - a)(x - b) - (x - a^2 - b^2 - 1)^2 = 0$ are (where $a, b \in \mathbb{R}$)
 (A) real & equal (B) real & different (C) imaginary (D) imaginary and equal
- 328 If α, β, γ are the roots of the equation $ax^3 + bx^2 + cx + d = 0$, then
 $a\left(\frac{\alpha^3}{\beta} + \frac{\beta^3}{\gamma} + \frac{\gamma^3}{\alpha}\right) + b\left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha}\right) + c\left(\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha}\right)$ is equal to-
 (A) a (B) b (C) c (D) d
- 329 If all roots of $x^4 - 8x^3 + ax^2 - bx + 16 = 0$ are positive real numbers, then $(a + b)$ is equal to -
 (A) 32 (B) 24 (C) 56 (D) 8
- 330 Given α, β be the roots of quadratic equation, $x^2 - 2ax + 7 = 0$ and γ, δ be the roots of the equation, $x^2 - 2bx + 1 = 0$. If $\alpha\delta = \beta\gamma$, then $\frac{a^2}{b^2}$ is equal to -
 (A) 5 (B) 6 (C) 7 (D) 8
- 331 Let α, β be roots of $x^2 - x + b = 0$ and $S_k = \alpha^k + \beta^k$, then the value(s) of b such that S_2, S_3 and S_4 are in arithmetic progression is -
 (A) 0 (B) 1 (C) $\frac{1}{5}$ (D) no real value(s) of b exists
- 332 Let α, β, γ be real numbers which satisfy the relation $\alpha + \frac{1}{\beta\gamma} = \frac{1}{2}, \beta + \frac{1}{\alpha\gamma} = -\frac{1}{10}, \gamma + \frac{1}{\alpha\beta} = \frac{1}{18}$, then the value $\frac{\beta - \alpha}{\alpha - \gamma}$ is equal to-
 (A) $-\frac{11}{40}$ (B) $\frac{11}{40}$ (C) $-\frac{27}{20}$ (D) none of these
- 333 There is only one real value of λ for which the quadratic equation $\lambda x^2 - 4x + 9 = 0$ has two positive integral solutions. The sum of these two solutions is-
 (A) 4 (B) 8 (C) 9 (D) 12
- 334 Let α, β, γ and δ be the roots (real or non real) of equations $x^4 - 3x + 1 = 0$. The value of $\alpha^3 + \beta^3 + \gamma^3 + \delta^3$ is
 (A) 3 (B) 12 (C) 9 (D) -9

Multiple Correct Answers Type

- 335 For the equation $\log_{x^2+6x+8}(\log_{2x^2+2x+3}(x^2 - 2x)) = 0$ -
 (A) number of solutions is two (B) sum of solutions is '-4'
 (C) number of solution is one (D) sum of solutions is '-1'
- 336 If equations $(a + 2)x^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root where $a, b, c \in \mathbb{N}$ then -
 (A) $b^2 - 4ac < 0$ (B) minimum value of $a + b + c$ is 16
 (C) $b^2 < 4ac + 8c$ (D) minimum value of $a + b + c = 7$

337 Graph of $f(x) = px^2 - qx - r$ is shown in the adjacent figure then -

- (A) $r(p - q - r) < 0$ (B) $q < 0$
 (C) $\sqrt{q^2 + 4pr} > |q|$ (D) $p < 0$



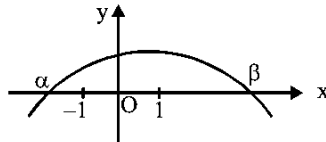
338 If α , β and γ are the roots of the equation $x^3 - 3x + 1 = 0$ then -

- (A) $\prod (\alpha + 1) = -3$ (B) $\sum (\alpha + 1) = 0$
 (C) $\sum (\alpha + 1)(\beta + 1) = -3$ (D) $\sum \alpha^2 = 6$

339 The values of k for which the system of equations $2x + ky = k$, $3x - y = 10$ has solution(s) satisfying $x > 0$, $y > 0$ are given by-

- (A) $k < 0$ (B) $k < -\frac{2}{3}$ (C) $k > 0$ (D) $k > \frac{20}{3}$

340 The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below.



Which of the following are correct ?

- (A) $\frac{c}{a} < -1$ (B) $|\beta - \alpha| > 2$
 (C) $f(x) > 0 \forall x \in (0, \beta)$ (D) $abc < 0$

341 If $x^2 + y^2 = 4$ and $E = x^2 + 2xy - y^2$, then-

- (A) Maximum value of E is $4\sqrt{2}$ (B) Minimum value of E is $-4\sqrt{2}$
 (C) $x \in [-2, 2]$ (D) $y \in [-2, 2]$

342 If the cubic polynomials $x^3 + ax^2 + 11x + 6$ and $x^3 + bx^2 + 14x + 8$ may have a common factor of the form $x^2 + px + q$, then -

- (A) $a + p = b + q$ (B) $ap < bq$ (C) pq divides ab (D) $p + q$ divides $a + b$

343 If the quadratic equation $x^2 - \frac{(\alpha^2 + 11)x}{9} + \frac{15}{4}(\alpha + \beta) + 16 = 0$ has two integral roots α & β , then -

- (A) $\alpha + \beta = 20$ (B) $|\alpha - \beta| = 120$ (C) $|\alpha - \beta| = 6$ (D) $\alpha + \beta = 128$

344 If $ax^2 + bx + c$ ($a > 1$ & a, b, c are integers) is equal to p for two distinct integral values of x , $p \in \text{prime}$, then $ax^2 + bx + c$ cannot be equal to $2p$ for -

- (A) $x = 0$ (B) $x = -1$ (C) $x = 1$ (D) $x = 3$

345 If $2y^3 + \ell y^2 + my + 4 = 0$ such that $\ell + 15 < 2m$ and $2\ell + m + 10 < 0$, then atleast one root of the equation must lie in -

- (A) $(-1, 0)$ (B) $(-2, -1)$ (C) $(0, 2)$ (D) $(1, 2)$

346 If a, b, c are non zero real numbers and $(ax^2 + bx + c)(ax^2 - bx - c) = 0$, then which of the following can be true -

- (A) equation has four real roots
- (B) equation has two real and two imaginary roots
- (C) equation has three real and one imaginary roots
- (D) equation has four imaginary roots

347 If equation $(x - a)(x - 3) = -1$ has integral roots, then

- (A) number of possible values of 'a' is 2.
- (B) number of possible values of 'a' is 4.
- (C) sum of all possible values of 'a' is 6.
- (D) product of all possible values of 'a' is 5.

348 Let α, δ are roots of equation $x^2 + k_1x - 5 = 0$ and β, γ are roots of equation $x^2 + k_2x + k_3 = 0$. If α, β, γ & δ are in A.P., then which of the following is/are correct -

- (A) $k_1 = k_2$
- (B) If $k_3 = 3$, then $|\alpha + \beta + \gamma + \delta| = 8$
- (C) $\alpha + \beta + \gamma + \delta = 2k_1$
- (D) $\alpha + \beta + \gamma + \delta = 2k_3$

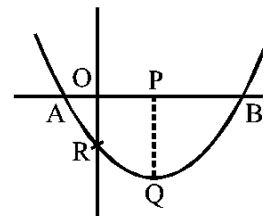
349 If equations $bx^2 + acx + b^2c = 0$ and $cx^2 + abx + b^2c = 0$ have a common root, (where a, b, c are non-zero distinct real numbers), then which of the following is/are correct -

- (A) $\frac{a}{c} + \frac{b}{a} + \frac{a}{b} = 0$
- (B) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 0$
- (C) $\frac{1}{bc^2} + \frac{1}{a^2c} + \frac{1}{cb^2} = 0$
- (D) $\frac{1}{ac^2} + \frac{1}{ba^2} + \frac{1}{cb^2} = 0$

350 Graph of $y = ax^2 + bx + c$ is shown in the figure and length of PQ & OR are 9 units & 5 units respectively (where Q is the vertex of the parabola)

and area of $\triangle OBQ$ is $\frac{45}{4}$ (unit)², then

- (A) length of AB is 3
- (B) length of AB is 4
- (C) $\left|\frac{b}{a}\right| > 1$
- (D) $\left|\frac{b}{a}\right| < 1$



351 If equation $x^2 + (2a + 4)x + (6a^2 + 4ab + b^2 + 8) = 0$ has real roots, then

- (A) $a^b \cdot b^a = 1$
- (B) product of roots is -4
- (C) product of roots is 16
- (D) $(\sin a)^a + \left(\sec\left(\frac{b}{2}\right)\right)^{b/2} = 1$

352 For the equation $(x)^4 + (x + 2)^4 = 16(x + 1)^4$

- (A) sum of real roots $= -2$
- (B) sum of imaginary roots $= -2$
- (C) product of real roots $= \frac{8}{7}$
- (D) product of imaginary roots $= \frac{8}{7}$

353 If roots of the equation $(4^a + 4^b + 4^c)x^2 - 2(2^{a+b} + 2^{b+c} + 2^{c+d})x + (4^b + 4^c + 4^d) = 0$ are real, then which of the following is/are always correct-

- (A) $a + b = c + d$ (B) $b = \frac{a+c}{2}$ (C) $a + d = b + c$ (D) $c = \frac{b+d}{2}$

354 If α and β are the roots of the equation $x^2 - 6x - 4 = 0$, then $\frac{\alpha^{17} + \beta^{17} - 4(\alpha^{15} + \beta^{15})}{(\alpha^8 + \beta^8)^2 - 2^{17}}$ is a

- (A) real number (B) composite number
(C) prime number (D) rational number

355 Let P_1 and P_2 are possible real values of P for which expression $3x^2 + 2Pxy + 2y^2 + 4x - 4y + 1$ can be resolved into linear factors, then -

- (A) $P_1 + P_2 = 8$ (B) $P_1 + P_2 = -8$
(C) P_1, P_2 both are positive (D) P_1, P_2 both are negative.

356 Let $E = (x + y)(y + z)$ & $x, y, z > 0$ also $xyz(x + y + z) = 1$, then $\lambda =$ minimum value of E , then λ is

- (A) Integer (B) Even
(C) Prime (D) Algebraic

357 Integral values of x for which $x^2 + 17x + 71$ is a perfect square of an integer is/are -

- (A) -10 (B) -7 (C) 10 (D) 7

358 Let $P(x) = x^2 + bx + c$ where $b, c \in \mathbb{R}$ $P(P(1)) = P(P(2)) = 0$ and $P(1) \neq P(2)$, then-

- (A) $2b + 1 = 0$ (B) $2c + 3 = 0$ (C) $2b + 3 = 0$ (D) $2c + 1 = 0$

359 For the quadratic equation $x^2 + (2 - m)x - 4 = 0$ in x identify the correct statement(s)-

- (A) if $m < 5$ then exactly one root is smaller than -1
(B) if $m < 5$ then both roots are smaller than -1
(C) if $m > 5$ then both roots are greater than -1
(D) roots are distinct real numbers for each $m \in \mathbb{R}$

360 If the equation $x^4 + 2ax^3 + x^2 + 2ax + 1 = 0$ have at least two distinct negative roots, then the possible values of a is/are-

- (A) 1 (B) 2 (C) 3 (D) 4

361 If $x^2 + 3x + 5$ is the greatest common divisor of $(x^3 + ax^2 + bx + 1)$ and $(2x^3 + 7x^2 + 13x + 5)$, then the value of $[a + b]$ is greater than or equal to-

- (A) 7 (B) 8 (C) 9 (D) 10

[note $[k]$ denotes greatest integer less than or equal to k]

362 If the equation $4x^4 + 2x^3 - 2x^2 + 3ax - a^2 = 0$ has four real roots, then a can be -

- (A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{16}$ (D) $\frac{1}{8}$

- 363 Suppose 'a' and 'b' are integers and $b \neq -1$. If the quadratic equation $x^2 + ax + b + 1 = 0$ has a positive integral root, then which of the following is/are true-
- (A) the other root is also a positive integer
 - (B) the other root is an integer
 - (C) $a^2 + b^2$ is a prime number
 - (D) $a^2 + b^2$ has a factor other than 1 and itself.

- 364 The equation $x^4 - (3m + 2)x^2 + m^2 = 0$ has four real roots in arithmetic progression. The possible value of m is/are -

- (A) $-\frac{6}{19}$ (B) $\frac{6}{19}$ (C) 6 (D) -6

- 365 If x, y, z are three real number such that $x + y + z = 4$ and $x^2 + y^2 + z^2 = 6$, then-

- (A) $x \in \left[\frac{2}{3}, 2\right]$ (B) $y \in \left[\frac{2}{3}, 2\right]$ (C) $z \in (2, 3)$ (D) $y \in (2, 3)$

Linked Comprehension Type

- 366 If α, β, γ are roots of the equation $x^3 + 2x^2 + 3x + 3 = 0$. Then

The value of $\frac{\alpha}{\alpha+1} + \frac{\beta}{\beta+1} + \frac{\gamma}{\gamma+1}$ is -

- (A) 5 (B) 6 (C) -5 (D) -6

- 367 If α, β, γ are roots of the equation $x^3 + 2x^2 + 3x + 3 = 0$. Then

Paragraph for Question 77 and 78

Let a, b, c be real numbers with $a > 0$ such that the quadratic equation $ax^2 + bcx + b^3 + c^3 - 4abc = 0$ has no real roots. Let $P(x) = ax^2 + bx + c$ and $Q(x) = ax^2 + cx + b$.

- 368 Let a, b, c be real numbers with $a > 0$ such that the quadratic equation $ax^2 + bcx + b^3 + c^3 - 4abc = 0$ has no real roots. Let $P(x) = ax^2 + bx + c$ and $Q(x) = ax^2 + cx + b$.

Which one of the following is true

- (A) $(c^2 - 4ab)(b^2 - 4ac) < 0$ (B) $(c^2 - 4ab)(b^2 - 4ac) > 0$
 (C) $(a^2 - 4bc)(b^2 - 4ac) < 0$ (D) $(a^2 - 4bc)(b^2 - 4ac) > 0$

- 369 Let a, b, c be real numbers with $a > 0$ such that the quadratic equation $ax^2 + bcx + b^3 + c^3 - 4abc = 0$ has no real roots. Let $P(x) = ax^2 + bx + c$ and $Q(x) = ax^2 + cx + b$.

Which of the following is true for P(x) and Q(x)

- (A) $P(x) > 0 \forall x \in \mathbb{R}$ and $Q(x) < 0 \forall x \in \mathbb{R}$
 (B) $P(x) < 0 \forall x \in \mathbb{R}$ and $Q(x) > 0 \forall x \in \mathbb{R}$
 (C) neither $P(x) > 0 \forall x \in \mathbb{R}$ nor $Q(x) > 0 \forall x \in \mathbb{R}$
 (D) exactly one of P(x) or Q(x) is positive for all real x.

- 370 Let α, β, γ be distinct real numbers such that

$$(2 \sin \theta + 1)\alpha^2 + (\sqrt{3} \tan \theta - 1)\alpha + (\cot \theta - \sqrt{3}) = 0$$

$$(2 \sin \theta + 1)\beta^2 + (\sqrt{3} \tan \theta - 1)\beta + (\cot \theta - \sqrt{3}) = 0$$

$$(2 \sin \theta + 1)\gamma^2 + (\sqrt{3} \tan \theta - 1)\gamma + (\cot \theta - \sqrt{3}) = 0$$

For the value of θ obtained from the given information, the number of values of θ in $[-\pi, 2\pi]$ is -

- (A) 2 (B) 3 (C) 4 (D) 5

- 371 Let α, β, γ be distinct real numbers such that

For the value of θ obtained from the given information, the value of k for which the equations $x^2 + 2 \sin \theta x + \cot^2 \theta = 0$, $x^2 + (1 - k)x + k = 0$ has one root in common -

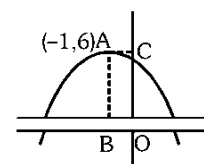
- (A) $k = 2$ (B) $k = 3$ (C) $k = \frac{3}{2}$ (D) can't have common root

- 372 Graph of $f(x) = ax^2 + bx + c$ is shown adjacently, for which $A(-1, 6)$ is the vertex and $b^2 - 4ac = 72$.

On the basis of above informations, answer the following questions :

The value of $a + b + c$ is equal to -

- (A) -6 (B) 6 (C) -10 (D) 10

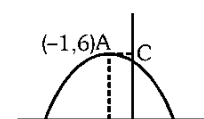


- 373 Graph of $f(x) = ax^2 + bx + c$ is shown adjacently, for which $A(-1, 6)$ is the vertex and $b^2 - 4ac = 72$.

On the basis of above informations, answer the following questions :

The quadratic equation with rational coefficients whose one of the roots is $a + b - \sqrt{c}$, is -

- (A) $x^2 - 18x + 78 = 0$ (B) $x^2 + 18x - 72 = 0$ (C) $x^2 + 18x + 78 = 0$ (D) $x^2 - 18x + 72 = 0$

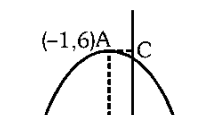


- 374 Graph of $f(x) = ax^2 + bx + c$ is shown adjacently, for which $A(-1, 6)$ is the vertex and $b^2 - 4ac = 72$.

On the basis of above informations, answer the following questions :

Range of $g(x) = (a + \frac{1}{2})x^2 + (b + 2)x + c$ when $x \in [-1, 0]$ is -

- (A) $[-\frac{3}{2}, \frac{9}{2}]$ (B) $[3, \frac{9}{2}]$ (C) $[0, \frac{23}{5}]$ (D) $[3, \frac{23}{5}]$



- 375 A polynomial equation is given by $2x^4 + kx^3 + 22x^2 + kx + 2 = 0$.

On the basis of above information, answer the following questions :

Interval of k for which the equation has no non real roots can be -

- (A) $k \in (12, 14)$ (B) $k \in (11, 12)$ (C) $[-12, -11]$ (D) $k \in [-13, -12]$

- 376 A polynomial equation is given by $2x^4 + kx^3 + 22x^2 + kx + 2 = 0$.

On the basis of above information, answer the following questions :

Interval of k for which exactly 2 roots of the equation are real can be -

- (A) $k \in (-10, -8)$ (B) $k \in (-14, -12)$ (C) $k \in \phi$ (D) $k \in (13, 16)$

377 A polynomial equation is given by $2x^4 + kx^3 + 22x^2 + kx + 2 = 0$.

On the basis of above information, answer the following questions :

Interval of k for which there are no real roots of the equation can be -

- (A) $k \in [-10, 10]$ (B) $k \in [3, 13]$ (C) $k \in (-13, -11]$ (D) $k \in [0, 13]$

378 Let graph of quadratic expression $ax^2 + bx + c$, $a > 0$ be symmetric about $x = -\frac{1}{2}$ and $4a + 2b + c < 0$.

Sum of roots of the equation $ax^2 + bx + c = 0$ is -

- (A) 1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) -1

379 Let graph of quadratic expression $ax^2 + bx + c$, $a > 0$ be symmetric about $x = -\frac{1}{2}$ and $4a + 2b + c < 0$.

Value of $(a + c)$ is -

- (A) positive (B) negative (C) zero (D) can't say

380 If α, β are the roots of equation $x^2 - 4x + 1 = 0$ and $P_n = \alpha^n + \beta^n$, $n \in \mathbb{N}$

Value of $\frac{P_6 + P_4}{P_5}$ is-

- (A) 1 (B) 2 (C) 4 (D) 6

381 If α, β are the roots of equation $x^2 - 4x + 1 = 0$ and $P_n = \alpha^n + \beta^n$, $n \in \mathbb{N}$

Value of P_5 is-

- (A) 724 (B) 1672 (C) 1364 (D) 1984

382 Let x_1, x_2, x_3 denotes three real roots of equation $x^3 - 3x - 1 = 0$ such that $x_1 < x_2 < x_3$

Value of $x_3^2 - x_2^2$ is equal to -

- (A) $x_3 - x_1$ (B) $x_1(x_2 - x_3)$ (C) $x_1 - x_3$ (D) $x_1(x_3 - x_2)$

383 Let x_1, x_2, x_3 denotes three real roots of equation $x^3 - 3x - 1 = 0$ such that $x_1 < x_2 < x_3$

The value of expression $(2 - x_1)(2 - x_2)(2 - x_3)$ is equal to -

- (A) 1 (B) -1
(C) $(1 + x_1)(1 + x_2)(1 + x_3)$ (D) $|(1 + x_1)(1 + x_2)(1 + x_3)|$

384 Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial function with leading coefficient one which satisfies following properties.

- (i) $P(x)$ is divisible by $x^2 + 1$.
- (ii) $P(x) + 1$ is divisible by $x^3 + x^2 + 1$.
- (iii) $P(x)$ of degree four.

If the equation $P(x) = \lambda$ (where λ is prime number) has atleast one integral solution, then sum of possible value(s) of λ is

- (A) 2 (B) 5 (C) 7 (D) 12

385 Let $P: \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial function with leading coefficient one which satisfies following properties.

- (i) $P(x)$ is divisible by $x^2 + 1$.
- (ii) $P(x) + 1$ is divisible by $x^3 + x^2 + 1$.
- (iii) $P(x)$ of degree four.

The value of $P(3)$ is -

- (A) 10 (B) 11 (C) 110 (D) 0

Paragraph for Question 95 and 96

Given that α, β are the roots of the equation $ax^2 + bx + c = 0$ such that $a, b, c, \alpha, \beta \in \mathbb{R}$ and $a \neq 0$,

386 Given that α, β are the roots of the equation $ax^2 + bx + c = 0$ such that $a, b, c, \alpha, \beta \in \mathbb{R}$ and $a \neq 0$,

- $1 < \alpha < 3 < \beta$.

Roots of the equation $a(x+3)^2 + b(x+3) + c = 0$ are -

- (A) both imaginary (B) both real and positive
(C) both negative (D) of different sign

387 Given that α, β are the roots of the equation $ax^2 + bx + c = 0$ such that $a, b, c, \alpha, \beta \in \mathbb{R}$ and $a \neq 0$,

Number of solutions of the equation $a(\sin x + 2)^2 + b(\sin x + 2) + c = 0$ in $(0, 2\pi]$ are -

- (A) 1 (B) 2 (C) 3 (D) 4

Matrix Match Type

388 Let $f(x) = \frac{x^2 - 11x + 18}{x^2 - 7x + 12}$

Column-I

- (A) Complete solution of $0 < f(x) < 1$
(B) Complete solution of $f(x) < 1$
(C) Complete solution of $f(x) < 0$
(D) Complete solution of $f(x) > 2$

Column-II

- (P) $(3, 4)$
(Q) $\left(\frac{3}{2}, 2\right) \cup (9, \infty)$
(R) $(2, 3) \cup (4, 9)$
(S) $(-\infty, 2) \cup (3, 4) \cup (9, \infty)$
(T) $\left(\frac{3}{2}, 3\right) \cup (4, \infty)$

389 **Column-I**

- (A) If $|x+2| + |x-3| = 5$, then sum of integral values of x satisfying the equation lies in the interval
(B) If $\sqrt{x+\sqrt{x}} - \sqrt{x-\sqrt{x}} = \frac{3}{2} \sqrt{\frac{x}{x+\sqrt{x}}}$, then the value/s of x satisfying the equation may lie in the interval
(C) If a, b, c are positive numbers such that $2a + 3b + 4c = 9$ then maximum value of $a^4 b^2 c^3$ lies in the interval
(D) If 6 lies between roots of the equation $x^2 + 2(m-3)x + 9 = 0$

Column-II

- (P) $(-3, -1)$
(Q) $[3, 5]$
(R) $(1, 3)$
(S) $(1, 2]$

390 then 'm' may belong to the interval (T) [5, 10]

• Match the set of values of x in column-II which satisfy the inequations in column-I.

Column-I**Column-II**

(A) $|x + 1| + |x - 2| + |x - 3| \geq 6$

(P) $x < 0$

(B) $|x| - 1 < 1 - x$

(Q) $x = 3$

(C) $|x - 4| < \sqrt{3 - x}$

(R) $x \leq 3$

(D) $\sqrt{x^3 - 4x^2 + 4x - 3} + \sqrt{3 - x} + x^2 \geq 9$

(S) $x \geq 4$

(T) $x \in \phi$

391 **Column-I**

Column-II

• (A) If x, y and z are real numbers such that

$$x^2 + y^2 + 2z^2 = 4x - 2z + 2yz - 5,$$

then the possible value of $(x - y - z)$ is

(P) 1

(Q) 2

(B) Value of $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^4}\right)\left(1 + \frac{1}{2^8}\right) \dots \infty$ is

(R) -3

(C) If ' α ' is root of $x^4 + x^2 - 1 = 0$

then value of $(\alpha^6 + 2\alpha^4)^7$ is

(S) -1

(D) The sum of all the real roots of the equation

392 $(x - 1)^2 - 2|x - 1| - 3 = 0$ is

(T) 4

Numerical Grid

The equation $x^2 + bx + c = 0$ has distinct roots. If 3 is subtracted from each root, the result are reciprocal of the original roots. Then the value of $(b^2 + 6c + bc)$, is

393 The roots of a quadratic equation $ax^2 - 45x + 100 = 0$ are two consecutive natural numbers, where $a \in$

394 N. The greater root is

• It is given that α, β are roots of equation $x^2 + 2ax + 3 = 0$ and γ, δ are roots of equation $x^2 + 7x + 5 = 0$. If $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ is minimum, then $[a]$ is equal to (where $[.]$ denotes greatest integer function)

395 If $\alpha = \sqrt[3]{45 + 29\sqrt{2}} + \sqrt[3]{45 - 29\sqrt{2}}$, then the value of $\alpha^3 - 21\alpha - 90$ is equal to

396 Number of solutions of the equation $(\sqrt{3} + \sqrt{2})^{x^2 - 15x} + (\sqrt{3} - \sqrt{2})^{x^2 - 15x} = 2\sqrt{3}$ is/are

397 Number of rational values of x satisfying equation $x^{(\log_5 x)^2 + 5 \log_5 x + 2} = 5^8$ is/are

398 If the three equations $x^2 + px + 12 = 0$

$$x^2 - qx + 15 = 0$$

$$x^2 + (p - q)x + 36 = 0$$

have a common negative root then the value of $|p + q|$ is

399 The number of negative integral solutions of the equation $x^{22x+1} + 2^{lx-3l+2} = x^{2 \cdot 2^{lx-3l+4}} + 2^{x-1}$ is

400 If the equations $x^2 - 4x + 1 = 0$ and $x^2 - (4\sin\theta)x - 7 = 0$ has one root common, then number of values

401 of θ in $(0, 2\pi)$ is

• Solve the system of equation ($x \geq y \geq z \geq 2$)

$$x^3 - 6z^2 + 12z - 8 = 0$$

$$y^3 - 6x^2 + 12x - 8 = 0$$

$$z^3 - 6y^2 + 12y - 8 = 0$$

402 Solve for x satisfying $\sqrt{x + 2\sqrt{x + 2\sqrt{x + \dots + 2\sqrt{x + 2\sqrt{3x}}}}} = x$ ($x > 0$)

403 Solve for x and y

• $y^2 = x^3 - 3x^2 + 2x$

$$x^2 = y^3 - 3y^2 + 2y \quad (x \neq y)$$

$$(7^4 + 64)(15^4 + 64)(23^4 + 64)(31^4 + 64)(39^4 + 64)$$

404 $\frac{(1^4 + 64)(9^4 + 64)(17^4 + 64)(25^4 + 64)(33^4 + 64)}{(3^4 + 64)(11^4 + 64)(19^4 + 64)(27^4 + 64)(35^4 + 64)} = ?$

405 Let $P(x)$ be a monic polynomial with degree 4 and $P(1) = 3$, $P(3) = 11$, $P(5) = 27$ find $P(-2) + 7P(6)$

406 If $x + \sqrt[4]{5 - x^4} = 2$, then find $x\sqrt[4]{5 - x^4}$ ($x \in \mathbb{R}$)

407 Find sum of all real solutions to the equation $(x + 1)(x^2 + 1)(x^3 + 1) = 30x^3$

408 If $a + b + c = 3$

• $a^2 + b^2 + c^2 = 29$

$$abc = 11$$

$$\text{find } a^5 + b^5 + c^5$$

409 Prove that sum of cubes of legs of right angled triangle is less than the cube of hypotenuse

410 Let $f(x) = x^2 - 2ax - a^2 - \frac{3}{4}$. Find all values of a such that $|f(x)| \leq 1 \quad \forall x \in [0, 1]$

411 Find the monic polynomial of eighth degree with rational coefficients having one root as

$$\sqrt{8 + \sqrt{6} + 4\sqrt{3} + 3\sqrt{2}}$$

412 Find number of real solutions to the equation $2x^4 - 3x^2 - 2x\sin x + 3 = 0$

413 If equations $2x^2 + bx + b = 0$ and $x^2 + 2x + 2b = 0$ have a common root, then number of integral values

• of 'b' is

414 If α, β, γ are roots of $x^3 + 2x^2 - x + 3 = 0$, then value of $\frac{(\alpha^3 + \alpha^2 + 3)(\beta^3 + \beta^2 + 3)(\gamma^3 + \gamma^2 + 3)}{\alpha\beta\gamma}$ is

- 415 Number of real solution(s) of the equation $3^{3x} - 3(3^{2x}) - 49(3^x) - 45 = 0$ is .
- 416 The sum of all real values of 'm' such that the equation $(x^2 - 2mx - 4(m^2 + 1))(x^2 - 4x - 2m(m^2 + 1)) = 0$ has exactly three different real roots is .
- 417 Let α, β are roots of $x^2 - bx - b = 0$, $b > 0$ and $|\alpha|, |\beta|$ are roots of $x^2 + px + q = 0$. If minimum value of $(p^2 - 8q)$ is k , then value of $|k|$ is .
- 418 For equation $(2x + 3)(3x + 4)(5x + 6)(30x + 31) = 10$. The value of greatest integer which is less than the product of real roots of this equation is .
- 419 If $x + \frac{1}{x} = 3$ and $x^5 + \frac{1}{x^5} = a$, then sum of digits of 'a' is .
- 420 If $a, b, c \in \mathbb{R} - \{0\}$, such that both roots of the equation $ax^2 + bx + c = 0$ lies in $(2, 3)$, then both roots of the equation $a(x + 1)^2 + b(x^2 - 1) + c(x - 1)^2 = 0$ are always greater than k , then maximum integral value of k is .
- 421 Let $P(x) = x^2 - x + 1$ and α be a root the equation $P(P(P(P(x)))) = 0$, then the absolute value of $(P(\alpha) - 1)P(\alpha)P(P(\alpha))P(P(P(\alpha)))$ is .
- 422 Let $f(x)$ be a polynomial of degree 3 with integer coefficients such that $f(0) = 3$ and $f(1) = 11$. If $f(x) = 0$ has exactly 2 integer roots, then number of such polynomials is .
- 423 The number of integral value(s) of $a \in (8, \infty)$ for which the expression $\log_{10}(4x^2 - ax + 3)$ is always non-negative for all $x \in [1, 3]$ is .
- 424 If each pair of $x^2 + ax + b = 0$, $x^2 + cx + d = 0$ and $x^2 + ex + f = 0$ has exactly one root in common and all equations do not have common roots, then the value of $\frac{(a + c + e)^2}{ac + ce + ea - b - d - f}$ is .
- 425 If α and β are roots of $x^2 + 2x + 2 = 0$, then the value of $\alpha^3 + 2\beta$ is .
- 426 Least positive integral value of k for which the equation $e^{2x} + (1 - 2k)e^x + k^2 - 1 = 0$ does not have any real solution, is (where e is Napier's constant)

427. The complete range of k for which the equation $x^4 - 2kx^2 + x + k^2 - k = 0$ has all real solutions is $\left[\frac{m}{n}, \infty\right)$

where m and n are relatively prime numbers. Then $m + n$ is

ANSWER KEY

- | | | | | |
|--|--|-------------|-------------|--------------------------------|
| 1. C | 2. C | 3. B | 4. A | 5. B |
| 6. D | 7. B | 8. B | 9. B | 10. B |
| 11. C | 12. A | 13. A | 14. D | 15. B |
| 16. A | 17. B | 18. C | 19. D | 20. D |
| 21. D | 22. B | 23. D | 24. A | 25. C |
| 26. C | 27. B | 28. A | 29. A | 30. A |
| 31. C | 32. B | 33. C | 34. B | 35. C |
| 36. B | 37. C | 38. C | 39. C | 40. A |
| 41. C | 42. D | 43. C | 44. C,D | 45. B,C |
| 46. A,B,C,D | 47. A,D | 48. B,D | 49. A,B,C,D | 50. A,B,C,D |
| 51. A,C | 52. A,C | 53. A,B,C,D | 54. A,C | 55. A,B |
| 56. A,C,D | 57. A,B | 58. A,C | 59. A,C | 60. A,C,D |
| 61. A,B,D | 62. B,C,D | 63. A,B,D | 64. B,D | 65. A,B,C,D |
| 66. A,B | 67. A,B | 68. A,C,D | 69. A,B,C,D | 70. A,B |
| 71. A,B,C,D | 72. B,D | 73. A,C | 74. A,B | 75. A |
| 76. B | 77. A | 78. D | 79. A | 80. D |
| 81. A | 82. C | 83. D | 84. D | 85. D |
| 86. A | 87. D | 88. B | 89. C | 90. A |
| 91. A,B | 92. A,D | 93. C | 94. C | 95. D |
| 96. B | 97. $(A) \rightarrow (Q); (B) \rightarrow (T); (C) \rightarrow (R); (D) \rightarrow (P)$ | | | |
| 98. $(A) \rightarrow (Q); (B) \rightarrow (R,S); (C) \rightarrow (Q); (D) \rightarrow (P)$ | | | | |
| 99. $(A) \rightarrow (P,S); (B) \rightarrow (P); (C) \rightarrow (T); (D) \rightarrow (Q)$ | | | | |
| 100. $(A) \rightarrow (T); (B) \rightarrow (Q); (C) \rightarrow (P); (D) \rightarrow (Q)$ | | | | |
| 101. 6 | 102. 5 | 103. 2 | 104. 0 | 105. 4 |
| 106. 3 | 107. 1 | 108. 0 | 109. 2 | 110. (2,2,2) |
| 111. 3 | 112. No solution | 113. 337 | 114. 1112 | 115. $\frac{8 - \sqrt{42}}{2}$ |
| 119. $\left[-\frac{1}{2}, \frac{\sqrt{2}}{4}\right]$ | 121. 0 | 122. 1 | 123. 5 | 124. 1 |
| 125. 3 | 126. 4 | 127. 1 | 128. 6 | 129. 2 |
| 130. 1 | 131. 0 | 132. 0 | 133. 4 | 134. 0 |
| 135. 2 | 136. 7 | | | |

- 428 If α, β, γ are the roots of the equation, $x^3 + P_0x^2 + P_1x + P_2 = 0$, then $(1-\alpha^2)(1-\beta^2)(1-\gamma^2)$ is equal to
- (A) $(1 + P_1)^2 - (P_0 + P_2)^2$ (B) $(1 + P_1)^2 + (P_0 + P_2)^2$
 (C) $(1 - P_1)^2 - (P_0 - P_2)^2$ (D) none of these
-
- 429 The equations $ax^2 + bx + a = 0$ and $x^3 - 2x^2 + 2x - 1 = 0$ have two roots in common. Then $a + b$ must be equal to
- (A) 1 (B) -1 (C) 0 (D) none of these
-
- 430 The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 respectively, then the ordered pair (x_1, x_2) is
- (A) $(-5, -7)$ (B) $(1, -1)$ (C) $(-1, 1)$ (D) $(5, 7)$
-
- 431 If all roots of $x^4 - 8x^3 + ax^2 - bx + 16 = 0$ are positive real numbers, then $(a + b)$ is equal to -
- (A) 32 (B) 24 (C) 56 (D) 8
-
- 432 Given α, β be the roots of quadratic equation, $x^2 - 2ax + 7 = 0$ and γ, δ be the roots of the equation, $x^2 - 2bx + 1 = 0$. If $\alpha\delta = \beta\gamma$, then $\frac{a^2}{b^2}$ is equal to -
- (A) 5 (B) 6 (C) 7 (D) 8
-
- 433 Let α, β be roots of $x^2 - x + b = 0$ and $S_k = \alpha^k + \beta^k$, then the value(s) of b such that S_2, S_3 and S_4 are in arithmetic progression is -
- (A) 0 (B) 1
 (C) $\frac{1}{5}$ (D) no real value(s) of b exists
-
- 434 There is only one real value of λ for which the quadratic equation $\lambda x^2 - 4x + 9 = 0$ has two positive integral solutions. The sum of these two solutions is-
- (A) 4 (B) 8 (C) 9 (D) 12
-
- 435 Let $f(x)$ is polynomial of two degree such that $f(1) = 6, f(-1) = 30$ and $f(3) > 0$. If one root is twice of the other root of the equation $f(x) = 0$, then the value of $f(3)$ is -
- (A) 54 (B) 78 (C) 96 (D) 110
-
- 436 The set of values of 'a' for which $f(x) = ax^2 + 2x(1 - a) - 4$ is negative for exactly three integral values of x , is -
- (A) $(0, 2)$ (B) $(0, 1]$ (C) $[1, 2)$ (D) $[2, \infty)$
-
- 437 The number of possible value of 'a' for which the expression $y = \frac{\alpha x^2 + 7x - 2}{\alpha + 7x - 2x^2}$ has atleast one common linear factor in numerator and denominator, is -
- (A) 0 (B) 1 (C) 2 (D) 3
-
- 438 If α, β are real numbers such that $\beta > \alpha$ & $|\alpha - \beta| < 2$, then the equation $(x - \alpha)(x - \beta) + 1 = 0$ has -
- (A) both roots in (α, β) (B) both roots in $(-\infty, \alpha)$ (C) both roots in (β, ∞) (D) no real roots

- 439 If α, β, γ are roots of equation $111x^3 - 11x - 1 = 0$, then $(\alpha\beta)^{-2} + (\beta\gamma)^{-2} + (\gamma\alpha)^{-2}$ equals
 (A) 2332 (B) 1331 (C) 1210 (D) 2442
- 440 If $a + b + c > \frac{9c}{4}$ and quadratic equation $ax^2 + 2bx - 5c = 0$ has non-real roots, then -
 (A) $a > 0, c > 0$ (B) $a > 0, c < 0$ (C) $a < 0, c < 0$ (D) $a < 0, c > 0$
- 441 Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then value of $P(9)$ is -
 (A) 98 (B) 899 (C) 80 (D) 898
- 442 The set of values of 'a' for which the inequality $x^2 - (a + 2)x - (a + 3) < 0$ is satisfied by atleast one positive real x, is -
 (A) $[-3, \infty)$ (B) $(-3, \infty)$ (C) $(-\infty, -3)$ (D) $(-\infty, 3]$
- 443 Set of the values of parameter 'm' for which every solution of the inequality $x^2 - 3x + 2 \leq 0$ is also a solution of the inequality $2x^2 - (m + 1)x - (2m + 3) < 0$, is -
 (A) $\left(\frac{3}{4}, \infty\right)$ (B) $\left(-\frac{2}{3}, \infty\right)$ (C) $\left(-\frac{2}{3}, \frac{3}{4}\right)$ (D) $(-\infty, \infty)$
- 444 Let $f(x) = x^2 + ax + b$ such that $f(2) + f(3) < 2$, then equation $f(x) = 1$ has $(a, b \in \mathbb{R})$
 (A) both roots real and distinct (B) both roots real and equal
 (C) non real roots (D) roots whose nature depends on value of a & b
- 445 If a, b, c are real numbers satisfying the condition $a + b + c = 0$ then the roots of the quadratic equation $3ax^2 + 5bx + 7c = 0$ are
 (A) positive (B) negative (C) real and distinct (D) imaginary
- 446 If x is real, the function, $\frac{(x-a)(x-b)}{(x-c)}$ will assume all real values, provided
 (A) $a > b > c$ (B) $a < b < c$ (C) $a > c > b$ (D) none of these
- 447 Let $p(x) = 0$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $p(x) = 0$ is
 (A) 7 (B) 49 (C) 56 (D) 63
- 448 The least value of the expression $x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$ is
 (A) 0 (B) 1 (C) no least value (D) none of these
- 449 If the equation $ax^2 + bx + c = 0$ has two positive and real root, then the equation $ax^2 + (b + 6a)x + (c + 3b) = 0$ has
 (A) no solution (B) atleast one positive solution
 (C) atleast one negative solution (D) none of the above
- 450 The set of real value(s) of p for which the equation, $|2x + 3| + |2x - 3| = px + 6$ has more than two solutions is

451 Consider the equation $x^2 + x - n = 0$, where n is an integer lying between 1 to 100. Total number of different values of 'n' so that the equation has integral roots, is

- (A) 6 (B) 4 (C) 9 (D) None of these

452 Number of positive integers n for which $n^2 + 96$ is a perfect square is

- (A) 4 (B) 8 (C) 12 (D) infinite

453 Which of the following statement is true

- (A) The only integral value of x for which $x^2 + 19x + 92$, is a perfect square is -8 .
 (B) The only integral value of x for which $x^2 + 19x + 92$, is a perfect square is -11 .
 (C) The number of integral values of x for which $x^2 + 19x + 92$ is a perfect square are two.
 (D) The number of integral values of x for which $x^2 + 19x + 92$ is infinite.

454 If α and β (where $\alpha, \beta \in \mathbb{R}$) are the roots of equation $x^2 + 4x + b = 0$ and $|\alpha - \beta| \leq 2$, then the complete range of b is-

- (A) (3,4) (B) (4,5) (C) [3,4] (D) (3,5)

455 The smallest positive number 'k' such that there exists $\alpha, \beta \in \mathbb{R}$ satisfying $\alpha + \beta = k$ and $\alpha\beta = k$, is

- (A) 4 (B) 1 (C) 2 (D) 8

456 Quadratic equation in x , $(k^2 - 2k + 2)x^2 - 2(\sqrt{k^4 + 4})x + (k^2 + 2k + 2) = 0 \forall k \in \mathbb{R}$ has

- (A) real & distinct roots (B) equal roots
 (C) imaginary roots (D) roots of opposite sign

457 For the quadratic equation $x^2 + (2 - m)x - 4 = 0$ in x identify the correct statement(s)-

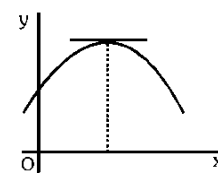
- (A) if $m < 5$ then exactly one root is smaller than -1
 (B) if $m < 5$ then both roots are smaller than -1
 (C) if $m > 5$ then both roots are greater than -1
 (D) roots are distinct real numbers for each $m \in \mathbb{R}$

458 Integral values of x for which $x^2 + 17x + 71$ is a perfect square of an integer is/are -

- (A) -10 (B) -7 (C) 10 (D) 7

459 Graph of $f(x) = px^2 - qx - r$ is shown in the adjacent figure then -

- (A) $r(p - q - r) < 0$ (B) $q < 0$
 (C) $\sqrt{q^2 + 4pr} > |q|$ (D) $p < 0$



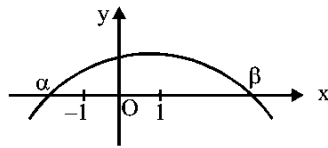
460 If α, β and γ are the roots of the equation $x^3 - 3x + 1 = 0$ then -

- (A) $\prod (\alpha + 1) = -3$ (B) $\sum (\alpha + 1) = 0$
 (C) $\sum (\alpha + 1)(\beta + 1) = -3$ (D) $\sum \alpha^2 = 6$

461 The values of k for which the system of equations $2x + ky = k$, $3x - y = 10$ has solution(s) satisfying $x > 0, y > 0$ are given by-

- (A) $k < 0$ (B) $k < -\frac{2}{3}$ (C) $k > 0$ (D) $k > \frac{20}{3}$

462 The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below.



Which of the following are correct ?

- (A) $\frac{c}{a} < -1$ (B) $|\beta - \alpha| > 2$ (C) $f(x) > 0 \forall x \in (0, \beta)$ (D) $abc < 0$

463 If $p, q, r, s \in \mathbb{R}$, then equation

$(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ must have-

- (A) 6 real roots
(B) at least two real roots
(C) 2 real & 4 imaginary roots
(D) 4 real & 2 imaginary roots

464 Roots of the equation $2(x - a)(x - b) - (x - a^2 - b^2 - 1)^2 = 0$ are (where $a, b \in \mathbb{R}$)

- (A) real & equal (B) real & different (C) imaginary (D) imaginary and equal

465 The number of solutions of equation $|2ax - 3| + |ax + 1| + |5 - ax| = \frac{1}{2}$ is/are -

- (A) 1 (B) 0 (C) 2 (D) 3

466 If sum of the roots of quadratic equation $ax^2 + bx + c = 0$ is equal to sum of the squares of their reciprocals,

then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to -

- (A) 2 (B) -2 (C) 1 (D) -1

467 The number of integers 'x' satisfying the equation $(x^2 - x - 1)^{x+2} = 1$ is-

- (A) 2 (B) 3 (C) 4 (D) 5

468 If equation $x^2 - px + p = 0$ has positive integral roots α, β , then value of $\alpha^{2016} - \beta^{2016}$ is (where $p \in \mathbb{R}$)

- (A) 4 (B) 2016 (C) $p^2 - 4$ (D) 0

469 If $\frac{9 - |2x + 3|}{2 - |x|} < 0$, then number of integral values of x is -

- (A) 3 (B) 4 (C) 5 (D) more than 5

470 If α and β are roots of the equation $ax^2 + bx + c = 0$ then roots of the equation

$a(2x + 1)^2 - b(2x + 1)(3 - x) + c(3 - x)^2 = 0$ are -

- (A) $\frac{2\alpha + 1}{\alpha - 3}, \frac{2\beta + 1}{\beta - 3}$ (B) $\frac{3\alpha + 1}{\alpha - 2}, \frac{3\beta + 1}{\beta - 2}$ (C) $\frac{2\alpha - 1}{\alpha - 2}, \frac{2\beta + 1}{\beta - 2}$ (D) none of these

471 Number of rational roots of the equation $x^3 - 3x + 1 = 0$

- (A) 0 (B) 1 (C) 2 (D) 3

472 The integral values of k for which the roots of $kx(x+2) + k - (x+2) = 0$ are rational is given by -

- (A) 5 (B) -3 (C) -6 (D) 12

473 Number of values of $x \in \mathbb{R}$ satisfying $\left|x - \frac{1}{x}\right| + \left|x + \frac{1}{x}\right| = 2$ are -

- (A) 1 (B) 2 (C) 3 (D) 4

474 If one of the roots of $ax^2 + bx + c = 0$ is greater than 1 and the other is less than -3 and if the roots of $cx^2 + bx + a = 0$ are α & β , then -

- (A) $\alpha < 1$ & $\beta > -\frac{1}{3}$ (B) $\alpha < 1$ & $\beta < -\frac{1}{3}$ (C) $\alpha > 1$ & $\beta > 1$ (D) $\alpha > 1$ & $\beta < -\frac{1}{3}$

475 origin ?

Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K + 5 = 0$ has atleast one positive root.

476 A quadratic polynomial $f(x) = x^2 + ax + b$ is formed with one of its zeros being $\frac{4+3\sqrt{3}}{2+\sqrt{3}}$ where a and b are integers. Also $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ is a biquadratic polynomial such that $g\left(\frac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$ where c and d are also integers. Find the values of a, b, c and d .

477 If the quadratic equations $x^2 + bx + ca = 0$ & $x^2 + cx + ab = 0$ (where $a \neq 0$; $b \neq c$) have a common root, prove that the equation containing their other root is $x^2 + ax + bc = 0$.

478 If the range of the function $f(x) = \frac{x^2 + ax + b}{x^2 + 2x + 3}$ is $[-5, 4]$, $a, b \in \mathbb{N}$, then find the value of $(a^2 + b^2)$.

479 If $x^2 + (a-b)x + (1-a-b) = 0$ where $a, b \in \mathbb{R}$ then find the values of 'a' for which equation has unequal real roots for all values of 'b'.

480 Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β .

481 For $a \leq 0$, determine all real roots of the equation $x^2 - 2a|x-a| - 3a^2 = 0$.

482 If α, β are the roots of the equation, $x^2 - 2x - a^2 + 1 = 0$ and γ, δ are the roots of the equation, $x^2 - 2(a+1)x + a(a-1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$ then find the values of 'a'.

Matrix Match Type :

483 Let $f(x) = \frac{x^2 - 11x + 18}{x^2 - 7x + 12}$

Column-I

- (A) Complete solution of $0 < f(x) < 1$
- (B) Complete solution of $f(x) < 1$
- (C) Complete solution of $f(x) < 0$
- (D) Complete solution of $f(x) > 2$

Column-II

- (P) $(3, 4)$
- (Q) $\left(\frac{3}{2}, 2\right) \cup (9, \infty)$
- (R) $(2, 3) \cup (4, 9)$
- (S) $(-\infty, 2) \cup (3, 4) \cup (9, \infty)$
- (T) $\left(3, 2\right) \cup (4, \infty)$

484 **Column-I**

- (A) If $|x + 2| + |x - 3| = 5$, then sum of integral values of x satisfying the equation lies in the interval

- (B) If $\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} = \frac{3}{2} \sqrt{\frac{x}{x + \sqrt{x}}}$, then the value/s of x

satisfying the equation may lie in the interval

- (C) If a, b, c are positive numbers such that $2a + 3b + 4c = 9$ then maximum value of $a^4 b^2 c^3$ lies in the interval

- (D) If 6 lies between roots of the equation $x^2 + 2(m - 3)x + 9 = 0$ then 'm' may belong to the interval

Column-II

- (P) $(-3, -1)$
- (Q) $[3, 5]$
- (R) $(1, 3)$
- (S) $(1, 2]$
- (T) $[5, 10]$

485 Match the set of values of x in column-II which satisfy the inequations in column-I.

Column-I

- (A) $|x + 1| + |x - 2| + |x - 3| \geq 6$
- (B) $|x| - 1 < 1 - x$
- (C) $|x - 4| < \sqrt{3 - x}$
- (D) $\sqrt{x^3 - 4x^2 + 4x - 3} + \sqrt{3 - x} + x^2 \geq 9$

Column-II

- (P) $x < 0$
- (Q) $x = 3$
- (R) $x \leq 3$
- (S) $x \geq 4$
- (T) $x \in \phi$

486	Column-I	Column-II
(A)	If x, y and z are real numbers such that $x^2 + y^2 + 2z^2 = 4x - 2z + 2yz - 5$, then the possible value of $(x - y - z)$ is	(P) 1
(B)	Value of $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^4}\right)\left(1 + \frac{1}{2^8}\right) \dots \infty$ is	(Q) 2
(C)	If ' α ' is root of $x^4 + x^2 - 1 = 0$ then value of $(\alpha^6 + 2\alpha^4)^7$ is	(R) -3
(D)	The sum of all the real roots of the equation $(x - 1)^2 - 2 x - 1 - 3 = 0$ is	(S) -1
		(T) 4

ANSWER KEY

1. A	2. C	3. A	4. C	5. C
6. A	7. D	8. D	9. C	10. D
11. D	12. D	13. A	14. B	15. B
16. A	17. A	18. C	19. C	20. C
21. B	22. B	23. D	24. C	25. A
26. C	27. C	28. A	29. B	30. A,C,D
31. A,B	32. A,B,C,D	33. A,D	34. B,D	35. A,B,C,D
36. B	37. B	38. B	39. A	40. C
41. D	42. A	43. B	44. A	45. D
46. B	47. A	48. $\{-4, -3, 3, 4\}$	49. $K \leq -1$	
50. $a = 2, b = -11, c = 4, d = -1$	52. 277	53. $a > 1$		
54. $\gamma = \alpha^2\beta$ and $\delta = \alpha\beta^2$ or $\gamma = \alpha\beta^2$ and $\delta = \alpha^2\beta$	55. $x = (1 - \sqrt{2})a$ or $(\sqrt{6} - 1)a$			
56. $a \in \left(-\frac{1}{4}, 1\right)$	57. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (T); (C) $\rightarrow$ (R); (D) $\rightarrow$ (P)			
58. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R,S); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (P)				
59. (A) $\rightarrow$ (P,S); (B) $\rightarrow$ (P); (C) $\rightarrow$ (T); (D) $\rightarrow$ (Q)				
60. (A) $\rightarrow$ (T); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)				

487 The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is

(A) $(-\infty, -2) \cup (2, \infty)$	(B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(C) $(-\infty, -1) \cup (1, \infty)$	(D) $(\sqrt{2}, \infty)$

488 If one root of the equation $x^2 + px + q = 0$ is the square of the other, then

(A) $p^3 + q^2 - q(3p + 1) = 0$	(B) $p^3 + q^2 + q(1 + 3p) = 0$
(C) $p^3 + q^2 + q(3p - 1) = 0$	(D) $p^3 + q^2 + q(1 - 3p) = 0$

- 489 Let a, b, c be the sides of a triangle. No two of them are equal and $\lambda \in \mathbb{R}$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then

(A) $\lambda < \frac{4}{3}$ (B) $\lambda > \frac{5}{3}$ (C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

- 490 Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of 'r' is

(A) $\frac{2}{9}(p-q)(2q-p)$ (B) $\frac{2}{9}(q-p)(2p-q)$
 (C) $\frac{2}{9}(q-2p)(2q-p)$ (D) $\frac{2}{9}(2p-q)(2q-p)$

- 491 Let p and q be real numbers such that $p \neq 0, p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

- 492 Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is

(A) 1 (B) 2 (C) 3 (D) 4

QUADRATIC

- 493 Let $-\frac{\pi}{6} < \theta < \frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals
- (A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

494 Quadratic Equation

Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

(A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

- 495 If $y = \log_{7-a}(2x^2 + 2x + a + 3)$ is defined $\forall x \in \mathbb{R}$, then possible integral value(s) of a is/are

(A) -3 (B) -2 (C) 4 (D) 5

496 Which of the following statement(s) is/are **True**?

- (A) The equation $\sqrt{x+1} + \sqrt{x-1} = 1$ has no real solution.
- (B) If $0 < p < \pi$ then the quadratic equation, $(\cos p - 1)x^2 + \cos px + \sin p = 0$ has real roots.
- (C) If $2a + b + c = 0$ ($c \neq 0$) then the quadratic equation, $ax^2 + bx + c = 0$ has no root in $(0, 2)$.
- (D) If x and y are positive real numbers and m, n are any positive integers then ; $\frac{x^n \cdot y^m}{(1+x^{2n})(1+y^{2m})} > \frac{1}{4}$.

497 Possible set of values of the parameter 'a' for which the inequality $a \cdot 9^x + 4(a-1)3^x + a - 1 > 0$ is satisfied for all real values of x .

- (A) $[1, 5]$ (B) $[3, \infty)$ (C) $[-3, -1]$ (D) $(-\infty, 0]$

498 Set of values of 't' for which $2 \sin t = \frac{1-2x+5x^2}{3x^2-2x-1}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, can be

- (A) $\left[-\frac{\pi}{2}, -\frac{\pi}{10}\right]$ (B) $\left[\frac{\pi}{10}, \frac{\pi}{2}\right]$ (C) $\left[-\frac{\pi}{2}, -\frac{3\pi}{10}\right]$ (D) $\left[\frac{3\pi}{10}, \frac{\pi}{2}\right]$

499 Two roots of a biquadratic $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ have their product equal to (-32) . Find the value of k .

500 If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d , then find the value of $\frac{a+b+c+d}{10}$ (a, b, c and d are distinct numbers)

501 Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $x^4 + 4x^2 + 28x + 5$, find the value of $P(1)$.

502 If $x^2 + (a-b)x + (1-a-b) = 0$ where $a, b \in \mathbb{R}$ then least integral value of 'a' for which equation has unequal real roots for all values of 'b', is

503 The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

504 Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. If equation $P(x) \cdot Q(y) = 28$ is satisfied for $x = \alpha$ and $y = \beta$, then $4(\alpha + \beta)$ is

- 505 Number of non positive integral solutions of equation $2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$ ($x \in \mathbb{R}$), is

REVISION ASSIGNMENT # 26**(QUADRATIC EQUATION)****MATHEMATICS**

- 506 The sum of the values of m for which the quadratic polynomial $P(x) = x^2 + (m+5)x + (5m+1)$ is a perfect square ($m \in \mathbb{R}$) is
 (A) 3 (B) 7 (C) 8
- 507 The value of c , for which $|\alpha^2 - \beta^2| = \frac{c}{4}$, where α and β are roots of $2x^2 - 7x + c = 0$, is
 (A) 2 (B) 6 (C) 5
- 508 The roots of $5x^2 - 7x + k = 0$ are $\sin A$ and $\cos A$ the value of k is -
 (A) $\frac{12}{5}$ (B) $\frac{49}{10}$ (C) 7
- 509 If $\frac{\alpha}{1} = \frac{\beta}{2} = \frac{\gamma}{3}$, then $\log_{\frac{7}{12}} \left(\frac{\gamma^2 + \beta^2 - \alpha^2}{\alpha\gamma + \beta\gamma - \alpha\beta} \right)$ is (where $\alpha, \beta, \gamma \neq 0$)
 (A) 5 (B) 0 (C) -1
- 510 The equations $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$, where $b^2 - 4ac > 0$, have a common root if
 (A) $a^3 + b^3 + c^3$ is equal to $3abc$ (where a, b, c are non zero distinct quantities)
 (A) $3abc$ (B) abc (C) 0

511 If sum of the roots of quadratic equation $ax^2 + bx + c = 0$ is equal to sum o

• then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to -

- (A) 2 (B) -2 (C) 1

512 If $x^{\log_{10} x} = 10000$, then number of rational values of x is -

- (A) 1 (B) 2 (C) 3

513 The value of $\log_{\sqrt{5}-\sqrt{3}} \sqrt{124-32\sqrt{3}\sqrt{5}} + 27^{\frac{1}{\log_{16} 9}}$ is equal to -

- (A) 68 (B) 62 (C) 66

514 If $a + b = 3 - \cos\theta$ and $a - b = \sin\theta$, then maximum value of 'ab' is -

- (A) 1 (B) 4 (C) 8

515 If $\tan(2\alpha - \beta) = \frac{1}{2}$ and $\tan(\alpha - 2\beta) = \frac{1}{3}$, then possible value of $(\alpha - \beta)$

- (A) 15° (B) 30° (C) 45°
(D) 10

$x^2 + 7x + c = 0$, is-

- (D) 4

- (D) 1

- (D) 2

$\neq 0$ have a common root, then

- (D) 1

f the squares of their reciprocals,

- (D) -1

- (D) 0

- (D) 64

- (D) 10

) is -

- (D) 60°

516 Least positive solution of equation $4\cos^\circ x + 3\sin^4 x = 4\sin^\circ x + 3\cos^4 x$ is

- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{6}$

517 The number of solutions for $x \in (-2\pi, 2\pi)$ of the equation $(\sqrt{3} \cos x + 1)$

- (A) 0 (B) 2 (C) 4

518 Number of solutions of equation $2\cos^2 \theta - \sqrt{2} \sin^2 \theta = 1$ in $[0, 2\pi]$ is

- (A) 0 (B) 1 (C) 2

519 . Number of integral solution of equation $\tan\left(\frac{\pi}{2}[x]\right) = 0$ in $x \in \left[-\frac{9\pi}{2}, \frac{11}{2}\right]$

integer function) is -

- (A) 20 (B) 15 (C) 16

520 . The sum of maximum & minimum values of θ satisfying all the inequa

$\sin\theta < 0, -1 < \cos\theta < -\frac{1}{2}$ and $\frac{1}{\sqrt{3}} \leq \tan\theta \leq 1$ in $\theta \in (0, 5\pi)$ is -

- (A) $\frac{33\pi}{12}$ (B) $\frac{49\pi}{12}$ (C) $\frac{53\pi}{12}$

521 . If x, y, z are three real number such that $x + y + z = 4$ and $x^2 + y^2 + z^2 =$

- (A) $x \in \left[\frac{2}{3}, 2\right]$ (B) $y \in \left[\frac{2}{3}, 2\right]$ (C) $z \in (2, 3)$

- 522 If for the quadratic equation $ax^2 + bx + c = 0$, $a > 0$, $a, b, c \in \mathbb{R}$, (a
 $\cdot$ $2a + b < 0$, then -
 (A) quadratic equation has two distinct real solutions.
 (B) exactly one root of the quadratic equation lies in $(-2, 1)$
 (C) atleast one root of the quadratic equation lies in $(-2, 2)$
 (D) exactly one root of the quadratic equation lies in $(1, \infty)$

Space for Rough Work

is -

(D) $\frac{\pi}{4}$

$\sin x \Big)^{\sqrt{3 \sin 2x + \cos 2x + 2}} = 4$ is -

(D) 8

which are solutions of equation

(D) 4

$\frac{\pi}{2} \Big]$ (where $[.]$ represents greatest

(D) 29

lities ;

(D) $\frac{73\pi}{12}$

rong answer)

6, then-

(D) $y \in (2, 3)$

$+ b + c)(4a - 2b + c) < 0$ and

-
- 523 Suppose 'a' and 'b' are integers and $b \neq -1$. If the quadratic equation x^2
 $\cdot$ integral root, then which of the following is/are true-
 (A) the other root is also a positive integer
 (B) the other root is an integer
 (C) $a^2 + b^2$ is a prime number
 (D) $a^2 + b^2$ has a factor other than 1 and itself.

- 524 Let $x^2 + b_1x + c_1 = 0$ has two real roots α, β and $x^2 + b_2x + c_2 = 0$ has two real roots γ, δ . b_1, b_2, c_1, c_2 are real numbers (where $c_1 \neq c_2 \neq 0$). Which of the following is true?
- (A) $b_1^2 - b_2^2 = 4(c_1 - c_2)$ (B) $c_2 - c_1 = 4 - 2b_1$
 (C) $b_1^2 - b_2^2 = 4(c_1 + c_2)$ (D) $c_2 - c_1 = 4 + 2b_1$

- 525 Value of $(\tan 11^\circ + \tan 13^\circ + \tan 11^\circ \tan 34^\circ + \tan 32^\circ + \tan 34^\circ + \tan 32^\circ \tan 34^\circ)$ is
- (A) prime number (B) composite number
 (C) even number (D) odd number

- 526 Let in ΔABC , $\cos A = \frac{3}{\sqrt{10}}$ and $\cos B = \frac{6}{\sqrt{61}}$, then -
- (A) ΔABC is an acute angled triangle (B) ΔABC is an obtuse angled triangle
 (C) $\tan A + \tan B + \tan C > 2$ (D) $\tan A + \tan B + \tan C < 2$

- 527 If the equation $a \cos x - \cos 2x = 2a - 7$ possess a solution, then a can be
- (A) 2 (B) 4 (C) 6

Linked Comprehension Type (Single Correct Answer Type) (3 Marks)

- 528 Consider an equation $(x - a)(x - b)(x - c) + (x - 1)(x^2 + x + 1) = 0$
- The roots of equation $2(x - \alpha)(x - \beta)(x - \gamma) + 1 - x^3 = 0$ are-
- (A) a, b, c (B) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ (C) $-a, -b, -c$

529 Consider an equation $(x - a)(x - b)(x - c) + (x - 1)(x^2 + x + 1) =$

- Sum of roots of the equation $2(x - \alpha)(x - \beta)(x - \gamma) - (x - a)(x - b)(x - c) = 0$ is-
- (A) -1 (B) 1 (C) 0

Space for Rough Work

$x^2 + ax + b + 1 = 0$ has a positive

3 real roots $\alpha + 2, \beta + 2$ where γ is/are true-

$\sin 13^\circ$ is

3 angled triangle

$C < 1$

3 equal to-

(D) 7

3 each, -1 for wrong answer)

0 whose roots are α, β, γ .

(D) $-\frac{1}{a}, -\frac{1}{b}, -\frac{1}{c}$

$(x - c) = 0$, is-

(D) -2

530 Match List-I with List-II and select the correct answer using the codes given below.

List-I

P. $(1 + \tan 1^\circ)(1 + \tan 7^\circ)(1 + \tan 38^\circ)(1 + \tan 44^\circ)$ is equal to

Q. If $a, b, c, d \in [0, \pi]$ and $\sin a + 7\sin b = 4(\sin c + 2\sin d)$

$$\cos a + 7\cos b = 4(\cos c + 2\cos d), \text{ then } \frac{2\cos(a-d)}{\cos(b-c)} + 1$$

is equal to (where $|b-c| \neq \frac{\pi}{2}$)

R. Number of integers in the range of $f(x) = \frac{18\cos^4(2x)}{\cos^4 2x + 5}$ is equal to

S. If $x^{\ell n 2} = y^{\ell n 5}$ and $25^{\ell n x} = 4^{\ell n y}$, then $x + y$ is equal to

Codes :

	P	Q	R	S
(A)	2	4	2	1
(B)	2	3	4	1
(C)	3	4	2	1
(D)	1	4	3	2

531 **List-I**

P. If $4\cos 25^\circ \cos 85^\circ \cos 35^\circ = \cos x$, $x \in \left(0, \frac{\pi}{2}\right)$ and $x = \frac{a\pi}{b}$

(where a & b are coprime numbers),
then value of $(5b - 9a)$ is

Q. Number of integers in range of $(8\sin x + 5\cos x)$ is

R. If $\sin x + \cos x = \frac{1}{3}$, then value of $9(\sin x - \cos x)^2$ is

S. Radius of two circles are 3 and 4 respectively
and distance between centres is 9. If length of
internal common tangent to these circles is k ,

then value of $\left(\frac{k^2}{2}\right)$ is

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	4	2	3	3
(C)	3	4	1	2
(D)	1	3	2	3

532 4

533 6

534 8

.

List-II

535 17

.

536 16

.

537 15

.

538 19

.

539 The value of $\tan^{-1} \frac{1}{16} + \tan^{-1} \frac{1}{16} + \tan^{-1} \frac{1}{16} + \dots + \tan^{-1} \frac{1}{16}$ is

.

540 If $15 \sin^4 \theta + 10 \cos^4 \theta = 6$ and $\left| \frac{\tan 3\theta}{\tan \theta} \right| = \frac{p}{q}$, where p & q are relatively

.

541 Number of solutions of $\sqrt{2 \sin^2 x + \sin x} + \sqrt{2 \sin^2 x + \sin x + 15} = 5$ is

.

542 If $b = \left(3 + \cot \frac{\pi}{8} + \cot \frac{11\pi}{24} - \cot \frac{5\pi}{24} \right)$, then the value of $|b\sqrt{2}|$ is

.

543 Least positive integral value of k for which the equation $e^{2x} + (1 - 2)$ any real solution, is (where e is Napier's constant)

.

544 If $(\alpha + 1)^2 x^2 + 2(\alpha + 1)(\alpha - 2)x + 4 > 0 \forall x \in \mathbb{R}$, then sum of all

.

545 The number of integral value(s) of $a \in (8, \infty)$ for which the expression non-negative for all $x \in [1, 3]$ is

.

- 546 If $x^{(\log_2 x)^2 - 6\log_2 x + 11} = 64$ and sum of values of x satisfying the equation is $\frac{a}{b}$ where a and b are coprime, then value of a+b is

REVISION ASSIGNMENT # 13 (GRP 1.0) (SOT AND SEQUENCE)

547. $\frac{-4}{-7)^6} \leq 0$ is -

(D) 4

2) $x - a - 1 = 0$ is least is -

(D) - 1

if they differ by atmost 4 then the

(D) 8

(D) none of these

then -

(D) $a > 4$

0 will lie between - 2 and 4 is -

(D) 1

than the roots of the equation,

if $A + B + C =$

(D) none

$\therefore$ is equal to -

(D) $-\frac{2}{5}$

equal to -

(D) 4

(D) $[-18, -2)$

E:\NODE6 (E)\Data\2014\Kota\JEE-Advanced\SMP\Maths\Unit#01\Eng\02. Quadratic\2 EXERCISES

$$b) \cup (1, \infty)$$

$$c) \quad (D) (-\infty, -4) \cup [1, \infty)$$

CORRECT ANSWERS)

(D) none of these

Equation will be real if -

$$a > 0$$

$$a < 0$$

and -

$$(D) a < 0$$

3

$$\frac{5 + \sqrt{13}}{2}$$

$$(D) 5$$

$$\frac{1}{100} \text{ and } \frac{1}{\sqrt{10}}$$

Value of x is $\frac{1}{100}$

INCORRECT ANSWERS)

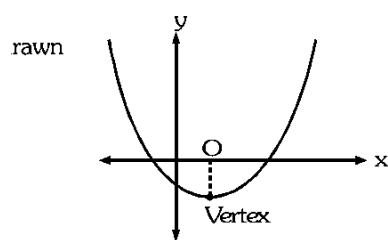
(D) $x^2 + 3x + 3 = 0$

∴ value of $\sin 2\alpha$ is -

(D) $24 / 25$

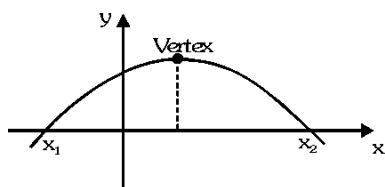
opposite in sign, then -

.



∴ then the roots of the quadratic

(D) imaginary



(D) none of these

and for all x in the interval $1 \leq x \leq 3$

(D) $(-2, 3)$

(x) form an arithmetic progression.

(D) $\frac{128}{1323}$

real roots, then the least value of

(D) 1

$\cdot p\} = (p - q)^2 + (x - p)^2 + (x - q)^2$

of x (D) infinite values of x

ect square, is -

$[\sqrt{3}, \infty)$

$1) = 0$ are -

is $K \in (-\infty, 0) \cup (1, \infty)$

2 is $K \in (2, \infty)$

$\frac{x_2}{x_1} < -2$

$c \in \mathbb{R}$, then $P(x).Q(x)$ has at least

E:\NODE6 (E)\Data\2014\Kota\JEE-Advanced\SMP\Maths\Unit#01\Eng\02. Quadratics\2. EXERCISES

finite then $a = 0$ & $b \in \mathbb{R}$

$(x - \alpha_4)(x - \alpha_6) = 0$ has -

$\infty, \alpha_1)$

$\alpha_5, \alpha_6)$

er root greater than 'a', if

(D) $a < -1$

$-4x - x^2$ has coincident roots,

12 (D) $b = -4$ or $b = 12$

$\rho^2 = 0$ such that unity lies between

(D) $(-1, 1)$

& $\tan C$ where A, B, C are the

(D) $\frac{p+r}{1+q-s}$

(D) $(10, \infty)$

en the roots of quadratic equation

0 has real roots.

$x^2 + bx + c$, to take both positive

.....

... ..

$+ \frac{a\beta^2}{b\beta + c}$ is equal to

o be matched. The statements in
abelled as p, q, r and s. Any given
ement(s) in **Column-II**.

meter. Match of the real values

Column-II

$$\left(-\infty, -\frac{8}{7}\right)$$

$$(-1, 4)$$

$$\left(-\frac{4}{3}, -\frac{8}{7}\right)$$

$$\left(-\infty, -\frac{4}{3}\right)$$

Quadratic\2 EXERCISES

i).

ination for Statement-I.

t explanation for statement-I

+ 4 = 0 have a common root,

ents then $p - iq$ will be the other

(D) D

-1 is a root of $f(x) = 0$ then the

$$f \text{ roots} = \frac{c}{a}$$

(D) D

the equation

ative.

st of roots > 0

(D) D

6, 7, -10). Then the equation

e sum of coefficients is zero) then

E:\NODE6 (E)\Data\2014\Kota\JEE-Advanced\SMP\Maths\Unit#01\Eng\02.

(D) D

ive integers, then $b^2 - 4c = 1$. $x^2 + (b^2 - 4ac)x - b = 0$ are real

(D) D

ne equation whose roots are $f(\alpha)$,
 $= g(y)$ (some function of y).
 in the desired equation which is

find equation whose roots are

ons :

of the equation

$$(D) \frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$$

ose roots are the reciprocals of

$$= 0$$

$$= 0$$

he equation whose roots are

)

)

of $\Pi\left(\frac{1+\alpha}{1-\alpha}\right)$ is equal to -

(D) -1

$$a^2 - b = 1$$

$$\sqrt{b} = (a + \sqrt{b})^{-1}$$

ons :

(D) $1 \pm \sqrt{2}$, 2

$(\sqrt{14})^t = 30$ are -

(D) 6

∴ $a = x^2 - 3$, then x is -

(D) 2

quadratic\2 EXERCISES

∴ roots are $\alpha^3 - 3\alpha^2 + 5\alpha - 2$,

ve that $b^3 + a^2c + ac^2 = 3abc$.

∴ of the equations $x^2 + px + q = 0$,

uation $x^2 - 2cx - 5d = 0$. If c and

nerical values of $a + b + c + d$.

$.8x + 45$

ays negative.

non root, prove that the equation

ve one common root & the second

satisfied for all $x \in \mathbb{R}$.

$\cdot 9a^2 = 0$ are greater than 3.

quation $x^2 - ax + 2 = 0$ belong to

$\delta = 0$ has atleast one positive root.

$-b \pm \sqrt{b^2 - 4ac}$ are real & distinct.

product equal to (-32) . Find the value

CISES

wer of the other, then show that

or of both $x^4 + 6x^2 + 25$ and

$x^2 - 2 = 0$ has real roots.

is then prove that the roots of the

$x^4 + 36 = 0$ have a common positive

the interval $(-2, 2)$ then prove that

$- 3.$

$= 0.$

$\alpha_2, \alpha_3, \dots, \alpha_n.$

$(-x^2)$ has only one solution.

the equation $x^2 + px + q = 0$ are

[AIEEE-2002]

6

56

en roots of the equation $(x - \alpha)$

[AIEEE-2002]

(4) $a + b$ and $b + c$

s-

[AIEEE-2002]

(4) none of these

 $x^2 + (3a - 1)x + 2 = 0$ is twice

[AIEEE-2003]

(4) $2/3$

ual to the sum of the square of

[AIEEE-2003]

on

.on

[AIEEE-2003]

(4) 2

f the sum at x equal to-

[AIEEE-2003]

(4) 2

se numbers are the roots of the

[AIEEE-2004]

0

)

its roots are-

[AIEEE-2004]

(4) $-1, 2$

ises

$x^2 + px + q = 0$ has equal roots,
[AIEEE-2004]

(4) 4

$x^2 - (a - 2)x - a - 1 = 0$ assume
[AIEEE-2005]

(4) 1

then $b^2 - 4c$ equals-

[AIEEE-2005]

(4) -2

less than 5, then k lies in the
[AIEEE-2005]

(4) (5, 6)

one root $x = \alpha$, then the equation
[AIEEE-2005]

is equal to α

$m^2 - 1 = 0$ are greater than -2 but
[AIEEE-2006]

(4) $m > 3$

is 15, respectively then the value
[AIEEE-2006]

(4) 3

[AIEEE-2006]

(4) 41

less than $\sqrt{5}$, then the set of
[AIEEE-2007]

E:\NODE6 (E)\Data\2014\Kota\JEE-Advanced\SMP\Maths\Unit#01\Eng\02. Quadratic\2 EXERC

1 common. The other roots of the
 ot is [AIEEE-2008]

(4) 2

real values of x , the expression
 [AIEEE-2009]

$109 =$ [AIEEE-2010]

(4) 2

$=f(x) - g(x)$. If $p(x) = 0$ only for
 [AIEEE-2011]

(4) 6

take in writing down the constant
 coefficient of x to get roots (3, 2).
 [AIEEE-2011]

(4) -6, -1

[AIEEE-2012]

real roots.

st.

he equation $ax^2 + bx + c = 0$.

[JEE 2001, Mains, 5 out of 100]

[JEE 2002 (screening), 3]

or which equation has unequal real

2003, Mains-4 out of 60]

then

$$= 0$$

$$= 0$$

$$2 < a < 5$$

[JEE 2004 (Screening)]

[JEE 2005(Mains), 2]

and $\lambda \in \mathbb{R}$. If the roots of the

$$(D) \lambda \in \left(\frac{4}{3}, \frac{5}{3} \right)$$

[JEE 2006, 3]

$$x^2 - 10ax - 11b = 0 \text{ are } c, d,$$

rs) [JEE 2006, 6]

2β be the roots of the equation

statements in Column II.

Advanced\SwP\Maths\Unit#01\Eng\02. Quadratic\2 EXERCISES

1

[JEE 2007, 3+6]

equation $x^2 + 2px + q = 0$ and
1}

nation for Statement-1

explanation for Statement-1

[JEE 2008, 3 (-1)]

$6(k^2 - k + 1) = 0$ are real, distinct

[JEE 2009, 4 (-1)]

β are nonzero complex numbers

$\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

[JEE 2010, 3]

$(p^3 + q) = 0$

$(p^3 - q) = 0$

≥ 1 , then the value of $\frac{a_{10} - 2a_8}{2a_9}$

[JEE 2011]

(D) 4

[JEE 2011]

(D) $\sqrt{2}$

E:\NODE6 (E)\Data\2014\Kota\JEE-

$$548 \quad x - 2 \geq 0 \Rightarrow x \geq 2$$

$$\cdot \quad x - 2 > 0$$

or

$$1)(x - 2) \geq 0 \Rightarrow x \leq 1$$

$$x - 2 < 0$$

inequation is $x \in \mathbb{R} - (1, 2]$



$$0 < x < a^p \quad (b) \log_a x > p \Rightarrow x > a^p$$

$$x > a^p \quad (b) \log_a x > p \Rightarrow 0 < x < a^p$$

$$0 < \log_4 (x^2 - 5) < 1$$

$$x^2 - 5 > 1$$

$$0 < x^2 - 5 < 4 \Rightarrow 1 < (x^2 - 5) < 4$$

45102. QUADRATIC THEORY

$$\in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

$$\text{tion : } x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

$$\frac{2}{\log_2 x - 1} \cdot$$

$$\leq 0$$

$$\frac{t(t+1)}{-1} \leq 0$$

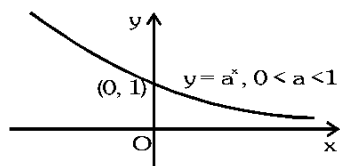
$$:]$$

$$\log_{2x+3}(2x+3)$$

collection of the systems

$$\begin{aligned} & < 0 \text{ \& } x \neq 0 & \Rightarrow \begin{cases} \begin{cases} x > -1 \\ -1 < x < 3 \end{cases} & \Rightarrow -1 < x < 3 \text{ \& } x \neq 0 \\ \text{or} & \text{or} \\ \begin{cases} -\frac{3}{2} < x < -1 \\ x < -1 \text{ or } x > 3 \end{cases} & \Rightarrow -\frac{3}{2} < x < -1 \end{cases} \\ & > 0 \end{aligned}$$

$$\text{tion is } x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$$



1

: $2 > 1$, we have $x + 2 > -\frac{2}{x}$ (the sign of the inequality is

$$\frac{x+2}{x} > 0$$

$$\equiv (0, \infty)$$

$$\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$$

ality is equivalent to the inequality $x - 1 > 4(1 + \sqrt{x})$

.....(i)

both sides

$$x > 0$$

$$x - 1 > 0$$

$$(b) \log_7 \left(\frac{2x-6}{2x-1} \right) > 0$$

RATIONAL ALGEBRAIC EXPRESSIONS :

$$\frac{b_1}{x + c_2}, \frac{a_1x^2 + b_1x + c_1}{a_2x + b_2} :$$

form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$. The following procedure is used :

$$y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$$

obtain a quadratic equation in x.

$$= 0$$

inequality for possible set of values of y.

$$x^2 - 3x + 4$$

NOR66 [E:\DATA\2014\KOTA\JEE-ADVANCED\SRP\MATHS\UNIT#01\ENG\02_ QUADRATIC\ THEORY

inable by $\frac{\quad}{x^2 + 3x + 4}$.

atic equation.

$$\Rightarrow 7y^2 - 50y + 7 \leq 0$$

$$\left[\frac{1}{7}, 7\right] - \{1\}$$

pression $\frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$ assumes all real values for real values

$$0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$$

)

$$16a \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$$

$$a \leq 7$$

549 of a

.

+

550 +

$$x + -1 \rightarrow y + -1$$

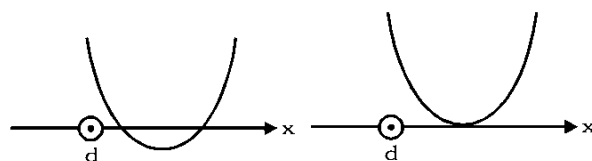
- some real values of x .

Solving problems on quadratic equations when the roots are
of constraints :

0 with $a > 0$ and let $f(x) = ax^2 + bx + c$

: greater than a specific number (say d).

:

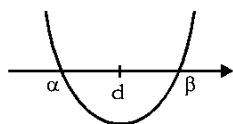


are less than a specific number d than the necessary and

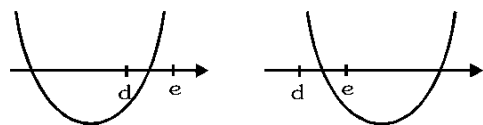
(d). Alternatively one root is greater than ' d ' and other root
equation.

: $f(d) < 0$

ed.

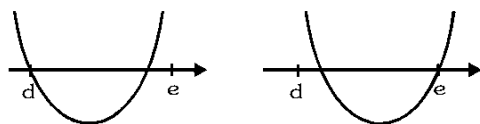


:



then

1.



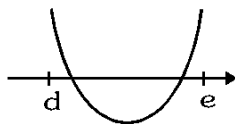
BOOK

$$f(d) = 0$$

$$f(e) = 0$$

for d and e ($d < e$).

:
0



less than d ($d < e$).

: $f(d) < 0$ and $f(e) < 0$

If $c = 0$ then we divide the whole equation by 'a'. Now assume

x^2 positive and hence above cases are applicable.

for 'a' for which the roots of the quadratic equation

(ii) equal

(iv) equal in magnitude but opposite in sign

(vi) negative

(viii) smaller than 3

the interval $(1, 3)$

$x^2 + Bx + C$ (say)

5.

$$+ 5) = 4(a + 1)(a - 4)$$

$$a \in (-\infty, -1) \cup (4, \infty).$$

$$a = -1, 4.$$

on the roots of the given equation.

$$a \in (-\infty, -1) \cup (4, \infty)$$

$$> a \in (-\infty, -5).$$

NOOR (B) DATA\2014\NOTA\EE-ADVANCED\SWP\MATHS\UNIT601\ENG\02. QUADRATIC\1.PP

the roots is zero

$$\text{i.e. } a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a = 1$$

$$-1) \cup (4, \infty)$$

roots are greater than zero

$$\text{i.} \quad \Rightarrow \quad -(a - 1) > 0, a + 5 > 0, a \in (-\infty, -1) \cup (4, \infty)$$

$$-1) \cup (4, \infty) \Rightarrow a \in (-5, -1].$$

roots are less than zero

$$\text{i.} \quad \Rightarrow \quad -(a - 1) < 0, a + 5 > 0, a \in (-\infty, -1) \cup (4, \infty)$$

$$-1) \cup (4, \infty) \Rightarrow a \in [4, \infty).$$

0.

$$\text{) and } a \in (-\infty, -1) \cup (4, \infty)$$

$$\in (-\infty, -1) \cup (4, \infty)$$

By these conditions simultaneously, there can be no value of a which is greater than 3.

0.

$$\in (-\infty, -1) \cup (4, \infty) \Rightarrow a \in (-8/7, -1) \cup (4, \infty)$$

$$3) > 0, D \geq 0.$$

$$4 > 0, 7a + 8 > 0, a \in (-\infty, -1) \cup (4, \infty)$$

$$a > -8/7, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in \left(-\frac{8}{7}, -1\right]$$

equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is

$$-k - 6 < 0$$

3

exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval $(0,3)$.

$$2a(-a + 6) < 0$$

$$> 6$$

$(0, 3)$

$\alpha + c = 0$ and β is a root of the equation $-ax^2 + bx + c = 0$,

if the equation $\frac{a}{2}x^2 + bx + c = 0$ lying between α and β .

$$-\frac{a}{2}\alpha^2$$

$$bx + c = 0)$$

$$b\beta + c + \frac{3a}{2}\beta^2$$

$$+ bx + c = 0)$$

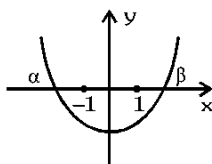
between α and β .

If 0 has two real roots α and β where $\alpha < -1$ and $\beta > 1$, then

$$< 0$$

$$\left) - \frac{b^2}{a^2} > 0$$

$$\{\alpha\beta < -1 \Rightarrow \frac{c}{a} < -1\}$$



NOB6 (E:\DATA\2011\NOTA\REF-ADVANCED\SAP\MATHS\UNIT10\1.ENG\02. QUADRATIC\1. THEORY

TWO VARIABLES :

may be resolved into two linear factors if ;

$$\begin{vmatrix} h & g \\ b & f \\ f & c \end{vmatrix} = 0$$

If linear factor then m is equal to -

$$(C) 6, -2$$

$$(D) -6, -2$$

$$n/2, c = -3$$

$$= 0$$

$$0 \Rightarrow -\frac{m^2}{4} + m + 3 = 0$$

$$m = -2, 6$$

Ans. (C)

Equation $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0,$$

$$f_1(x - \alpha_2) \dots (x - \alpha_n)$$

we get

$$\frac{1}{x^n}$$

taken k at a time.

quadratic equation $ax^2 + bx + c = 0$, then

equation $ax^3 + bx^2 + cx + d = 0$, then

then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$

actly n root & if the equation has more than n roots, it is an

all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e.
pairs.

real & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root

uch that $f(a)$ & $f(b)$ are of opposite signs, then $f(x)=0$ must
'b'.

least one real root of a sign opposite to that of its last term.

ynomial equation $f(x) = 0$ is the number of changes of signs

has atmost one positive real root.

polynomial equation $f(x) = 0$ is the number of changes of

ts.

of $4x^3 + 20x^2 - 23x + 6 = 0$.

$$2\alpha + \beta = -5 \quad \dots\dots\dots (i)$$

$$\alpha^2 + 2\alpha\beta = -\frac{23}{4} \quad \& \quad \alpha^2\beta = -\frac{6}{4}$$

$$\Rightarrow \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \Rightarrow 12\alpha^2 + 40\alpha - 23 = 0$$

NCERT (B) DATA 2014/2015

$$5 - 2x\left(-\frac{23}{6}\right) \neq -\frac{3}{2} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -6$$

6

Ans.

$px - r = 0$, find :

$$3 + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$$

$$\gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\}$$

$$3pq$$

$$) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma)$$

$$pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$$

rove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

$$+ cx + d = 0$$

$$\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

possible if all α, β, γ are real. So atleast one root is non-real,

ice given cubic equation has two non-real and one real roots.

equation $f(x) \equiv x^4 + qx^2 + rx - s = 0$ has one positive, one

are positive

s is positive

e & two roots imaginary.

the equation has atmost one positive root.

gn, the equation has atmost only one negative root.

st, one positive root & two imaginary roots.

wo roots α and β . If we have to find an equation whose roots
in this equation can be found by finding α in terms of y . Now
if y directly in the equation.

α and β , then find the equation whose roots are :

$$\frac{\beta}{1}, \frac{\beta}{\beta+1} \quad (c) \quad \alpha^2, \beta^2$$

$$\Rightarrow cy^2 - 2by + 4a = 0$$

$$ax + 4a = 0$$

$$c = 0 \Rightarrow (a + c - b)y^2 + (-2c + b)y + c = 0$$

$$a) x^2 + (b - 2c)x + c = 0$$

$$= 0$$

$$(ac - b^2)x + c^2 = 0$$

are α, β, γ then find equation whose roots are $\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}$.

$$x\beta\gamma = -\frac{d}{a}$$

than three, leaves remainders 2, 1 and -1 when divided, $(x + 1)$. What will be the remainder when it is divided by

N0066 [E:\DATA\2014\KOTA\EE-ADVANCED\SRM\MAHS\UNIT01\ENG\02. QUADRATIC\ THEORY

$$p(x) = (x-1)(x+2)(x+1) + a_0x^2 + a_1x + a_2$$

$$f(-2) = 1, f(-1) = -1.$$

$$a_2 = \frac{2}{3}$$

$$(x-1)(x+2)(x+1)$$

$q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$,
 $-\delta$ in terms of p, q, r and s . Deduce the condition that

$\therefore$

$$= 0$$

$$\dots\dots\dots(1)$$

$$+ s = 0$$

$$\dots\dots\dots(2)$$

)

$$\beta(\gamma + \delta) + \gamma\delta]$$

i)

$$+ s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2$$

$$+ s((\alpha + \beta)^2 - 2\alpha\beta) + sr(\alpha + \beta) + s^2$$

$$1) + sr(-p) + s^2$$

$$- prs$$

$$[p - r)$$

$$\beta = \delta) \dots\dots\dots(3)$$

$$i) = 0 \dots\dots\dots(4)$$

$$0$$

which is the required condition.

all $x \in \mathbb{R}$, then find the interval in which y lies.

$\forall x \in \mathbb{R}$

$$4p^2 \geq 0$$

$(x^2 + x + 1)$ is -2 . So,

$$y^2 - 5y + 5 < 0$$

DO YOURSELF

(ii) $a, \frac{1}{a}$; (iii) $\frac{7}{3}$ (iv) $3, -\frac{1}{5}$

(b) real & distinct; (c) real & coincident

negative, $c \rightarrow$ negative

$(2, -1)$; (c) $\left(-\frac{1}{4}, \frac{1}{3}\right]$

(e) $[3, 7]$; (f) ϕ

≥ 2

> 0 (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$

> 0 (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$

$= 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$

(ii) $x \in \mathbb{R} - (0, 1]$

(ii) $x \in (-1, 1)$

(iii) $a < 2$; (iv) $a < 2$; (v) $12 < \lambda < 16$

(b) $-\frac{c}{d}$, (c) $\frac{1}{a^2}(3ad - bc)$

$cx^2 - bx + 1 = 0$; (c) $acx^2 + (a + c)bx + (a + c)^2 = 0$

$p(p^2 - q) = 0$

551 INTRODUCTION :

The algebraic expression of the form $ax^2 + bx + c$, $a \neq 0$ is called a quadratic expression, because the highest order term in it is of second degree. Quadratic equation means, $ax^2 + bx + c = 0$. In general whenever one says zeroes of the expression $ax^2 + bx + c$, it implies roots of the equation $ax^2 + bx + c = 0$, unless specified otherwise.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

552 SOLUTION OF QUADRATIC EQUATION & RELATION BETWEEN ROOTS & CO-EFFICIENTS :

(a) The general form of quadratic equation is $ax^2 + bx + c = 0$, $a \neq 0$.

The roots can be found in following manner :

$$a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = 0 \Rightarrow \left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$$

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

This expression can be directly used to find the two roots of a quadratic equation.

(b) The expression $b^2 - 4ac \equiv D$ is called the discriminant of the quadratic equation.

(c) If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then :

$$(i) \alpha + \beta = -b/a \quad (ii) \alpha\beta = c/a \quad (iii) |\alpha - \beta| = \sqrt{D}/|a|$$

(d) A quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e.

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \text{ i.e. } x^2 - (\text{sum of roots})x + \text{product of roots} = 0.$$

Illustration 1 : If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is -

$$(A) x^2 + 4x + 1 = 0 \quad (B) x^2 - 4x + 4 = 0 \quad (C) x^2 - 4x - 1 = 0 \quad (D) x^2 + 2x + 3 = 0$$

Solution : Since α, β are the roots of equation $x^2 - 3x + 5 = 0$

$$\text{So } \alpha^2 - 3\alpha + 5 = 0$$

$$\beta^2 - 3\beta + 5 = 0$$

$$\therefore \alpha^2 - 3\alpha = -5$$

$$\beta^2 - 3\beta = -5$$

Putting in $(\alpha^2 - 3\alpha + 7)$ & $(\beta^2 - 3\beta + 7)$ (i)

$$-5 + 7, -5 + 7$$

$\therefore$ 2 and 2 are the roots.

$\therefore$ The required equation is

$$x^2 - 4x + 4 = 0.$$

Ans. (B)

Illustration 2 : If α and β are the roots of $ax^2 + bx + c = 0$, find the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$.

Solution : We know that $\alpha + \beta = -\frac{b}{a}$ & $\alpha\beta = \frac{c}{a}$

$$\begin{aligned} (a\alpha + b)^{-2} + (a\beta + b)^{-2} &= \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \\ &= \frac{a^2\beta^2 + b^2 + 2ab\beta + a^2\alpha^2 + b^2 + 2ab\alpha}{(a^2\alpha\beta + ba\beta + ba\alpha + b^2)^2} = \frac{a^2(\alpha^2 + \beta^2) + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} \\ (\alpha^2 + \beta^2 \text{ can always be written as } (\alpha + \beta)^2 - 2\alpha\beta) \\ &= \frac{a^2[(\alpha + \beta)^2 - 2\alpha\beta] + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} = \frac{a^2\left[\frac{b^2 - 2ac}{a^2}\right] + 2ab\left(-\frac{b}{a}\right) + 2b^2}{\left(a^2\frac{c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^2} = \frac{b^2 - 2ac}{a^2c^2} \end{aligned}$$

Alternatively :

If α & β are roots of $ax^2 + bx + c = 0$

then, $a\alpha^2 + b\alpha + c = 0$

$$\Rightarrow a\alpha + b = -\frac{c}{\alpha}$$

$$\text{same as } a\beta + b = -\frac{c}{\beta}$$

$$\begin{aligned} \therefore (a\alpha + b)^{-2} &= (\alpha\beta + b)^{-2} = \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} \\ &= \frac{(-b/a)^2 - 2(c/a)}{c^2} \\ &= \frac{b^2 - 2ac}{a^2c^2} \end{aligned}$$

553 **NATURE OF ROOTS :**

(a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then ;

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

- (i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
- (ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal)
- (iii) $D < 0 \Leftrightarrow$ roots are imaginary.
- (iv) If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa. ($p, q \in \mathbb{R}$ & $i = \sqrt{-1}$).

(b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then ;

- (i) If D is a perfect square, then roots are rational.
- (ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd) then other root will be $p - \sqrt{q}$.

Illustration 3 : If the coefficient of the quadratic equation are rational & the coefficient of x^2 is 1, then find the equation one of whose roots is $\tan \frac{\pi}{8}$.

Solution : We know that $\tan \frac{\pi}{8} = \sqrt{2} - 1$

Irrational roots always occur in conjugational pairs.

Hence if one root is $(-1 + \sqrt{2})$, the other root will be $(-1 - \sqrt{2})$. Equation is

$$(x - (-1 + \sqrt{2}))(x - (-1 - \sqrt{2})) = 0 \quad \Rightarrow \quad x^2 + 2x - 1 = 0$$

Illustration 4 : Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Solution : Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. Since integral roots will always be rational it means D should be a perfect square.

From (i) $D = a^2 - 20a + 96$.

$$\Rightarrow D = (a - 10)^2 - 4 \Rightarrow 4 = (a - 10)^2 - D$$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$$\Rightarrow (a - 10) = \pm 2 \quad \Rightarrow \quad a = 12, 8$$

Ans.

554 **ROOTS UNDER PARTICULAR CASES :**

Let the quadratic equation $ax^2 + bx + c = 0$ has real roots and

- (a) If $b = 0 \Rightarrow$ roots are equal in magnitude but opposite in sign
- (b) If $c = 0 \Rightarrow$ one root is zero other is $-b/a$
- (c) If $a = c \Rightarrow$ roots are reciprocal to each other
- (d) If $\begin{cases} a > 0, c < 0 \\ a < 0, c > 0 \end{cases} \Rightarrow$ roots are of opposite signs
- (e) If $\begin{cases} a > 0, b > 0, c > 0 \\ a < 0, b < 0, c < 0 \end{cases} \Rightarrow$ both roots are negative.
- (f) If $\begin{cases} a > 0, b < 0, c > 0 \\ a < 0, b > 0, c < 0 \end{cases} \Rightarrow$ both roots are positive.
- (g) If sign of $a = \text{sign of } b \neq \text{sign of } c \Rightarrow$ Greater root in magnitude is negative.
- (h) If sign of $b = \text{sign of } c \neq \text{sign of } a \Rightarrow$ Greater root in magnitude is positive.
- (i) If $a + b + c = 0 \Rightarrow$ one root is 1 and second root is c/a or $(-b-a)/a$.

Illustration 5 : If equation $\frac{x^2 - bx}{ax - c} = \frac{k-1}{k+1}$ has roots equal in magnitude & opposite in sign, then the value of k is -

(A) $\frac{a+b}{a-b}$

(B) $\frac{a-b}{a+b}$

(C) $\frac{a}{b} + 1$

(D) $\frac{a}{b} - 1$

Solution : Let the roots are α & $-\alpha$.

given equation is

$$(x^2 - bx)(k+1) = (k-1)(ax - c) \quad \{\text{Considering, } x \neq c/a \text{ \& } k \neq -1\}$$

$$\Rightarrow x^2(k+1) - bx(k+1) = ax(k-1) - c(k-1)$$

$$\Rightarrow x^2(k+1) - bx(k+1) - ax(k-1) + c(k-1) = 0$$

$$\text{Now sum of roots} = 0 \quad (\because \alpha - \alpha = 0)$$

$$\therefore b(k+1) + a(k-1) = 0 \Rightarrow k = \frac{a-b}{a+b}$$

Ans. (B)

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***Illustration 6 :** If roots of the equation $(a-b)x^2 + (c-a)x + (b-c) = 0$ are equal, then a, b, c are in

- (A) A.P. (B) H.P. (C) G.P. (D) none of these

Solution : $(a-b)x^2 + (c-a)x + (b-c) = 0$

As roots are equal so

$$B^2 - 4AC = 0 \Rightarrow (c-a)^2 - 4(a-b)(b-c) = 0 \Rightarrow (c-a)^2 - 4ab + 4b^2 + 4ac - 4bc = 0$$

$$\Rightarrow (c-a)^2 + 4ac - 4b(c+a) + 4b^2 = 0 \Rightarrow (c+a)^2 - 2 \cdot (2b)(c+a) + (2b)^2 = 0$$

$$\Rightarrow [c+a-2b]^2 = 0 \Rightarrow c+a-2b = 0 \Rightarrow c+a = 2b$$

Hence a, b, c are in A. P.

Alternative method :

$$\because \text{Sum of the coefficients} = 0$$

$$\text{Hence one root is 1 and other root is } \frac{b-c}{a-b}.$$

Given that both roots are equal, so

$$1 = \frac{b-c}{a-b} \Rightarrow a-b = b-c \Rightarrow 2b = a+c$$

Hence a, b, c are in A.P.

Ans. (A)

Do yourself - 3 :

(i) Consider $f(x) = x^2 + bx + c$.

(a) Find c if $x = 0$ is a root of $f(x) = 0$.

(b) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.

(c) Comment on sign of b & c, if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$.

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IDENTITY :

An equation which is true for every value of the variable within the domain is called an identity, for example : $5(a-3) = 5a - 15$, $(a+b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in \mathbb{R}$.

Note : A quadratic equation cannot have three or more roots & if it has, it becomes an identity.

$$\text{If } ax^2 + bx + c = 0 \text{ is an identity} \Leftrightarrow a = b = c = 0$$

556 COMMON ROOTS OF TWO QUADRATIC EQUATIONS :

(a) Only one common root.

Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$ then

$$a\alpha^2 + b\alpha + c = 0 \text{ \& } a'\alpha^2 + b'\alpha + c' = 0. \text{ By Cramer's Rule } \frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$$

$$\text{Therefore, } \alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$$

So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$.

(b) If both roots are same then $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$.

Illustration 8: Find p and q such that $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots in common.

Solution : $a_1 = p, b_1 = 5, c_1 = 2$

$$a_2 = 3, b_2 = 10, c_2 = q$$

We know that :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{p}{3} = \frac{5}{10} = \frac{2}{q} \Rightarrow p = \frac{3}{2}; q = 4$$

***Illustration 9:** Find the possible value(s) of a for which the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have atleast one common root.

Solution : Let α is a common root

$$\text{then } \alpha^2 + a\alpha + 1 = 0$$

$$\& \quad \alpha^2 + \alpha + a = 0$$

by cramer's rule

$$\frac{\alpha^2}{a^2 - 1} = \frac{\alpha}{1 - a} = \frac{1}{1 - a}$$

$$\Rightarrow (1 - a)^2 = (a^2 - 1)(1 - a)$$

$$\Rightarrow a = 1, -2$$

557 Illustrations 10: A polynomial in x of degree greater than three, leaves remainders 2, 1 and -1 when divided, respectively, by $(x-1)$, $(x+2)$ and $(x+1)$. What will be the remainder when it is divided by $(x-1)(x+2)(x+1)$.

Solution : Let required polynomial be $f(x) = p(x)(x-1)(x+2)(x+1) + a_0x^2 + a_1x + a_2$

By remainder theorem, $f(1) = 2$, $f(-2) = 1$, $f(-1) = -1$.

$$\Rightarrow a_0 + a_1 + a_2 = 2$$

$$4a_0 - 2a_1 + a_2 = 1$$

$$a_0 - a_1 + a_2 = -1$$

$$\text{Solving we get, } a_0 = \frac{7}{6}, a_1 = \frac{3}{2}, a_2 = \frac{2}{3}$$

Remainder when $f(x)$ is divided by $(x-1)(x+2)(x+1)$

$$\text{will be } \frac{7}{6}x^2 + \frac{3}{2}x + \frac{2}{3}.$$

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SOLUTION OF RATIONAL INEQUALITIES :

Let $y = \frac{f(x)}{g(x)}$ be an expression in x where $f(x)$ & $g(x)$ are polynomials in x . Now, if it is given that

$y > 0$ (or < 0 or ≥ 0 or ≤ 0), this calls for all the values of x for which y satisfies the constraint. This solution set can be found by following steps :

Step I : Factorize $f(x)$ & $g(x)$ and generate the form :

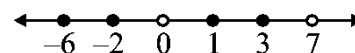
$$y = \frac{(x-a_1)^{n_1}(x-a_2)^{n_2}\dots\dots(x-a_k)^{n_k}}{(x-b_1)^{m_1}(x-b_2)^{m_2}\dots\dots(x-b_p)^{m_p}}$$

where $n_1, n_2, \dots, n_k, m_1, m_2, \dots, m_p$ are natural numbers and $a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_p$ are real numbers. Clearly, here $a_1, a_2, \dots, a_k$ are roots of $f(x) = 0$ & $b_1, b_2, \dots, b_p$ are roots of $g(x) = 0$.

Step II : Here y vanishes (becomes zero) for $a_1, a_2, \dots, a_k$. These points are marked on the number line with a black dot. They are solution of $y = 0$.

If $g(x) = 0$, $y = \frac{f(x)}{g(x)}$ attains an undefined form, hence $b_1, b_2, \dots, b_k$ are excluded from the solution. These points are marked with white dots.

$$\text{e.g. } f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$$



Step-III : Check the value of y for any real number greater than the right most marked number on the number line. If it is positive, then y is positive for all the real numbers greater than the right most marked number and vice versa.

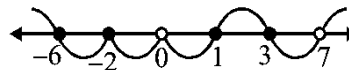
Step-IV : If the exponent of a factor is odd, then the point is called simple point and if the exponent of a factor is even, then the point is called double point

$$\frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$$

Here 1, 3, -6 and 7 are simple points and -2 & 0 are double points.

From right to left, beginning above the number line (if y is positive in step 3 otherwise from below the line), a wavy curve should be drawn which passes through all the marked points so that when passing through a simple point, the curve intersects the number line and when passing through a double point, the curve remains on the same side of number line.

$$f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$$



As exponents of $(x+2)$ and x are even, the curve does not cross the number line. This method is called wavy curve method.

Step-V : The intervals where the curve is above number line, y will be positive and the intervals where the curve is below the number line, y will be negative. The appropriate intervals are chosen in accordance with the sign of inequality & their union represents the solution of inequality.

Note :

- (i) Points where denominator is zero will never be included in the answer.
- (ii) If you are asked to find the intervals where $f(x)$ is non-negative or non-positive then make the intervals closed corresponding to the roots of the numerator and let it remain open corresponding to the roots of denominator.
- (iii) Normally we cannot cross-multiply in inequalities. But we cross multiply if we are sure that

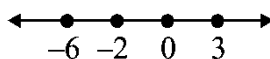
quantity in denominator is always positive.

- (iv) Normally we cannot square in inequalities. But we can square if we are sure that both sides are non negative.
- (v) We can multiply both sides with a negative number by changing the sign of inequality.
- (vi) We can add or subtract equal quantity to both sides of inequalities without changing the sign of inequality.

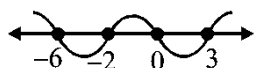
Illustration 12: $(x^2 - x - 6)(x^2 + 6x) \geq 0$

Solution : $(x-3)(x+2)(x)(x+6) \geq 0$

Consider $E = x(x-3)(x+2)(x+6)$, $E = 0 \Rightarrow x = 0, 3, -2, -6$ (all are simple points)



For $x \geq 3$ $E = \underbrace{x}_{+ve} \underbrace{(x-3)}_{+ve} \underbrace{(x+2)}_{+ve} \underbrace{(x+6)}_{+ve}$
 $= \text{positive}$



Hence for $x(x-3)(x+2)(x+6) \geq 0$

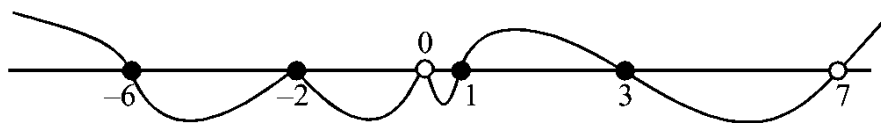
$$x \in (-\infty, -6] \cup [-2, 0] \cup [3, \infty)$$

Illustration 13: Let $f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$. Solve the following inequality

- (i) $f(x) > 0$ (ii) $f(x) \geq 0$ (iii) $f(x) < 0$ (iv) $f(x) \leq 0$

Solution :

We mark on the number line zeros of the function : 1, -2, 3 and -6 (with black circles) and the points of discontinuity 0 and 7 (with white circles), isolate the double points : -2 and 0 and draw the wavy curve :



From graph, we get

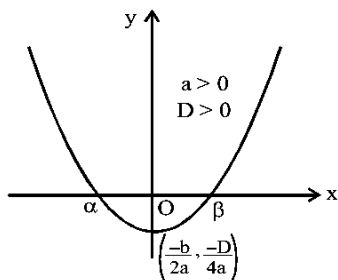
- (i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$
- (ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$
- (iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$
- (iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

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QUADRATIC EXPRESSION AND ITS GRAPHS :

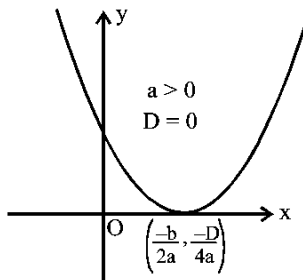
Consider the quadratic expression, $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in \mathbb{R}$ then ;

- (a) The graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.
- (b) The graph of $y = ax^2 + bx + c$ can be divided in 6 broad categories which are as follows :
(Let the real roots of quadratic equation $ax^2 + bx + c = 0$ be α & β where $\alpha \leq \beta$).

Fig. 1**Roots are real & distinct**

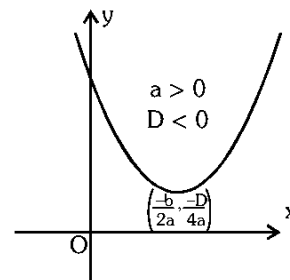
$$ax^2 + bx + c > 0 \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$ax^2 + bx + c < 0 \forall x \in (\alpha, \beta)$$

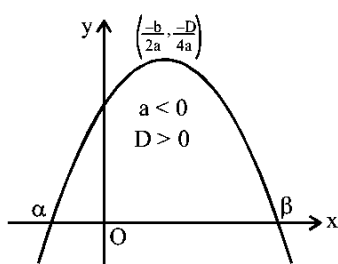
Fig. 2**Roots are coincident**

$$ax^2 + bx + c > 0 \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \text{ for } x = \alpha = \beta$$

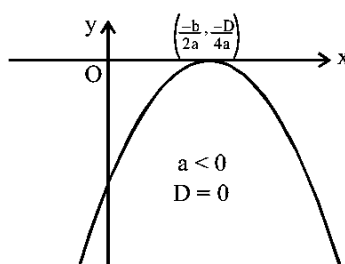
Fig.3**Roots are complex conjugate**

$$ax^2 + bx + c > 0 \forall x \in \mathbb{R}$$

Fig. 4**Roots are real & distinct**

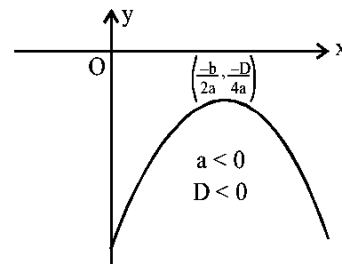
$$ax^2 + bx + c > 0 \forall x \in (\alpha, \beta)$$

$$ax^2 + bx + c < 0 \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

Fig. 5**Roots are coincident**

$$ax^2 + bx + c < 0 \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \text{ for } x = \alpha = \beta$$

Fig.6**Roots are complex conjugate**

$$ax^2 + bx + c < 0 \forall x \in \mathbb{R}$$

Important Note :

- (i) The quadratic expression $ax^2 + bx + c > 0$ for each $x \in \mathbb{R} \Rightarrow a > 0, D < 0$ & vice-versa (Fig. 3)
- (ii) The quadratic expression $ax^2 + bx + c < 0$ for each $x \in \mathbb{R} \Rightarrow a < 0, D < 0$ & vice-versa (Fig. 6)

560 MAXIMUM & MINIMUM VALUES OF QUADRATIC EXPRESSIONS : $y = ax^2 + bx + c$:

We know that $y = ax^2 + bx + c$ takes following form : $y = a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a^2} \right]$,

which is a parabola. $\therefore$ vertex $= \left(\frac{-b}{2a}, \frac{-D}{4a} \right)$

When $a > 0$, y will take a minimum value at vertex ; $x = \frac{-b}{2a}$; $y_{\min} = \frac{-D}{4a}$

When $a < 0$, y will take a maximum value at vertex ; $x = \frac{-b}{2a}$; $y_{\max} = \frac{-D}{4a}$.

If quadratic expression $ax^2 + bx + c$ is a perfect square, then $a > 0$ and $D = 0$

Illustration 14 : The value of the expression $x^2 + 2bx + c$ will be positive for all real x if -

- (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

Solution : As $a > 0$, so this expression will be positive if $D < 0$

$$\text{so } 4b^2 - 4c < 0$$

$$b^2 < c$$



Ans. (D)

Illustration 15 : The minimum value of the expression $4x^2 + 2x + 1$ is -

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1

Solution : Since $a = 4 > 0$

$$\text{therefore its minimum value} = -\frac{D}{4a} = \frac{4(4)(1) - (2)^2}{4(4)} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4}$$

Ans. (C)

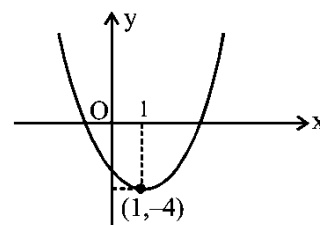
***Illustration 16 :** If $y = x^2 - 2x - 3$, then find the range of y when :

- (i) $x \in \mathbb{R}$ (ii) $x \in [0, 3]$ (iii) $x \in [-2, 0]$

Solution : We know that minimum value of y will occur at

$$x = -\frac{b}{2a} = -\frac{(-2)}{2 \times 1} = 1$$

$$y_{\min} = -\frac{D}{4a} = \frac{-(4 + 3 \times 4)}{4} = -4$$



(i) $x \in \mathbb{R};$

$y \in [-4, \infty)$

Ans.

(ii) $x \in [0, 3]$

$f(0) = -3, f(1) = -4, f(3) = 0$

$\therefore f(3) > f(0)$

 $\therefore y$ will take all the values from minimum to $f(3)$.

$y \in [-4, 0]$

Ans.

(iii) $x \in [-2, 0]$

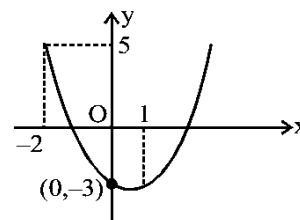
This interval does not contain the minimum value of y for $x \in \mathbb{R}$. y will take values from $f(0)$ to $f(-2)$

$f(0) = -3$

$f(-2) = 5$

$y \in [-3, 5]$

Ans.

**Illustration 17 :** If $ax^2 + bx + 10 = 0$ does not have real & distinct roots, find the minimum value of $5a - b$.**Solution :** Either $f(x) \geq 0 \forall x \in \mathbb{R}$ or $f(x) \leq 0 \forall x \in \mathbb{R}$

$\therefore f(0) = 10 > 0 \Rightarrow f(x) \geq 0 \forall x \in \mathbb{R}$

$\Rightarrow f(-5) = 25a - 5b + 10 \geq 0$

$\Rightarrow 5a - b \geq -2$

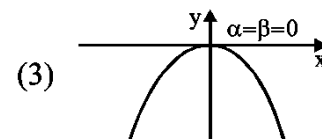
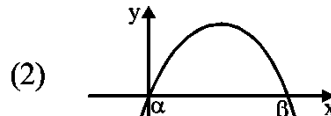
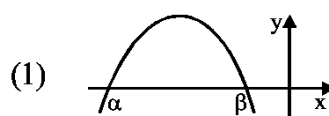
Ans.

Do yourself - 6

(i) Find the minimum value of:

(a) $y = x^2 + 2x + 2$

(b) $y = 4x^2 - 16x + 15$

(ii) For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, comment on the sign of:(i) a (ii) b (iii) c (iv) D (v) $\alpha + \beta$ (vi) $\alpha\beta$ (iii) Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in \mathbb{R}^+$, then the vertex of the graph will lie in which quadrant.

*(iv) Find the range of 'a' for which :

(a) $ax^2 + 3x + 4 > 0 \quad \forall x \in \mathbb{R}$

(b) $ax^2 + 4x - 2 < 0 \quad \forall x \in \mathbb{R}$

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INEQUALITIES INVOLVING MODULUS FUNCTION :

Properties of modulus function :

- (i) $|x| \geq a \Rightarrow x \geq a \text{ or } x \leq -a$, where a is positive.
- (ii) $|x| \leq a \Rightarrow x \in [-a, a]$, where a is positive
- (iii) $|x| > |y| \Rightarrow x^2 > y^2$
- (iv) $||a| - |b|| \leq |a \pm b| \leq |a| + |b|$
- (v) $|x + y| = |x| + |y| \Rightarrow xy \geq 0$
- (vi) $|x - y| = |x| + |y| \Rightarrow xy \leq 0$

Illustration 18: If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then

- (A) $0 \leq x \leq 4$ (B) $x \leq -2$ or $x \geq 4$
(C) $x \leq 0$ or $x \geq 4$ (D) none of these

Solution :

Case I : $x \leq 1$, then

$$1 - x + 2 - x + 3 - x \geq 6 \Rightarrow x \leq 0$$

Hence $x \leq 0$... (i)

Case II : $1 < x \leq 2$, then

$$x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow x \leq -2$$

But $1 < x \leq 2 \Rightarrow$ No solution. ... (ii)

Case III : $2 < x \leq 3$, then

$$x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$$

But $2 < x \leq 3 \Rightarrow$ No solution. ... (iii)

Case IV : $x > 3$, then

$$x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow x \geq 4$$

Hence $x \geq 4$... (iv)

From (i), (ii), (iii) and (iv) the given inequality holds for $x \leq 0$ or $x \geq 4$.

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Illustration 19 : Solve for x : (a) $\|x-1\|+2\leq 4$. (b) $\frac{x-4}{x+2} \leq \left\| \frac{x-2}{x-1} \right\|$

Solution : (a) $\|x-1\|+2\leq 4 \Rightarrow -4 \leq x-1+2 \leq 4$
 $\Rightarrow -6 \leq x-1 \leq 2$
 $\Rightarrow |x-1| \leq 2 \Rightarrow -2 \leq x-1 \leq 2$
 $\Rightarrow -1 \leq x \leq 3 \Rightarrow x \in [-1, 3]$

(b) **Case 1 :** Given inequation will be satisfied for all x such that

$$\frac{x-4}{x+2} \leq 0 \Rightarrow x \in (-2, 4] - \{1\} \quad \text{.....(i)}$$

(Note : $\{1\}$ is not in domain of RHS)

$$\text{Case 2 : } \frac{x-4}{x+2} > 0 \Rightarrow x \in (-\infty, -2) \cup (4, \infty) \quad \text{.....(ii)}$$

Given inequation becomes

$$\frac{x-2}{x-1} \geq \frac{x-4}{x+2} \quad \text{or} \quad \frac{x-2}{x-1} \leq -\frac{x-4}{x+2}$$

on solving we get

$$x \in (-2, 4/5) \cup (1, \infty)$$

taking intersection with (ii) we get

$$x \in (4, \infty)$$

.....(iii)

on solving we get

$$x \in (-2, 0] \cup (1, 5/2]$$

taking intersection with (ii) we get

$$x \in \phi$$

Hence, solution of the original inequation : $x \in (-2, \infty) - \{1\}$ (taking union of (i) & (iii))

Illustration 20 : The equation $|x| + \left| \frac{x}{x-1} \right| = \frac{x^2}{|x-1|}$ is always true for x belongs to

- (A) $\{0\}$ (B) $(1, \infty)$ (C) $(-1, 1)$ (D) $(-\infty, \infty)$

Solution : $\frac{x^2}{|x-1|} = \left| x + \frac{x}{x-1} \right|$

$$\therefore |x| + \left| \frac{x}{x-1} \right| = \left| x + \frac{x}{x-1} \right| \text{ is true only if } \left(x \cdot \frac{x}{x-1} \right) \geq 0 \Rightarrow x \in \{0\} \cup (1, \infty). \text{ Ans (A,B)}$$

IRRATIONAL INEQUALITIES :

Illustration 21 : Solve for x , if $\sqrt{x^2-3x+2} > x-2$

Solution :
$$\left[\begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 \geq 0 \\ (x^2-3x+2) > (x-2)^2 \end{array} \right] \quad \text{or} \quad \Rightarrow \quad \left[\begin{array}{l} (x-1)(x-2) \geq 0 \\ (x-2) \geq 0 \Rightarrow x > 2 \\ x-2 > 0 \end{array} \right]$$

$$\left[\begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 < 0 \end{array} \right] \quad \text{or} \quad \Rightarrow \quad \left[\begin{array}{l} (x-1)(x-2) \geq 0 \\ x-2 < 0 \Rightarrow x \leq 1 \end{array} \right]$$

Hence, solution set of the original inequation is $x \in \mathbb{R} - (1, 2]$

563 LOGARITHMIC INEQUALITIES :

Points to remember :

$$(i) \log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$$

$$(ii) \text{ If } a > 1, \text{ then} \quad (a) \log_a x < p \Rightarrow 0 < x < a^p \quad (b) \log_a x > p \Rightarrow x > a^p$$

$$(iii) \text{ If } 0 < a < 1, \text{ then} \quad (a) \log_a x < p \Rightarrow x > a^p \quad (b) \log_a x > p \Rightarrow 0 < x < a^p$$

Illustration 22 : Solve for x : (a) $\log_{0.5}(x^2 - 5x + 6) \geq -1$ (b) $\log_{1/3}(\log_4(x^2 - 5)) > 0$

Solution : (a) $\log_{0.5}(x^2 - 5x + 6) \geq -1 \Rightarrow 0 < x^2 - 5x + 6 \leq (0.5)^{-1}$

$$\Rightarrow 0 < x^2 - 5x + 6 \leq 2$$

$$\begin{cases} x^2 - 5x + 6 > 0 \\ x^2 - 5x + 6 \leq 2 \end{cases} \Rightarrow x \in [1, 2) \cup (3, 4]$$

Hence, solution set of original inequation : $x \in [1, 2) \cup (3, 4]$

$$(b) \log_{1/3}(\log_4(x^2 - 5)) > 0 \Rightarrow 0 < \log_4(x^2 - 5) < 1$$

$$\begin{cases} 0 < \log_4(x^2 - 5) \Rightarrow x^2 - 5 > 1 \\ \log_4(x^2 - 5) < 1 \Rightarrow 0 < x^2 - 5 < 4 \end{cases} \Rightarrow 1 < (x^2 - 5) < 4$$

$$\Rightarrow 6 < x^2 < 9 \Rightarrow x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

Hence, solution set of original inequation : $x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$

Illustration 23 : Solve for x : $\log_2 x \leq \frac{2}{\log_2 x - 1}$.

Solution : Let $\log_2 x = t$

$$t \leq \frac{2}{t-1} \Rightarrow t - \frac{2}{t-1} \leq 0$$

$$\Rightarrow \frac{t^2 - t - 2}{t-1} \leq 0 \Rightarrow \frac{(t-2)(t+1)}{(t-1)} \leq 0$$

$$\Rightarrow t \in (-\infty, -1] \cup (1, 2]$$

or $\log_2 x \in (-\infty, -1] \cup (1, 2]$

or $x \in \left(0, \frac{1}{2}\right] \cup (2, 4]$

Illustration 24 : Solve the inequation : $\log_{2x+3} x^2 < \log_{2x+3} (2x+3)$

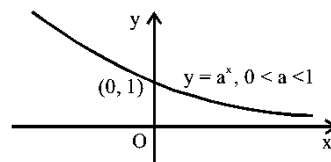
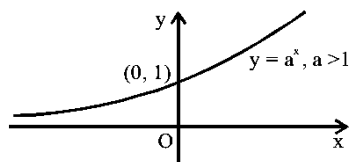
Solution : This inequation is equivalent to the collection of the systems

$$\left[\begin{array}{l} 2x+3 > 1 \\ 0 < x^2 < 2x+3 \end{array} \right] \Rightarrow \left[\begin{array}{l} x > -1 \\ (x-3)(x+1) < 0 \text{ \& } x \neq 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} x > -1 \\ -1 < x < 3 \end{array} \right] \Rightarrow -1 < x < 3 \text{ \& } x \neq 0$$

$$\text{or} \left[\begin{array}{l} 0 < 2x+3 < 1 \\ x^2 > 2x+3 > 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} -\frac{3}{2} < x < -1 \\ (x-3)(x+1) > 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} -\frac{3}{2} < x < -1 \\ x < -1 \text{ or } x > 3 \end{array} \right] \Rightarrow -\frac{3}{2} < x < -1$$

Hence, solution of the original inequation is $x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$

564 EXPONENTIAL INEQUATIONS :



$$\text{If } a^{f(x)} > b \Rightarrow \begin{cases} f(x) > \log_a b & \text{when } a > 1 \\ f(x) < \log_a b & \text{when } 0 < a < 1 \end{cases}$$

Illustration 25 : Solve for x : $2^{x+2} > \left(\frac{1}{4}\right)^{\frac{1}{x}}$

Solution : We have $2^{x+2} > 2^{-2/x}$. Since the base $2 > 1$, we have $x + 2 > -\frac{2}{x}$ (the sign of the inequality is retained).

$$\begin{aligned} \text{Now } x + 2 + \frac{2}{x} > 0 &\Rightarrow \frac{x^2 + 2x + 2}{x} > 0 \\ \Rightarrow \frac{(x+1)^2 + 1}{x} > 0 &\Rightarrow x \in (0, \infty) \end{aligned}$$

Illustration 26 : Solve for x : $(1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$

Solution : We have $\left(\frac{5}{4}\right)^{1-x} < \left(\frac{16}{25}\right)^{2(1+\sqrt{x})}$ or $\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$

Since the base $0 < \frac{4}{5} < 1$, the inequality is equivalent to the inequality $x - 1 > 4(1 + \sqrt{x})$

$$\Rightarrow \frac{x-5}{4} > \sqrt{x}$$

Now, R.H.S. is positive

$$\Rightarrow \frac{x-5}{4} > 0 \Rightarrow x > 5 \quad \text{.....(i)}$$

$$\text{we have } \frac{x-5}{4} > \sqrt{x}$$

both sides are positive, so squaring both sides

$$\Rightarrow \frac{(x-5)^2}{16} > x \quad \text{or} \quad \frac{(x-5)^2}{16} - x > 0$$

$$\text{or } x^2 - 26x + 25 > 0 \quad \text{or } (x-25)(x-1) > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (25, \infty) \quad \text{.....(ii)}$$

intersection (i) & (ii) gives $x \in (25, \infty)$

Do yourself-8 :

(i) Solve for x : (a) $\log_{0.3}(x^2 + 8) > \log_{0.3}(9x)$ (b) $\log_7\left(\frac{2x-6}{2x-1}\right) > 0$

(ii) Solve for x : $\left(\frac{2}{3}\right)^{\frac{|x|-1}{|x|+1}} > 1$

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MAXIMUM & MINIMUM VALUES OF RATIONAL ALGEBRAIC EXPRESSIONS :

$$y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}, \frac{1}{ax^2 + bx + c}, \frac{a_1x + b_1}{a_2x^2 + b_2x + c_2}, \frac{a_1x^2 + b_1x + c_1}{a_2x + b_2} :$$

Sometime we have to find range of expression of form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$.

The following procedure is used :

Step 1 : Equate the given expression to y i.e. $y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$

Step 2 : By cross multiplying and simplifying, obtain a quadratic equation in x.

$$(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$$

Step 3 : Put Discriminant ≥ 0 and solve the inequality for possible set of values of y.

Illustration 27 : For $x \in \mathbb{R}$, find the set of values attainable by $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$.

Solution : Let $y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$

$$x^2(y - 1) + 3x(y + 1) + 4(y - 1) = 0$$

Case- I : $y \neq 1$

For $y \neq 1$ above equation is a quadratic equation.

So for $x \in \mathbb{R}$, $D \geq 0$

$$\Rightarrow 9(y + 1)^2 - 16(y - 1)^2 \geq 0 \Rightarrow 7y^2 - 50y + 7 \leq 0$$

$$\Rightarrow (7y - 1)(y - 7) \leq 0 \Rightarrow y \in \left[\frac{1}{7}, 7 \right] - \{1\}$$

Case II : when $y = 1$

$$\Rightarrow 1 = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$$

$$\Rightarrow x^2 + 3x + 4 = x^2 - 3x + 4$$

$$\Rightarrow x = 0$$

Hence $y = 1$ for real value of x .

so range of y is $\left[\frac{1}{7}, 7\right]$

Illustration 28 : Find the values of a for which the expression $\frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$ assumes all real values for real values of x .

Solution : Let $y = \frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$

$$x^2(a + 4y) + 3(1 - y)x - (4 + ay) = 0$$

If $x \in \mathbb{R}$, $D \geq 0$

$$\Rightarrow 9(1 - y)^2 + 4(a + 4y)(4 + ay) \geq 0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$$

for all $y \in \mathbb{R}$, $(9 + 16a) > 0$ & $D \leq 0$

$$\Rightarrow (4a^2 + 46)^2 - 4(9 + 16a)(9 + 16a) \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$$

$$\Rightarrow a^2 - 8a + 7 \leq 0 \Rightarrow 1 \leq a \leq 7$$

$$9 + 16a > 0 \text{ \& } 1 \leq a \leq 7$$

Taking intersection, $a \in [1, 7]$

Now, checking the boundary values of a

For $a = 1$

$$y = \frac{x^2 + 3x - 4}{3x - 4x^2 + 1} = -\frac{(x-1)(x+4)}{(x-1)(4x+1)}$$

$$\therefore x \neq 1 \Rightarrow y \neq -1$$

$$\Rightarrow a = 1 \text{ is not possible.}$$

if $a = 7$

$$y = \frac{7x^2 + 3x - 4}{3x - 4x^2 + 7} = \frac{(7x-4)(x+1)}{(7-4x)(x+1)} \quad \therefore x \neq -1 \Rightarrow y \neq -1$$

So y will assume all real values for some real values of x .

So $a \in (1, 7)$

566 LOCATION OF ROOTS :

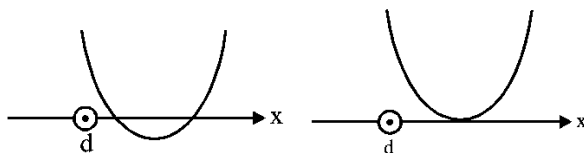
- This article deals with an elegant approach of solving problems on quadratic equations when the roots are located / specified on the number line with variety of constraints :

Consider the quadratic equation $ax^2 + bx + c = 0$ with $a > 0$ and let $f(x) = ax^2 + bx + c$

Type-1 : Both roots of the quadratic equation are greater than a specific number (say d).

The necessary and sufficient condition for this are :

(i) $D \geq 0$; (ii) $f(d) > 0$; (iii) $-\frac{b}{2a} > d$



Note : When both roots of the quadratic equation are less than a specific number d then the necessary and sufficient condition will be :

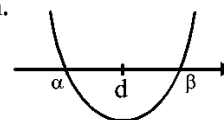
(i) $D \geq 0$; (ii) $f(d) > 0$; (iii) $-\frac{b}{2a} < d$

Type-2 :

Both roots lie on either side of a fixed number say (d). Alternatively one root is greater than ' d ' and other root less than ' d ' or ' d ' lies between the roots of the given equation.

The necessary and sufficient condition for this are : $f(d) < 0$

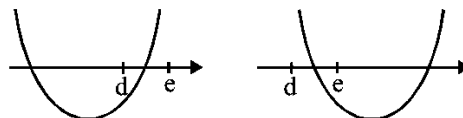
Note : Consideration of discriminant is not needed.

**Type-3 :**

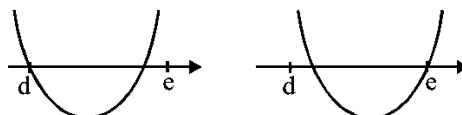
Exactly one root lies in the interval (d, e).

The necessary and sufficient condition for this are :

$f(d) \cdot f(e) < 0$



Note : The extremes of the intervals found by given conditions give ' d ' or ' e ' as the root of the equation.



Hence in this case also check for end points.

$$f(d) = 0$$

$$f(e) = 0$$

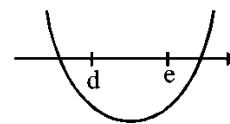
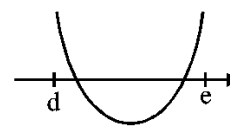
Type-4 :

When both roots are confined between the number d and e ($d < e$).

The necessary and sufficient condition for this are :

$$(i) D \geq 0; \quad (ii) f(d) > 0 \quad ; \quad (iii) f(e) > 0$$

$$(iv) d < -\frac{b}{2a} < e$$



Type-5 :

One root is greater than e and the other root is less than d ($d < e$).

The necessary and sufficient condition for this are : $f(d) < 0$ and $f(e) < 0$

Note : If $a < 0$ in the quadratic equation $ax^2 + bx + c = 0$ then we divide the whole equation by ' a '.

Now assume $x^2 + \frac{b}{a}x + \frac{c}{a}$ as $f(x)$. This makes the coefficient of x^2 positive and hence above cases are applicable.

Illustration 29 : Find the values of the parameter ' a ' for which the roots of the quadratic equation $x^2 + 2(a-1)x + a + 5 = 0$ are

- | | |
|--|--|
| (i) real and distinct | (ii) equal |
| (iii) opposite in sign | (iv) equal in magnitude but opposite in sign |
| (v) positive | (vi) negative |
| (vii) greater than 3 | (viii) smaller than 3 |
| (ix) such that both the roots lie in the interval (1, 3) | |

Solution :

$$\text{Let } f(x) = x^2 + 2(a-1)x + a + 5 = Ax^2 + Bx + C \text{ (say)}$$

$$\Rightarrow A = 1, B = 2(a-1), C = a + 5.$$

$$\text{Also } D = B^2 - 4AC = 4(a-1)^2 - 4(a+5) = 4(a+1)(a-4)$$

- (i) $D > 0$
 $\Rightarrow (a+1)(a-4) > 0 \Rightarrow a \in (-\infty, -1) \cup (4, \infty).$
- (ii) $D = 0$
 $\Rightarrow (a+1)(a-4) = 0 \Rightarrow a = -1, 4.$
- (iii) This means that 0 lies between the roots of the given equation.
 $\Rightarrow f(0) < 0$ and $D > 0$ i.e. $a \in (-\infty, -1) \cup (4, \infty)$
 $\Rightarrow a+5 < 0 \Rightarrow a < -5 \Rightarrow a \in (-\infty, -5).$
- (iv) This means that the sum of the roots is zero
 $\Rightarrow -2(a-1) = 0$ and $D > 0$ i.e. $a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a = 1$
 which does not belong to $(-\infty, -1) \cup (4, \infty)$
 $\Rightarrow a \in \phi.$
- (v) This implies that both the roots are greater than zero
 $\Rightarrow -\frac{B}{A} > 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) > 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$
 $\Rightarrow a < 1, -5 < a, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-5, -1].$
- (vi) This implies that both the roots are less than zero
 $\Rightarrow -\frac{B}{A} < 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) < 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$
 $\Rightarrow a > 1, a > -5, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in [4, \infty).$
- (vii) In this case
 $-\frac{B}{2a} > 3, A.f(3) > 0$ and $D \geq 0.$

$$\Rightarrow -(a-1) > 3, 7a+8 > 0 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

Since no value of 'a' can satisfy these conditions simultaneously, there can be no value of a for which both the roots will be greater than 3.

(viii) In this case

$$-\frac{B}{2a} < 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow a > -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-8/7, -1] \cup [4, \infty)$$

(ix) In this case

$$1 < -\frac{B}{2A} < 3, A.f(1) > 0, A.f(3) > 0, D \geq 0.$$

$$\Rightarrow 1 < -1(a-1) < 3, 3a+4 > 0, 7a+8 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow -2 < a < 0, a > -4/3, a > -8/7, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in \left(-\frac{8}{7}, -1\right]$$

Illustration 30 : Find value of k for which one root of equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is less than 2.

Solution : $4 - 2(k+1) + k^2 + k - 8 < 0 \Rightarrow k^2 - k - 6 < 0$

$$(k-3)(k+2) < 0 \Rightarrow -2 < k < 3$$

Taking intersection, $k \in (-2, 3)$.

Illustration 31 : Find all possible values of a for which exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval (0,3).

Solution : $f(0) \cdot f(3) < 0$

$$\Rightarrow 2a(9 - 3(a+1) + 2a) < 0 \Rightarrow 2a(-a+6) < 0$$

$$\Rightarrow a(a-6) > 0 \Rightarrow a < 0 \text{ or } a > 6$$

Checking the extremes.

$$\text{If } a = 0, \quad x^2 - x = 0$$

$$x = 0, 1$$

$$1 \in (0, 3)$$

$$\text{If } a = 6, \quad x^2 - 7x + 12 = 0$$

$$x = 3, 4 \quad \text{But } 4 \notin (0, 3)$$

Hence solution set is

$$a \in (-\infty, 0] \cup (6, \infty)$$

Do yourself - 10 :

(i) If α, β are roots of $7x^2 + 9x - 2 = 0$, find their position with respect to following ($\alpha < \beta$) :

(a) -3 (b) 0 (c) 1

(ii) If $a > 1$, roots of the equation $(1 - a)x^2 + 3ax - 1 = 0$ are -

(A) one positive one negative (B) both negative

(C) both positive (D) both non-real

(iii) Find the set of value of a for which the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3.

(iv) If α, β are the roots of $x^2 - 3x + a = 0$, $a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a .

(v) If α, β are roots of $4x^2 - 16x + \lambda = 0$, $\lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then find the range of λ .

567 **Illustration 32 :** If $x^2 + 2xy + 2x + my - 3$ have two linear factor then m is equal to -

(A) 6, 2 (B) $-6, 2$ (C) 6, -2 (D) $-6, -2$

Solution : Here $a=1, h=1, b=0, g=1, f=m/2, c=-3$

$$\text{So } \Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & m/2 \\ 1 & m/2 & -3 \end{vmatrix} = 0$$

$$\Rightarrow -\frac{m^2}{4} - (-3 - m/2) + m/2 = 0 \quad \Rightarrow -\frac{m^2}{4} + m + 3 = 0$$

$$\Rightarrow m^2 - 4m - 12 = 0 \quad \Rightarrow m = -2, 6 \quad \text{Ans. (C)}$$

Do yourself - 11 :

(i) Find the value of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into two linear factors.

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THEORY OF EQUATIONS :

Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are roots of the equation, $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$, where $a_0, a_1, \dots, a_n$ are constants and $a_0 \neq 0$.

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)$$

$$\therefore a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = a_0(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

Comparing the coefficients of like powers of x , we get

$$\sum \alpha_i = -\frac{a_1}{a_0} = S_1 \text{ (say)}$$

$$\text{or } S_1 = -\frac{\text{coefficient of } x^{n-1}}{\text{coefficient of } x^n}$$

$$S_2 = \sum_{i \neq j} \alpha_i \alpha_j = (-1)^2 \frac{a_2}{a_0}$$

$$S_3 = \sum_{i \neq j \neq k} \alpha_i \alpha_j \alpha_k = (-1)^3 \frac{a_3}{a_0}$$

$$\vdots$$

$$S_n = \alpha_1 \alpha_2 \dots \alpha_n = (-1)^n \frac{a_n}{a_0} = (-1)^n \frac{\text{constant term}}{\text{coefficient of } x^n}$$

where S_k denotes the sum of the product of root taken k at a time.

Quadratic equation : If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$$\text{then } \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Cubic equation : If α, β, γ are roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \quad \text{and} \quad \alpha\beta\gamma = -\frac{d}{a}$$

Note :

- (i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely.
- (ii) Every equation of n th degree ($n \geq 1$) has exactly n roots & if the equation has more than n roots, it is an identity.
- (iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs.**
- (iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not a perfect square.
- (v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have at least one real root between 'a' and 'b'.
- (vi) Every equation $f(x) = 0$ of degree odd has at least one real root of a sign opposite to that of its last term.

Illustration 33 : If two roots are equal, find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$.

Solution : Let roots be α, α and β

$$\therefore \alpha + \alpha + \beta = -\frac{20}{4} \quad \Rightarrow \quad 2\alpha + \beta = -5 \quad \dots\dots\dots (i)$$

$$\therefore \alpha \cdot \alpha + \alpha\beta + \alpha\beta = -\frac{23}{4} \quad \Rightarrow \quad \alpha^2 + 2\alpha\beta = -\frac{23}{4} \quad \& \quad \alpha^2\beta = -\frac{6}{4}$$

from equation (i)

$$\alpha^2 + 2\alpha(-5 - 2\alpha) = -\frac{23}{4} \Rightarrow \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \Rightarrow 12\alpha^2 + 40\alpha - 23 = 0$$

$$\therefore \alpha = 1/2, -\frac{23}{6}$$

$$\text{when } \alpha = \frac{1}{2}$$

$$\alpha^2\beta = \frac{1}{4}(-5-1) = -\frac{3}{2}$$

$$\text{when } \alpha = -\frac{23}{6} \Rightarrow \alpha^2\beta = \frac{23 \times 23}{36} \left(-5 - 2 \times \left(-\frac{23}{6} \right) \right) \neq -\frac{3}{2} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -6$$

$$\text{Hence roots of equation} = \frac{1}{2}, \frac{1}{2}, -6$$

Ans.

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Illustration 34 : If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, find :

$$(i) \sum \alpha^3 \quad (ii) \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$$

Solution :

We know that $\alpha + \beta + \gamma = p$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = r$$

$$(i) \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma + (\alpha + \beta + \gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\} \\ = 3r + p\{p^2 - 3q\} = 3r + p^3 - 3pq$$

$$(ii) \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma) \\ = p(\alpha^2 + \beta^2 + \gamma^2) - 3r - p^3 + 3pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$$

Illustration 35 : If $b^2 < 2ac$ and $a, b, c, d \in \mathbb{R}$, then prove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

Solution :

Let α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$

$$\text{Then } \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$, which is not possible if all α, β, γ are real. So at least one root is non-real, but complex roots occurs in pair. Hence given cubic equation has two non-real and one real roots.

Do yourself - 12 :

- (i) Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$
- (ii) If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then find the value of c .
- (iii) If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

$$(a) \sum \alpha^2 \quad (b) \sum \frac{1}{\alpha} \quad (c) \sum \alpha^2(\beta + \gamma)$$

569 TRANSFORMATION OF THE EQUATION :

Let $ax^2 + bx + c = 0$ be a quadratic equation with two roots α and β . If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α & β , then this equation can be found by finding α in terms of y . Now as α satisfies given equation, put this α in terms of y directly in the equation.

$$y = f(\alpha)$$

By transformation, $\alpha = g(y)$

$$a(g(y))^2 + b(g(y)) + c = 0$$

This is the required equation in y .

Illustration 36 : If the roots of $ax^2 + bx + c = 0$ are α and β , then find the equation whose roots are :

$$(a) \quad \frac{-2}{\alpha}, \frac{-2}{\beta} \quad (b) \quad \frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1} \quad (c) \quad \alpha^2, \beta^2$$

Solution :

$$(a) \quad \frac{-2}{\alpha}, \frac{-2}{\beta}$$

$$\text{put, } y = \frac{-2}{\alpha} \Rightarrow \alpha = \frac{-2}{y}$$

$$a\left(\frac{-2}{y}\right)^2 + b\left(\frac{-2}{y}\right) + c = 0 \Rightarrow cy^2 - 2by + 4a = 0$$

$$\text{Required equation is } cx^2 - 2bx + 4a = 0$$

$$(b) \quad \frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$$

$$\text{put, } y = \frac{\alpha}{\alpha+1} \Rightarrow \alpha = \frac{y}{1-y}$$

$$\Rightarrow a\left(\frac{y}{1-y}\right)^2 + b\left(\frac{y}{1-y}\right) + c = 0 \Rightarrow (a+c-b)y^2 + (-2c+b)y + c = 0$$

Required equation is $(a+c-b)x^2 + (b-2c)x + c = 0$

(c) α^2, β^2

$$\text{put } y = \alpha^2 \Rightarrow \alpha = \sqrt{y}$$

$$ay + b\sqrt{y} + c = 0$$

$$b^2y = a^2y^2 + c^2 + 2acy$$

$$\Rightarrow a^2y^2 + (2ac - b^2)y + c^2 = 0$$

Required equation is $a^2x^2 + (2ac - b^2)x + c^2 = 0$

Illustration 37 : If the roots of $ax^3 + bx^2 + cx + d = 0$ are α, β, γ then find equation whose roots are

$$\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}.$$

Solution : Put $y = \frac{1}{\alpha\beta} = \frac{\gamma}{\alpha\beta\gamma} = -\frac{a\gamma}{d}$ ($\because \alpha\beta\gamma = -\frac{d}{a}$)

$$\text{Put } x = -\frac{dy}{a}$$

$$\Rightarrow a\left(-\frac{dy}{a}\right)^3 + b\left(-\frac{dy}{a}\right)^2 + c\left(-\frac{dy}{a}\right) + d = 0$$

Required equation is $d^2x^3 - bdx^2 + acx - a^2 = 0$

Miscellaneous Illustrations :

Illustrations 38 : If α, β are the roots of $x^2 + px + q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s . Deduce the condition that the equations have a common root.

Solution : α, β are the roots of $x^2 + px + q = 0$

$$\therefore \alpha + \beta = -p, \alpha\beta = q \quad \dots\dots\dots(1)$$

and γ, δ are the roots of $x^2 + rx + s = 0$

$$\therefore \gamma + \delta = -r, \gamma\delta = s \quad \dots\dots\dots(2)$$

Now, $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

$$\begin{aligned} &= [\alpha^2 - \alpha(\gamma + \delta) + \gamma\delta] [\beta^2 - \beta(\gamma + \delta) + \gamma\delta] \\ &= (\alpha^2 + r\alpha + s) (\beta^2 + r\beta + s) \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2 \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s((\alpha + \beta)^2 - 2\alpha\beta) + sr(\alpha + \beta) + s^2 \\ &= q^2 - pqr + r^2q + s(p^2 - 2q) + sr(-p) + s^2 \\ &= (q - s)^2 - rpq + r^2q + sp^2 - prs \\ &= (q - s)^2 - rq(p - r) + sp(p - r) \\ &= (q - s)^2 + (p - r)(sp - rq) \end{aligned}$$

For a common root (Let $\alpha = \gamma$ or $\beta = \delta$) $\dots\dots\dots(3)$

then $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = 0 \quad \dots\dots\dots(4)$

from (3) and (4), we get

$$(q - s)^2 + (p - r)(sp - rq) = 0$$

$\Rightarrow (q - s)^2 = (p - r)(rq - sp)$, which is the required condition.

Illustrations 39: If $(y^2 - 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in \mathbb{R}$, then find the interval in which y lies.

Solution : $(y^2 - 5y + 3)(x^2 + x + 1) < 2x, \forall x \in \mathbb{R}$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1}$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = P$$

$$\Rightarrow px^2 + (p-2)x + p = 0$$

$$(1) \text{ Since } x \text{ is real, } (p-2)^2 - 4p^2 \geq 0$$

$$\Rightarrow -2 \leq p \leq \frac{2}{3}$$

$$(2) \text{ The minimum value of } 2x/(x^2 + x + 1) \text{ is } -2.$$

$$\text{So, } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0$$

$$\Rightarrow y \in \left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2} \right)$$

ANSWERS FOR DO YOURSELF

1: (i) (a) $-1, -2$; (b) 4 ; (c) $1 \pm \sqrt{2}$; (ii) $a, \frac{1}{a}$; (iii) $\frac{7}{3}$ (iv) $3, -\frac{1}{5}$

2: (i) $b = -4, c = 1$; (ii) (a) imaginary; (b) real & distinct; (c) real & coincident

3: (i) (a) $c = 0$; (b) $c = 1$; (c) $b \rightarrow \text{negative}, c \rightarrow \text{negative}$

4: (i) $b = \frac{9}{2}, c = 5$; (ii) $c = 0, 6$

5: (i) (a) $x \in (-\infty, -3] \cup [2, \infty)$; (b) $x \in (-2, -1)$; (c) $\left(-\frac{1}{4}, \frac{1}{3}\right]$;

(d) $x \in (-6, -3) \cup \left(\frac{1}{2}, 2\right) - \{1\} \cup (9, \infty)$; (e) $[3, 7]$; (f) ϕ

6: (i) (a) 1 (b) -1

(ii) (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$

(2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$

(3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$

(iii) Third quadrant

(iv) (a) $a > 9/16$ (b) $a < -2$

7: (i) $x \in (-\infty, -2) \cup (1, 3/2)$ (ii) $x \in \mathbb{R} - (0, 1]$

8: (i) (a) $x \in (1, 8)$ (b) $x \in (-\infty, 1/2)$ (ii) $x \in (-1, 1)$

9: (ii) least value $= 0$, greatest value $= 1$.

10: (i) $-3 < \alpha < 0 < \beta < 1$; (ii) C ; (iii) $a < 2$; (iv) $a < 2$; (v) $12 < \lambda < 16$

11: (i) $0, 2$

12: (ii) $-27, 5$; (iii) (a) $\frac{1}{a^2}(b^2 - 2ac)$, (b) $-\frac{c}{d}$, (c) $\frac{1}{a^2}(3ad - bc)$

13: (i) (a) $c^2y^2 + y(2ac - b^2) + a^2 = 0$; (b) $acx^2 - bx + 1 = 0$; (c) $acx^2 + (a+c)bx + (a+c)^2 = 0$

(ii) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

EXERCISE (E-1)

570 For what values of a does the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots?

571 Find the value of k for which the equation $(k-1)x^2 + (k+4)x + k+7 = 0$ has equal roots.

572 Find the values of a for which the roots of the equation $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ are equal.

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573 Form a quadratic equation whose roots are the numbers $\frac{1}{10 - \sqrt{72}}$ and $\frac{1}{10 + 6\sqrt{2}}$.

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574 For what values of a is the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero ?

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575 For what values of a is the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5 ?

.

576 The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p .

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577 Find k in the equation $5x^2 - kx + 1 = 0$ such that the difference between the roots of the equation is unity.

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578 Find p in the equation $x^2 - 4x + p = 0$ if it is known that the sum of the squares of its roots is equal to 16.

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579 For what values of a is the difference between the roots of the equation $2x^2 - (a + 1)x + (a - 1) = 0$ equal to their product ?

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580 Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.

.

581 Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x_1^3 + x_2^3$.

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582 Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.

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583 For what values of m does the equation $x^2 - x + m = 0$ possess no real roots ?

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584 For what values of c does the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots ?

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585 Find integral values of k for which the quadratic equation $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ possess no real roots ?

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586 For what values of k is the inequality $x^2 - (k - 3)x - k + 6 > 0$ valid for all real x ?

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587 For what integral k is the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ valid for any real x ?

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588 Find the least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots.

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589 Find all values of a for which the inequality $(a + 4)x^2 - 2ax + 2a - 6 < 0$ is satisfied for all $x \in \mathbb{R}$.

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590 Find all values of a for which the inequality $(a - 3)x^2 - 2ax + 3a - 6 > 0$ is satisfied for all values of x .

.

- 591 Find all values of a for which the inequality $(a - 1)x^2 - (a + 1)x + a + 1 > 0$ is satisfied for all real x .
- 592 For what values of a do the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have exactly one root in common ?
- 593 The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c .
- 594 For what values of a do the graphs of the functions $y = 2ax + 1$ and $y = (a - 6)x^2 - 2$ not intersect?

EXERCISE (E-2)

- 595 For what values of ' a ' does the quadratic equation $x^2 + (2a\sqrt{a^2 - 3})x + 4 = 0$ possess equal roots ?
- 596 For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a + 2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$? Find the roots.
- 597 Find a such that one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other.
- 598 For what value of a is the difference between the roots of the equation $(a - 2)x^2 - (a - 4)x - 2 = 0$ equal to 3 ?
- 599 Find all the values of a for which the sum of the roots of the equation $x^2 - 2a(x - 1) - 1 = 0$ is equal to the sum of the squares of its roots.
- 600 Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q .
- 601 Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.
- 602 For what values of m does the equation $x^2 - x + m^2 = 0$ possess no real roots ?
- 603 For what values of m does the equation $mx^2 - (m + 1)x + 2m - 1 = 0$ possess no real roots ?
- 604 For what values of a is the inequality $ax^2 + 2ax + 0.5 > 0$ valid throughout the entire number axis ?
- 605 For what values of a does the equation $(2 - x)(x + 1) = a$ possess real and positive roots ?
- 606 For what least integral k is the quadratic trinomial $(k - 2)x^2 + 8x + k + 4$ positive for all values of x ?
- 607 Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0$, $m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.
- 608 For what values of p does the vertex of the parabola $y = x^2 + 2px + 13$ lie at a distance of 5 from the origin ?

609 If the roots of the equation $x^2 - 5x + 16 = 0$ are α, β and the roots of the equation $x^2 + px + q = 0$ are $(\alpha^2 + \beta^2)$ and $\frac{\alpha\beta}{2}$, then- [AIEEE-2002]

- (1) $p = 1$ and $q = 56$ (2) $p = 1$ and $q = -56$
 (3) $p = -1$ and $q = 56$ (4) $p = -1$ and $q = -56$

610 If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are- [AIEEE-2002]

(1) a and c (2) b and c (3) a and b (4) $a + b$ and $b + c$

611 If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is- [AIEEE-2002]

(1) $19/3$ (2) $25/3$ (3) $-19/3$ (4) none of these

612 The value of a for which one roots of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is [AIEEE-2003]

(1) $-2/3$ (2) $1/3$ (3) $-1/3$ (4) $2/3$

613 The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$, is- [AIEEE-2003]

(1) 4 (2) 1 (3) 3 (4) 2

614 If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are- [AIEEE-2004]

(1) $0, -1$ (2) $-1, 1$ (3) $0, 1$ (4) $-1, 2$

615 If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is- [AIEEE-2004]

(1) 3 (2) 12 (3) $49/4$ (4) 4

616 If value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assume the least value is- [AIEEE-2005]

(1) 2 (2) 3 (3) 0 (4) 1

617 If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals- [AIEEE-2005]

(1) 1 (2) 2 (3) 3 (4) -2

618 If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval- [AIEEE-2005]

(1) $[4, 5]$ (2) $(-\infty, 4)$ (3) $(6, \infty)$ (4) $(5, 6)$

619 If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then [JEE 2004 (Screening)]

(1) $-5 < a < 2$ (2) $a < -5$ (3) $a > 5$ (4) $2 < a < 5$

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EXERCISE (S-1)**Solve following Inequalities over the set of real numbers -**

(i) $\frac{x^2 + 2x - 3}{x^2 + 1} < 0$

(ii) $\frac{(x-1)(x+2)^2}{-1-x} < 0$

(iii) $x^4 - 2x^2 - 63 \leq 0$

(iv) $\frac{x+1}{(x-1)^2} < 1$

(v) $\frac{x^2 - 7x + 12}{2x^2 + 4x + 5} > 0$

(vi) $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$

(vii) $\frac{x^4 + x^2 + 1}{x^2 - 4x - 5} < 0$

(viii) $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$

(ix) $\frac{1}{x+2} < \frac{3}{x-3}$

(x) $\frac{14x}{x+1} - \frac{9x-30}{x-4} < 0$

(xi) $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} > 3$

(xii) $\frac{x^2 + 2}{x^2 - 1} < -2$

(xiii) $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \geq 0$

(xiv) $\frac{5-4x}{3x^2-x-4} < 4$

(xv) $\frac{(x+2)(x^2-2x+1)}{4+3x-x^2} \geq 0$

(xvi) $\frac{x^4-3x^3+2x^2}{x^2-x-30} > 0$

(xvii) $\frac{2x}{x^2-9} \leq \frac{1}{x+2}$

(xviii) $\frac{1}{x-2} + \frac{1}{x-1} > \frac{1}{x}$

(xix) $\frac{20}{(x-3)(x-4)} + \frac{10}{x-4} + 1 > 0$

(xx) $\frac{(x-2)(x-4)(x-7)}{(x+2)(x+4)(x+7)} > 1$

(xxi) $(x^2-2x)(2x-2) - 9\frac{2x-2}{x^2-2x} \leq 0$

(xxii) $\frac{|x+2|-x}{x} < 2$

(xxiii) $\left| \frac{x^2-5x+4}{x^2-4} \right| \leq 1$

(xxiv) $\frac{1}{x-2} - \frac{1}{x} \leq \frac{2}{x+2}$

621 α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

622 Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a+b)x + (a-b+8) = 0$ has unequal real roots for all $a \in \mathbb{R}$.

623 Find the range of values of a , such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a+4}{x^2 - 8x + 32}$ is always negative.

624 Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as m/n then find the value of $(m + n)$.

625 When $y^2 + my + 2$ is divided by $(y - 1)$ then the quotient is $f(y)$ and the remainder is R_1 . When $y^2 + my + 2$ is divided by $(y + 1)$ then quotient is $g(y)$ and the remainder is R_2 . If $R_1 = R_2$ then find the value of m .

626 If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b + c + 1 = 0$ or $b^2 + c^2 + 1 = bc + b + c$.

627 Find the value of m for which the quadratic equations $x^2 - 11x + m = 0$ and $x^2 - 14x + 2m = 0$ may have common root.

628 If α, β are the roots of $ax^2 + bx + c = 0$, ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, ($A \neq 0$) for some constant δ , then prove that, $\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$.

629 If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$ then which of the following expressions in α, β will denote the symmetric functions of roots. Give proper reasoning.

(i) $f(\alpha, \beta) = \alpha^2 - \beta$ (ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$

(iii) $f(\alpha, \beta) = \ln \frac{\alpha}{\beta}$ (iv) $f(\alpha, \beta) = \cos(\alpha - \beta)$

630 Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.

631 Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$

632 We call 'p' a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ is satisfied for any real x . Find the smallest integral good number.

633 Find the values of 'a' for which $-3 < [(x^2 + ax - 2)/(x^2 + x + 1)] < 2$ is valid for all real x .

634 Suppose $a, b, c \in I$ such that greatest common divisor of $x^2 + ax + b$ and $x^2 + bx + c$ is $(x + 1)$ and the least common multiple of $x^2 + ax + b$ and $x^2 + bx + c$ is $(x^3 - 4x^2 + x + 6)$. Find the value of $(a + b + c)$.

635 If roots of the equation $(x - \alpha)(x - 4 + \beta) + (x - 2 + \alpha)(x + 2 - \beta) = 0$ are p and q then find the absolute value of the sum of the roots of the equations $2(x - p)(x - q) - (x - \alpha)(x - 4 + \beta) = 0$ and $2(x - p)(x - q) - (x - 2 + \alpha)(x + 2 - \beta) = 0$.

636 If the roots of $x^2 - ax + b = 0$ are real & differ by a quantity which is less than c ($c > 0$), prove that b lies between $(1/4)(a^2 - c^2)$ & $(1/4)a^2$.

637 Find all values of p for which the roots of the equation $(p - 3)x^2 - 2px + 5p = 0$ are real and positive.

638 Find all the values of the parameter 'a' for which both roots of the quadratic equation

$$x^2 - ax + 2 = 0 \text{ belong to the interval } (0, 3).$$

639 At what values of 'a' do all the zeroes of the function $f(x) = (a - 2)x^2 + 2ax + a + 3$ lie on the interval

$$(-2, 1)?$$

640 Consider the quadratic polynomial $f(x) = x^2 - 4ax + 5a^2 - 6a$

- (a) Find the smallest positive integral value of 'a' for which $f(x)$ positive for every real x.
- (b) Find the largest distance between the roots of the equation $f(x) = 0$
- (c) Find the set of values of 'a' for which range of $f(x)$ is $[-8, \infty)$

641 Solve the inequality. Where ever base is not given take it as 10.

$$(i) \quad (\log_2 x)^4 - \left(\log_{\frac{1}{2}} \frac{x^5}{4} \right)^2 - 20 \log_2 x + 148 < 0. \quad (ii) \quad (\log 100x)^2 + (\log 10x)^2 + \log x \leq 14$$

$$(iii) \quad \log_{1/2} (x+1) > \log_2 (2-x).$$

$$(iv) \quad \log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1.$$

$$(v) \quad \log_{1/5} (2x^2 + 5x + 1) < 0.$$

$$(vi) \quad \log_{1/2} x + \log_3 x > 1.$$

$$(vii) \quad \log_{x^2} (2+x) < 1$$

$$(viii) \quad (\log_{|x+6|} 2) \cdot \log_2 (x^2 - x - 2) \geq 1$$

$$(ix) \quad \log_x \frac{4x+5}{6-5x} < -1$$

$$(x) \quad \log_3 \frac{|x^2 - 4x| + 3}{x^2 + |x - 5|} \geq 0$$

642

EXERCISE (S-2)

Solve the following where $x \in \mathbb{R}$.

$$(a) \quad (x-1) |x^2 - 4x + 3| + 2x^2 + 3x - 5 = 0$$

$$(b) \quad 3 |x^2 - 4x + 2| = 5x - 4$$

$$(c) \quad |x^3 + 1| + x^2 - x - 2 = 0$$

$$(d) \quad 2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$$

643 Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$.

644 Suppose a cubic polynomial $f(x) = x^3 + px^2 + qx + 72$ is divisible by both $x^2 + ax + b$ and $x^2 + bx + a$ (where a, b, p, q are constants and $a \neq b$). Find the sum of the squares of the roots of the cubic polynomial.

645 Given $x, y \in \mathbb{R}, x^2 + y^2 > 0$. If the maximum and minimum value of the expression $E = \frac{x^2 + y^2}{x^2 + xy + 4y^2}$ are M and m , and A denotes the average value of M and m , compute (2016)A.

646 Find the complete set of real values of 'a' for which both roots of the quadratic equation

$$(a^2 - 6a + 5)x^2 - \sqrt{a^2 + 2a}x + (6a - a^2 - 8) = 0 \text{ lie on either side of the origin.}$$

647 Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K + 5 = 0$ has atleast one positive root.

648 Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and

649 $3x^4 + 4x^2 + 28x + 5$, find the value of $P(1)$.

Let α, β, γ be distinct real numbers such that

$$a\alpha^2 + b\alpha + c = (\sin\theta)\alpha^2 + (\cos\theta)\alpha$$

$$a\beta^2 + b\beta + c = (\sin\theta)\beta^2 + (\cos\theta)\beta$$

$$a\gamma^2 + b\gamma + c = (\sin\theta)\gamma^2 + (\cos\theta)\gamma$$

(where $a, b, c \in \mathbb{R}$)

(a) Find the maximum value of the expression $\frac{a^2 + b^2}{a^2 + 3ab + 5b^2}$

(b) If $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$ makes an angle $\frac{\pi}{3}$ with $\vec{V}_2 = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, then find the number of values of $\theta \in [0, 2\pi]$

650 Find the product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and

651 $Q(x) = x^3 + 4x^2 - x - 10$.

If $ax^{17} + bx^{16} + 1$ is divisible by $x^2 - x - 1$, then find the integral value of a .

EXERCISE (JM)

652 If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is :- [AIEEE-2009]

(1) Greater than $-4ab$ (2) Less than $-4ab$ (3) Greater than $4ab$ (4) Less than $4ab$

653 Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$.

If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is: [AIEEE-2011]

(1) 18 (2) 3 (3) 9 (4) 6

654 Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are: [AIEEE-2011]

(1) $-4, -3$ (2) $6, 1$ (3) $4, 3$ (4) $-6, -1$

655 The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has :

[AIEEE-2012]

(1) exactly four real roots. (2) infinite number of real roots.
(3) no real roots. (4) exactly one real root.

656 If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is : [JEE-MAIN-2013]

(1) $1 : 2 : 3$ (2) $3 : 2 : 1$ (3) $1 : 3 : 2$ (4) $3 : 1 : 2$

657 Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE-MAIN-2015]

- (1) 3 (2) -3 (3) 6 (4) -6

658 The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is :- [JEE-MAIN-2016]

- (1) 5 (2) 3 (3) -4 (4) 6

EXERCISE (JA)

659 The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is [JEE 2009, 4 (-1)]

660 Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [JEE 2010, 3]

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
(C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

661 Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is [JEE 2011]

- (A) 1 (B) 2 (C) 3 (D) 4

662 A value of b for which the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$, have one root in common is - [JEE 2011]

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

663 Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? [JEE 2015, 4M, -0M]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

- 664 Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

[JEE(Advanced)-2016, 3(-1)]

- (A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

- 665 If $a_4 = 28$, then $p + 2q =$ [JEE(Advanced)-2017, 3(-1)]

- (A) 14 (B) 7 (C) 12 (D) 21

- 666 $a_{12} =$ [JEE(Advanced)-2017, 3(-1)]

- (A) $2a_{11} + a_{10}$ (B) $a_{11} - a_{10}$ (C) $a_{11} + a_{10}$ (D) $a_{11} + 2a_{10}$

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ANSWER KEY

EXERCISE (E-1)

1. $a = 20 \pm 6\sqrt{5}$ 2. $k_1 = \frac{-22}{\alpha}, k_2 = 2$ 3. $a = 4$ 4. $28x^2 - 20x + 1 = 0$

- 5
5. $a_1 = -2, a_2 = 1$ 6. $a_1 = 2, a_2 = 9/2$ 7. $p = \pm 7$ 8. $k = \pm 3\sqrt{5}$ 9. $p = 0$
10. $a = 2$ 11. $x_1^3 + x_2^3 = 3pq - p^3$ 12. $x_1^3 + x_2^3 = \frac{a(a^2 - 18a + 9)}{27}$ 13. $\frac{215}{27}$
14. For all $m \in \left(\frac{1}{4}, +\infty\right)$ 15. For all $c \in [2, 4)$ 16. $k = 13$ 17. $(-3, 5)$
18. $k = 3$ 19. $k = 3$ 20. For all $a \in (-\infty, -6)$ 21. For all $a \in (6, +\infty)$
22. For all $a \in \left(\frac{5}{3}, +\infty\right)$ 23. $a = -2$ 24. $c < 0$ 25. For all $a \in (-6, 3)$

EXERCISE (E-2)

1. $a = \pm 2$ 2. $a = -4$ 3. $a_1 = -\frac{3}{2}, a_2 = 6$ 4. $\{2, 18\}$ for $a = 6$, $\left\{\frac{2}{19}, \frac{18}{19}\right\}$ for $a = -\frac{6}{19}$
5. $a_1 = -\frac{125}{8}, a_2 = \frac{27}{8}$ 6. $a_1 = 3/2, a_2 = 3$ 7. $a_1 = 1/2, a_2 = 1$
8. $p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2$ 9. $\frac{1}{x_1^3} + \frac{1}{x_2^3} = -\frac{27a^3 + 36a}{8}$
10. For all $m \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$ 11. For all $m \in \left(-\infty, -\frac{1}{7}\right) \cup (1, +\infty)$ 12. $\left[0, \frac{1}{2}\right)$
13. For all $a \in \left[2, \frac{9}{4}\right]$ 14. $k = 5$ 15. $m = -2$ 16. $\{-4, -3, 3, 4\}$
17. 4 18. 3 19. 1 20. 4 21. 1 22. 1 23. 3 24. 4
25. 1 26. 2 27. 1

EXERCISE (S-1)

1. (i) $(-3, 1)$ (ii) $(-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$ (iii) $[-3, 3]$ (iv) $(-\infty, 0) \cup (3, +\infty)$
 (v) $(-\infty, 3) \cup (4, +\infty)$ (vi) $(-\infty, +\infty)$ (vii) $(-1, 5)$ (viii) $[1, 3] \cup (5, +\infty)$
 (ix) $(-9/2, -2) \cup (3, +\infty)$ (x) $(-1, 1) \cup (4, 6)$ (xi) $(1/2, 3)$ (xii) $(-1, 0) \cup (0, 1)$
 (xiii) $[-\sqrt{2}, -1) \cup (-1, \sqrt{2}] \cup [3, 4)$ (xiv) $(-\infty, -\sqrt{7}/2) \cup (-1, \sqrt{7}/2) \cup (4/3, +\infty)$
 (xv) $(-\infty, -2] \cup (-1, 4)$ (xvi) $(-\infty, -5) \cup (1, 2) \cup (6, +\infty)$ (xvii) $(-\infty, -3) \cup (-2, 3)$
 (xviii) $(-\sqrt{2}, 0) \cup (1, \sqrt{2}) \cup (2, +\infty)$ (xix) $(-\infty, -2) \cup (-1, 3) \cup (4, +\infty)$
 (xx) $(-\infty, -7) \cup (-4, -2)$ (xxi) $(-\infty, -1] \cup (0, 1] \cup (2, 3]$ (xxii) $(-\infty, 0) \cup (1, +\infty)$
 (xxiii) $[0, 8/5] \cup [5/2, +\infty)$ (xxiv) $\left(-2, \frac{(3-\sqrt{17})}{2}\right] \cup (0, 2) \cup \left[\frac{(3+\sqrt{17})}{2}, +\infty\right)$
2. 254 3. 5 4. $a \in \left(-\infty, -\frac{1}{2}\right)$ 5. 191 6. 0 8. 0 or 24
9. $2x^2 + 2x \cos(A - B) - 2$ 10. (ii) and (iv) 11. $\left(-\frac{3}{4}, \frac{3}{2}\right)$ 12. 20 13. 4
14. $-2 < a < 1$ 15. -6 16. 4 18. For all $p \in \left[3, \frac{15}{4}\right]$ 19. $2\sqrt{2} \leq a < \frac{11}{3}$
20. $\left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6]$ 21. (a) 7, (b) 6, (c) 2 or 4
22. (i) $x \in \left(\frac{1}{16}, \frac{1}{8}\right) \cup (8, 16)$ (ii) $\frac{1}{\sqrt{10^9}} \leq x \leq 10$ (iii) $-1 < x < \frac{1-\sqrt{5}}{2}$ or $\frac{1+\sqrt{5}}{2} < x < 2$

- (iv) $2^{-\sqrt{2}} < x < 2^{-1}$; $1 < x < 2^{\sqrt{2}}$ (v) $(-\infty, -2.5) \cup (0, \infty)$
 (vi) $0 < x < 3^{1/1-\log 3}$ (where base of log is 2) (vii) $-2 < x < -1, -1 < x < 0, 0 < x < 1, x > 2$
 (viii) $x < -7, -5 < x \leq -2, x \geq 4$ (ix) $\frac{1}{2} < x < 1$ (x) $x \leq -\frac{2}{3}$; $\frac{1}{2} \leq x \leq 2$

EXERCISE (S-2)

1. (a) $x = 1$; (b) $x = 2$ or 5 ; (c) $x = -1$ or 1 ; (d) $x \geq -1$ or $x = -3$
 2. $3y^3 - 9y^2 - 3y + 1 = 0$; $(\alpha - 2)(\beta - 2)(\gamma - 2) = 3$ 3. 146 4. 1344
 5. $(-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$ 6. $K \leq -1$ 7. $P(1) = 4$ 8. (a) 2, (b) 3
 9. 6 10. 987

EXERCISE (JM)

1. 1 2. 1 3. 2 4. 3 5. 1 6. 1 7. 2

EXERCISE (JA)

1. 2 2. B 3. C 4. B 5. A,D 6. C 7. C 8. C

667 For what value of 'a', the equation $(a^2 - a - 2)x^2 + (a^2 - 4)x + (a^2 - 3a + 2) = 0$, will have more than two solutions ? Does there exist a real value of 'x' for which the above equation will be an identity in 'a' ?

668 If α and β are the roots of the equation $2x^2 + 3x + 4 = 0$, then find the values of

(i) $\alpha^2 + \beta^2$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

669 If α and β are the roots of the equation $ax^2 + bx + c = 0$, then find the equation whose roots are given by

(i) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$ (ii) $\alpha^2 + 2, \beta^2 + 2$

670 If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3, \beta^2 = 5\beta - 3$, then find the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

671 In copying a quadratic equation of the form $x^2 + px + q = 0$, the coefficient of x was wrongly written as -10 in place of -11 and the roots were found to be 4 and 6. Find the roots of the correct equation.

672

(i) Find the value of the expression $2x^3 + 2x^2 - 7x + 72$ when $x = \frac{3+5\sqrt{-1}}{2}$.

(ii) Find the value of the expression $2x^3 + 2x^2 - 7x + 72$ when $x = \frac{-1+\sqrt{15}}{2}$.

(iii) Solve the following equation $2^{2x} + 2^{x+2} - 32 = 0$

673 Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β

674 If α, β are roots of $x^2 - px + q = 0$ and $\alpha - 2, \beta + 2$ are roots of $x^2 - px + r = 0$, then prove that $16q + (r + 4 - q)^2 = 4p^2$.

675 If one root of the equation $ax^2 + bx + c = 0$ is equal to n^{th} power of the other root, then show that $(ac^n)^{1/(n+1)} + (a^n c)^{1/(n+1)} + b = 0$.

676 If the sum of the roots of quadratic equation $(a + 1)x^2 + (2a + 3)x + (3a + 4) = 0$ is -1 , then find the product of the roots.

677 Find the least prime integral value of ' $2a$ ' such that the roots α, β of the equation $2x^2 + 6x + a = 0$ satisfy the inequality $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < 2$.

678 If α and β be two real roots of the equation $x^3 + px^2 + qx + r = 0$ ($r \neq 0$) satisfying the relation $\alpha\beta + 1 = 0$, then prove that $r^2 + pr + q + 1 = 0$.

679 If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then find the value of

$$\left(\alpha - \frac{1}{\beta\gamma}\right)\left(\beta - \frac{1}{\gamma\alpha}\right)\left(\gamma - \frac{1}{\alpha\beta}\right).$$

680 (i) Solve the equation $24x^3 - 14x^2 - 63x + \lambda = 0$, one root being double of another. Hence find the value(s) of λ .

(ii) Solve the equation $18x^3 + 81x^2 + \lambda x + 60 = 0$, one root being half the sum of the other two. Hence find the value of λ .

681 If α, β, γ are roots of equation $x^3 - 6x^2 + 10x - 3 = 0$, then find cubic equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$.

682 If α, β and γ are roots of $2x^3 + x^2 - 7 = 0$, then find the value of $\sum_{\alpha, \beta, \gamma} \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$.

683 Find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$ if two of its roots are equal.

684 If $2 + i\sqrt{3}$ is a root of the equation $x^2 + px + q = 0$ (where $p, q \in \mathbb{R}$ and $i^2 = -1$), then find the ordered pair (p, q) .

685 If the roots of the equation $x^2 - 2cx + ab = 0$ are real and unequal, then prove that the roots of $x^2 - 2(a + b)x + a^2 + b^2 + 2c^2 = 0$ will be imaginary.

686 For what values of k the expression $kx^2 + (k + 1)x + 2$ will be a perfect square of a linear polynomial.

687 Show that if roots of equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + c^2 - ab = 0$ are equal, then either $b = 0$ or $a^3 + b^3 + c^3 = 3abc$

688 If $a, b, c \in \mathbb{R}$, then prove that the roots of the equation $\frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c} = 0$ are always real and cannot have roots if $a = b = c$.

689 If the roots of the equation $\frac{1}{(x+p)} + \frac{1}{(x+q)} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then show that $p+q = 2r$ and that the product of the roots is equal to $(-1/2)(p^2 + q^2)$.

690 (i) If $-2 + i\beta$ is a root of $x^3 + 63x + \lambda = 0$ (where $\beta \in \mathbb{R} - \{0\}$, $\lambda \in \mathbb{R}$ and $i^2 = -1$), then find roots of equation.

(ii) If $\frac{-1}{2} + i\beta$, is a root of $2x^3 + bx^2 + 3x + 1 = 0$ (where $b, \beta \in \mathbb{R} - \{0\}$ and $i^2 = -1$), then find the value(s) of b .

691 Solve the equation $x^4 + 4x^3 + 5x^2 + 2x - 2 = 0$, one root being $-1 + \sqrt{-1}$.

692 Draw graph of $y = 12x^3 - 4x^2 - 3x + 1$. Hence find number of positive zeroes.

693 Draw the graph of the following expressions :

(i) $y = x^2 + 4x + 3$ (ii) $y = 9x^2 + 6x + 1$ (iii) $y = -2x^2 + x - 1$

694 Find the range of following quadratic expressions :

(i) $f(x) = -x^2 + 2x + 3 \quad \forall x \in \mathbb{R}$

(ii) $f(x) = x^2 - 2x + 3 \quad \forall x \in [0, 3]$

(iii) $f(x) = x^2 - 4x + 6 \quad \forall x \in (0, 1]$

695 If x be real, then find the range of the following rational expressions :

(i) $y = \frac{x^2 + x + 1}{x^2 + 1}$ (ii) $y = \frac{x^2 - 2x + 9}{x^2 - 2x - 9}$

696 Find the range of values of k , such that $f(x) = \frac{kx^2 + 2(k+1)x + (9k+4)}{x^2 - 8x + 17}$ is always negative.

697 $x^2 + (a-b)x + (1-a-b) = 0$, $a, b \in \mathbb{R}$. Find the condition on 'a' for which

(i) Both roots of the equation are real and unequal $\forall b \in \mathbb{R}$.

(ii) Roots are imaginary $\forall b \in \mathbb{R}$

698 If both roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, then show that $a > 11/9$.

699 Find all the values of 'K' for which one root of the equation $x^2 - (K+1)x + K^2 + K - 8 = 0$, exceeds 2 & the other root is smaller than 2.

700 Find all the real values of 'a', so that the roots of the equation $(a^2 - a + 2)x^2 + 2(a-3)x + 9(a^4 - 16) = 0$ are of opposite sign.

701 Find all the values of 'a', so that exactly one root of the equation $x^2 - 2ax + a^2 - 1 = 0$, lies between the numbers 2 and 4, and no root of the equation is either equal to 2 or equal to 4.

702 If α & β are the two distinct roots of $x^2 + 2(K-3)x + 9 = 0$, then find the values of K such that $\alpha, \beta \in (-6, 1)$.

703 If one of the roots of the equation $ax^2 + bx + c = 0$ be reciprocal of one of the roots of $a_1x^2 + b_1x + c_1 = 0$, then prove that $(aa_1 - cc_1)^2 = (bc_1 - ab_1)(b_1c - a_1b)$.

704 Find the value of 'a' so that $x^2 - 11x + a = 0$ and $x^2 - 14x + 2a = 0$ have a common root.

.

705 If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root and a, b, c are non-zero real numbers, then

.

find the value of $\frac{a^3 + b^3 + c^3}{abc}$.

706 If $x^2 + px + q = 0$ and $x^2 + qx + p = 0$, ($p \neq q$) have a common root, show that $1 + p + q = 0$; show that their other roots are the roots of the equation $x^2 + x + pq = 0$.

.

707 Draw the graphs of following :

.

(i) $y = 2x^3 + 9x^2 - 24x + 15$ (ii) $y = -3x^4 + 4x^3 + 12x^2 - 2$

708 Find values of 'k' if equation $x^3 - 3x^2 + 2 = k$ has

.

(i) 3 real roots (ii) 1 real root

709 The roots of the equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are

.

(A) $\frac{c-a}{b-c}, 1$ (B) $\frac{a-b}{b-c}, 1$ (C) $\frac{b-c}{a-b}, 1$ (D) $\frac{c-a}{a-b}, 1$

710 If α, β are the roots of quadratic equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$, then $(\alpha - \gamma) \cdot (\alpha - \delta)$ is equal to :

.

(A) $q + r$ (B) $q - r$ (C) $-(q + r)$ (D) $-(p + q + r)$

711 Two real numbers α & β are such that $\alpha + \beta = 3$, $\alpha - \beta = 4$, then α & β are the roots of the quadratic equation:

.

(A) $4x^2 - 12x - 7 = 0$ (B) $4x^2 - 12x + 7 = 0$ (C) $4x^2 - 12x + 25 = 0$ (D) none of these

712 For the equation $3x^2 + px + 3 = 0$, $p > 0$ if one of the roots is square of the other, then p is equal to:

.

(A) $1/3$ (B) 1 (C) 3 (D) $2/3$

713 Consider the following statements :

.

S_1 : If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then value of $b^2 - 4c$ is equal to 1.

S_2 : If α, β are roots of $x^2 - x + 3 = 0$ then value of $\alpha^4 + \beta^4$ is equal 7.

S_3 : If α, β, γ are the roots of $x^3 - 7x^2 + 16x - 12 = 0$ then value of $\alpha^2 + \beta^2 + \gamma^2$ is equal to 17.

State, in order, whether S_1, S_2, S_3 are true or false

(A) TTT (B) FTF (C) TFT (D) FTT

714 If two roots of the equation $x^3 - px^2 + qx - r = 0$, ($r \neq 0$) are equal in magnitude but opposite in sign, then:

.

(A) $pr = q$ (B) $qr = p$ (C) $pq = r$ (D) None of these

715 If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:

.

(A) zero (B) -1 (C) -7 (D) 1

716 Let α, β, γ be the roots of $(x - a)(x - b)(x - c) = d$, $d \neq 0$, then the roots of the equation

.

$(x - \alpha)(x - \beta)(x - \gamma) + d = 0$ are :

(A) $a + 1, b + 1, c + 1$ (B) a, b, c (C) $a - 1, b - 1, c - 1$ (D) $\frac{a}{b}, \frac{b}{c}, \frac{c}{a}$

717 If α, β, γ are the roots of the equation $x^3 + ax + b = 0$ then value of $\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2}$ is equal to :

.

(A) $\frac{3b}{2a}$ (B) $\frac{-3b}{2a}$ (C) $3b$ (D) $2b$

718 If two of the roots of equation $x^4 - 2x^3 + ax^2 + 8x + b = 0$ are equal in magnitude but opposite in sign, then value of $4a + b$ is equal to :

- (A) 16 (B) 8 (C) -16 (D) -8

719 If one roots of equation $x^2 - \sqrt{3}x + \lambda = 0$, $\lambda \in \mathbb{R}$ is $\sqrt{3} + 2$ then other root is

- (A) $\sqrt{3} - 2$ (B) -2 (C) $2 - \sqrt{3}$ (D) 2

720 If roots of equation $2x^2 + bx + c = 0$; $b, c \in \mathbb{R}$, are real & distinct then the roots of equation $2cx^2 + (b - 4c)x + 2c - b + 1 = 0$ are

- (A) imaginary (B) equal (C) real and distinct (D) can't say

721 Let one root of the equation $x^2 + \ell x + m = 0$ is square of other root. If $m \in \mathbb{R}$ then

- (A) $\ell \in \left(-\infty, \frac{1}{4}\right] \cup \{1\}$ (B) $\ell \in (-\infty, 0]$ (C) $\ell \in \left(-\infty, \frac{1}{9}\right]$ (D) $\ell \in \left[\frac{1}{4}, 1\right]$

722 If a, b, c are integers and $b^2 = 4(ac + 5d^2)$, $d \in \mathbb{N}$, then roots of the quadratic equation $ax^2 + bx + c = 0$ are

- (A) Irrational (B) Rational & different (C) Complex conjugate (D) Rational & equal

723 Let a and b be real numbers such that $4a + 2b + c = 0$ and $ab > 0$. Then the equation $ax^2 + bx + c = 0$ has

- (A) real roots (B) imaginary roots (C) exactly one root (D) none of these

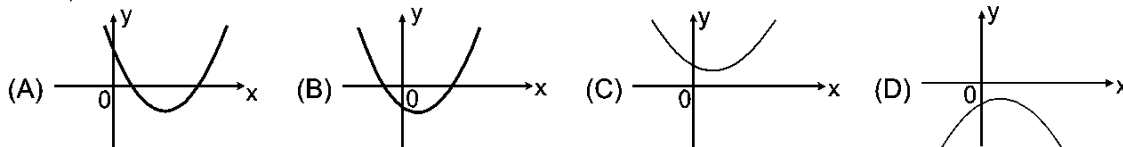
724 Consider the equation $x^2 + 2x - n = 0$, where $n \in \mathbb{N}$ and $n \in [5, 100]$. Total number of different values of 'n' so that the given equation has integral roots, is

- (A) 4 (B) 6 (C) 8 (D) 3

725 If α & β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then

- (A) $0 < \alpha < \beta$ (B) $\alpha < 0 < \beta^2 < \alpha^2$ (C) $\alpha < \beta < 0$ (D) $\alpha < 0 < \alpha^2 < \beta^2$

726 Which of the following graph represents the expression $f(x) = ax^2 + bx + c$ ($a \neq 0$) when $a > 0, b < 0$ & $c < 0$?



727 The expression $y = ax^2 + bx + c$ has always the same sign as of 'a' if :

- (A) $4ac < b^2$ (B) $4ac > b^2$ (C) $4ac = b^2$ (D) $ac < b^2$

728 The entire graph of the expression $y = x^2 + kx - x + 9$ is strictly above the x-axis if and only if

- (A) $k < 7$ (B) $-5 < k < 7$ (C) $k > -5$ (D) none of these

729 If $a, b \in \mathbb{R}$, $a \neq 0$ and the quadratic equation $ax^2 - bx + 1 = 0$ has imaginary roots then $a + b + 1$ is:

- (A) positive (B) negative (C) zero (D) depends on the sign of b

730 If a and b are the non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is

- (A) $\frac{3}{2}$ (B) $\frac{9}{4}$ (C) $-\frac{9}{4}$ (D) 1

731 If $y = -2x^2 - 6x + 9$, then

- (A) maximum value of y is -11 and it occurs at $x = 2$
 (B) minimum value of y is -11 and it occurs at $x = 2$
 (C) maximum value of y is 13.5 and it occurs at $x = -1.5$
 (D) minimum value of y is 13.5 and it occurs at $x = -1.5$

732 If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ are such that $\min f(x) > \max g(x)$, then the relation between b and c , is

- (A) no relation (B) $0 < c < b/2$ (C) $c^2 < 2b$ (D) $c^2 > 2b^2$

733 If $b > a$, then the equation $(x - a)(x - b) - 1 = 0$, has:

- (A) both roots in $[a, b]$ (B) both roots in $(-\infty, a)$
(C) both roots in $[b, \infty)$ (D) one root in $(-\infty, a)$ & other in (b, ∞)

734 If α, β are the roots of the quadratic equation $x^2 - 2p(x - 4) - 15 = 0$, then the set of values of 'p' for which one root is less than 1 & the other root is greater than 2 is:

- (A) $(7/3, \infty)$ (B) $(-\infty, 7/3)$ (C) $x \in \mathbb{R}$ (D) none of these

735 If α, β be the roots of $4x^2 - 16x + \lambda = 0$, where $\lambda \in \mathbb{R}$, such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral solutions of λ is

- (A) 5 (B) 6 (C) 2 (D) 3

736 Set of real values of k if the equation $x^2 - (k-1)x + k^2 = 0$ has atleast one root in $(1, 2)$ is

- (A) $(2, 4)$ (B) $[-1, 1/3]$ (C) $\{3\}$ (D) ϕ

737 If the equations $k(6x^2 + 3) + rx + 2x^2 - 1 = 0$ and $6k(2x^2 + 1) + px + 4x^2 - 2 = 0$ have both roots common, then the value of $(2r - p)$ is

- (A) 0 (B) $1/2$ (C) 1 (D) none of these

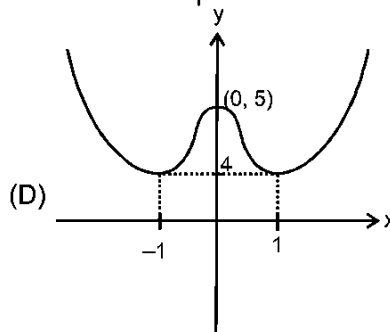
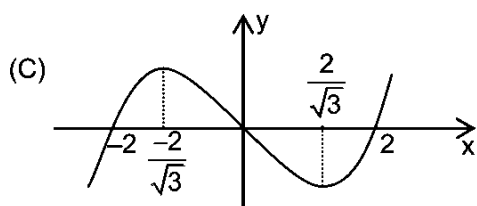
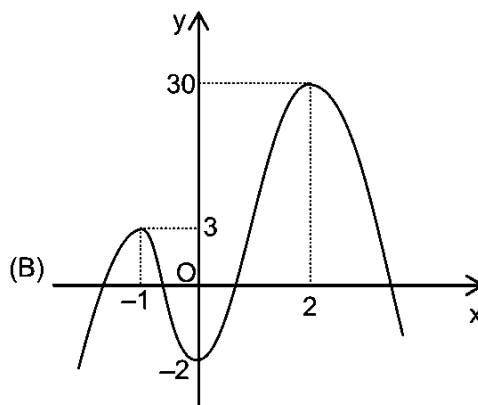
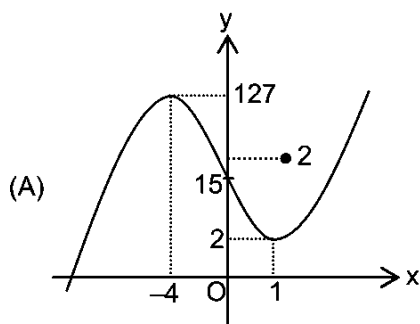
738 If $3x^2 - 17x + 10 = 0$ and $x^2 - 5x + \lambda = 0$ has a common root, then sum of all possible real values of λ is

- (A) 0 (B) $-\frac{29}{9}$ (C) $\frac{26}{9}$ (D) $\frac{29}{3}$

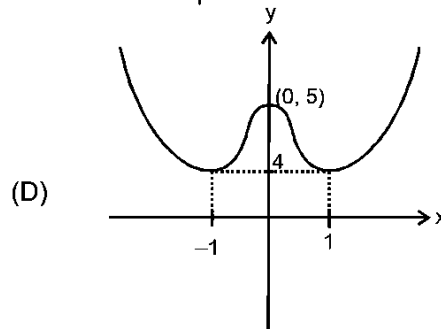
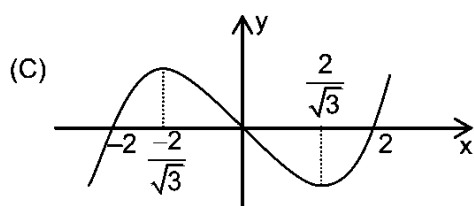
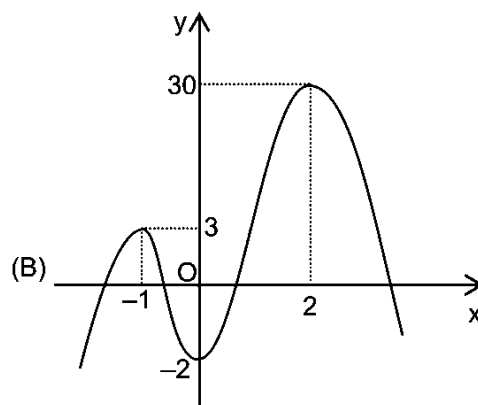
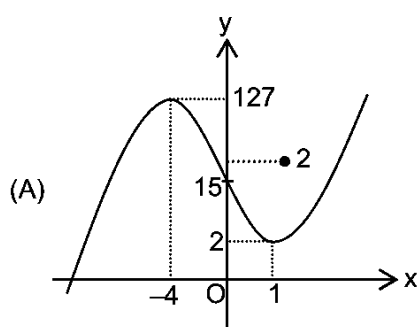
739 If a, b, p, q are non-zero real numbers, then two equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have :

- (A) no common root (B) one common root if $2a^2 + b^2 = p^2 + q^2$
(C) two common roots if $3pq = 2ab$ (D) two common roots if $3qb = 2ap$

740 The graphs of $y = \frac{x^3 - 4x}{4}$ is



741 The graphs of $y = x^4 - 2x^2 + 5$ is



PART - III : MATCH THE COLUMN

742 Column - I

- (A) If $\alpha, \alpha + 4$ are two roots of $x^2 - 8x + k = 0$, then possible value of k is
- (B) If α, β are roots of $x^2 + 2x - 4 = 0$ and $\frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $x^2 + qx + r = 0$ then value of $\frac{-3}{q+r}$ is
- (C) If α, β are roots of $ax^2 + c = 0, ac \neq 0$, then $\alpha^3 + \beta^3$ is equal to
- (D) If roots of $x^2 - kx + 36 = 0$ are Integers then number of values of $k =$

Column - II

- (p) 4
- (q) 0
- (r) 12
- (s) 10

743 If graph of the expression $f(x) = ax^2 + bx + c$ ($a \neq 0$) are given in column-II, then Match the items in column-I with in column-II (where $D = b^2 - 4ac$)

Column-I

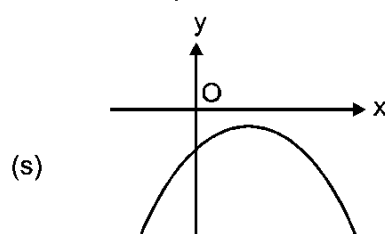
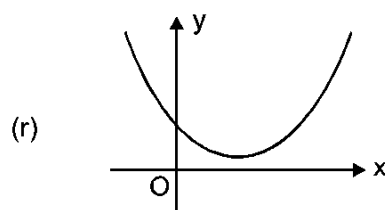
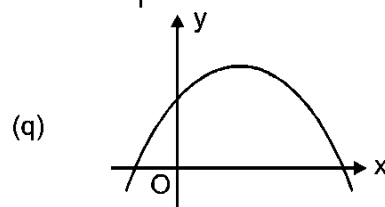
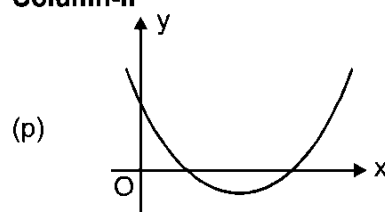
(A) $\frac{abc}{D} > 0$

(B) $\frac{abc}{D} < 0$

(C) $abc > 0$

(D) $abc < 0$

Column-II



744

- Let $y = Q(x) = ax^2 + bx + c$ be a quadratic expression. Match the inequalities in **Column-I** with possible graphs in **Column-II**.

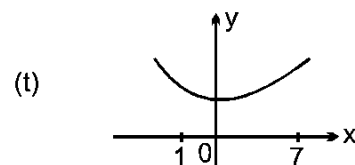
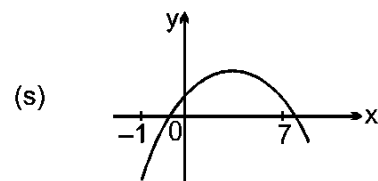
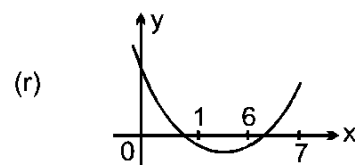
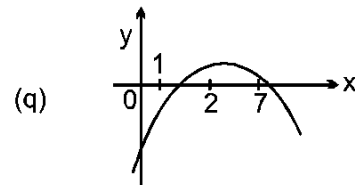
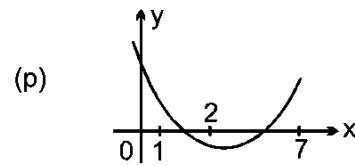
Column-I

(A) $Q(x) > 0, \forall x \in (2, 7)$

(B) $Q(x) > 0, \forall x \in (-\infty, 1)$

(C) $Q(x) < 0, \forall x \in (1, 6)$

(D) $Q(x) < 0, \forall x \in (-\infty, -1)$

Column-II

745 Let $a > 0, b > 0$ & $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$

- (A) are real & negative (B) have negative real parts
(C) are rational numbers (D) have positive real parts

746 If the roots of the equation $x^2 + 2ax + b = 0$ are real and distinct and they differ by at most $2m$, then b lies in the interval

- (A) $(a^2 - m^2, a^2)$ (B) $[a^2 - m^2, a^2)$ (C) $(a^2, a^2 + m^2)$ (D) none of these

747 The set of possible values of λ for which $x^2 - (\lambda^2 - 5\lambda + 5)x + (2\lambda^2 - 3\lambda - 4) = 0$ has roots, whose sum and product are both less than 1, is

- (A) $\left(-1, \frac{5}{2}\right)$ (B) $(1, 4)$ (C) $\left[1, \frac{5}{2}\right]$ (D) $\left(1, \frac{5}{2}\right)$

748 If $p, q, r, s \in \mathbb{R}$, then equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ has

- (A) 6 real roots (B) atleast two real roots
(C) 2 real and 4 imaginary roots (D) 4 real and 2 imaginary roots

- 749 If coefficients of biquadratic equation are all distinct and belong to the set $\{-9, -5, 3, 4, 7\}$, then equation has
- (A) atleast two real roots
 - (B) four real roots, two are conjugate surds and other two are also conjugate surds
 - (C) four imaginary roots
 - (D) None of these

- 750 Let $p, q, r, s \in \mathbb{R}$, $x^2 + px + q = 0$, $x^2 + rx + s = 0$ such that $2(q + s) = pr$ then
- (A) atleast one of the equation have real roots.
 - (B) either both equations have imaginary roots or both equations have real roots.
 - (C) one of equations have real roots and other equation have imaginary roots
 - (D) atleast one of the equations have imaginary roots.

- 751 The equation, $\pi^x = -2x^2 + 6x - 9$ has:
- (A) no solution
 - (B) one solution
 - (C) two solutions
 - (D) infinite solutions

- 752 If $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1$ for all $x \in \mathbb{R}$, then λ belongs to the interval
- (A) $(-2, 1)$
 - (B) $\left[-2, \frac{2}{5}\right)$
 - (C) $\left(\frac{2}{5}, 1\right)$
 - (D) none of these

- 753 Let conditions C_1 and C_2 be defined as follows : $C_1 : b^2 - 4ac \geq 0$, $C_2 : a, -b, c$ are of same sign. The roots of $ax^2 + bx + c = 0$ are real and positive, if
- (A) both C_1 and C_2 are satisfied
 - (B) only C_2 is satisfied
 - (C) only C_1 is satisfied
 - (D) none of these

- 754 If 'x' is real, then $\frac{x^2 - x + c}{x^2 + x + 2c}$ can take all real values if :
- (A) $c \in [0, 6]$
 - (B) $c \in [-6, 0]$
 - (C) $c \in (-\infty, -6) \cup (0, \infty)$
 - (D) $c \in (-6, 0)$

- 755 If both roots of the quadratic equation $(2 - x)(x + 1) = p$ are distinct & positive, then complet set of values of p is:
- (A) $(2, \infty)$
 - (B) $(2, 9/4)$
 - (C) $(-\infty, -2)$
 - (D) $(-\infty, \infty)$

- 756 If two roots of the equation $(a - 1)(x^2 + x + 1)^2 - (a + 1)(x^4 + x^2 + 1) = 0$ are real and distinct, then 'a' lies in the interval
- (A) $(-2, 2)$
 - (B) $(-\infty, -2) \cup (2, \infty)$
 - (C) $(2, \infty)$
 - (D) $(-\infty, -2)$

- 757 The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 respectively, then the ordered pair (x_1, x_2) is:
- (A) $(-5, -7)$
 - (B) $(1, -1)$
 - (C) $(-1, 1)$
 - (D) $(5, 7)$

758

If a, b, c are real and $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in the interval:

(A) $\left[\frac{1}{2}, 2\right]$

(B) $[0, 2]$

(C) $\left[-\frac{1}{2}, 1\right]$

(D) $\left[-1, \frac{1}{2}\right]$

PART - II : NUMERICAL VALUE QUESTIONS**INSTRUCTION :**

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. Find sum of square of real roots of equation $x(x+1)(x+2)(x+3) = 120$
2. Find product of all real values of x satisfying $(5+2\sqrt{6})^{x^2-3} + (5-2\sqrt{6})^{x^2-3} = 10$
3. If a, b are the roots of $x^2 + px + 1 = 0$ and c, d are the roots of $x^2 + qx + 1 = 0$. Then find the value of $(a-c)(b-c)(a+d)(b+d)/(q^2 - p^2)$.
4. α, β are roots of the equation $\lambda(x^2 - x) + x + 5 = 0$. If λ_1 and λ_2 are the two values of λ for which the roots α, β are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 4$, then the value of $\left(\frac{\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}}{15}\right)$ is
5. Let one root of equation $(\ell - m)x^2 + \ell x + 1 = 0$ be double of the other. If ℓ be real and $m \leq k$ then find the least value of k .
6. Let α, β be the roots of the equation $x^2 + ax + b = 0$ and γ, δ be the roots of $x^2 - ax + b - 2 = 0$. If $\alpha\beta\gamma\delta = 24$ and $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{11}{5}$, then find the value of a .
7. If $a > b > 0$ and $a^3 + b^3 + 27ab = 729$ then the quadratic equation $ax^2 + bx - 9 = 0$ has roots α, β ($\alpha < \beta$). Find the value of $4\beta - \alpha\alpha$.
8. Let α and β be roots of $x^2 - 6(5t^2 - 3t + 7)x - 2 = 0$ with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then find the minimum value of $\frac{a_{100} - 2a_{98}}{a_{99}}$ (where $t \in \mathbb{R}$)
9. If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - Kx^3 + Kx^2 + Lx + M = 0$, where K, L & M are real numbers, then the minimum value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is $-n$. Find the value of n .
10. Consider $y = \frac{2x}{1+x^2}$, where x is real, then the range of expression $y^2 + y - 2$ is $[a, b]$. Find the value of $(b-a)$.
11. If the roots of the equation $3x^3 + Px^2 + Qx - 37 = 0$ are each one more than the roots of the equation $x^3 - Ax^2 + Bx - C = 0$, where A, B, C, P & Q are constants, then the value of $A + B + C$ is equal to :
12. If one root of the equation $t^2 - (12x)t - (f(x) + 64x) = 0$ is twice of other, then find the maximum value of the function $f(x)$, where $x \in \mathbb{R}$.
13. The values of k , for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ possess atleast one positive root, are $(-\infty, -b]$. Find value of b .
14. Find the least value of 'a' for which atleast one of the roots of the equation $x^2 - (a-3)x + a = 0$ is greater than 2.

759 If the quadratic equations $3x^2 + ax + 1 = 0$ & $2x^2 + bx + 1 = 0$ have a common root, then the value of the expression $5ab - 2a^2 - 3b^2$ is

760 The equations $3x^2 - 7ax + b = 0$, $x^3 - px^2 + qx = 0$, where $a, b, p, q \in \mathbb{R} - \{0\}$ have one common root & the second equation has two equal roots. Find value of $\frac{3q+b}{aq}$.

761 If $x - y$ and $y - 2x$ are two factors of the expression $x^3 - 3x^2y + \lambda xy^2 + \mu y^3$, then $\lambda^2 + \mu^2$ is

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

762 Possible values of 'p' for which the equation $(p^2 - 3p + 2)x^2 - (p^2 - 5p + 4)x + p - p^2 = 0$ does not possess more than two roots is/are
(A) 0 (B) 1 (C) 2 (D) 4

763 If a, b are non-zero real numbers and α, β the roots of $x^2 + ax + b = 0$, then

- (A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$
- (B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$
- (C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$
- (D) $(\alpha - 1), (\beta - 1)$ are the roots of the equation $x^2 + x(a + 2) + 1 + a + b = 0$

764 If α, β are the roots of $ax^2 + bx + c = 0$ ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of, $Ax^2 + Bx + C = 0$ ($A \neq 0$) for some constant δ , then

- (A) $\delta = \frac{1}{2} \left(\frac{B}{A} - \frac{b}{a} \right)$
- (B) $\delta = \frac{1}{2} \left(\frac{b}{a} - \frac{B}{A} \right)$
- (C) $\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$
- (D) $\frac{b^2 + 4ac}{a^2} = \frac{B^2 + 4AC}{A^2}$

765 If one root of the equation $4x^2 + 2x - 1 = 0$ is ' α ', then

- (A) α can be equal to $\frac{-1 + \sqrt{5}}{4}$
- (B) α can be equal to $\frac{1 + \sqrt{5}}{4}$
- (C) other root is $4\alpha^3 - 3\alpha$.
- (D) other root is $4\alpha^3 + 3\alpha$

766 If α, β are roots of $x^2 + 3x + 1 = 0$, then

- (A) $(7 - \alpha)(7 - \beta) = 0$
- (B) $(2 - \alpha)(2 - \beta) = 11$
- (C) $\frac{\alpha^2}{3\alpha + 1} + \frac{\beta^2}{3\beta + 1} = -2$
- (D) $\left(\frac{\alpha}{1 + \beta} \right)^2 + \left(\frac{\beta}{\alpha + 1} \right)^2 = 18$

767 If both roots of $x^2 - 32x + c = 0$ are prime numbers then possible values of c are
(A) 60 (B) 87 (C) 247 (D) 231

7. Let $f(x) = x^2 - a(x + 1) - b = 0$, $a, b \in \mathbb{R} - \{0\}$, $a + b \neq 0$. If α and β are roots of equation $f(x) = 0$, then the value of $\frac{1}{\alpha^2 - a\alpha} + \frac{1}{\beta^2 - a\beta} - \frac{2}{a + b}$ is equal to

- (A) 0
- (B) $f(a) + a + b$
- (C) $f(b) + a + b$
- (D) $f\left(\frac{a}{2}\right) + \frac{a^2}{4} + a + b$

768 If $f(x)$ is a polynomial of degree three with leading coefficient 1 such that $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, then

- (A) $f(4) = 22$
- (B) $f\left(\frac{6}{5}\right) = \left(\frac{6}{5}\right)^3$
- (C) $f(x) = x^3$ holds for exactly two values of x .
- (D) $f(x) = 0$ has a root in interval $(0, 1)$.

769 Let $P(x) = x^{32} - x^{25} + x^{18} - x^{11} + x^4 - x^3 + 1$. Which of the following are **CORRECT** ?

- (A) Number of real roots of $P(x) = 0$ are zero.
- (B) Number of imaginary roots of $P(x) = 0$ are 32.
- (C) Number of negative roots of $P(x) = 0$ are zero.
- (D) Number of imaginary roots of $P(x) + P(-x) = 0$ are 32.

770 If α, β are the real and distinct roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always (given $\alpha \neq -\beta$)

- (A) two real roots
- (B) two negative roots
- (C) two positive roots
- (D) one positive root and one negative root

771 $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is

- (a, b, c, d $\in \mathbb{R}$)
- (A) $-d/a$
- (B) d/a
- (C) $(b-a)/a$
- (D) $(a-b)/a$

772 If $-5 + i\beta, -5 + i\gamma$ (where $\beta^2 \neq \gamma^2$; $\beta, \gamma \in \mathbb{R}$ and $i^2 = -1$) are roots of $x^3 + 15x^2 + cx + 860 = 0$, $c \in \mathbb{R}$, then

- (A) $c = 222$
- (B) all the three roots are imaginary
- (C) two roots are imaginary but not complex conjugate of each other.
- (D) $-5 + 7i\sqrt{3}, -5 - 7i\sqrt{3}$ are imaginary roots.

773 Let $f(x) = ax^2 + bx + c > 0, \forall x \in \mathbb{R}$ or $f(x) < 0, \forall x \in \mathbb{R}$. Which of the following is/are **CORRECT** ?

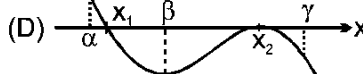
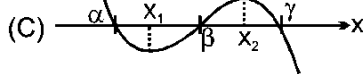
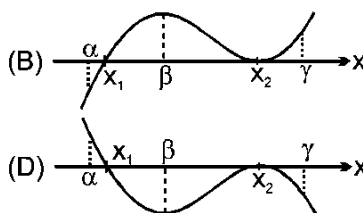
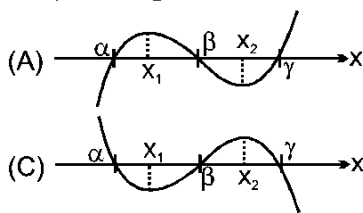
- (A) If $a + b + c > 0$ then $f(x) > 0, \forall x \in \mathbb{R}$
- (B) If $a + c < b$ then $f(x) < 0, \forall x \in \mathbb{R}$
- (C) If $a + 4c > 2b$ then $f(x) < 0, \forall x \in \mathbb{R}$
- (D) $ac > 0$.

774 Let $x_1 < \alpha < \beta < \gamma < x_4, x_1 < x_2 < x_3$. If $f(x)$ is a cubic polynomial with real coefficients such that

- $(f(\alpha))^2 + (f(\beta))^2 + (f(\gamma))^2 = 0, f(x_1)f(x_2) < 0, f(x_2)f(x_3) < 0$ and $f(x_1)f(x_3) > 0$ then which of the following are **CORRECT** ?
- (A) $\alpha \in (x_1, x_2), \beta \in (x_2, x_3)$ and $\gamma \in (x_3, x_4)$
- (B) $\alpha \in (x_1, x_3), \beta, \gamma \in (x_3, x_4)$
- (C) $\alpha, \beta \in (x_1, x_2)$ and $\gamma \in (x_4, \infty)$
- (D) $\alpha \in (x_1, x_3), \beta \in (x_2, x_3)$ and $\gamma \in (x_2, x_4)$

775 If $f(x)$ is cubic polynomial with real coefficients, $\alpha < \beta < \gamma$ and $x_1 < x_2$ be such that $f(\alpha) = f(\beta) = f(\gamma) =$

- $f'(x_1) = f'(x_2) = 0$ then possible graph of $y = f(x)$ is (assuming y-axis vertical)



776 Let $f(x) = \frac{3}{x-2} + \frac{4}{x-3} + \frac{5}{x-4}$, then $f(x) = 0$ has

- (A) exactly one real root in (2, 3)
- (B) exactly one real root in (3, 4)
- (C) 3 different roots
- (D) atleast one negative root

777 If the quadratic equations $ax^2 + bx + c = 0$ ($a, b, c \in \mathbb{R}, a \neq 0$) and $x^2 + 4x + 5 = 0$ have a common root, then a, b, c must satisfy the relations:

- (A) $a > b > c$
- (B) $a < b < c$
- (C) $a = k; b = 4k; c = 5k$ ($k \in \mathbb{R}, k \neq 0$)
- (D) $b^2 - 4ac$ is negative.

778 If the quadratic equations $x^2 + abx + c = 0$ and $x^2 + acx + b = 0$ have a common root, then the equation containing their other roots is/are :

- (A) $x^2 + a(b+c)x - a^2bc = 0$
- (B) $x^2 - a(b+c)x + a^2bc = 0$
- (C) $a(b+c)x^2 - (b+c)x + abc = 0$
- (D) $a(b+c)x^2 + (b+c)x - abc = 0$

779 Consider the following statements.

- **S₁ :** The equation $2x^2 + 3x + 1 = 0$ has irrational roots.
S₂ : If $a < b < c < d$, then the roots of the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are real and distinct.
S₃ : If $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root and $a, b, c \in \mathbb{N}$, then the minimum value of $(a + b + c)$ is 10.
S₄ : The value of the biquadratic expression $x^4 - 8x^3 + 18x^2 - 8x + 2$, when $x = 2 + \sqrt{3}$, is 1
 Which of the following are **CORRECT** ?
 (A) S_2 and S_4 are true. (B) S_1 and S_3 are false.
 (C) S_1 and S_2 are true. (D) S_3 and S_4 are false.

780 If the equations $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ & $x^2 + (a + b)x + 36 = 0$ have a common positive root, then which of the following are true ?
 (A) $ab = 56$ (B) common positive root is 3
 (C) sum of uncommon roots is 21. (D) $a + b = 15$.

781 If $x^2 + \lambda x + 1 = 0$, $\lambda \in (-2, 2)$ and $4x^3 + 3x + 2c = 0$ have common root then $c + \lambda$ can be

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) $\frac{3}{2}$

782 Let quadratic equation $p(x) = 0$ (where $p(x) = x^2 + bx + c$) and equation $p(p(p(x))) = 0$ has a common root, then which of the following statement is/are correct.

- (A) If $b, c \in \mathbb{R}$, then $b^2 - 4c \geq 0$
 (B) If $P(0) = 1$, then $p(1) = 0$
 (C) equations $p(p(p(x))) = 0$ and $p(p(p(p(p(x)))) = 0$ has at least two common root.
 (D) zero is root of equation $p(p(p(p(p(p(x)))))) = 0$

783 Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

[IIT-JEE 2010, Paper-1, (3, -1)/ 84]

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

784 Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of

• $\frac{a_{10} - 2a_8}{2a_9}$ is [IIT-JEE 2011, Paper-1, (3, -1), 80]

- (A) 1 (B) 2 (C) 3 (D) 4

785 A value of b for which the equations

[IIT-JEE 2011, Paper-2, (3, -1), 80]

• $x^2 + bx - 1 = 0$

$x^2 + x + b = 0$

have one root in common is

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

786 The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has

[JEE (Advanced) 2014, Paper-2, (3, -1)/60]

- (A) only purely imaginary roots (B) all real roots
 (C) two real and two purely imaginary roots (D) neither real nor purely imaginary roots

5*. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

[JEE (Advanced) 2015, P-2 (4, -2)/ 80]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

- 787 Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x \sec \theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x \tan \theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

[JEE (Advanced) 2016, Paper-1, (3, -1)/62]

- (A) $2(\sec \theta - \tan \theta)$ (B) $2 \sec \theta$ (C) $-2 \tan \theta$ (D) 0

- 788 $a_{12} =$ [JEE(Advanced) 2017, Paper-2, (3, 0)/61]

- (A) $a_{11} + 2a_{10}$ (B) $2a_{11} + a_{10}$ (C) $a_{11} - a_{10}$ (D) $a_{11} + a_{10}$

- 789 If $a_4 = 28$, then $p + 2q =$ [JEE(Advanced) 2017, Paper-2, (3, 0)/61]
(A) 14 (B) 7 (C) 21 (D) 12

- 9*. Let α and β be the roots of $x^2 - x - 1 = 0$ with $\alpha > \beta$. For all positive integers n , define

$$a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, n \geq 1$$

$$b_1 = 1 \text{ and } b_n = a_{n-1} + a_{n+1}, n \geq 2$$

the which of the following options is/are correct ?

[JEE(Advanced) 2019, Paper-1, (4, -1)/62]

$$(1) \sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$$

$$(2) b_n = \alpha^n + \beta^n \text{ for all } n \geq 1$$

$$(3) a_1 + a_2 + \dots + a_n = a_{n+2} - 1 \text{ for all } n \geq 1$$

$$(4) \sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$$

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

- 790 Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are :

[AIEEE- 2011, II, (4, -1), 120]

- (1) 6, 1 (2) 4, 3 (3) -6, -1 (4) -4, -3

- 791 Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is :

[AIEEE- 2011, II, (4, -1), 120]

- (1) 3 (2) 9 (3) 6 (4) 18

- 792 The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has :

[AIEEE- 2012 (4, -1), 120]

- (1) infinite number of real roots (2) no real roots
(3) exactly one real root (4) exactly four real roots

- 793 If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is

[AIEEE - 2013, (4, -1), 120]

- (1) 1 : 2 : 3 (2) 3 : 2 : 1 (3) 1 : 3 : 2 (4) 3 : 1 : 2

- 794 If $a \in \mathbb{R}$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

[JEE(Main) 2014, (4, -1), 120]

- (1) $(-2, -1)$ (2) $(-\infty, -2) \cup (2, \infty)$ (3) $(-1, 0) \cup (0, 1)$ (4) $(1, 2)$

- 795 Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r are in the A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is : [JEE(Main) 2014, (4, -1), 120]
- (1) $\frac{\sqrt{34}}{9}$ (2) $\frac{2\sqrt{13}}{9}$ (3) $\frac{\sqrt{61}}{9}$ (4) $\frac{2\sqrt{17}}{9}$
- 796 Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE(Main) 2015, (4, -1), 120]
- (1) 6 (2) -6 (3) 3 (4) -3
- 796 The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is : [JEE(Main) 2019, Online (09-01-19), P-2 (4, -1), 120]
- (1) 3 (2) 4 (3) 5 (4) 2
- 797 If λ be the ratio of the roots of the quadratic equation in x , $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is : [JEE(Main) 2019, Online (12-01-19), P-1 (4, -1), 120]
- (1) $-2 + \sqrt{2}$ (2) $4 - 3\sqrt{2}$ (3) $2 - \sqrt{3}$ (4) $4 - 2\sqrt{3}$
- 798 If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is : [JEE(Main) 2019, Online (08-04-19), P-1 (4, -1), 120]
- (1) 3 (2) 4 (3) 2 (4) 5
- 799 If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r \cdot \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to : [JEE(Main) 2019, Online (12-04-19), P-1 (4, -1), 120]
- (1) $\frac{29}{358}$ (2) $\frac{21}{346}$ (3) $\frac{7}{116}$ (4) $\frac{1}{12}$
- 800 If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to - [JEE(Main) 2019, Online (12-04-19), P-2 (4, -1), 120]
- (1) 0 (2) $\alpha\gamma$ (3) $\beta\gamma$ (4) $\alpha\beta$
- 801 Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true? [JEE(Main) 2020, Online (07-01-20), P-2 (4, -1), 120]
- (1) $p_5 = p_2 \cdot p_3$ (2) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
 (3) $p_3 = p_5 - p_4$ (4) $p_5 = 11$
- 802 The number of real roots of the equation, $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is : [JEE(Main) 2020, Online (09-01-20), P-1 (4, -1), 120]
- (1) 3 (2) 1 (3) 4 (4) 2
- 803 Let $a, b \in \mathbb{R}$, $a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to : [JEE(Main) 2020, Online (09-01-20), P-2 (4, -1), 120]
- (1) 25 (2) 26 (3) 24 (4) 28

804 Find the number of values of x satisfying the relation

$$\alpha_1^3 \left(\frac{\prod_{i=2}^n (x - \alpha_i)}{\prod_{i=2}^n (\alpha_1 - \alpha_i)} \right) + \sum_{j=2}^{n-1} \left(\frac{\prod_{i=1}^{j-1} (x - \alpha_i) \prod_{i=j+1}^n (x - \alpha_i)}{\prod_{i=1}^{j-1} (\alpha_j - \alpha_i) \prod_{i=j+1}^n (\alpha_j - \alpha_i)} \right) \alpha_j^3 + \left(\frac{\prod_{i=1}^{n-1} (x - \alpha_i)}{\prod_{i=1}^{n-1} (\alpha_n - \alpha_i)} \right) \alpha_n^3 - x^3 = 0 \text{ (where } n \geq 5).$$

805 Prove that roots of $a^2x^2 + (b^2 + a^2 - c^2)x + b^2 = 0$ are not real, if $a + b > c$ and $|a - b| < c$.
(where a, b, c are positive real numbers)

806 Solve the inequality, $\frac{1}{x-1} - \frac{4}{x-2} + \frac{4}{x-3} - \frac{1}{x-4} < \frac{1}{30}$.

807 If three real and distinct numbers a, b, c are in G.P. (i.e., $b^2 = ac$) and $a + b + c = x b$, then prove that $x < -1$ or $x > 3$.

808 If $V_n = \alpha^n + \beta^n$, where α, β are roots of equation $x^2 + x - 1 = 0$. Then prove that $V_n + V_{n-3} = 2V_{n-2}$ and hence evaluate V_7 (n is a whole number)

809 Find all 'm' for which $f(x) \equiv x^2 - (m-3)x + m > 0$ for all values of 'x' in $[1, 2]$.

810 Find the values of a , for which the quadratic expression $ax^2 + (a-2)x - 2$ is negative for exactly two integral values of x .

811 Find the number of real roots of $\left(x + \frac{1}{x}\right)^3 + \left(x + \frac{1}{x}\right) = 0$

812 If α, β are roots of the equation $x^2 - 34x + 1 = 0$, evaluate $\sqrt[4]{\alpha} - \sqrt[4]{\beta}$, where $\sqrt[4]{\cdot}$ denotes the principal value.

813 Find the values of 'a' for which the equation

$$(x^2 + x + 2)^2 - (a-3)(x^2 + x + 2)(x^2 + x + 1) + (a-4)(x^2 + x + 1)^2 = 0 \text{ has atleast one real root.}$$

814 Show that the quadratic equation $x^2 + 7x - 14(q^2 + 1) = 0$ where q is an integer, has no integral roots.

815 Find the integral values of 'a' for which the equation $x^4 - (a^2 - 5a + 6)x^2 - (a^2 - 3a + 2) = 0$ has only real roots.

816 If α, β, γ and γ, α are the roots of $a_i x^2 + b_i x + c_i = 0$; $i = 1, 2, 3$ then show that

$$(\alpha + \beta + \gamma) + (\alpha\beta + \beta\gamma + \alpha\gamma) + \alpha\beta\gamma = \pm \left\{ \prod_{i=1}^3 \left(\frac{a_i - b_i + c_i}{a_i} \right) \right\}^{\frac{1}{2}} - 1$$

817 Suppose that $a_1 > a_2 > a_3 > a_4 > a_5 > a_6$ and

$$\begin{aligned} p &= a_1 + a_2 + a_3 + a_4 + a_5 + a_6 \\ q &= a_1 a_3 + a_3 a_5 + a_5 a_1 + a_2 a_4 + a_4 a_6 + a_6 a_2 \\ r &= a_1 a_3 a_5 + a_2 a_4 a_6, \end{aligned}$$

then show that roots of the equation $2x^3 - px^2 + qx - r = 0$ are real.



818 If $\beta + \cos^2\alpha$, $\beta + \sin^2\alpha$ are the roots of $x^2 + 2bx + c = 0$ and $\gamma + \cos^4\alpha$, $\gamma + \sin^4\alpha$ are the roots of $X^2 + 2BX + C = 0$, then prove that $b^2 - B^2 = c - C$.

819 Find the set of values of 'a' if $(x^2 + x)^2 + a(x^2 + x) + 4 = 0$ has

- (i) all four real & distinct roots.
- (ii) four roots in which only two roots are real and distinct.
- (iii) all four imaginary roots.
- (iv) four real roots in which only two are equal.

820 $f(x) = x^2 + bx + c$, where $b, c \in \mathbb{R}$, if $f(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$ then find $f(x)$.

821 Let $ax^4 + bx^3 + x^2 + (3-a)x + 3 = 0$ and $x^2 + (2-a)x + 3 = 0$ have common roots. If $a \in (-1, 5)$ then find $|a+12b|$

822 How many quadratic equations are there which are unchanged by squaring their roots ?

823 Let $P(x) = x^5 + x^2 + 1$ have zeros $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ and $Q(x) = x^2 - 2$, then find

- (i) $\prod_{i=1}^5 Q(\alpha_i)$ (ii) $\sum_{i=1}^5 Q(\alpha_i)$ (iii) $\sum_{1 \leq i < j \leq 5} Q(\alpha_i) Q(\alpha_j)$ (iv) $\sum_{i=1}^5 Q^2(\alpha_i)$

824 If a, b, c are non-zero, unequal rational numbers then prove that the roots of the equation $(abc^2)x^2 + 3a^2cx + b^2cx - 6a^2 - ab + 2b^2 = 0$ are rational.

825 If a, b, c represents sides of a Δ then prove that equation $x^2 - (a^2 + b^2 + c^2)x + a^2b^2 + b^2c^2 + c^2a^2 = 0$ has imaginary roots.

826 If x_1 is a root of $ax^2 + bx + c = 0$, x_2 is a root of $-ax^2 + bx + c = 0$ where $0 < x_1 < x_2$, show that the equation $ax^2 + 2bx + 2c = 0$ has a root x_3 satisfying $0 < x_1 < x_3 < x_2$.

827 Find the number of positive real roots of $x^4 - 4x - 1 = 0$.

828 If $(1+k)\tan^2x - 4\tan x - 1 + k = 0$ has real roots $\tan x_1$ and $\tan x_2$, where $\tan x_1 \neq \tan x_2$, then find k .

829 Let Δ^2 be the discriminant and α, β be the roots of the equation $ax^2 + bx + c = 0$. Then find equation whose roots are $2a\alpha + \Delta$ and $2a\beta - \Delta$.

830 Prove that $\frac{\pi^e}{x-e} + \frac{e^\pi}{x-\pi} + \frac{\pi^\pi + e^e}{x-\pi-e} = 0$ has one real root in (e, π) and other in $(\pi, \pi + e)$.

831 If α, β^2 are integers, β^2 is non-zero multiple of 3 and $\alpha + i\beta, -2\alpha$ are roots of $x^3 + ax^2 + bx - 316 = 0$, $a, b, \beta \in \mathbb{R}$, then find a, b .

- 832 Let polynomial $f(x) = ax^4 + bx^3 + cx^2 + dx + e$ have integral coefficient (where $a > 0$) If there exist four distinct integer $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ ($\alpha_1 < \alpha_2 < \alpha_3 < \alpha_4$) such that $f(\alpha_1) = f(\alpha_2) = f(\alpha_3) = f(\alpha_4) = 5$ and equation $f(x) = 9$ has atleast one integral roots then find

(i) $f\left(\frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{4}\right)$ (ii) $f'\left(\frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{4}\right)$

(iii) Range of $f(x)$ in $[\alpha_2, \alpha_3]$

(iv) Difference of largest and smallest root of equation $f(x) = 9$

Quadratic Equation



- 833 If x and y both are non-negative integral values for which $(xy - 7)^2 = x^2 + y^2$, then find the sum of all possible values of x .

- 834 Find the set of all real values of λ such that the root of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are always real for any choice of a, b, c (where a, b, c represents sides of scalene triangle).

(A) $\left(-\infty, \frac{4}{3}\right)$ (B) $\left(\frac{4}{3}, \infty\right)$ (C) $\left(\frac{1}{3}, \frac{5}{3}\right)$ (D) $\left(\frac{4}{3}, \frac{5}{3}\right)$

- 32_. Let $P(x) = x^2 + bx + c$ ($b, c \in \mathbb{R}$), then which of the following statement implies that $P(P(x)) = 0$ has atleast one negative root.

(A) $P(x) = 0$ has root of opposite sign

(B) $P(x) = 0$ has both roots positive

(C) $P(x) = 0$ has both roots negative

(D) $\left(c - \frac{b^2}{4}\right)^2 + b\left(c - \frac{b^2}{4}\right) + c < 0$ & $b > 0$

835 **ANSWERS**

- Infinite
5. -29
7. [1, 2)
9. ± 2
12. $a \in \{1, 2\}$
16. (i) $a \in (-\infty, -4)$ (ii) $a \in \left(\frac{65}{4}, \infty\right)$ (iii) $a \in \left(-4, \frac{65}{4}\right)$ (iv) $a \in \phi$
17. $x^2 - 2x + 5$ 18. 3 19. 4
20. (i) -23 (ii) -10 (iii) 40 (iv) 20
24. 1 25. $(-\sqrt{5}, -1) \cup (-1, \sqrt{5})$
26. $x^2 + 2bx + b^2 = 0$ or $x^2 + 2bx - 3b^2 + 16ac = 0$
28. $a = 0, b = 63$
29. (i) 9 (ii) 0 (iii) [5, 9] (iv) $2\sqrt{5}$
30. 14 31. (A) 32. (AD)

- 836
- If a, b, c are positive then both roots of the equation $ax^2 + bx + c = 0$
- 1) are real and negative
 - 2) are real and positive
 - 3) have negative real parts
 - 4) have positive real parts

- 837
- In a quadratic equation $ax^2 + bx + c = 0$ if 'a' and 'c' are of opposite signs and 'b' is real, then roots of the equation are
- 1) real and distinct
 - 2) real and equal
 - 3) imaginary
 - 4) both roots positive

- 838
- If one root of a quadratic equation is real and the other is imaginary, then the coefficients of the equation are
- 1) real numbers
 - 2) rational numbers
 - 3) irrational numbers
 - 4) complex numbers

- 839
- If $a \neq b$ the roots of the equation $(x-a)(x-b) = b^2$ are
- 1) real and distinct
 - 2) real and equal
 - 3) real
 - 4) imaginary

840 If a, b, c are positive numbers in G.P. then

the roots of the equation $ax^2 + bx + c = 0$

- 1) are real and negative
- 2) have negative real parts
- 3) are equal
- 4) have negative imaginary parts

841 If the ratio of the roots of the equation

$ax^2 + bx + c = 0$ is $m : n$ then

- 1) $\frac{m}{n} + \frac{n}{m} = \frac{b^2}{ac}$
- 2) $\sqrt{\left(\frac{m}{n}\right)} + \sqrt{\left(\frac{n}{m}\right)} = \frac{|b|}{\sqrt{ac}}$
- 3) $\sqrt{\left(\frac{m}{n}\right)} + \sqrt{\left(\frac{n}{m}\right)} = \frac{b^2}{ac}$
- 4) $\frac{m}{n} + \frac{n}{m} = \frac{a^2}{b^2}$

842 If α, β are the roots of the equation

$ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b} =$

- 1) $\frac{a}{bc}$
- 2) $\frac{b}{ac}$
- 3) $\frac{c}{ab}$
- 4) $\frac{ab}{c}$

QUADRATIC EQUATIONS & EXPRESSIONS

843 If the roots of $ax^2 + bx + c = 0$ and

$px^2 + qx + r = 0$ differ by the same quantity,

then $\frac{b^2 - 4ac}{q^2 - 4pr} =$

- 1) $\left(\frac{p}{a}\right)^2$
- 2) $\left(\frac{c}{p}\right)^2$
- 3) $\left(\frac{a}{p}\right)^2$
- 4) $\left(\frac{p}{c}\right)^2$

844 If one root of the equation $ax^2 + bx + c = 0$ is equal to the n^{th} power of the other, then

$(ac^n)^{1/n+1} + (a^n c)^{1/n+1} + b =$

- 1) 0
- 2) 1
- 3) -1
- 4) 2

845 If $ax^2 + 2bx + c = 0$ and $px^2 + 2qx + r = 0$ have one and only one root in common and a, b, c, p, q, r being rational, then $b^2 - ac$ and $q^2 - pr$ are

- 1) both are perfect squares
- 2) $b^2 - ac$ is a perfect square but $q^2 - pr$ is not a perfect square
- 3) $q^2 - pr$ is a perfect square but $b^2 - ac$ is not a perfect square
- 4) both are not perfect squares

846 If both the roots of $ax^2 + bx + c = 0$ are positive then

- 1) $\Delta > 0, ab > 0, ac > 0$
- 2) $\Delta < 0, ab < 0, ac < 0$
- 3) $\Delta > 0, ab < 0, ac > 0$
- 4) $\Delta > 0, ab > 0, bc > 0$

847 If both the roots of $ax^2 + bx + c = 0$ are negative then

- 1) $\Delta > 0, ab > 0, bc < 0$
- 2) $\Delta > 0, a, b, c$, have the same signs
- 3) $\Delta < 0, ab > 0, ac < 0$
- 4) $\Delta < 0, ab > 0, bc > 0$

848 If both the roots of $ax^2 + bx + c = 0$ are negative and $b < 0$ then

- 1) $a < 0, c < 0$
- 2) $a < 0, c > 0$
- 3) $a > 0, c < 0$
- 4) $a > 0, c > 0$

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JEE-MAIN-SR-MATHS VOL-I

849 If $a > 0$, then the expression $ax^2 + bx + c$ is positive for all values of 'x' provided.

- 1) $b^2 - 4ac > 0$
- 2) $b^2 - 4ac < 0$
- 3) $b^2 - 4ac = 0$
- 4) $b^2 - ac < 0$

$$(x - a)(x - b)$$

850 For real x, the expression $\frac{(x-a)(x-b)}{(x-c)}$ will assume real values provided

- 1) $a > b > c$
- 2) $a < b < c$
- 3) $c > a > b$
- 4) $a < c < b$

851 .	If α, β are the roots of $ax^2 + bx + c = 0$ and $k \in R$ then the condition so that $\alpha < k < \beta$ is 1) $ac > 0$ 2) $ak^2 + bk + c > 0$ 3) $ac < 0$ 4) $a^2k^2 + abk + ac < 0$
852 .	If $a > 0$ and $b^2 - 4ac < 0$, then the graph of $y = ax^2 + bx + c$ 1) lies entirely below the x-axis 2) lies entirely above the x-axis 3) cuts the x-axis 4) touches the x-axis and lies below it
853 .	If $a > 0$ and $b^2 - 4ac = 0$, then the graph of $y = ax^2 + bx + c$ 1) lies entirely above the x-axis 2) touches the x-axis and lies above it 3) touches the x-axis and lies below it 4) cuts the x-axis
854 .	If $a < 0$ and $b^2 - 4ac < 0$, then the graph of $y = ax^2 + bx + c$ 1) lies entirely below the x-axis 2) lies entirely above the x-axis 3) cut the x-axis 4) touches the x-axis
855 .	If $b^2 - 4ac > 0$ then the graph of $y = ax^2 + bx + c$ 1) cuts x-axis in two real points 2) touches the x-axis 3) lies entirely above the x-axis 4) can not be determined
856 .	If a, b, c are positive real numbers, then the number of real roots of the equation $ax^2 + b x + c = 0$ is 1) 2 2) 4 3) 0 4) -1
857 .	Product of real roots of the equation $t^2x^2 + x + 9 = 0$ 1) is always positive 2) is always negative 3) does not exist 4) $\frac{9}{t^2}$
858 .	If α and $\beta, (\alpha < \beta)$, are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then 1) $0 < \alpha < \beta$ 2) $\alpha < 0 < \beta < \alpha$ 3) $\alpha < \beta < 0$ 4) $\alpha < 0 < \alpha < \beta$

859 If $0 < a < b < c$, and the roots α, β of the equation $ax^2 + bx + c = 0$ are non real complex roots, then

- 1) $|\alpha| = |\beta|$ 2) $|\alpha| > 1$
 3) $|\beta| < 1$ 4) $|\alpha| > 1, |\beta| < 1$

860 If $b^2 \geq 4ac$ for the equation $ax^4 + bx^2 + c = 0$, then all roots of the equation will be real, if

- 1) $b > 0, a > 0, c > 0$ 2) $b < 0, a > 0, c > 0$
 3) $b > 0, a > 0, c < 0$ 4) $b < 0, a < 0, c < 0$

CUQ KEY

- 1) 3 2) 1 3) 4 4) 1 5) 2 6) 2
 7) 2 8) 3 9) 1 10) 1 11) 3 12) 2
 13) 1 14) 2 15) 4 16) 4 17) 2 18) 2
 19) 1 20) 1 21) 3 22) 3 23) 2 24) 1 25) 2

CUQ HINTS

- $\alpha + \beta = -\frac{b}{a} < 0, \alpha\beta = \frac{c}{a} > 0$
- $ac < 0 \Rightarrow \Delta = b^2 - 4ac > 0$
- If 1 root is real and other is complex number then the coefficients will be complex numbers.
- $\Delta = (a-b)^2 + 4b^2 > 0$
- $b^2 = ac \therefore b^2 - 4ac < 0 \therefore$ roots are imaginary
 $\therefore a, b, c$ are +ve $\Rightarrow$ roots have -ve real parts.
- $\frac{b^2}{ac} = \frac{(m+n)^2}{mn} \Rightarrow \frac{m+n}{\sqrt{mn}} = \sqrt{\frac{b^2}{ac}}$
- $a\alpha + b = -c/\alpha; a\beta + b = -c/\beta$
- $|\alpha - \beta| = |\alpha_1 - \beta_1|$

$$\sqrt{\frac{\Delta}{a^2}} = \sqrt{\frac{\Delta}{p^2}} \Rightarrow \frac{b^2 - 4ac}{q^2 - 4pr} = \frac{a^2}{p^2}$$

9. roots are α, α^n ; $\alpha + \alpha^n = \frac{-b}{a}, \alpha^{n+1} = \frac{c}{a}$.
10. $ax^2 + 2bx + c = 0$ and $px^2 + 2qx + r = 0$ have one root common then it is rational other is also rational then $b^2 - ac = \text{perfect square}$ and $q^2 - pr = \text{perfect square}$
11. $\Delta > 0$, $ab < 0$ and $ac > 0$
12. $\Delta > 0$ and $ab > 0$, $ac > 0$
13. a, b and c will have same sign
14. roots are non real complex roots.

15. put $y = \frac{(x-a)(x-b)}{(x-c)}$

$$x^2 - (a+b+y)x + ab + cy = 0$$

$$x \in R \Rightarrow \Delta = (a+b+y)^2 - 4(ab+cy) > 0$$

$$\Rightarrow y^2 + 2(a+b-2c)y + (a-b)^2 > 0$$

true for all $y \in R$

$$\Rightarrow \Delta = 4(a+b-2c)^2 - 4(a-b)^2 < 0$$

$$\Rightarrow (c-a)(c-b) < 0$$

16. $f(x) = ax^2 + bx + c$
 $a f(k) < 0 \Rightarrow a^2 k^2 + abk + ac < 0$.
17. $b^2 - 4ac < 0$ and $a > 0$ then the graph lies entirely above the x-axis.
18. $b^2 - 4ac = 0$ and $a > 0$ then the graph of y touches the x-axis and lies above x-axis.
19. $a < 0$ and $b^2 - 4ac < 0$ the roots are imaginary then the graph lies entirely below the x - axis.
20. $b^2 - 4ac > 0 \Rightarrow$ roots are real and distinct.
 $\therefore$ graph cuts the x-axis in two distinct points.
21. $ax^2 + b|x| + c > 0, \forall x$
22. real roots does not exist.
23. $\alpha < \beta, c < 0 < b$; $\alpha + \beta = -b, \alpha\beta = C$
 C is negative and b is positive
 $\alpha + \beta$ and $\alpha\beta$ are negative
 $\therefore$ one root is positive and other root is negative
 $\alpha < 0 < \beta$ $\alpha + \beta = -b$ $\beta = -b - \alpha$ $\beta < |\alpha|$

QUADRATIC EQUATIONS & EXPRESSIONS

24. $0 < a < b < c, \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

For non real complex roots $b^2 - 4ac < 0$

$$\Rightarrow \frac{b^2}{a^2} - \frac{4c}{a} < 0$$

$$a^2 - 4\alpha\beta < 0$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta < 0$$

$$\Rightarrow (\alpha - \beta)^2 < 0$$

$$\therefore 0 < a < b < c$$

$\therefore$ roots are conjugate, then $|\alpha| = |\beta|$

25. All roots of equation $ax^4 + bx^2 + c = 0$ will be real if both roots of $ay^2 + by + c = 0$ will be +ve

(let $x^2 = y$), ie, sum of roots $= -\frac{b}{a} > 0$ and

product of roots $= \frac{c}{a} > 0$.

Hence, a and b are of opposite sign, while a and c are of same sign.



LEVEL-I (C.W)

RELATION BETWEEN ROOTS AND COEFFICIENTS

- 861 If α, β are roots of $ax^2 + bx + c = 0$ then

$$\frac{1}{\alpha^3} + \frac{1}{\beta^3} =$$

1) $\frac{3abc - b^3}{a^3}$

2) $\frac{3ab - b^3}{a^2c}$

3) $\frac{3abc - b^3}{c^3}$

4) $\frac{b^2 - 2ac}{ac}$

- 862 If α, β are roots of the equation

$2x^2 + 6x + b = 0 (b < 0)$ then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is less

than

1) 2

2) -2

3) 18

4) 0

863 If α, β are the roots of $x^2 - p(x+1) + C = 0$

then $(\alpha+1)(\beta+1) =$

- 1) $1-C$ 2) $1+C$ 3) $C-1$ 4) C

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ROOTS AND/OR CONDITIONS GIVEN

4. If ω is an imaginary cube root of unity. Then the equation whose roots are $2\omega + 3\omega^2, 2\omega^2 + 3\omega$ is

- 1) $x^2 + 5x + 7 = 0$ 2) $x^2 + 5x - 7 = 0$
 3) $x^2 - 5x + 7 = 0$ 4) $x^2 - 5x - 7 = 0$

5. If p and q are the roots of $x^2 + px + q = 0$ then

- 1) $p = 1$ 2) $p = 1$ or 0
 3) $p = -2$ 4) $p = -2$ or 0

NATURE OF THE ROOTS AND PROPERTIES

6. If one root of $x^2 - (3+2i)x + (1+3i) = 0$ is $1+i$ then the other root is

- 1) $1-i$ 2) $2+i$ 3) $3+i$ 4) $1+3i$

7. If the expression

$x^2 - (5m-2)x + (4m^2 + 10m + 25)$ can be expressed as a perfect square, then $m =$

- 1) $\frac{8}{3}$ or 4 2) $-\frac{8}{3}$ or 4

3) $\frac{4}{3}$ or 8 4) $-\frac{4}{3}$ or 8

8. If the roots of the quadratic equation $x^2 - 4x - \log_3 a = 0$ are real, then the least value of a is

1) 81 2) $\frac{1}{81}$ 3) $\frac{1}{64}$ 4) 9

9. Roots of the equation $2x^2 - 5x + 1 = 0$ and $x^2 + 5x + 2 = 0$ are

- 1) Reciprocal and of the same sign
2) Reciprocal and of opposite sign
3) Equal in magnitude 4) Imaginary

SOLVING EQUATIONS

10. If $3^{1+x} + 3^{1-x} = 10$ then the values of x are

1) 1, -1 2) 1, 0 3) 1, 2 4) -1, -2

11. If $(x+1)$ is a factor of

$$x^4 + (p-3)x^3 - (3p-5)x^2 + (2p-9)x + 6$$

then the value of p is

1) -4 2) 0 3) 4 4) 2

12. The number of real solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is (are)

1) 0 2) 1 3) 2 4) Infinitely many

13. The roots of the equation $x^{\sqrt{x}} = (\sqrt{x})^x$ are

- 1) 0 and 1 2) 0 and 4
3) 1 and 4 4) 0,1 and 4

COMMON ROOTS

- 14. If $(x-2)$ is a common factor of the expressions $x^2 + ax + b$ and $x^2 + cx + d$,**

then $\frac{b-d}{c-a} =$

- 1) -2 2) -1 3) 1 4) 2

SIGN OF THE EXPRESSIONS AND INEQUALITIES

- 15. If $x < 5$, then the sign of the expression**

$2x + 7 - 5x^2$ is

- 1) positive 2) negative
3) non negative 4) can not say

- 16. If x is real and $5x^2 + 2x + 9 > 3x^2 + 10x + 7$, then x lies in the interval**

- 1) $(2 - \sqrt{3}, 2 + \sqrt{3})$
2) $(-\infty, 2 - \sqrt{3}) \cup (2 + \sqrt{3}, \infty)$
3) $(\sqrt{2} - 1, \sqrt{2} + 1)$
4) $(2 + \sqrt{3}, \infty)$

MAXIMUM AND MINIMUM

- 17. Maximum value of $-5x^2 + 2x + 3$ is**

- 1) $\frac{14}{5}$ 2) $\frac{13}{5}$ 3) $\frac{12}{5}$ 4) $\frac{16}{5}$

18. The maximum value of $\frac{1}{4x^2 + 2x + 1}$ is

- 1) $\frac{4}{3}$ 2) $\frac{2}{3}$ 3) 1 4) $\frac{3}{4}$

19. If $x \in [2, 4]$ then for the expression

$$x^2 - 6x + 5$$

- 1) the least value = -4 2) the greatest value = 4
3) the least value = 3
4) the greatest value = -5

MODULUS FUNCTIONS

20. Number of solutions of the equation

$$|x|^2 - 3|x| + 2 = 0 \text{ is.....}$$

- 1) 4 2) 2 3) 0 4) 1

21. For the equation $|x|^2 + |x| - 6 = 0$ the roots are

- 1) one and only one real number
2) real with sum one
3) real with sum zero
4) real with product zero

TRANSFORMED EQUATIONS

22. The equation formed by increasing each root of $ax^2 + bx + c = 0$ by 1 is $2x^2 + 8x + 2 = 0$ then

- 1) $a+b=0$ 2) $b+c=0$ 3) $b=c$ 4) $a=b$

QUADRATIC EXPRESSION IN TWO VARIABLES

23. The condition for $ax^2 + 2cxy + by^2 + 2bx + 2ay + c$ is resolvable into two linear factors is
 1) $a^3+b^3+c^3=3abc$ 2) $a^3+b^3+c^3=abc$
 3) $a^3+b^3+c^3=ab+bc+ca$ 4) $a^3+b^3+c^3+abc=0$
24. If $x^2 + 4xy + 4y^2 + 4x + cy + 3$ can be written as the product of two linear factors, then $c =$
 1) 2 2) 4 3) 6 4) 8

LEVEL-I (C.W.) - KEY

- 1) 3 2) 2 3) 2 4) 1 5) 2 6) 2
 7) 4 8) 2 9) 2 10) 1 11) 3 12) 1
 13) 3 14) 4 15) 4 16) 2 17) 4 18) 1
 19) 1 20) 1 21) 3 22) 3 23) 1 24) 4

LEVEL-I (C.W.) - HINTS

$$1. \quad \frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{(\alpha\beta)^3}$$

$$2. \quad \alpha + \beta = -3, \alpha\beta = b/2$$

$$D = 36 - 8b > 0 (\because b < 0)$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$\frac{(\alpha + \beta)^2}{\alpha\beta} - 2 = \frac{18}{b} - 2 < -2$$

QUADRATIC EQUATIONS & EXPRESSIONS

3. $\alpha\beta + \alpha + \beta + 1 = -p + c + p + 1$
4. $\alpha + \beta = 2(\omega + \omega^2) + 3(\omega^2 + \omega) = -5$
 $\alpha\beta = (2\omega + 3\omega^2)(2\omega^2 + 3\omega) = 7$
5. $p + q = -p, pq = q$
 $\Rightarrow q = 0, p = 1 \Rightarrow p = 0, q = -2$
6. $\alpha + \beta = 3 + 2i$
7. $\Delta = 0$
8. $D \geq 0 \Rightarrow \log_3 a \geq -4 \Rightarrow a \geq 3^{-4} \Rightarrow a \geq \frac{1}{81}$
9. $f(x) = 2x^2 - 5x + 1 = 0$
 $f\left(-\frac{1}{x}\right) = 0 \Rightarrow x^2 + 5x + 2 = 0$
10. Put $3^x = t$
11. Since $x+1$ is a factor of $f(x)$,
 -1 must be a root of $f(x)=0$
 $\therefore f(-1) = 0 \Rightarrow -6p + 24 = 0 \Rightarrow p = 4$
12. $5^x + 5^{-x} \geq 2$ but $\sin(e^x) \leq 1$
14. $x - 2$ is a common factor $\Rightarrow x = 2$ is common root
 $\therefore 4 + 2a + b = 0$ and $4 + 2c + d = 0$
 $\Rightarrow 2(a - c) + (b - d) = 0 \Rightarrow \frac{b - d}{c - a} = 2$

15. $5x^2 - 2x - 7 = 0$;
 $5x^2 + 5x - 7x - 7 = 0 \quad (x+1)(5x-7) = 0$

16. $2x^2 - 8x + 2 > 0$; $x^2 - 4x + 1 > 0$

$$\text{root} = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$$

$$= (-\infty, 2 - \sqrt{3}) \cup (2 + \sqrt{3}, \infty)$$

17. $\frac{4ac - b^2}{4a}$

18. Maximum value of $\left(\frac{1}{4x^2 + 2x + 1} \right)$

$$= \frac{1}{\min \text{ of } (4x^2 + 2x + 1)} = \frac{1}{\frac{4ac - b^2}{4a}}$$

19. $x^2 - 6x + 5 = (x-3)^2 - 4 \geq -4$

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JEE-MAIN-SR-MATHS VOL-I

20. $(|x| - 2)(|x| - 1) = 0 \Rightarrow x = \pm 1, \pm 2$

21. $(|x| + 3)(|x| - 2) = 0 \Rightarrow x = \pm 2$

22. $(\alpha + 1)(\beta + 1) = 1 \Rightarrow \alpha\beta = -(\alpha + \beta)$

$$\Rightarrow \frac{c}{a} = \frac{b}{a} \Rightarrow b = c$$

23. $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

24. $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

**LEVEL-I (H.W)****RELATION BETWEEN ROOTS AND COEFFICIENTS**

864

If α, β are the roots of $x^2 + 3x - 2 = 0$ then

$$\frac{\alpha+1}{\beta} + \frac{\beta+1}{\alpha} =$$

1) $\frac{-15}{2}$ 2) $\frac{6}{5}$ 3) $\frac{-5}{2}$ 4) -5

865 If α, β are the roots of $x^2 + bx + c = 0$ and $\alpha + h, \beta + h$ are the roots of $x^2 + qx + r = 0$ then $h =$

- 1) $b + q$ 2) $b - q$ 3) $\frac{1}{2}(b + q)$ 4) $\frac{1}{2}(b - q)$

866 If the sum of the roots of the equation $5x^2 - 4x + 2 + k(4x^2 - 2x - 1) = 0$ is 6, then $k =$

- 1) $13/17$ 2) $17/13$ 3) $-17/13$ 4) $-13/11$

867 If α, β are the roots of the equation $x^2 + 3x + 2 = 0$. Then $\alpha^5 + \beta^5 =$

- 1) 33 2) -32 3) 35 4) -33

868 The quadratic equation whose roots are $\frac{\sqrt{2} + \sqrt{3}i}{\sqrt{2} - \sqrt{3}i}$ and $\frac{\sqrt{2} - \sqrt{3}i}{\sqrt{2} + \sqrt{3}i}$

- 1) $5x^2 - 2x + 5 = 0$ 2) $5x^2 + 2x + 5 = 0$
3) $5x^2 + 2x - 5 = 0$ 4) $5x^2 - 2x - 5 = 0$

869 Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

- 1) $x^2 + 18x + 16 = 0$ 2) $x^2 - 18x - 16 = 0$
3) $x^2 + 18x - 16 = 0$ 4) $x^2 - 18x + 16 = 0$

870 If the ratio of the roots of $x^2 + bx + c = 0$ and $x^2 + qx + r = 0$ is same then

- 1) $r^2c = qb^2$ 2) $r^2b = qc^2$
3) $rb^2 = cq^2$ 4) $rc^2 = bq^2$

NATURE OF THE ROOTS AND PROPERTIES

871 If the roots of the equation

$$(a^2 + b^2)x^2 + 2(bc + ad)x + (c^2 + d^2) = 0$$

are real then a^2, bd, c^2 are in

- 1) A.P 2) G.P 3) H.P 4) A.G.P

872 The roots of the equation $x^2 - 2\sqrt{3}x + 3 = 0$ are

- 1) rational and equal 2) rational and not equal
3) irrational 4) imaginary

- 873 If '1' is one of the roots of $ax^2 + 3x + 5 = 0$
 then the second root is

1) $\frac{5}{3}$ 2) $\frac{-5}{3}$ 3) $\frac{5}{16}$ 4) $\frac{-5}{8}$

- 874 If '3' is root of $x^2 + kx - 24 = 0$ then it is also
 root of

1) $x^2 + 5x + k = 0$ 2) $x^2 + kx + 24 = 0$
 3) $x^2 - kx + 6 = 0$ 4) $x^2 - 5x + k = 0$

- 875 If $4 - i\sqrt{3}$ is a root of quadratic equation then
 the equation is

1) $x^2 - 8x + 13 = 0$ 2) $x^2 - 8x + 19 = 0$
 3) $x^2 - 8x - 13 = 0$ 4) $x^2 - 8x - 19 = 0$

- 876 If the product of the roots of the equation

$5x^2 - 4x + 2 + k(4x^2 - 2x - 1) = 0$ is 2 then
 $k =$

1) $\frac{-8}{9}$ 2) $\frac{8}{9}$ 3) $\frac{4}{9}$ 4) $\frac{-4}{9}$

COMMON ROOTS

14. If the equations $x^2 + ax + b = 0$ and
 $x^2 + bx + a = 0 (a \neq b)$ have a common root
 then $a + b =$
 1) -1 2) 0 3) 1 4) 2

MAXIMUM AND MINIMUM

15. If x is real then the maximum value of the
 expression $5 + 4x - 4x^2$ is
 1) 6 2) -6 3) 12 4) -12
16. The minimum value of the quadratic
 expression $x^2 + bx + c$ is
 1) cb^2 2) c^2b 3) $c + b^2$ 4) $c - \frac{b^2}{4}$
17. If $2x - 7 - 5x^2$ has maximum value at $x = a$
 then $a =$
 1) $\frac{-1}{5}$ 2) $\frac{1}{5}$ 3) $\frac{34}{5}$ 4) $\frac{-34}{5}$

MODULUS FUNCTIONS

18. If a, b, c are positive real numbers then the number of real roots of the equation

$$ax^2 + b|x| + c = 0 \text{ is}$$

- 1) 2 2) 4 3) 0 4) -1

19. Number of real roots of the equation

$$3x^2 + 4|x| + 5 = 0 \text{ are}$$

- 1) 2 2) 4 3) 0 4) -1

TRANSFORMED EQUATIONS

20. The equation formed by decreasing each root of the equation $ax^2 + bx + c = 0$ by 1 is

$$2x^2 + 8x + 2 = 0 \text{ then}$$

- 1) $a = -b$ 2) $b = -c$ 3) $c = -a$ 4) $b = a + c$

21. If α and β are the roots of $x^2 - 2x + 4 = 0$ then the value of $\alpha^6 + \beta^6$ is (EAM - 2009)

- 1) 32 2) 64 3) 128 4) 256

LEVEL-I (H.W.) - KEY

- 1) 4 2) 4 3) 4 4) 4 5) 2 6) 4
 7) 3 8) 2 9) 3 10) 4 11) 3 12) 2
 13) 1 14) 1 15) 1 16) 4 17) 2 18) 3
 19) 3 20) 2 21) 3

LEVEL-I (H.W.) - HINTS

1. $\alpha + \beta = -3, \alpha\beta = -2$
2. $\alpha + h + \beta + h = -q \Rightarrow 2h = -q - (\alpha + \beta)$
3. Sum of the roots $= 6 \frac{2k+4}{5+4k} = 6 \Rightarrow k = \frac{-13}{11}$

QUADRATIC EQUATIONS & EXPRESSIONS

4. $\alpha = -2, \beta = -1$

5. $\alpha + \beta = \frac{2(2-3)}{2+3}; \alpha\beta = 1$

The equation is $5x^2 + 2x + 5 = 0$

6. Let the two numbers be α and β . Given that

$$\sqrt{\alpha\beta} = 4, \alpha + \beta = 18, \alpha\beta = 16$$

$\therefore$ Required equation is $x^2 - 18x + 16 = 0$

7. $\frac{b_1^2}{a_1c_1} = \frac{b_2^2}{a_2c_2}$

8. $(ax+d)^2 + (bx+c)^2 = 0$

$$ax+d=0 \text{ and } bx+c=0$$

$$\frac{-d}{a} = \frac{-c}{b} = x \Rightarrow ac = bd$$

$$\Rightarrow a^2c^2 = (bd)^2$$

9. $\Delta = 12 - 12 = 0$ roots are $-\sqrt{3}, -\sqrt{3}$

10. $a+3+5=0$, replace a value find roots

11. $9+3k-24=0 \Rightarrow K=5$

12. $4-i\sqrt{3}, 4+i\sqrt{3}$ are the roots $\alpha + \beta = 8$ and
 $\alpha\beta = 19$

13. $\frac{2-k}{5+4k} = 2; 2-10 = 8k+k; k = \frac{-8}{9}$

877 $\alpha^2 + a\alpha + b = 0$ and $\alpha^2 + ba + a = 0$

$$\Rightarrow \alpha(a-b) + (b-a) = 0 \Rightarrow \alpha = 1$$

878 $\frac{4ac-b^2}{4a} = \frac{-80-16}{4(-4)} = \frac{-96}{-16} = 6$

879 Use formula $= \frac{4ac-b^2}{4a}$

880 $x = -\frac{h}{2a} = -\frac{2}{2(-5)} = \frac{1}{5}$

881 $\alpha x^2 + b|x| + c > 0, \forall x \in R$

882 L.H.S is positive for all $x \in R$

883 $f(x+1) = 0$

884 $\alpha = -2\omega^2$

$\beta = -2\omega$

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LEVEL-II (C.W)

RELATION BETWEEN ROOTS AND COEFFICIENTS

885 If α, β are the roots of $ax^2 + bx + c = 0$ then

the value $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$ is

1) $\frac{b^2(b^2 - 4ac)}{c^2a^2}$ 2) $\frac{b^2(b^2 + 4ac)}{c^2a^2}$

3) $\frac{b^2(b^2 - 2ac)}{c^2a^2}$ 4) $\frac{b^2(b^2 + 2ac)}{c^2a^2}$

886 Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

1) $x^2 + 18x + 16 = 0$ 2) $x^2 - 18x - 16 = 0$

3) $x^2 + 18x - 16 = 0$ 4) $x^2 - 18x + 16 = 0$

887 If $\alpha \neq \beta$ but $\alpha^2 = 2\alpha - 3$; $\beta^2 = 2\beta - 3$ then

the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

1) $2x^2 + 3x + 2 = 0$ 2) $3x^2 + 2x + 3 = 0$

3) $2x^2 - 3x + 2 = 0$ 4) $3x^2 - 2x + 3 = 0$

888 If $\alpha + \beta = -2$ and $\alpha^3 + \beta^3 = -56$ then the quadratic equation whose roots are α, β is

[EAM 2008]

1) $x^2 + 2x - 16 = 0$ 2) $x^2 + 2x - 15 = 0$

3) $x^2 + 2x - 12 = 0$ 4) $x^2 + 2x - 8 = 0$

889 If the sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to sum of the squares of their reciprocals, then bc^2, ca^2, ab^2 are in

1) A.P 2) G.P 3) H.P 4) A.G.P

- 890 $\tan 22^\circ$ and $\tan 23^\circ$ are roots of $x^2 + ax + b = 0$ then
- 1) $a + b + 1 = 0$ 2) $a - b + 1 = 0$
 3) $b - a + 1 = 0$ 4) $a + b = 1$

- 891 If the difference of the squares of the roots of the equation $x^2 - 6x + q = 0$ is 24, then the value of q is
- 1) -7 2) 8 3) 5 4) 4

- 892 Let α, β be the roots of the equation $(x - a)(x - b) = c, c \neq 0$, then the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are
- 1) a, c 2) b, c
 3) a, b 4) $a + b, b + c$

- 893 If α, β are the roots of the equation $x^2 - 15x + 1 = 0$, then the value of $\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2}$ is
- 1) 225 2) 900 3) 223 4) 0

ROOTS ARE GIVEN WITH CONDITIONS

- 894 The quadratic equation whose roots are the x and y intercepts of the line passing through $(1, 1)$ and making a triangle of area A with the axes, may be
- 1) $x^2 + Ax + 2A = 0$ 2) $x^2 - 2Ax + 2A = 0$
 3) $x^2 - Ax + 2A = 0$ 4) $x^2 - 2Ax - 2A = 0$

- 895 If one root of $x^2 + px + 12 = 0$ is 4 while the equation $x^2 + px + q = 0$ has equal roots then $q =$
- 1) -7 2) 4 3) 42 4) $\frac{49}{4}$

- 896 If $k > 0$ and the product of the roots of the equation $x^2 - 3kx + 2e^{2\log k} - 1 = 0$ is 7 then the sum of the roots is
- 1) 2 2) 4 3) 6 4) 8

- 897 The value of 'a' for which one root of the quadratic equation

$(a^2 - 5a + 3)x^2 - (3a - 1)x + 2 = 0$ is twice as large as other is

- 1) $-2/3$ 2) $1/3$ 3) $-1/3$ 4) $2/3$

- 898 Two students while solving a quadratic equation in x, one copied the constant term incorrectly and got the roots as 3 and 2. The other copied the constant term and coefficient of x^2 as -6 and 1 respectively. The correct roots are

- 1) 3, -2 2) -3, 2
3) -6, -1 4) 6, -1

- 899 Let A, G and H be the A.M, G.M and H.M of two positive numbers a and b. The quadratic equation whose roots are A and H is

- 1) $Ax^2 - (A^2 + G^2)x + AG^2 = 0$
2) $Ax^2 - (A^2 + H^2)x + AH^2 = 0$
3) $Hx^2 + (H^2 + G^2)x + HG^2 = 0$
4) $Gx^2 - (H^2 + G^2)x + GH^2 = 0$

- 900 If one root of the equation $ax^2 + bx + c = 0$ is the square of the other, then

- 1) $b^2 + ac^2 + a^2c = 3abc$
2) $b^3 + ac^2 + a^2c = 3abc$
3) $b^2 + ac^2 + a^2c + 3abc = 0$
4) $b^3 + ac^2 + a^2c + 3abc = 0$

- 901 The roots of the equation $ax^2 + bx + c = 0, a \in R^+$, are two consecutive odd positive integers. Then

- 1) $|b| \leq 4a$ 2) $|b| \geq 4a$ 3) $|b| \geq 2a$ 4) $|b| \leq a$

- 902 The number of real solution of the equation

$\left(\frac{9}{10}\right)^x = -3 + x - x^2$ is

- 1) 2 2) 0 3) 1 4) 3

- 903 If the roots of $x^2 - bx + c = 0$ are two consecutive integers then $b^2 - 4c =$

- 1) 0 2) 1 3) 2 4) 3

- 904 If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively then the value of $2 + q - p$ is
 1) 1 2) 2 3) 3 4) 0

NATURE OF THE ROOTS AND PROPERTIES

- 905 If $a + b + c = 0$ then the equation $3ax^2 + 2bx + c = 0$ has
 1) imaginary roots 2) real and equal roots
 3) real and different roots 4) rational roots

- 906 If the roots of the equation $(a^2 + b^2)x^2 + 2(bc + ad)x + (c^2 + d^2) = 0$ are real then a^2, bd, c^2 are in
 1) AP 2) GP 3) HP 4) AGP

QUADRATIC EQUATIONS & EXPRESSIONS

- 907 Both the roots of the equation

$$(x-b)(x-c) + (x-c)(x-a) + (x-a)(x-b) = 0$$

are
 1) positive 2) negative
 3) real 4) imaginary

- 908 If a, b, c are in A.P. then the roots of the equation $ax^2 + 2bx + c = 0$ are
 1) real and distinct 2) real and equal
 3) real 4) imaginary

- 909 If $ax^2 - (2a+3)x + (3+5a) = 0$ has no real roots, then 'a' lies in the interval

1) $\left(-\frac{3}{4}, \frac{3}{4}\right)$ 2) $\left(-\infty, -\frac{3}{4}\right)$
 3) $\left(-\infty, \frac{3}{4}\right)$ 4) $\left(-\infty, -\frac{3}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$

- 910 If the roots of

$$(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$$

are equal then

which of the following is not possible

1) $a + b + c = 0$ 2) $a = b = c$
 3) $a + bw + cw^2 = 0$ 4) $a + bw^2 + cw = 0$

- 911 If the roots of $ax^2 - bx - c = 0$ are changed by the same quantity then the expression in a, b, c that does not change is

1) $\frac{b^2 - 4ac}{a^2}$ 2) $\frac{b^2 - 4ac}{a}$
 3) $\frac{b^2 + 4ac}{a^2}$ 4) $\frac{b^2 + 4ac}{a}$

- 912 If the roots of the equation $bx^2 + cx + a = 0$ be imaginary then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is

- 1) greater than $4ab$ 2) less than $4ab$
 3) greater than $-4ab$ 4) less than $-4ab$

SOLVING EQUATIONS

- 913 Let $f(x) = x^2 - 3x + 4$, the value(s) of x which satisfies $f(1) + f(x) = f(1)f(x)$ is
- 1) 1 2) 2
 3) 1 or 2 4) 1 and 0



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- 914 The number of non-zero solutions of the equation $x^2 - 5x - (\operatorname{sgn} x)6 = 0$ is
- 1) 1 2) 2 3) 3 4) 4

- 915 A root of equation $\frac{a+c}{x+a} + \frac{b+c}{x+b} = \frac{2(a+b+c)}{x+a+b}$ is
- 1) a 2) b 3) c 4) $a+b+c$

- 916 The equation $\log_2(3-x) + \log_2(1-x) = 3$ has
- 1) One root 2) Two roots
 3) Infinite roots 4) No root

- 917 The number of real roots of $\sin(2^x)\cos(2^x) = \frac{1}{4}(2^x + 2^{-x})$ is
- 1) 1 2) 2 3) 3 4) No solution

- 918 The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
- 1) infinite number of real roots
 2) no real roots
 3) exactly one real root
 4) exactly four real roots

- 919 The real value of 'a' for which $x^2 + i(a-1)x + 5 = 0$ will have a pair of conjugate complex roots is
- 1) values satisfying $a^2 - 2a + 21 > 0$
 2) 1 3) all values 4) no value

COMMON ROOTS

- 920 If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root and $a \neq 0$ then $\frac{a^3 + b^3 + c^3}{abc} =$
- 1) 1 2) 2 3) 3 4) 9

- 921 The equations $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root. If $a, b, c \in \mathbb{N}$ then the least possible values of $a + b + c$ is equal to
- 1) 3 2) 6 3) 9 4) 12

- 922 If the equations $ax^2 + 2bx + 3c = 0$ and $3x^2 + 8x + 15 = 0$ have a common root where a, b, c are the length of the sides of $\triangle ABC$ then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to
- 1) 1 2) $\frac{3}{2}$ 3) $\sqrt{2}$ 4) 2

LOCATION OF ROOTS

- 923 The set of values of p for which 6 lie between the roots of the equation $x^2 + 2(p-3)x + 9 = 0$ is

1) $\left(\frac{3}{4}, \infty\right)$ 2) $\left(-\infty, -\frac{3}{4}\right)$ 3) $\left(-\frac{3}{4}, \frac{3}{4}\right)$ 4) $\left[-\frac{3}{4}, \frac{3}{4}\right]$

- 924 All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4, lie in the interval
- 1) $-1 < m < 3$ 2) $1 < m < 4$
 3) $-2 < m < 0$ 4) $m > 3$

- 925 If $a \in R$ and the roots of $x^2 - 2x - a^2 + 1 = 0$ lies between the roots of $x^2 - 2(a+1)x + a(a-1) = 0$ then 'a' belongs to

1) $(1, \infty)$ 2) $\left(-\frac{1}{4}, 1\right)$ 3) $(-\infty, 0)$ 4) $\left(-\infty, -\frac{1}{4}\right)$

SIGN OF THE EXPRESSIONS AND INEQUALITIES

- 926 If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is
- 1) $(-3, 3)$ 2) $(-3, \infty)$ 3) $(3, \infty)$ 4) $(-\infty, -3)$

- 927 The greatest negative integer satisfying $x^2 + 4x - 77 < 0$ and $x^2 > 4$ is
- 1) -1 2) -2 3) -3 4) -10

- 928 The solution set contained in R of the inequality $3^x + 3^{1-x} - 4 < 0$ is
- 1) $(1, 3)$ 2) $(0, 1)$ 3) $(1, 2)$ 4) $(0, 2)$

- 929 If the equation $ax^2 + 2bx - 3c = 0$ has non-real roots and $\frac{3c}{4} < a + b$ then c is always
- 1) < 0 2) > 0 3) ≥ 0 4) zero

- 930 The equation $(\cos p - 1)x^2 + (\cos p)x + (\sin p) = 0$ in the variable x has real roots. Then p can take any value in the interval

1) $(0, 2\pi)$ 2) $(-\pi, 0)$ 3) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 4) $(0, \pi)$

MAXIMUM AND MINIMUM OF QUADRATIC EXPRESSION

- 931 If a and b ($\neq 0$) are the roots of the equation $x^2 + ax + b = 0$ then the least value of $x^2 + ax + b$ ($x \in R$) is

1) $\frac{9}{4}$ 2) $\frac{-9}{4}$ 3) $\frac{-1}{4}$ 4) $\frac{1}{4}$

- 932 . **If $x^2 + \frac{1}{x^2} = A$ and $x - \frac{1}{x} = B$ ($|x| > 1$) then least value of $\frac{A}{B}$ is**
 1) 2 2) $\sqrt{2}$ 3) $-\sqrt{2}$ 4) $2\sqrt{2}$

- 933 . **If α, β are the roots of $x^2 - (a-2)x - (a+1) = 0$ where 'a' is a variable then the least value of $\alpha^2 + \beta^2$ is**
 1) 2 2) 3 3) 5 4) 7

MODULUS FUNCTIONS

- 934 . **The minimum value of $|x| + \left|x + \frac{1}{2}\right| + |x-3| + \left|x - \frac{5}{2}\right|$ is**
 1) 2 2) 4 3) 6 4) 4

- 935 . **Sum of the roots of the equation $(x-2)^2 - 2|x-2| - 15 = 0$ is**
 1) 4 2) 0 3) -4 4) 8

- 936 . **The real roots of the equation $|x^2 + 4x + 3| + 2x + 5 = 0$ are**
 1) 4; $-1 + \sqrt{3}$ 2) -4; $-1 - \sqrt{3}$
 3) -6, -1 4) 6, -1

- 937 . **If x satisfies $|x-1| + |x-2| + |x-3| \geq 6$ then**
 1) $0 \leq x \leq 4$ 2) $x \leq -2$ (or) $x \geq 4$
 3) $x \leq 0$ (or) $x \geq 4$ 4) R

MISCELLANEOUS

- 938 . **If $a = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$, $\alpha = a + a^2 + a^4$ and $\beta = a^3 + a^5 + a^6$ then α, β are roots of The equation**
 1) $x^2 + x + 1 = 0$ 2) $x^2 + x + 2 = 0$
 3) $x^2 + 2x + 2 = 0$ 4) $x^2 + 2x + 3 = 0$

QUADRATIC EQUATIONS & EXPRESSIONS

939 The set of values of x for which the inequality

- $[x]^2 - 5[x] + 6 \leq 0$ (where $[.]$ denote the greatest integral function) hold good is

- 1) $2 \leq [x] < 3$ 2) $2 \leq x < 4$
 3) $2 \leq x < 3$ 2) $2 \leq x \leq 4$

940 The number of quadratic equations which are unchanged by squaring their roots is

- 1) 2 2) 4 3) 6 4) 8

941 If α is a root of the equation $4x^2 + 2x - 1 = 0$ then the other root is given by

- 1) $-2\alpha, -1$ 2) $4\alpha^2 + \alpha - 1$
 3) $4\alpha^3 - 3\alpha$ 4) $4\alpha^2 - 3\alpha$

942 The least value of expression

- $x^2 + 4y^2 + 9z^2 - 2x + 8y + 27z + 15$ is

- 1) 15 2) 5 3) 0 4) $\frac{-41}{4}$

943 Let $f(x) = ax^2 + bx + c$, if

- $f(-1) < 1, f(1) > -1, f(3) < -4$ and $a \neq 0$ then

- 1) $a > 0$ 2) $a > 1$ 3) $a < -\frac{1}{8}$ 4) $-\frac{1}{8} < a < 0$

944 The number of distinct pairs (x, y) of real numbers that satisfy the equation

$$4x^2 + 4xy + 2y^2 - 2y + 1 = 0$$

- 1) 0 2) 1 3) 4 4) infinitely many

EQUATIONS REDUCIBLE TO QUADRATIC

945 The number of solutions of the equation

- $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$ is

- 1) 1 2) 2 3) 3 4) 4

$$y = 2 + \frac{1}{4 + \frac{1}{\dots}}$$

946 If $\dots$ then

- 1) $y = 6$ 2) $y = 5$ 3) $y = \sqrt{6}$ 4) $y = \sqrt{5}$

TRANSFORMED EQUATIONS

- 947 If the equation $ax^2 + bx + c = 0$ is not altered when each of the coefficient is increased by the same quantity then $x^2 + x + 1 =$
 1) 1 2) 0 3) 3 4) 2

S

JEE-MAIN-SR-MATHS VOL-I

- 948 If α and β are the roots of the equation $ax^2 + bx + c = 0$ and if $px^2 + qx + r = 0$ has roots $\frac{1-\alpha}{\alpha}$ and $\frac{1-\beta}{\beta}$ then $r =$ (EAM - 2007)
 1) $a + 2b$ 2) $a + b + c$ 3) $ab + bc + ca$ 4) abc

RANGE

- 949 Given that, for all real x , the expression $\frac{x^2 - 2x + 4}{x^2 + 2x + 4}$ lies between $\frac{1}{3}$ and 3. The values between which the expression $\frac{9 \cdot 3^{2x} + 6 \cdot 3^x + 4}{9 \cdot 3^{2x} - 6 \cdot 3^x + 4}$ lies are
 1) $\frac{1}{3}$ and 3 2) -2 and 0
 3) -1 and 1 4) 0 and 2

- 950 Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is

- 1) $[0, 1]$ 2) $\left[0, \frac{1}{2}\right]$ 3) $\left[\frac{1}{2}, 1\right]$ 4) $(0, 1]$

INEQUALITIES

- 951 If x is real and $\frac{x-1}{4x+5} < \frac{x-3}{4x-3}$ then x lies in the interval

- 1) $\left(-\frac{3}{4}, \frac{5}{4}\right)$ 2) $\left(-\frac{5}{4}, -\frac{3}{4}\right)$
 3) $\left(-\frac{5}{4}, \frac{3}{4}\right)$ 4) $\left(-\frac{3}{4}, \frac{7}{5}\right)$

952 . If $(\log_5 x)^2 + \log_5 x < 2$ then x belongs to the interval

- 1) $\left(\frac{1}{25}, 5\right)$ 2) $\left(\frac{1}{5}, \frac{1}{\sqrt{5}}\right)$ 3) $(1, \infty)$ 4) $(5, 25)$

953 . If $(x-1)(x-2)(x+5) < 0$ then

- 1) $x < -3; 0 < x < 2$ 2) $x < -5; 1 < x < 2$
3) $x > 2; -5 < x < 1$ 4) $x < 1; -5 < x < 2$

OTHER MODELS

954 . If $x^3 + ax + 1 = 0$ and $x^4 + ax^2 + 1 = 0$ have common root then the exhaustive set of values of a is (EAM - 2011)

- 1) $(-\infty, -2)$ 2) $[-2, \infty)$ 3) $\{-2\}$ 4) $[-2, 2]$

955 . If x is real then the value of $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$ lies in the interval (EAM - 2011)

- 1) $\left[\frac{1}{3}, 3\right]$ 2) $\left[\frac{1}{5}, 5\right]$ 3) $\left[\frac{1}{6}, 6\right]$ 4) $\left[\frac{1}{7}, 7\right]$

956 . In ΔPQR , $\angle R = \frac{\pi}{4}$, $\tan\left(\frac{\angle P}{3}\right)$, $\tan\left(\frac{\angle Q}{3}\right)$ are the roots of the equation $ax^2 + bx + c = 0$, then (EAM - 2012)

- 1) $a + b = c$ 2) $b + c = 0$
3) $a + c = 0$ 4) $b = c$

- 957 If the harmonic mean between the roots of $(5 + \sqrt{2})x^2 - bx + (8 + 2\sqrt{5}) = 0$ is 4, then the value of b is (EAM - 2013)

1) 2 2) 3 3) $4 - \sqrt{5}$ 4) $4 + \sqrt{5}$

LEVEL-II (C.W.) - KEY

- 1) 1 2) 4 3) 2 4) 4 5) 1 6) 2
 7) 3 8) 3 9) 3 10) 2 11) 4 12) 3
 13) 4 14) 4 15) 1 16) 2 17) 2 18) 2
 19) 2 20) 3 21) 3 22) 2 23) 3 24) 3
 25) 4 26) 1 27) 3 28) 3 29) 3 30) 1
 31) 3 32) 1 33) 4 34) 2 35) 2 36) 3
 37) 3 38) 4 39) 2 40) 1 41) 2 42) 1
 43) 3 44) 2 45) 1 46) 4 47) 2 48) 4
 49) 3 50) 3 51) 1 52) 2 53) 3 54) 2
 55) 2 56) 2 57) 3 58) 4 59) 3 60) 2
 61) 2 62) 4 63) 2 64) 2 65) 1 66) 4
 67) 3 68) 1 69) 2 70) 3 71) 4 72) 1 73) 4

LEVEL-II (C.W.) - HINTS

1. $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2 = \frac{(\alpha - \beta)^2 (\alpha + \beta)^2}{(\alpha\beta)^2}$
2. $\sqrt{\alpha\beta} = 4, \alpha + \beta = 18$
 $\therefore$ the required equation is $x^2 - 18x + 16 = 0$
3. Roots of $x^2 - 2x + 3 = 0$ are α, β
 The required quadratic equation is

$$x^2 - \left(\frac{\alpha^2 + \beta^2}{\alpha\beta} \right)x + 1 = 0 \Rightarrow 3x^2 + 2x + 3 = 0$$

$$4. \quad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \\ \Rightarrow \alpha\beta = -8$$

$$5. \quad \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$6. \quad \alpha = \tan 22^\circ, \beta = \tan 23^\circ \\ \tan 45^\circ = \frac{\tan 22^\circ + \tan 23^\circ}{1 - \tan 22^\circ \tan 23^\circ}$$

$$7. \quad |\alpha^2 - \beta^2| = 24$$

$$\frac{|b|}{a^2} \sqrt{b^2 - 4ac} = 24; \quad 36 - 4q = 16; \quad q = 5$$

$$8. \quad \alpha + \beta = a + b, \alpha\beta = ab - c \Rightarrow ab = \alpha\beta + c$$

$$9. \quad \alpha + \beta = 15, \alpha\beta = 1$$

$$\left(\frac{1}{\alpha} - 15 \right)^{-2} + \left(\frac{1}{\beta} - 15 \right)^{-2} ; \quad \frac{1}{\alpha} = \beta, \frac{1}{\beta} = \alpha \\ = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = (\alpha + \beta)^2 - 2$$

$$10. \quad \text{Let equation of line be } \frac{x}{a} + \frac{y}{b} = 1 \text{ passing through}$$

$$(1,1) \Rightarrow a + b = ab \dots (1)$$

$$\text{Area of the triangle} = \frac{1}{2} ab = A$$

$$\therefore ab = 2A, a + b = 2A$$

$$11. \quad \text{sub } x = 4 \text{ in first then } p = -7 \text{ and } q = \frac{p^2}{4}$$

$$12. \quad \text{Product} = 2e^{2 \log k} - 1 = 7 ; \quad 2k^2 - 1 = 7$$

$$13. \quad \frac{(m+n)^2}{mn} = \frac{b^2}{ac} \text{ where } m:n=1:2$$

$$14. \quad \text{Let original eq be } x^2 + ax + b = 0 \text{ from the given} \\ -a = 5, b = -6$$

$$15. \quad x^2 - (A+H)x + AH = 0$$

$$\text{put } H = \frac{G^2}{A} (AH = G^2)$$

QUADRATIC EQUATIONS & EXPRESSIONS

16. $\alpha + \alpha^2 = -b/a; \alpha^3 = c/a$

$$\alpha^3 + (\alpha^3)^2 + 3\alpha^3(-b/a) = \frac{-b^3}{c^3};$$

$$\frac{c}{a} + \frac{c^2}{a^2} - \frac{3bc}{a^2} = \frac{-b^3}{a^3}; a^2c + ac^2 = 3abc - b^3$$

17. Let $\alpha = 2k - 1, \beta = 2k + 1, k \in \mathbb{N}$

$$\alpha + \beta = -b/a = 4k > 0 \Rightarrow b < 0 \text{ and}$$

$$-b = 4ak \geq 4a \Rightarrow |b| \geq 4a$$

18. LHS always positive RHS always negative

$$LHS \neq RHS$$

19. $b^2 - 4c = 1$

20. $\frac{1}{\sqrt{3}} + 2 - \sqrt{3} = -p; \frac{2}{\sqrt{3}} - 1 = q; 2 + q - p = 3$

21. $\Delta = 4(a^2 + c^2 - ac) = 2(a^2 + c^2 + (a - c^2)) > 0$

22. $(ax + d)^2 + (bx + c)^2 = 0$

$$ax + d = 0 \text{ and } bx + c = 0$$

$$\frac{-d}{a} = \frac{-c}{b} = x \Rightarrow ac = bd \Rightarrow a^2c^2 = (bd)^2$$

23. $3x^2 - 2(a + b + c)x + (ab + bc + ca) = 0$

$$B^2 - 4AC = 4[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$= 4[a^2 + b^2 + c^2 - ab - bc - ca] \geq 0$$

$\therefore$ roots are equal.

$$\Delta = 4b^2 - 4ac = (a+c)^2 - 4ac$$

$$24. \quad = (a-c)^2 \geq 0 \quad (\because 2b = a+c)$$

$$25. \quad \Delta < 0 \Rightarrow (2a+3)^2 - 4a(3+5a) < 0$$

$$\Rightarrow 4a^2 + 12a + 9 - 12a - 20a^2 < 0$$

$$\therefore a \in \left(-\infty, -\frac{3}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$$

$$26. \quad \Delta = 0 \Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$$

$$\Rightarrow (a+b\omega+c\omega^2)(a+b\omega^2+c\omega) = 0$$

$$(\text{or}) (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$$\Rightarrow a = b = c$$

5

JEE-MAIN-SR-MATHS VOL-I

27. If the roots are changed by same quantity, then the difference of the roots will not be change.

28. As $bx^2 + cx + a = 0$ has imaginary roots So,

$$c^2 < 4ab \quad \text{Now, } 3b^2x^2 + 6bcx + 2c^2$$

$$= 3(bx+c)^2 - c^2 \geq -c^2 > -4ab$$

$$29. \quad f(1) = 2 \Rightarrow f(1) + f(x) = f(1)f(x)$$

$$\Rightarrow \therefore 2 + x^2 - 3x + 4 = 2(x^2 - 3x + 4)$$

$$\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow x = 1, x = 2$$

30. **CASE I:** When $x < 0$, $\text{sgn } x = -1$,

$$\Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3$$

But $x < 0$. Hence the equation has no solution in this case.

CASE II: When $x > 0$, $\text{sgn } x = 1$

$$\Rightarrow x^2 - 5x - 6 = 0 \Rightarrow x = -1, 6$$

$$\Rightarrow x = 6 [\because x > 0]$$

Hence the given equation has only one solution.

31. Verify from options

32. $\log_2(3-x)(1-x) = 3 \Rightarrow (3-x)(1-x) = 8$

$$\Rightarrow x = 5 \text{ \& } x = -1, x = 5 \text{ fail } \therefore x = -1$$

33. $\sin(2 \cdot 2^x) = \frac{2^x + 2^{-x}}{2} \geq 1$ and $\sin(2 \cdot 2^x) \neq 1$

34. Let $e^{\sin x} = t \Rightarrow t^2 - 4t - 1 = 0$

35. Let roots be $p + iq, p - iq$

$$\text{sum of roots} = 2p = (1-a)i \Rightarrow 1-a = 0$$

$$a = 1 (\because p \text{ is real})$$

36. $a\alpha^2 + b\alpha + c = 0$ and

$$b\alpha^2 + c\alpha + a = 0 \Rightarrow \alpha = 1 \therefore a + b + c = 0$$

37. $\frac{a}{1} = \frac{b}{3} = \frac{c}{5} = \lambda (\text{say}) \Rightarrow a + b + c = 9\lambda$

38.. $\Delta = -116 < 0$ Roots are imaginary and conjugate to each other. Both roots are common

$$\therefore \frac{a}{3} = \frac{2b}{8} = \frac{3c}{15}$$

$$a = 3, b = 4, c = 5$$

$\triangle ABC$ is right angle triangle $c^2 = a^2 + b^2$

39. $af(k) < 0$ and $\Delta > 0$

40. The roots of $x^2 - 2mx + m^2 - 1 = 0$ are

$$\frac{2m \pm \sqrt{4m^2 - 4(m^2 - 1)}}{2} = m \pm 1$$

$$\text{Given } -2 < m - 1, m + 1 < 4 \Rightarrow m > -1, m < 3$$

41. $(x-1)^2 = a^2 \Rightarrow x = 1 \pm a$

$$\text{let } f(x) = x^2 - 2(a+1)x + a(a-1)$$

$$\text{then } f(1+a) < 0 \text{ and } f(1-a) < 0$$

42. $|\alpha - \beta| < \sqrt{5} \Rightarrow \sqrt{a^2 - 4} < \sqrt{5}$

43. $(x+11)(x-7) < 0$ and $(x+2)(x-2) > 0$

$$x \in (-11, -2) \cup (2, 7)$$

44. Let $3^x = t \quad \therefore 3^x + 3^{1-x} - 4 < 0$

$$\Rightarrow t^2 - 4t + 3 < 0 \Rightarrow (t-1)(t-3) < 0$$

$$\therefore 1 < t < 3 \Rightarrow 1 < 3^x < 3 \Rightarrow x \in (0, 1)$$

45. $4a + 4b - 3c > 0 \Rightarrow f(2) > 0$

$$\therefore f(x) > 0 \forall x \in R \quad \therefore f(0) > 0$$

46. $\Delta = \cos^2 p - 4(\cos p - 1)(\sin p)$

$$= (\cos p - 2\sin p)^2 + 4(1 - \sin p)\sin p$$

$$\Delta \geq 0 \Rightarrow \sin p \geq 0 \Rightarrow p \in (0, \pi)$$

47. $a + b = -a, ab = b \Rightarrow a = 1, b = -2$

$$\min = \frac{4b - a^2}{4}$$

48. $\frac{A}{B} = \frac{x^2 + \frac{1}{x^2}}{x - \frac{1}{x}} = \frac{\left(x - \frac{1}{x}\right)^2 + 2}{x - \frac{1}{x}}$

$$= \left(x - \frac{1}{x}\right) + \frac{2}{x - \frac{1}{x}} \text{ where } p = x - \frac{1}{x}$$

$$= p + \frac{2}{p} \geq 2\sqrt{2} \quad (A.M \geq G.M)$$

49. $\alpha^2 + \beta^2 = (a-2)^2 + 2(a+1)$

$$= (a-1)^2 + 5 \geq 5$$

50. The minimum of $\left\{f(0), f\left(\frac{-1}{2}\right), f(3), f\left(\frac{5}{2}\right)\right\}$

51. $(|x-2|-5)(|x-2|+3) = 0$

$$|x-2| = 5, |x-2| + 3 \neq 0$$

$$x = \pm 5 + 2 \Rightarrow x = 7, -3$$

52. -4 satisfies the equation

53. suppose $x \geq 3$ eq becomes

$$(x-1) + (x-2) + (x-3) \geq 6 \Rightarrow x \geq 4$$

suppose $2 \leq x \leq 3$ eq becomes

$$(x-1) + (x-2) - (x-3) \geq 6 \Rightarrow x \geq 6$$

suppose $1 \leq x \leq 2$ eq becomes

$$(x-1) + (x-2) - (x-3) \geq 6 \Rightarrow x \leq -2$$

suppose $x \leq 1$ eq becomes

$$-(x-1) - (x-2) - (x-3) \geq 6 \Rightarrow x \leq 0$$

from the above cases $x \leq 0$ or $x \geq 4$

$$54. \quad \alpha = a + a^2 + a^4, \quad \beta = a^3 + a^5 + a^6$$

$1, a, a^2, a^3, a^4, a^5, a^6$ are 7th roots of unity.

$$1 + a + a^2 + a^3 + a^4 + a^5 + a^6 = 0$$

$$\alpha + \beta = -1$$

$$\alpha\beta = 3 + a + a^2 + a^3 + a^4 + a^5 + a^6 = 2$$

$$55. \quad [x^2] - 5[x] + 6 \leq 0; \quad ([x] - 2)([x] - 3) \leq 0$$

$$2 \leq [x] \leq 3; \quad 2 \leq x < 4$$

$$56. \quad x^2 = 0, x^2 - x = 0 \quad x^2 + x + 1 = 0, x^2 - 2x + 1 = 0$$

$$57. \quad 4x^2 + 2x - 1 = 0 \quad \alpha = \frac{\sqrt{5}-1}{4} = \sin 18^\circ;$$

$$\beta = -\left(\frac{\sqrt{5}+1}{4}\right) = -\sin 54^\circ$$

$$= -(3\alpha - 4\alpha^3) = 4\alpha^3 - 3\alpha$$

$$58. \quad x^2 + 4y^2 + 9z^2 - 2x + 8y + 27z + 15$$

$$= (x-1)^2 + 4(y+1)^2 + 9(z+3/2)^2 - \frac{41}{4} \geq -41/4$$

$$59. \quad a - b + c < 1, \quad a + b + c > -1; \quad 9a + 3b + c < -4$$

$$\Rightarrow 12a + 4c < -1 \text{ and}$$

$$6a - 2c < -1 \Rightarrow 24a < -3$$

$$60. \quad \text{The equation is } (2x+y)^2 + (y-1) = 0$$

$$958. \quad \text{min value} = \frac{4ac - b^2}{4a} = \frac{1}{1+b^2} \therefore m(b) = \frac{1}{1+b^2}$$

$$\text{Range} = (0, 1]$$

$$(x-1)(4x-3) - (x-3)(4x+5)$$

959 $\frac{18}{(4x+5)(4x-3)} < 0$

$$\Rightarrow \frac{18}{(4x+5)(4x-3)} < 0$$

$$\therefore (4x+5)(4x-3) < 0; \therefore x \in \left(-\frac{5}{4}, \frac{3}{4} \right)$$

960 We have, $(\log_5 x)^2 + \log_5 x < 2$

Put $\log_5 x = a$, then

$$a^2 + a < 2 \Rightarrow (a+2)(a-1) < 0$$

$$\Rightarrow -2 < a < 1 \text{ or } -2 < \log_5 x < 1$$

$$\therefore 5^{-2} < x < 5 \text{ or } \frac{1}{25} < x < 5$$

69. $\therefore x \in (-\infty, -5) \cup (1, 2)$

From the diagram of wave nature of the curve

70. solve $x(x^3 + ax + 1) = 0$ i.e.,

$$x^4 + ax^2 + x = 0 \text{ and } x^4 + ax^2 + 1 = 0$$

71. range = $\left(f\left(+\sqrt{\frac{c}{a}}\right), f\left(-\sqrt{\frac{c}{a}}\right) \right)$

72. $\angle R = \frac{\pi}{4} \therefore P + Q = \frac{3\pi}{4}$

$$\tan\left(\frac{P}{3} + \frac{Q}{3}\right) = \tan\left(\frac{\pi}{4}\right) \Rightarrow a + b = c$$

73. $\alpha + \beta = \frac{b}{5 + \sqrt{2}}$ and $\alpha\beta = \frac{8 + 2\sqrt{5}}{5 + \sqrt{2}}$ given that

harmonic mean between the roots of the given equation is 4.

$$\frac{2\alpha\beta}{\alpha + \beta} = 4; b = \frac{8 + 2\sqrt{5}}{2} = 4 + \sqrt{5}$$



LEVEL-II (H.W)

961

If α, β are the roots of the equation

$8x^2 - 3x + 27 = 0$, then the value of

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} \text{ is}$$

- 1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{7}{2}$ 4) 4

962 If α, β are the roots of $x^2 - px + q = 0$ then the equation whose roots are $\alpha\beta + \alpha + \beta$, $\alpha\beta - \alpha - \beta$ is

- 1) $x^2 + 2qx + p^2 + q^2 = 0$
- 2) $x^2 - 2qx + p^2 + q^2 = 0$
- 3) $x^2 - 2qx + q^2 - p^2 = 0$
- 4) $x^2 + 2qx + q^2 - p^2 = 0$

963 If $\sin \theta, \cos \theta$ are the roots of the equation $ax^2 + bx + c = 0$ then

- 1) $a^2 - b^2 + 2ac = 0$ 2) $a^2 + b^2 + 2ac = 0$
- 3) $a - b + 2ac = 0$ 4) $a + b + 2c = 0$

964 If α, β are the roots of $x^2 - x + 1 = 0$ then

$\alpha^5 + \beta^5 =$

- 1) 2 2) 1 3) 12 4) 4

965 If α, β are the roots of $3x^2 + 5x - 7 = 0$ then

the value of $\left(\frac{1}{3\alpha+5}\right)^2 + \left(\frac{1}{3\beta+5}\right)^2$ is

- 1) $\frac{67}{63}$ 2) $\frac{67}{441}$ 3) $\frac{109}{441}$ 4) $\frac{109}{63}$

966 If the sum of the roots of the equation

$ax^2 + bx + c = 0$ is equal to sum of their squares, then

- 1) $ab + b^2 + 2ac = 0$ 2) $ab + a^2 + 2ac = 0$
- 3) $ab + b^2 - 2ac = 0$ 4) $ab + a^2 - 2ac = 0$

967 If α, β are the roots of $6x^2 - 6x + 1 = 0$ then

$\frac{1}{2}(a + ba + c\alpha^2 + d\alpha^3) + \frac{1}{2}(a + b\beta + c\beta^2 + d\beta^3) =$

- 1) $a + b + c + d$
- 2) $a + 2b + 3c + 4d$
- 3) $a + (b/2) + (c/3) + (d/4)$ 4) 0

968 If α, β are the roots of $x^2 + ax - b = 0$ and

γ, δ are the roots of $x^2 + ax + b = 0$ then

$(\alpha - \gamma)(\alpha - \delta)(\beta - \delta)(\beta - \gamma) =$

- 1) $4b^2$ 2) b^2 3) $2b^2$ 4) $3b^2$

969 If the arithmetic mean of the roots of a quadratic equation is $\frac{8}{5}$ and the arithmetic mean of their reciprocal is $\frac{8}{7}$ then the equation is

- 1) $5x^2 + 16x + 7 = 0$ 2) $5x^2 - 16x + 7 = 0$
 3) $7x^2 + 16x + 5 = 0$ 4) $7x^2 - 16x + 5 = 0$

970 If one root of the equation $8x^2 - 6x + k = 0$ is the square of the other, then $k =$

- 1) 0,3 2) -1,27 3) 0,-2 4) 1,-27

971 If the ratio of the roots of $ax^2 + 2bx + c = 0$ is same as the ratio of the roots of $px^2 + 2qx + r = 0$ then

- 1) $\frac{b^2}{ac} = \frac{p^2}{qr}$ 2) $\frac{b}{ac} = \frac{q}{pr}$
 3) $\frac{b^2}{ac} = \frac{q^2}{pr}$ 4) $\frac{b}{ac} = \frac{q^2}{pr}$

972 If the difference of the roots of the equation $x^2 - bx + c = 0$ is equal to the difference of the roots of the equation $x^2 - cx + b = 0$ and $b \neq c$ then $b + c =$

- 1) 0 2) 2 3) 4 4) -4

973 Which one of the following Means of the roots of the equation $x^2 - 2bx + a^2 = 0$ is the A.M. of the roots of $x^2 - 2ax + b^2 = 0$?

- 1) A.M 2) G.M 3) H.M 4) A.G.P

974 If $\tan A, \tan B$ are the roots of $x^2 - 2x + 2 = 0$ then $\sin^2(A + B) =$

- 1) $\frac{4}{5}$ 2) $\frac{1}{2}$ 3) $\frac{3}{5}$ 4) $\frac{1}{4}$

975 If the roots of $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$ are numerically equal but opposite in sign then $k =$

- 1) c 2) $\frac{1}{c}$ 3) $\frac{a+b}{a-b}$ 4) $\frac{a-b}{a+b}$

**NATURE OF THE ROOTS AND
PROPERTIES**

976 If $pr = 2(q + s)$ then among the equations

$$x^2 + px + q = 0 \text{ and } x^2 + rx + s = 0$$

- 1) both have real roots
- 2) both have imaginary roots
- 3) at least one has real roots
- 4) at least one has imaginary roots.

977 The roots of the equation

$$(b - c)x^2 + 2(c - a)x + (a - b) = 0 \text{ are always}$$

- 1) real and distinct
- 2) real and equal
- 3) real
- 4) imaginary

978 The roots of the equation

$$(a + c - b)x^2 + 2cx + (b + c - a) = 0 (a \neq b) \text{ are}$$

- 1) real and distinct
- 2) real and equal
- 3) real
- 4) imaginary

979 If the roots of the equation $x^2 - 2cx + ab = 0$ be real and unequal, the roots of the equation

$$x^2 - 2(a + b)x + (a^2 + b^2 + 2c^2) = 0 \text{ are}$$

- 1) real and distinct
- 2) real and equal
- 3) real
- 4) imaginary

980 If a, b, c, d are real and no two of them are simultaneously zero, then the equation

$$(x^2 + ax - 3b)(x^2 - cx + b) \\ (x^2 - dx + 2b) = 0 \quad \text{has}$$

- 1) at least two real roots
- 2) at least four real roots
- 3) all roots real
- 4) at least two imaginary roots

981 a, b, c are rational. Then the roots of

$$(a + b + c)x^2 - 2(a + c)x + (a - b + c) = 0 \text{ are}$$

- 1) real
- 2) equal
- 3) rational
- 4) imaginary

SOLVING EQUATIONS

982 If the equation

$$(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2) + (\lambda^2 - 4) = 0 \text{ is satisfied}$$

by more than two values of x then $\lambda =$

- 1) 1
- 2) -2
- 3) 3
- 4) 2

983 The number of real roots of the equation

- $(\sqrt{3}+1)^{2x} + (\sqrt{3}-1)^{2x} = 2^{3x}$ is equal to
 1) 0 2) 1 3) 2 4) more than 2

984 If $a \in Z$ and the equation

- $(x-a)(x-10)+1=0$ has integral roots, then values of 'a' are
 1) 10,8 2) 12,10 3) 12,8 4) 10,12

985 The number of real values of x for which

- $\left(\frac{a}{b}\right)^x = x - 2x^2 - 4$, where $a, b > 0$
 1) 1 2) 2 3) 0 4) infinite

986 Number of solutions of the equation

- $|x| = \cos x$
 1) 1 2) 2 3) 0 4) 3

COMMON ROOTS**987 The value of λ in order that the equations**

- $2x^2 + 5\lambda x + 2 = 0$ and $4x^2 + 8\lambda x + 3 = 0$ have a common root is given by
 1) 1 2) -1 3) ± 1 4) 3

988 If every pair from among the equations

- $x^2 + ax + bc = 0$, $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ has common root, then the sum of three common roots is
 1) abc 2) 2abc
 3) $3(a+b+c)$ 4) $(a+b+c)$

- 989** If $x^2 - hx - 21 = 0$, $x^2 - 3hx + 35 = 0$ ($h > 0$)
 • have a common root, then the value of h is
 1) ± 8 2) ± 4 3) 4 4) 2

- 990** The conditions that a root of the equation $ax^2 + bx + c = 0$ may be reciprocal to a root of $a_1x^2 + b_1x + c_1 = 0$ is

- 1) $(bb_1 - aa_1)^2 = (ab_1 - bc_1)(ba_1 - b_1c)$
 2) $(cb_1 - ba_1)^2 = (ac_1 - cd_1)(ab_1 - bc_1)$
 3) $(cc_1 - aa_1)^2 = (ab_1 - bc_1)(ba_1 - b_1c)$
 4) $a + b + c = 0$

LOCATION OF ROOTS

991 | **If both roots of the equation $x^2 + x + a = 0$ exceed 'a' then**
 .
 1) $2 < a < 3$ 2) $a > 3$ 3) $-3 < a < 3$ 4) $a < -2$

992 | **If both roots of the equation $x^2 - 2ax + a^2 - 1 = 0$ lie in the interval $(-3, 4)$ then sum of the integral parts of 'a' is**
 .
 1) 0 2) 2 3) 4 4) -1

993 | **The expression $(a-2)x^2 + 2(2a-3)x + (5a-6)$ is positive for all real values of x, then**
 .
 1) 'a' can be any real number
 2) $a > 1$ 3) $a > 3$ 4) $a = 3$
SIGN OF THE EXPRESSIONS AND INEQUALITIES

994 | **If $x^2 + 2ax + b \geq c \forall x \in R$ then**
 .
 1) $b - c \geq a^2$ 2) $c - a \geq b^2$
 3) $a - b \geq c^2$ 4) $a + b + c \geq 0$

995 | **The roots of the equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are real and equal if**
 .
 1) $a > b > c$ 2) $a = b = c$
 3) $a < b < c$ 4) $a + b + c = 0$

996 | **The largest negative integer for which $\frac{(x-4)(x-2)}{(x-1)(x-5)} > 0$ is**
 .
 1) -2 2) -1 3) -3 4) -4

MAXIMUM AND MINIMUM OF EXPRESSION

997 | **Let $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$, then $\min\{f(x)\} > \max\{g(x)\}$ holds for**
 .
 1) $0 < c < b\sqrt{2}$ 2) $|c| > |b|\sqrt{2}$
 3) $|c| < |b|\sqrt{2}$ 4) no real value of b and c

MODULUS FUNCTIONS

998 Number of rational roots of the equation

- $|x^2 - 2x - 3| + 4x = 0$ is
 1) 1 2) 2 3) 3 4) 4

999 The roots of the equation $|x^2 - x - 6| = x + 2$ are

- 1) 1, -2, 2 2) 1, 2, 4 3) -2, 2, 4 4) -2, 2, 3

100 For $a > 0$, all the real roots of the equation

0. $x^2 - 3a|x - a| - 7a^2 = 0$ are
 1) $4a, 5a$ 2) $-4a, 5a$ 3) $-4a, -5a$ 4) $4a, -5a$

100 The set of real values of x satisfying $|x - 1| \leq 3$

1. and $|x - 1| \geq 1$
 1) $[2, 4]$ 2) $(-\infty, 2) \cup (4, \infty)$
 3) $[-2, 0] \cup [2, 4]$ 4) $[0, 2]$

OTHER MODELS**100 If $\sin \theta, \sin k\theta$ are the roots of**

2. $4x^2 + 2x - 1 = 0$, then $k =$
 1) -2 2) -3 3) 3 4) 2

100 If α, β are the roots of $x^2 - 3x + a = 0$ and γ, δ that of $x^2 - 12x + b = 0$ and $\alpha, \beta, \gamma, \delta$ form an increasing G.P. then

- 1) $a = 3, b = 12$ 2) $a = 4, b = 16$
 3) $a = 2, b = 32$ 4) $a = 12, b = 3$

100 Let $f(x)$ be a polynomial for which the remainders when divided by $x - 1, x - 2, x - 3$ respectively are 3, 7, 13. Then the remainder of $f(x)$ when divided by $(x - 1)(x - 2)(x - 3)$ is

- 1) $f(x)$ 2) $x^2 + x + 1$ 3) $x^2 + 1$ 4) $x + 2$

100 Let a, b, c be the real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$, has a root γ that always satisfies

- 1) $\gamma = (\alpha + \beta)/2$ 2) $\gamma = (\alpha + \beta/2)$
 3) $\gamma = \alpha$ 4) $\alpha < \gamma < \beta$

- 100
6. **The equation $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0$ has**
- 1) all the roots are equal
 - 2) one real and two imaginary
 - 3) 3 real roots namely $x = a, x = b, x = c$
 - 4) no real roots

TRANSFORMED EQUATIONS

- 100
7. **The value of 'a' for which the quadratic equation $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possesses roots of opposite signs lies on**
- 1) $(-\infty, 1)$
 - 2) $(-\infty, 0)$
 - 3) $(1, 2)$
 - 4) $[1, 2]$

- 100
8. **If the roots of the equation $ax^2 + bx + c = 0$ are the reciprocals of the roots of the equation $px^2 + qx + r = 0$ then**
- 1) $acq^2 = b^2 pr$
 - 2) $ac = pr$
 - 3) $b^2 ac = q^2 pr$
 - 4) $ab = pq$

RANGE

- 100
9. **If $y = \tan x \cot 3x, x \in R$ then**
- 1) $\frac{1}{3} < y < 1$
 - 2) $\frac{1}{3} \leq y \leq 1$
 - 3) $\frac{1}{3} \leq y \leq 3$
 - 4) $y \leq \frac{1}{3} \text{ or } y \geq 3$

- 101
0. **If $5^x + (2\sqrt{3})^{-x} \geq 13^x$ then the solution set for x is**
- 1) $[2, +\infty)$
 - 2) $\{2\}$
 - 3) $(-\infty, 2]$
 - 4) $[0, 2]$

INEQUALITIES

- 101
1. **If $x^2 + 6x - 27 > 0; -x^2 + 3x + 4 > 0$ then x lies in the interval**
- 1) $(3, 4)$
 - 2) $\{3, 4\}$
 - 3) $(-\infty, 3) \cup (4, \infty)$
 - 4) $(-9, 4)$

- 101
2. **The range of values of x which satisfy $5x + 2 < 3x + 8$ and $\frac{x+2}{x-1} < 4$ are**
- 1) $(2, 3)$
 - 2) $(-\infty, 1) \cup (2, 3)$
 - 3) $(2, \infty)$
 - 4) R

- 101 The set of exhaustive values of 'a' for which
 3. the inequality $(a^2 + 3)x^2 + (a + 2)x - 4 < 2$
 holds of at least one negative x, is

- 1) $a > 0$ 2) $a < 0$
 3) $-\infty < a < \infty$ 4) $a \in \phi$

QUADRATIC EXPRESSION IN TWO VARIABLES

- 101 If the expression
 4. $3x^2 + 2pxy + 2y^2 + 2ax - 4y + 1$ can be
 resolved into two linear factors then p must
 be a root of the equation

- 1) $x^2 + ax + 6 = 0$ 2) $x^2 + 4ax + 6 = 0$
 3) $x^2 + 4ax + 2a^2 + 6 = 0$ 4) $x^2 - 4ax + 6 = 0$

55. The set of values of x for which the
 inequalities $x^2 - 3x - 10 < 0$, $10x - x^2 - 16 > 0$
 hold simultaneously is (EAM - 2007)

- 1) $(-2, 5)$ 2) $(2, 8)$ 3) $(-2, 8)$ 4) $(2, 5)$

56. If p and q are distinct prime numbers and if
 the equation $x^2 - px + q = 0$ has positive
 integers as its roots, then the roots of the
 equation are (Eamcet-2014)

- 1) 1, -1 2) 2, 3 3) 1, 2 4) 3, 1

57. Let $f(x) = x^2 + ax + b$, where $a, b \in R$. If
 $f(x) = 0$ has all its roots imaginary then the
 roots of $f(x) + f'(x) + f''(x) = 0$ are (EAM
 - 2009)

- 1) real and distinct 2) imaginary
 3) equal 4) rational and equal

58. For $x \in R$, the least value of $\frac{x^2 - 6x + 5}{x^2 + 2x + 1}$ is
 (EAM - 2010)

$$1) -1 \quad 2) -\frac{1}{2} \quad 3) -\frac{1}{4} \quad 4) -\frac{1}{3}$$

59. The product of real roots of the equation

$$|x|^{6/5} - 26|x|^{3/5} - 27 = 0 \text{ is (EAM - 2012)}$$

$$1) -3^{10} \quad 2) -3^{12} \quad 3) -3^{12/5} \quad 4) -3^{21/5}$$

60. The set of solutions satisfying both

$$x^2 + 5x + 6 \geq 0 \text{ and } x^2 + 3x - 4 < 0 \text{ is}$$

(EAM - 2013)

$$1) (-4, 1) \quad 2) (-4, -3] \cup [-2, 1)$$

$$3) (-4, -3) \cup (-2, 1) \quad 4) [-4, -3] \cup [-2, 1]$$

61. If α and β are the roots of the equation

$$x^2 - 2x + 4 = 0, \text{ then } \alpha^9 + \beta^9 \text{ is equal to}$$

(EAM - 2013)

$$1) -2^8 \quad 2) 2^9 \quad 3) -2^{10} \quad 4) 2^{10}$$

LEVEL-II (H.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 2 | 2) 3 | 3) 1 | 4) 2 | 5) 2 | 6) 3 |
| 7) 3 | 8) 1 | 9) 2 | 10) 4 | 11) 3 | 12) 4 |
| 13) 2 | 14) 1 | 15) 4 | 16) 3 | 17) 1 | 18) 1 |
| 19) 4 | 20) 1 | 21) 3 | 22) 4 | 23) 2 | 24) 3 |
| 25) 3 | 26) 2 | 27) 3 | 28) 4 | 29) 3 | 30) 3 |
| 31) 4 | 32) 1 | 33) 3 | 34) 1 | 35) 2 | 36) 2 |
| 37) 2 | 38) 1 | 39) 3 | 40) 4 | 41) 3 | 42) 2 |
| 43) 3 | 44) 2 | 45) 4 | 46) 2 | 47) 3 | 48) 1 |
| 49) 4 | 50) 3 | 51) 1 | 52) 1 | 53) 3 | 54) 3 |
| 55) 4 | 56) 3 | 57) 2 | 58) 4 | 59) 1 | 60) 2 |
| 61) 3 | | | | | |

LEVEL-II (H.W.) - HINTS

$$1. \quad \alpha + \beta = \frac{3}{8}, \quad \alpha\beta = \frac{27}{8}$$

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{\alpha + \beta}{(\alpha\beta)^{\frac{1}{3}}} = \frac{\frac{3}{8}}{\frac{3}{2}} = \frac{1}{4}$$

$$2. \quad m = \alpha\beta + \alpha + \beta = q + p,$$

$$n = \alpha\beta - \alpha - \beta = q - p$$

The quadratic equation which roots m and n

$$\text{is } x^2 - 2qx + q^2 - p^2 = 0$$

$$3. \quad \sin \theta + \cos \theta = -b/a; \quad \sin \theta \cos \theta = c/a$$

$$1 + 2\frac{c}{a} = \frac{b^2}{a^2}; \quad a^2 + 2ac = b^2$$

$$4. \quad \alpha = -\omega, \quad \beta = -\omega^2; \quad \alpha^5 + \beta^5 = -\omega^2 - \omega = 1$$

$$5. \quad \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} = \frac{b^2 - 2ac}{a^2 c^2}$$

$$= \frac{25 + 42}{9 \times 49} = \frac{67}{441}$$

$$6. \quad \alpha + \beta = \alpha^2 + \beta^2; \quad \frac{-b}{a} = \frac{b^2 - 2ac}{a^2}$$

$$\therefore b^2 + ab = 2ac$$

7. given expression =

$$= a + \frac{b}{2}(\alpha + \beta) + \frac{c}{2}(\alpha^2 + \beta^2) + \frac{d}{2}(\alpha^3 + \beta^3)$$

$$= a + \frac{b}{2}(\alpha + \beta) + \frac{c}{2}[(\alpha + \beta)^2$$

$$- 2\alpha\beta + \frac{d}{2}(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)]$$

$$= a + \frac{b}{2}(1) + \frac{c}{2}[(1)^2 - 2(1/6)]$$

$$+ \frac{d}{2}[(1)^3 - 3(1/6)(1)] = a + \frac{b}{2} + \frac{c}{3} + \frac{d}{4}$$

8. $\alpha + \beta = -a$, $\alpha\beta = -b$; $\gamma + \delta = -a$, $\gamma\delta = b$

$$(\alpha^2 + a\alpha + b)(\beta^2 + a\beta + b) = 4b^2$$

9. $\frac{\alpha + \beta}{2} = 8/5$; $\alpha + \beta = 16/5$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{16}{7}; \frac{16}{5} \times \frac{7}{16} = \alpha\beta \quad \alpha\beta = 7/5$$

The equation is $5x^2 - 16x + 7 = 0$

10. $a^2c + ac^2 = 3abc - b^3$

$$11. \frac{(m+n)^2}{mn} = \frac{4b^2}{ac} = \frac{4q^2}{pr} \Rightarrow \frac{b^2}{ac} = \frac{q^2}{pr}$$

$$12. |\alpha - \beta|^2 = |\gamma - \delta|^2$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$$

$$\Rightarrow b^2 - 4c = c^2 - 4b$$

$$b^2 - c^2 = 4c - 4b \Rightarrow b + c = -4$$

$$13. \quad \frac{\alpha + \beta}{2} = a; \alpha_1\beta_1 = a^2 \Rightarrow \sqrt{\alpha_1\beta_1} = a$$

$\therefore$ A.M of roots of 2nd equation = G.M of roots of 1st equation

$$14. \quad \tan A + \tan B = 2; \tan A \tan B = 2$$

$$\tan(A + B) = \frac{2}{1-2} = -2, \sin^2(A + B) = \frac{4}{5}$$

$$15. \quad \text{Coefficient of } x=0$$

$$16. \quad \Delta_1 + \Delta_2 = p^2 - 4q + r^2 - 4s$$

$$= p^2 + r^2 - 4(q + s) = p^2 + r^2 - 2pr = (p - r)^2 \geq 0$$

at least one of $\Delta_1 \geq 0, \Delta_2 \geq 0$

$\therefore$ at least one has real roots.

$$17. \quad \Delta = 4(c - a)^2 - 4(b - c)(a - b)$$

$$= 4[c^2 + a^2 - 2ac - ab + b^2 + ac - bc]$$

$$= 4[a^2 + b^2 + c^2 - ab - bc - ca] > 0$$

$$18. \quad \Delta = 4c^2 - 4(b + c - a)(a + c - b)$$

$$= 4[c^2 - [c - (a - b)][c + a - b]]$$

$$= 4(a - b)^2 \geq 0 \text{ and } a \neq b \text{ roots are real and distinct}$$

19. $c^2 - ab > 0$ then

$$\Delta = 4 \left[(a+b)^2 - (a^2 + b^2 + 2c^2) \right]$$

$$= 4 \left[2ab - 2c^2 \right] < 0$$

20. If Δ_1, Δ_2 and Δ_3 be the discriminants of individual quadratics, then $D_1 + D_2 + D_3 > 0$

21. $\Delta = 4(a+c)^2 - 4(a+c+b)(a+c-b)$

$$\Delta = 4(a+c)^2 - 4 \left[(a+c)^2 - b^2 \right] ; \Delta = 4b^2$$

22. $a=b=c=0$

23. The equation is $\left(\frac{\sqrt{3}+1}{2\sqrt{2}} \right)^{2x} + \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right)^{2x} = 1$

$$\Rightarrow (\sin \theta)^{2x} + (\cos \theta)^{2x} = 1$$

24. Since 'a' and 'x' are integers

$$\therefore (x-a)(x-10)+1=0 \Rightarrow (x-a)(x-10)=-1$$

$$\Rightarrow (x-a=1, x-10=-1)$$

$$\text{or } (x-a=-1 \text{ and } x-10=1)$$

$$\Rightarrow (x=9, a=8) \text{ or } (x=11 \text{ and } a=12)$$

Thus $a = 8$ or 12

25. $LHS > 0 ; RHS = -2 \left(x^2 - \frac{x}{2} \right) - 4$

$$= -2 \left[\left(x - \frac{1}{4} \right)^2 - \frac{1}{16} \right] - 4$$

$$= -2 \left[\left(x - \frac{1}{4} \right)^2 \right] - \frac{31}{8} < 0$$

given equation has no solution.

Number of solutions is 0

26. draw the graph

27. $(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$

$$\therefore 4\lambda^2 = 4 \Rightarrow \lambda = \pm 1$$

28. Common root of $x^2 + ax + bc = 0$ and

$x^2 + bx + ca = 0$ is 'c' common root of
 $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ is 'a'
 common root of $x^2 + cx + ab = 0$ and
 $x^2 + ax + bc = 0$ is 'b'.

29. $\alpha^2 - h\alpha - 21 = 0$ and $\alpha^2 - 3h\alpha + 35 = 0$

$$\Rightarrow 2h\alpha - 56 = 0 \Rightarrow \alpha = \frac{28}{h}$$

Substitute in either of the equations

30. $ax^2 + bx + c = 0$ and $c_1x^2 + b_1x + a_1 = 0$ have a common root then find the condition

31. $f(a) > 0$ and $\alpha + \beta > 2a$ and $\Delta > 0$

32. $x^2 - 2ax + a^2 - 1 = 0 \Rightarrow (x - a)^2 = 1$

Roots are $a - 1$ and $a + 1$

Both roots lie in $(-3, 4) \therefore a-1 > -3, a+1 < 4$

ie $-2 < a < 3 \Rightarrow [a] = -2, -1, 0, 1, 2$ whose sum is zero.

$$33. \Delta < 0 \Rightarrow (2a-3)^2 - (a-2)(5a-6) < 0$$

$$4a^2 - 12a + 9 - 5a^2 + 16a - 12 < 0$$

$$\Rightarrow -a^2 + 4a - 3 < 0 \Rightarrow a^2 - 4a + 3 > 0$$

$$\Rightarrow a \in (-\infty, 1) \cup (3, \infty) \text{ and } a-2 > 0$$

$$34. x^2 + 2ax - b - c \geq 0 \forall x \in R \Rightarrow \Delta \leq 0$$

$$35. \Delta = 4((a+b+c)^2 - (ab+bc+ca)) \\ = 2((a-b)^2 + (b-c)^2 + (c-a)^2) = 0$$

$$36. \text{ Let } y = \frac{(x-4)(x-2)}{(x-1)(x-5)}$$

$$\therefore x < 1 \text{ or } 2 < x < 4 \text{ or } x > 5$$

$$37. \min\{f(x)\} = 2c^2 - b^2 \text{ and}$$

$$\max\{g(x)\} = b^2 + c^2$$

38. use extreme value formula

39. by verification

40. by verification

$$41. |x-1| \leq 3 \Rightarrow -3 \leq x-1 \leq 3$$

$$\Rightarrow -2 \leq x \leq 4 \text{ and } |x-1| \geq 1$$

$$\Rightarrow x-1 \leq -1 \text{ or } x-1 \geq +1$$

$$\Rightarrow x \leq 0 \text{ or } x \geq 2 \text{ combining } x \in [-2, 0] \cup [2, 4]$$

$$42. \quad \alpha, \beta = \frac{-2 \pm \sqrt{4+16}}{8} = \frac{\pm\sqrt{5}-1}{4}$$

$$\alpha = \sin 18^\circ, \beta = -\left(\frac{\sqrt{5}+1}{4}\right) = \sin 54^\circ = \sin(-54^\circ) \Rightarrow$$

$$\theta = 18^\circ; k = -3$$

43. by verification

44. Verification

45. α is a root of $a^2x^2 + bx + c = 0$

$$\Rightarrow a^2\alpha^2 + b\alpha + c = 0$$

β is a root of $a^2x^2 - bx - c = 0$

$$\Rightarrow a^2\beta^2 - b\beta - c = 0$$

$$\text{Let } f(x) = a^2x^2 + 2bx + 2c$$

$$f(\alpha) = a^2\alpha^2 + 2b\alpha + 2c = -a^2\alpha^2 < 0$$

$$f(\beta) = a^2\beta^2 + 2b\beta + 2c = 3a^2\beta^2 > 0$$

by rolles theorem $\exists \gamma \in (\alpha, \beta) \ni f(\gamma) = 0$

$$46. \quad f(x) = (x-a)^3 - (x-b)^3 + (x-c)^3$$

$$f'(x) = 3(x-a)^2 + 3(x-b)^2 + 3(x-c)^2 > 0$$

$\forall x \in R$ $f(x)$ is strictly increasing

$\therefore f(x) = 0$ has one real and two imaginary

$$47. \quad \alpha\beta < 0 \Rightarrow a^2 - 3a + 2 < 0; a \in (1, 2)$$

48. $cx^2 + bx + a = 0$; $px^2 + qx + r = 0$ have same roots

$$\therefore \frac{c}{p} = \frac{b}{q} = \frac{a}{r}$$

$$\therefore ac / pr = b^2 / q^2$$

49. $y = \frac{\tan x (1 - 3 \tan^2 x)}{3 \tan x - \tan^3 x} = \frac{1 - 3t^2}{3 - t^2}$

$$(y - 3)t^2 - 3y + 1 = 0 \Rightarrow \Delta \geq 0$$

50. $\therefore \cos^x \alpha + \sin^x \alpha \geq 1$,

where $\cos \alpha = 5/13$ equality holds for $x=2$,

if $x < 2$ both $\cos \alpha$ and $\sin \alpha$ increase

$$\cos^x \alpha + \sin^x \alpha > 1 \text{ if } x < 2$$

51. $x^2 + 6x - 27 > 0 \Rightarrow (x + 9)(x - 3) > 0$

$$\Rightarrow x < -9 \text{ or } x > 3$$

$$\text{also, } -x^2 + 3x + 4 > 0 \Rightarrow (x - 4)(x + 1) < 0$$

$$\Rightarrow -1 < x < 4 \dots (2)$$

From (1) and (2) $x \in (3, 4)$

$$52. \quad 5x + 2 < 3x + 8 \Rightarrow x < 3 \dots (1)$$

$$\frac{x+2}{x-1} < 4 \Rightarrow \frac{x+2}{x-1} - 4 < 0$$

$$\Rightarrow \frac{x-2}{x-1} > 0 \Rightarrow x < 1 \text{ or } x > 2 \dots (2)$$

$$\text{From (1) and (2) } x \in (-\infty, 1) \cup (2, 3)$$

$$53. \quad \text{If } f(x) = (a^2 + 3)x^2 + (a + 2)x - 6,$$

$$\text{then } f(0) < 0$$

$$54. \quad \Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$55. \quad \text{Solve the inequalities find the common interval.}$$

$$56. \quad \text{let } p = 3 \text{ and } q = 2, \text{ then given equation becomes}$$

$$x^2 - 3x + 2 = 0$$

$$57. \quad f(x) = x^2 + ax + b \text{ has imaginary roots}$$

$$\Delta < 0 \Rightarrow a^2 - 4b < 0$$

$$f'(x) = 2x + a;$$

$$f''(x) = 2$$

$$\Rightarrow x^2 + (a+2)x + b + a + 2 = 0$$

$$\therefore x = \frac{-(a+2) \pm \sqrt{(a+2)^2 - 4(a+b+2)}}{2}$$

$$\because a^2 - 4b < 0$$

$$a^2 - 4a - 4 < 0$$

$\therefore$ equation has imaginary roots

58. Let $y = \frac{x^2 - 6x + 5}{x^2 + 2x + 1}$

$$(y-1)x^2 + (2y+6)x + (y-5) = 0$$

$$x \in R, \Delta \geq 0$$

59. Put $|x|^{3/5} = t$

$$\therefore t^2 - 26t - 27 = 0 \Rightarrow t = 27 \text{ or } -1$$

$$|x|^{3/5} = 27$$

$$\because |x|^{3/5} \text{ cannot be negative so } x = 3^5 \text{ or } -3^5$$

$$\text{Product } x = 3^5 \times (-3)^5 = -3^{10}$$

60. Verification

61. $\alpha + \beta = 2;$

$$\alpha\beta = 4$$

$$\alpha^9 + \beta^9 = (\alpha^3)^3 + (\beta^3)^3$$



LEVEL-III

101 If $0 < \alpha < \beta < \gamma < \frac{\pi}{2}$, then the equation
5.

$$\frac{1}{x - \sin \alpha} + \frac{1}{x - \sin \beta} + \frac{1}{x - \sin \gamma} = 0 \text{ has}$$

- 1) imaginary roots 2) real and equal roots
3) real and unequal roots 4) rational roots

101 If n is a multiple of 6 and α, β are the roots

6. **of $x^2 + x + 1 = 0$ then $(1 + \alpha)^{-n} + (1 + \beta)^{-n} =$**
 1) 2 2) -2 3) 0 4) -1

101 If α, β are real and $\alpha^2, -\beta^2$ are the roots of

7. **the equation $a^2x^2 + x + (1 - a^2) = 0 (a > 1)$ then $\beta^2 =$**
 1) a^2 2) 1 3) $1 - a^2$ 4) $1 + a^2$

101 If α, β are the roots of $x^2 + px - q = 0$ and

8. **γ, δ are that of $x^2 + px + r = 0$ then**
 $(\alpha - \gamma)(\beta - \gamma)(\alpha - \delta)(\beta - \delta) =$
 1) $(q - r)^2$ 2) $(q + r)^2$
 3) $-(q + r)^2$ 4) $-(q - r)^2$

101 If α, β are the roots of $ax^2 + bx + c = 0$ and

9. **$S_n = \alpha^n + \beta^n$ then $aS_{n+1} + bS_n + cS_{n-1} =$**
 1) 0 2) 1 3) 2 4) 5

102 α and β are the roots of the equation

0. **$x^2 + px + p^3 = 0, (p \neq 0)$. If the point (α, β)**
lies on the curve $x = y^2$ then the roots of
the given equation are
 1) 4, -2 2) 4, 2 3) 1, -1 4) 1, 1

102 If a and b are distinct positive real numbers

1. **such that $a, a_1, a_2, a_3, a_4, a_5, b$ are in A.P;**
 $a, b_1, b_2, b_3, b_4, b_5, b$ are in GP; and
 $a, c_1, c_2, c_3, c_4, c_5, b$ are in HP; Then the roots
of $a_3x^2 + b_3x + c_3 = 0$ are
 1) real and distinct 2) real and equal
 3) imaginary 4) rational

102 p, q, r and s are integers. If the A.M. of the

2. **roots of $x^2 - px + q^2 = 0$ and G.M. of the roots**
of $x^2 - rx + s^2 = 0$ are equal then

- 1) q is an odd integer 2) r is an even integer
 3) p is an even integer 4) s is an odd integer

9. If $x^2 + x + 1 = 0$ then the value of

$$\left(x + \frac{1}{x}\right) + \left(x^2 + \frac{1}{x^2}\right) + \dots + \left(x^{52} + \frac{1}{x^{52}}\right) \text{ equals}$$

- 1) 1 2) 0 3) -1 4) -52

102 3. **The value of the continued fraction**

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots \infty}}} \quad \text{is}$$

- 1) $\frac{\sqrt{5}-1}{2}$ 2) $\frac{\sqrt{5}+1}{2}$ 3) $\frac{\sqrt{5}-1}{4}$ 4) $\frac{\sqrt{5}+1}{4}$

102 4. **Number of real roots of the equation**

$$\sqrt{x} + \sqrt{x - \sqrt{1-x}} = 1 \quad \text{is}$$

- 1) 0 2) 1 3) 2 4) 3

102 5. **The equation**

$$2 \cos^2 \frac{x}{2} \sin^2 x = x^2 + \frac{1}{x^2}, 0 \leq x \leq \frac{\pi}{2} \quad \text{has}$$

- 1) One real solution
2) No real solution
3) More than one real solution
4) two real solutions

102 6. **The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root common. The other roots of the first and second equations are integers in the ratio 4:3. Then the common root is (AIEEE 2008)**

- 1) 3 2) 2 3) 1 4) 4

102 7. **If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have a common root and the second equation has equal roots then**

- 1) $b+d = ac$ 2) $2(b+d) = ac$
3) $b+d = 2ac$ 4) $(b+d)^2 = a+c$

102 8. **If $x^2 + 2ax + a < 0 \quad \forall x \in [1, 2]$ then**

- 1) $a \in \left(-\infty, -\frac{4}{5}\right)$ 2) $a \in (0, \infty)$
3) $a \in \left(-\frac{4}{5}, 0\right)$ 4) $a \in \left(\frac{4}{5}, \infty\right)$

- 102 If $f(x)$ is a quadratic expression which is
9. positive for all real values of x and

$g(x) = f(x) + f'(x) + f''(x)$ then for any
real value of x

- 1) $g(x) < 0$ 2) $g(x) > 0$ 3) $g(x) = 0$ 4) $g(x) = 2$

- 103 If the roots of the equation $x^2 + 2ax + b = 0$
0. are real and distinct and they differ by at most
2m then 'b' lies in the interval

- 1) $(a^2 - m^2, a^2)$ 2) $[a^2 - m^2, a^2)$
3) $(a^2, a^2 + m^2)$ 4) $[a^2 + m^2, a^2)$

- 103 The solution of the equation $(3|x| - 3)^x = |x| + 7$
1.

which belongs to the domain of $\sqrt{x(x-3)}$ are
given by

- 1) $\pm \frac{1}{9}, \pm 2$ 2) $\frac{1}{9}, -2$ 3) $-\frac{1}{9}, -2$ 4) $-\frac{1}{9}, 2$

- 103 For $a \leq 0$, the roots of the equation
2.

$x^2 - 2a|x - a| - 3a^2 = 0$ are IIT-1987

- 1) $(1 - \sqrt{2})a, (-1 + \sqrt{6})a$
2) $(1 + \sqrt{2})a, (-1 + \sqrt{6})a$
3) $(1 + \sqrt{2})a, (-1 - \sqrt{6})a$
4) $(1 - \sqrt{2})a, (-1 - \sqrt{6})a$

- 103 If n is a positive integer and $n \in [5, 100]$ then
3. the number of positive integral roots of the
equation $x^2 + 2x - n = 0$ is

- 1) 4 2) 6 3) 8 4) 10

- 103 The number of solutions of the system of
4. equations given below is

$$|x| + |y| = 1; x^2 + y^2 = a^2; \left(\frac{1}{\sqrt{2}} < a < 1 \right)$$

- 1) infinite 2) 2 3) 4 4) 8

103 5. If α, β are the roots of $x^2 + x + 1 = 0$ and

$$S_n = \alpha^n + \beta^n \text{ then } \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix} =$$

- 1) 27 2) -27 3) -3 4) 9

103 6. If α, β are the roots of the equation $\lambda(x^2 - x) + x + 5 = 0$ and if λ_1 and λ_2 are the two values of λ for which the roots α, β are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$, then

the value of $\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} =$

- 1) 250 2) 1022 3) 254 4) 0

103 7. If $0 < x < 1000$ and $\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x}{3} \right\rfloor + \left\lfloor \frac{x}{5} \right\rfloor = \frac{x}{30}$ where $[x]$ is the greatest Integer less than or equal to x , then the number of possible values of x is

- 1) 34 2) 32 3) 33 4) 35

103 8. If α, β are the roots of the equation $ax^2 + bx + c = 0$ then the value of $(1 + \alpha + \alpha^2)(1 + \beta + \beta^2)$ is

- 1) positive 2) negative
3) non-negative 4) non positive

103 9. The equation $(x-3)^9 + (x-3^2)^9 + (x-3^3)^9 + \dots + (x-3^9)^9 = 0$ has

- 1) all the roots are real
2) one real and 8 imaginary roots
3) real roots namely $x = 3, 3^2, \dots, 3^9$
4) five real and 4 imaginary roots

104 0. The root of the equation $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$, where $i = \sqrt{-1}$, which has greater modulus is

- 1) $(3-5i)/2$ 2) $(5-3i)/2$
3) $(3+i)/2$ 4) $(1+3i)/2$

- 104
1. If $xy = 2(x + y)$, $x \leq y$ and $x, y \in N$, then
number of solutions of the equation
1) two 2) three
3) no solution 4) infinitely many solution

- 104
2. If $c > 0$ and the equation $3ax^2 + 4bx + c = 0$
has no real root, then
1) $2a + c > b$ 2) $a + 2c > b$
3) $3a + c > 4b$ 4) $a + 3c < b$

104 The polynomial

3. $(ax^2 + bx + c)(ax^2 - dx - c)$, $ac \neq 0$, has
1) four real zeroes
2) at least two real zeroes
3) at most two real zeros 4) all the above

104 The range of a for which the equation

4. $x^2 + ax - 4 = 0$ has its smaller root in the
interval $(-1, 2)$ is
1) $(-\infty, -3)$ 2) $(0, 3)$
3) $(0, \infty)$ 4) $(-\infty, -3) \cup (0, \infty)$

104 If x, y, z are real and distinct, then

5. $x^2 + 4y^2 + 9z^2 - 6yz - 3zx - 2xy$ is always
1) positive 2) non negative 3) negative 4) zero

104 If $\sqrt{9x^2 + 6x + 1} < 2 - x$ then

6.
1) $x \in \left(-\frac{3}{2}, \frac{1}{4}\right)$ 2) $x \in \left[-\frac{3}{2}, \frac{1}{4}\right]$
3) $x \in \left[-\frac{3}{2}, \frac{1}{4}\right)$ 4) $x < \frac{1}{4}$

104 For the quadratic equation

7. $4x^2 - 2(a + c - 1)x + ac - b = 0$ ($a > b > c$)
1) both roots are greater than a
2) both roots are less than c
3) both roots lie between $c/2$ and $a/2$
4) exactly one of the roots lies between $c/2$ and $a/2$

104 If $\frac{2x}{2x^2 + 5x + 2} > \frac{1}{x+1}$, then

8.
1) $x < 2$ 2) $-1 < x < 1$
3) $-2 < x < 0$ 4) $-\frac{2}{3} < x < -\frac{1}{2}$

104 If x, y, z are real such that $x + y + z = 4$,

9. $x^2 + y^2 + z^2 = 6$, then $x \in$

- 1) $(-1, 1)$ 2) $[0, 2]$ 3) $[2, 3]$ 4) $\left[\frac{2}{3}, 2\right]$

105 The equation $2^{2x} + (a-1)2^{x+1} + a = 0$ has roots of opposite signs then exhaustive set of values of a is

- 1) $a \in (-1, 0)$ 2) $a < 0$
3) $a \in (-\infty, 1/3)$ 4) $a \in (0, 1/3)$

105 1. If the equation $|x^2 + bx + c| = k$ has four real roots, then

- 1) $b^2 - 4c > 0$ and $0 < k < \frac{b^2 - 4c}{4}$
2) $b^2 - 4c < 0$ and $0 < k < \frac{b^2 - 4c}{4}$
3) $b^2 - 4c > 0$ and $k > \frac{b^2 - 4c}{4}$
4) $b^2 - 4c < 0$ and $k > \frac{b^2 - 4c}{4}$

105 2. The set of values of a for which $(a-1)x^2 - (a+1)x + a - 1 \geq 0$ is true for all $x \geq 2$

- 1) $(-\infty, 1)$ 2) $\left(1, \frac{7}{3}\right]$ 3) $\left(\frac{7}{3}, \infty\right)$ 4) $(1, \infty)$

105 3. If α, β are the roots of $x^2 + px + q = 0$ and

$x^{2n} + p^n x^n + q = 0$ and if $\left(\frac{\alpha}{\beta}\right), \left(\frac{\beta}{\alpha}\right)$ are the

roots of $x^n + 1 + (x+1)^n = 0$, ($n \in N$) then

- 1) must be an odd integer
2) may be any integer
3) must be an even integer
4) cannot say anything

- 105
4. If $(x^2+x+2)^2 - (a-3)(x^2+x+1)(x^2+x+2) + (a-4)(x^2+x+1)^2 = 0$ has at least one real root, then the complete set of values of a is.
- 1) $\left(5, \frac{19}{3}\right]$ 2) $\left(\frac{19}{3}, \infty\right)$ 3) $(-\infty, 5)$ 4) $(1, 10)$

- 105
5. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then, the value of r is (AIE - 2007)
- 1) $\frac{2}{9}(p-q)(2q-p)$ 2) $\frac{2}{9}(q-p)(2p-q)$
- 3) $\frac{2}{9}(q-2p)(2q-p)$ 4) $\frac{2}{9}(2p-q)(2q-p)$

- 105
6. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is (AIE - 2010)
- 1) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
- 2) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
- 3) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
- 4) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

- 105
7. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then

the value of $\frac{a_{10} - 2a_8}{2a_9}$ is ((AIE - 2011)

- 1) 1 2) 2 3) 3 4) 4

45. A value of b for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$ have one root in common is

- 1) $-\sqrt{2}$ 2) $-i\sqrt{3}$ 3) $i\sqrt{5}$ 4) $\sqrt{2}$

46. The real number k for which the equation $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$ (JEE MAIN - 2013)

- 1) lies between 1 and 2 2) lies between 2 and 3
3) lies between -1 and 0 4) does not exist

47. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in R$, have a common root then $a : b : c$ is (JEE MAIN- 2013)

- 1) 1:2:3 2) 3:2:1 3) 1:3:2 4) 3:1:2

48. If $a \in R$ and the equation

$-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where, $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval (JEE - 2014)

- 1) $(-1, 0) \cup (0, 1)$ 2) $(1, 2)$

3) $(-2, -1)$ 4) $(-\infty, -2) \cup (2, \infty)$

49. Let α and β be the roots of the equation

$px^2 + qx + r = 0$, $p \neq 0$. If p, q and r are in

AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$

is ((JEE MAIN - 2014)

1) $\frac{\sqrt{61}}{9}$ 2) $\frac{2\sqrt{17}}{9}$ 3) $\frac{\sqrt{34}}{9}$ 4) $\frac{2\sqrt{13}}{9}$

50. $\left\{ x \in R : \frac{14x}{x+1} - \frac{9x-30}{x-4} < 0 \right\}$ is equal to

(EAM - 2010)

1) $(-1, 4)$ 2) $(1, 4) \cup (5, 7)$
3) $(1, 7)$ 4) $(-1, 1) \cup (4, 6)$

LEVEL - III - KEY

- 1) 3 2) 1 3) 2 4) 2 5) 1 6) 1
7) 3 8) 3 9) 3 10) 2 11) 2 12) 2
13) 2 14) 2 15) 1 16) 2 17) 2 18) 3
19) 1 20) 3 21) 4 22) 2 23) 3 24) 3
25) 3 26) 2 27) 1 28) 1 29) 3 30) 2
31) 1 32) 2 33) 1 34) 4 35) 4 36) 4
37) 3 38) 1 39) 2 40) 3 41) 1 42) 4
43) 2 44) 3 45) 2 46) 4 47) 1 48) 1
49) 4 50) 4

LEVEL - III - HINTS

$$1. \quad 0 < \alpha < \beta < \gamma < \frac{\pi}{2} \Rightarrow \sin \alpha < \sin \beta < \sin \gamma$$

given equation is $f(x) = (x - \sin \beta)(x - \sin \gamma)$

$$+ (x - \sin \alpha)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \beta)$$

$$f(\sin \alpha) = (\sin \alpha - \sin \beta)(\sin \alpha - \sin \gamma) > 0$$

$$f(\sin \beta) = (\sin \beta - \sin \alpha)(\sin \beta - \sin \gamma) < 0$$

$$f(\sin \gamma) = (\sin \gamma - \sin \alpha)(\sin \gamma - \sin \beta) > 0$$

$\therefore f(x) = 0$ has one root between $\sin \alpha$ and $\sin \beta$

and another root between $\sin \beta$ and $\sin \gamma$

$$2. \quad \alpha = \omega, \beta = \omega^2$$

$$3. \quad \alpha^2 + (-\beta^2) = -\frac{1}{a^2}, \alpha^2(-\beta^2) = \frac{1-a^2}{a^2} = \frac{1}{a^2} - 1$$

$$\alpha^2 \beta^2 = 1 - \frac{1}{a^2}; \quad \alpha^2 \beta^2 = 1 + \alpha^2 - \beta^2$$

$$\therefore (1 + \alpha^2)(1 - \beta^2) = 0$$

$$4. \quad (\gamma^2 - (\alpha + \beta)\gamma + \alpha\beta)(\delta^2 - (\alpha + \beta)\delta + \alpha\beta) \\ = (\gamma^2 + p\gamma - q)(\delta^2 + p\delta - q)$$

$$5. \quad ax^2 + bx + c = 0$$

$$\Rightarrow ax^{n+1} + bx^n + cx^{n-1} = 0$$

$$a\alpha^{n+1} + b\alpha^n + c\alpha^{n-1} = 0$$

$$a\beta^{n+1} + b\beta^n + c\beta^{n-1} = 0$$

adding above equations

$$\Rightarrow aS_{n+1} + bS_n + cS_{n-1} = 0$$

$$6. \quad \alpha + \beta = -p, \alpha\beta = p^3$$

$$(\alpha, \beta) \text{ lies on } x = y^2 \Rightarrow \alpha = \beta^2$$

$$\Rightarrow \alpha\beta = \beta^3 \Rightarrow \beta^3 = p^3$$

$$\Rightarrow \beta = p \text{ and } \alpha = -2p$$

$$7. \quad (GM)^2 = (AM) \cdot (HM) \Rightarrow b_3^2 = a_3 \cdot c_3$$

when a_3, b_3, c_3 are AM, GM, HM of a and b

$$\text{respectively } \Delta = b_3^2 - 4a_3c_3 = b_3^2 - 4b_3^2 < 0$$

$$8. \quad \text{By given condition } \frac{\alpha + \beta}{2} = \sqrt{\gamma\delta} \Rightarrow p = 2s.$$

$\therefore p$ is even

$$9. \quad 1 + x + x^2 = 0 \Rightarrow x = w, \frac{1}{w}$$

$$10. \quad x = 1 + \frac{1}{x}$$

$$11. \quad x - \sqrt{1-x} = 1 + x - 2\sqrt{x} ; 1 - x = 1 + 4x - 4\sqrt{x}$$

$$-5x = -4\sqrt{x} \Rightarrow 25x^2 = 16x \Rightarrow x = \frac{16}{25}$$

12. $L.H.S < 2$ and $R.H.S. \geq 2$;

13. $\alpha, 4\beta$ are roots of $x^2 - 6x + a = 0$ and

$\alpha, 3\beta$ are roots of $x^2 - cx + 6 = 0$

$\alpha + 4\beta = 6$ and $4\alpha\beta = a$

$\alpha + 3\beta = c$ and $3\alpha\beta = 6$

$\Rightarrow a = 8 \Rightarrow \alpha = 2$

14. $x^2 - ax + b = 0$; $\alpha = \frac{a}{2}, \alpha^2 = b$; put α in

$$x^2 - cx + d = 0 ; \alpha^2 - c\alpha + d = 0 ; b + d = \frac{ac}{2}$$

15. $f(1) < 0$ and $f(2) < 0$

16. $f(x) = x^2 + 1, g(x) = x^2 + 2x + 3$

$$g(x) = (x+1)^2 + 2 > 0$$

17. $\Delta > 0 \Rightarrow a^2 > b \Rightarrow b < a^2 ; |\alpha - \beta| \leq 2m$

$$a^2 - b \leq m^2 \Rightarrow a^2 - m^2 \leq b ; b \geq a^2 - m^2$$

18. By verification.

19. (i) suppose $x - a > 0$

$$x^2 - 2a|x-a| - 3a^2 = 0 \Rightarrow x^2 - 2a(x-a) - 3a^2 = 0$$

$$\Rightarrow x = (1 - \sqrt{2})a$$

(ii) suppose $x - a < 0$

$$x^2 - 2a|x-a| - 3a^2 = 0 \Rightarrow x^2 + 2a(x-a) - 3a^2 = 0$$

$$\Rightarrow x = (-1 + \sqrt{6})a$$

20. $x = -1 \pm \sqrt{n+1} \Rightarrow n+1$ must be a perfect square

21. every line intersect the circle in two distinct points.

22. $\alpha = w, \beta = w^2$

23. $\lambda x^2 + (1-\lambda)x + 5 = 0$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{4}{5} \Rightarrow \frac{(1-\lambda)^2 - 10\lambda}{\lambda^2 5 / \lambda} = \frac{4}{5}$$

$$\lambda^2 - 12\lambda + 1 = 4\lambda ; \lambda^2 - 16\lambda + 1 = 0$$

$$\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} = \frac{b^2 - 2ac}{ac} = 256 - 2 = 254$$

24. $\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x}{3} \right\rfloor + \left\lfloor \frac{x}{5} \right\rfloor = \frac{31}{30}x = \frac{x}{2} + \frac{x}{3} + \frac{x}{5}$

$$\Rightarrow \frac{x}{2}, \frac{x}{3}, \frac{x}{5} \text{ are all integers.}$$

$$\Rightarrow x \text{ is a multiple of '30'}$$

25. $1 + \alpha + \alpha^2 \geq 0$ because $\Delta < 0$

$$1 + \beta + \beta^2 \geq 0 \text{ because } \Delta_1 < 0$$

$$\therefore (1 + \alpha + \alpha^2)(1 + \beta + \beta^2) \geq 0$$

26. Let $f(x) = (x-3)^9 + (x-3^2)^9 + (x-3^3)^9$

$$+ \dots + (x - 3^9)^9$$

$$\therefore f'(x) = 9 \left\{ (x - 3)^8 + (x - 3^2)^8 + (x - 3^3)^8 + \dots + (x - 3^9)^8 \right\} > 0$$

So, $f'(x) = 0$ has no real roots.

So, $f(x) = 0$ has one real and 8 imaginary roots

$$27. \text{ Roots} = \frac{4(2-i) \pm \sqrt{16(2-i)^2 + 8(1+i)(5+3i)}}{4(1+i)}$$

$$= -\frac{i}{1+i} \text{ or } \frac{4-i}{1+i} = \frac{-1-i}{2} \text{ or } \frac{3-5i}{2}$$

$$28. \quad xy = 2(x+y) \Rightarrow y(x-2) = 2x$$

$$\therefore y = \frac{2x}{x-2} \text{ but } x, y \in \mathbb{N}$$

By trial, $x = 3, 4, 6$

then $y = 6, 4, 3$ but $x \leq y$

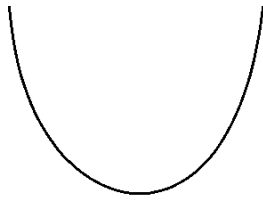
$\therefore x = 3, 4$ and $y = 6, 4$ two solutions.

29. $\therefore$ Equation has no real roots

$$\therefore B^2 - 4AC < 0$$

$$\Rightarrow 16b^2 - 12ac < 0$$

$$\quad | \quad |$$



$$\Rightarrow ac > \frac{4b^2}{3} > 0 \quad \therefore a > 0$$

If $f(x) = 3ax^2 + 4bx + c \therefore f(x) > 0$

or $3ax^2 + 4bx + c > 0, \forall x \in R$

$$\therefore 3a(-1)^2 + 4b(-1) + c > 0$$

$$\Rightarrow 3a + c > 4b$$

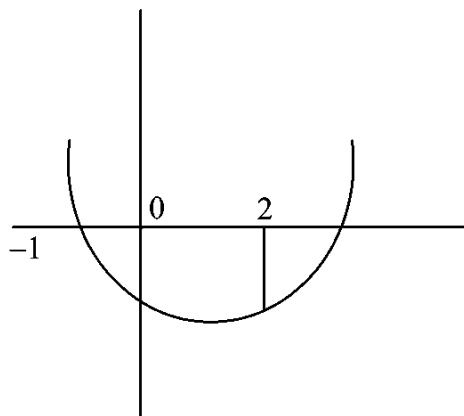
30. Let D_1 and D_2 be the discriminants of $ax^2 + bx + c = 0$ and $ax^2 - dx - c = 0$, respectively. Then,

$D_1 = b^2 - 4ac$ and $D_2 = d^2 + 4ac$

Now, $D_1 + D_2 = b^2 + d^2 > 0$

$\Rightarrow$ At least one of D_1 and D_2 is positive. Hence, at least one of the equation $ax^2 + bx + c = 0$ and $ax^2 - dx - c = 0$ has real roots. Thus, the given polynomial has at least two real roots.

31. Clearly, $f(-1) > 0, f(2) < 0$. Now,



$$f(0) = -4 < 0$$

$$\Rightarrow f(-1) = 1 - a - 4 > 0$$

$$\text{and } f(2) = 4 + 2a - 4 < 0$$

$$\Rightarrow a < -3 \text{ and } a < 0 \Rightarrow a \in (-\infty, -3)$$

$$32. (x-2y)^2 + (2y-3z)^2 + (3z-x)^2 \geq 0$$

$$33. 9x^2 + 6x + 1 \geq 0 \text{ and } 2 - x > 0 \text{ and}$$

$$9x^2 + 6x + 1 < (2-x)^2$$

$$34. f\left(\frac{c}{2}\right) = c^2 - ac - c^2 + c + ac - b = c - b < 0$$

$$f\left(\frac{a}{5}\right) = a^2 - a^2 - ac + a + ac - b = a - b > 0$$

(2)

Hence a root lies between $\frac{c}{2}$ & $\frac{a}{2}$

35. $\frac{2x}{2x^2+5x+2} - \frac{1}{x+1} > 0$

$$\Rightarrow \frac{3x+2}{(2x+1)(x+2)(x+1)} < 0$$

36. Eliminate 'z' and convert quadratic in y and use $\Delta \geq 0$

37. Let $2^x = t$; $\Rightarrow t^2 + 2(a-1)t + a = 0$

One root is positive $\Rightarrow x > 0 \Rightarrow t > 1$

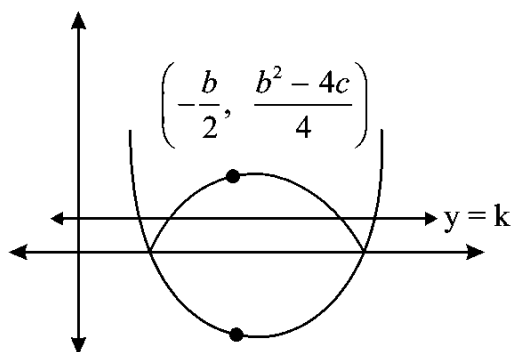
Another root is -ve $\Rightarrow x < 0 \Rightarrow t < 1$

$\therefore$ 1 is a value between roots of t.

$$\therefore f(1) < 0 \Rightarrow a < \frac{1}{3}$$

38. $x^2 + bx + c = 0$ should have distinct roots
 $\Rightarrow \Delta > 0$

$$\Rightarrow b^2 - 4c > 0, k \in \left(0, \frac{b^2 - 4c}{4}\right)$$



$$\left(-\frac{b}{2}, \frac{4c - h^2}{4} \right)$$

$$39. a(x^2 - x + 1) - (x^2 + x + 1) \geq 0$$

$$\Rightarrow a \geq \frac{x^2 + x - 1}{x^2 - x + 1} \Rightarrow a \geq 1 + \frac{2}{x - 1 + \frac{1}{x}}$$

But given $x \geq 2$

$$\Rightarrow a \geq 1 + \frac{2}{\left[\frac{3}{2}, \infty \right)} \Rightarrow a \in \left(1, \frac{7}{3} \right]$$

$$40. \alpha + \beta = -p; \alpha\beta = q$$

$$t = x^n \Rightarrow t^2 + p^n t + q = 0$$

Its roots are α^n, β^n ; $\therefore \alpha^n + \beta^n = -p^n$ (1)

Given $\frac{\alpha}{\beta}$ is a root of $x^n + 1 + (x+1)^n = 0$

$$\Rightarrow \frac{\alpha^n}{\beta^n} + 1 + \frac{(\alpha + \beta)^n}{\beta^n} = 0 \Rightarrow \alpha^n + \beta^n + (-p)^n = 0$$

$$\Rightarrow -p^n + (-p)^n = 0 \quad (\text{from (1)})$$

$\Rightarrow n$ must be even integer

$$41. x^2 + x + 1 = t \Rightarrow t = \left[\frac{3}{4}, \infty \right)$$

$$(t+1)^2 - (a-3)t(t+1) + (a-4)t^2 = 0$$

$$t^2[1-a+3+a-4] + t[2-a+3] + [1] = 0$$

$$\Rightarrow t(5-a)+1=0 \Rightarrow 5-a=-\frac{1}{t}$$

$$\Rightarrow a=5+\frac{1}{t}=5+\left(0,\frac{4}{3}\right]=\left(5,\frac{19}{3}\right]$$

42. The equation $x^2 - px + r = 0$ has roots α, β and

the equation $x^2 - qx + r = 0$ has roots $\frac{\alpha}{2}, 2\beta$

$$\Rightarrow r = \alpha\beta \text{ and } \alpha + \beta = p \text{ and } \frac{\alpha}{2} + 2\beta = q$$

$$\beta = \frac{2q-p}{3} \text{ and } \alpha = \frac{2(2p-q)}{3}$$

$$\Rightarrow \alpha\beta = r = \frac{2}{9}(2p-q)2p-q$$

43. Sum of roots $= \frac{\alpha^2 + \beta^2}{\alpha\beta}$ and product $= 1$

given $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$

$$\Rightarrow (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) = q$$

$$\therefore \alpha^2 + \beta^2 - \alpha\beta = \frac{-q}{p} \text{ and } (\alpha + \beta)^2 = p^2$$

$$\Rightarrow \alpha^2 + \beta^2 = \frac{p^3 - 2q}{3p}$$

$$\therefore \text{required equation is } x^2 - \frac{(p^3 - 2q)x}{(p^3 + q)} + 1 = 0$$

$$\Rightarrow (p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$$

$$\begin{aligned} 44. \frac{a_{10} - 2a_8}{2a_9} &= \frac{(\alpha^{10} - \beta^{10}) - 2(\alpha^8 - \beta^8)}{2(\alpha^9 - \beta^9)} \\ &= \frac{\alpha^8(\alpha^2 - 2) - \beta^8(\beta^2 - 2)}{2(\alpha^9 - \beta^9)} \\ &= \frac{\alpha^8(6\alpha) - \beta^8(6\beta)}{2(\alpha^9 - \beta^9)} = \frac{6(\alpha^9 - \beta^9)}{2(\alpha^9 - \beta^9)} = 3 \end{aligned}$$

45. $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a

common root $\Rightarrow (1+b)^2 = (b^2 + 1)(1-b)$

$$\Rightarrow b = 0, \pm\sqrt{3}i$$

105 Let $f(x) = 2x^3 + 3x + k$

8.

$$f'(x) = 6x^2 + 3 > 0, \forall x \in \mathbb{R}$$

$\Rightarrow f(x)$ is strictly increasing function.

$\Rightarrow f(x) = 0$ has only real root, so two roots are not possible.

105 Given equations are $x^2 + 2x + 3 = 0 \rightarrow (1)$

9.

$$ax^2 + bx + c = 0 \rightarrow (2) \text{ equations (1) and (2)}$$

both have same roots, if $\frac{a}{1} = \frac{b}{2} = \frac{c}{3} \quad \therefore$

$$a:b:c = 1:2:3$$

106 Let $t = x - [x]$, then the given equations is

0.

$$-3t^2 + 2t + a^2 = 0 \Rightarrow t = \frac{1 \pm \sqrt{1+3a^2}}{3}$$

$$\therefore 0 \leq t \leq 1, \text{ we have } 0 \leq \frac{1 \pm \sqrt{1+3a^2}}{3} < 1$$

$$\text{Taking positive sign, we get } 0 \leq \frac{1 + \sqrt{1+3a^2}}{3} < 1$$

$$\Rightarrow (a+1)(a-1) < 0 \quad ; \quad a \in (-1, 1)$$

For no integral solution of a , we consider the interval

$$(-1, 0) \cup (0, 1)$$

$$106 \quad \alpha + \beta = \frac{-a}{p}, \quad \alpha\beta = \frac{r}{p} \quad \text{..(i)}$$

1.

since, p, q and r are in AP $2q = p + r$..(ii)

$$\frac{1}{\alpha} + \frac{1}{\beta} = 4 \Rightarrow \frac{\alpha + \beta}{\alpha\beta} = 4 \Rightarrow \alpha + \beta = 4\alpha\beta$$

$$\Rightarrow q = -4r$$

On putting the value of q in in Eq(ii) we get

$$p = -9r \quad \text{use } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$106 \quad \text{by simplyfying we get } \frac{(x-1)(x-6)}{(x+1)(x-4)} < 0$$

2.

$$\text{Hence, } x \in (-1, 1) \cup (4, 6)$$



LEVEL-IV

ASSERTION & REASON TYPE

106
3.

Assertion (A): If $2x^2 + 3x + 4 = 0$ and $ax^2 + bx + c = 0$ have a common root, then $a : b : c = 2 : 3 : 4$

Reason(R): For a quadratic equation in x , complex roots occur in conjugate pairs.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

106
4.

Assertion (A): The roots of $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are real.

Reason(R): Discriminant value above Quadratic equation is positive or equal to zero.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

106
5.

(i) If α, β are the roots of $ax^2 + bx + c = 0$. If K is a real number [$K \neq 0, 1$] then the equation whose roots are $k\alpha, k\beta$ is $ax^2 + kbx + k^2c = 0$.

(ii) 8, 2 are roots of $x^2 + ax + \beta = 0$ and 3, 3 are the roots of $x^2 + \alpha x + b = 0$ then the roots of $x^2 + ax + b = 0$ are 9, 1.

Which of the above statement is correct

- 1) only (i) 2) only (ii)
- 3) Both (i) and (ii) 4) Neither (i) nor (ii)

106
6.

Assertion(A): $x^2 + x + 1$ is greater than zero for all real x .

Reason (R): When $b^2 - 4ac < 0$. Then $a, ax^2 + bx + c$ have same sign for all real values of x .

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

- 106 **Assertion (A):** The curve $y = x^2 - 6x + 9$ touches the x-axis.

Reason(R): $y = ax^2 + bx + c$ touches the x-axis if $\Delta = b^2 - 4ac = 0$.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

- 106 **Assertion (A)** If $x^2 - 7x + 10 < 0$ then $x \in (2, 5)$.

8. **Reason (R):** If 'a' and $ax^2 + bx + c$ have opposite signs and $b^2 - 4ac > 0$ then x lies between the roots of $ax^2 + bx + c = 0$.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

- 106 **Assertion (A):** The equation $x - \frac{x}{x-1} = 1 - \frac{x}{x-1}$ has no root

Reason(R): $x-1 \neq 0$, then only above equation defined.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

- 107 **Assertion (A):**

0. $f(x) = ax^2 + bx + c$, for $a(\neq 0), b, c \in R$.
When $a > 0$ and $b^2 - 4ac < 0$, $f(x) = 0$ has complex roots.

Reason(R): The graph of $f(x) = ax^2 + bx + c$ cuts x-axis.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

107
1.

Assertion (A): If $f(x)$ is a quadratic expression such that $f(1) + f(2) = 0$, if -1 is a root of $f(x) = 0$ then the other root is $\frac{8}{5}$.

Reason (R): If $f(x) = ax^2 + bx + c$ then $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

107
2.

Assertion (A): If $a > 0, b > 0$ and $c > 0$, then

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

Reason (R): For positive numbers a, b, c, $AM \geq GM$

- 1) Both A and R are true and R is a consequence of A.
- 2) Both A and R are true and R is not a consequence of A.
- 3) A is true but R is false
- 4) Both A and R are false

107
3.

Assertion (A): If both roots of equation

$$4x^2 - 2x + a = 0, a \in R, \text{ lie in the interval } (-1, 1),$$

$$\text{then } -2 < a \leq \frac{1}{4}.$$

Reason (R): If $f(x) = 4x^2 - 2x + a$, then

$$D \geq 0, f(-1) > 0 \text{ and } f(1) > 0 \Rightarrow -2 < a \leq \frac{1}{4}.$$

- 1) Both A and R are true and R is a consequence of A.
- 2) Both A and R are true and R is not a consequence of A.
- 3) A is true but R is false
- 4) Both A and R are false

- 107 I. If $f(x)$ be a quadratic expression which is
 4. positive for all real x and
 $g(x) = f(x) + f'(x) + f''(x)$ for any real x ,
 then $g(x) > 0$.

II. If the equations $px^2 - 7x + 3p = 0$ and
 $2x^2 + qx + 6 = 0$ have the same roots then
 $pq = -14$. Which of the above
 statement(s) is(are) true?

- 1) only I 2) Only II
 3) both I and II 4) neither I nor II

- 107 The arrangement of the following quadratic
 5. equations in the ascending order of their
 number of real roots.

- A: $x^2 - 5x + 6 = 0$ B: $x^2 - 7|x| = 0$
 C: $x^2 - 4x + 5 = 0$ D: $x^2 - 5|x| + 6 = 0$
 1) C, A, B, D 2) D, C, B, A
 3) A, D, B, C 4) D, A, B, C

- 107 The arrangement of following quadratic
 6. equations in the ascending order of their
 product of roots.

- A: $3x^2 - 5x + 7 = 0$ B: $\sqrt{3}x^2 - 10x - 8\sqrt{3} = 0$
 C: $6x^2 - 5x + 1 = 0$ D: $x^2 - 5x + 6 = 0$
 1) B, C, A, D 2) C, A, B, D
 3) C, D, B, A 4) D, A, B, C

MATRIX - MATCH TYPE

- 107 Let α and β be the roots of the quadratic
 7. equation $ax^2 + bx + c = 0$. Match the following
 columns and choose the correct answer:

Column - I

Column - II

(A) $\alpha = \beta$ (1) $(ac^2)^{1/3} + (a^2c)^{1/3} + b = 0$

(B) $\alpha = 2\beta$ (2) $2b^2 = 9ac$

(C) $\alpha = 3\beta$ (3) $b^2 = 6ac$

(D) $\alpha = \beta^2$ (4) $3b^2 = 16ac$

(5) $b^2 = 4ac$

(Eamcet-2008) (6) $(ac^2)^{1/3} + (a^2c)^{1/3} = b$

	A	B	C	D
(1)	5	2	4	6
(2)	5	2	1	4
(3)	5	4	2	6
(4)	5	2	4	1

- 107
8. If α, β are the roots of $ax^2 + bx + c = 0$ observe the following lists.

List-I**List-II**

A: a, b, c have same sign

1. $\alpha\beta = 1$ B: If $a=c$

2. -1

C: If $a+b+c=0$, then one3. α and β are

root is

negative

D: If $a+c=b$, then one root is

4. 0

5. 1

The correct match for list-I from list-II is

A B C D**A B C D**

1) 1 3 2 5

2) 1 3 4 5

3) 3 1 5 4

4) 3 1 5 2

- 107
9. Observe the following lists

List-I**List-II****(Inequation) (Solution set)**A: $x^2 - 4x + 3 > 0$

1. (3, 4)

B: $x^2 - 5x + 6 \leq 0$ 2. $(-1, 1) \cup (2, 4)$ C: $x^2 + 6x - 27 > 0$,3. $(-\infty, 1) \cup (3, \infty)$ $-x^2 + 3x + 4 > 0$ D: $x^2 - 3x - 4 < 0$,

4. [3, 4]

 $x^2 - 3x + 2 > 0$

5. [2, 3]

Matching List-I, from List-II

A B C D**A B C D**

1) 1 2 3 4

2) 3 5 1 2

3) 1 3 5 4

4) 3 2 4 1

- 108
0. Let α and β be the roots of the equation $ax^2 + bx + c = 0$. Observe the lists given below:

EAMCET-2008**List-I****List-II****(i) $\alpha = \beta$** **A) $(ac^2)^{1/3} + (a^2c)^{1/3} + b = 0$** **(ii) $\alpha = 2\beta$** **B) $2b^2 = 9ac$** **(iii) $\alpha = 3\beta$** **C) $b^2 = 6ac$** **(iv) $\alpha = \beta^2$** **D) $3b^2 = 16ac$** **E) $b^2 = 4ac$** **F) $(ac^2)^{1/3} + (a^2c)^{1/3} = b$**

The correct match of List-I from List-II is

(i)**(ii)****(iii)****(iv)****(1)****E****B****D F****(2)****E****B****A D****(3)****E****D****B F****(4)****E****B****D A**

108 **Observe the following lists:**

1. **List-I** **List-II**

A. $\sqrt{x^2 - 5x + 6} < -3 \Rightarrow x \in$ 1. $\left[-2, \frac{1}{2}\right]$

B. $2-3x-2x^2 \geq 0 \Rightarrow x \in$ 2. $(-1, 2)$

C. $x^2 - 3x + 2 > 0,$ 3. $[-1, 2)$

$x^2 - 3x - 4 \leq 0 \Rightarrow x \in$

D. $\frac{x+1}{x-2} < 0 \Rightarrow x \in$ 4. ϕ

5. $[-1, 1) \cup (2, 4]$

The correct match for List-I from List-II is

A B C D

A B C D

1) 4 1 5 2

2) 4 5 1 2

3) 1 2 3 4

4) 2 3 4 5

20. Let $x^2 - (m-3)x + m = 0, (m \in R)$ be a quadratic equation. Match the following.

List - I

List - II

1) Both roots are greater than 2 A) $m \in (10, \infty)$

2) Both roots are smaller than 2 B) $m \in \phi$

3) Exactly one root lies in $(1, 2)$ C) $m \in [9, 10)$

4) Both roots lie in the interval $(1, 2)$ D) $m \in (-\infty, 1]$

	I	II	III	IV
1)	C	D	A	B
2)	D	C	A	B
3)	C	D	B	A
4)	C	B	D	A

INTEGER ANSWER TYPE QUESTIONS

21. The number of solutions of the equation

$$\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1} \text{ is } \underline{\hspace{2cm}}$$

22. The number of solution of

$$2^{\sin(|x|)} = 4^{|\cos x|} \text{ in } [-\pi, \pi] \text{ is equal to } \underline{\hspace{2cm}}$$

LEVEL - IV - KEY

1) 1	2) 1	3) 3	4) 1
5) 1	6) 1	7) 1	8) 3
9) 2	10) 1	11) 1	12) 3
13) 1	14) 1	15) 4	16) 4
17) 2	18) 4	19) 1	20) 1
21) 0	22) 4		

LEVEL - IV - HINTS

1. for $2x^2 + 3x + 4 = 0$

$$\Delta = 9 - 16 < 0$$

roots are complex conjugate

$$\Rightarrow ax^2 + bx + c = 0, \text{ and}$$

$$2x^2 + 3x + 4 = 0$$

have both roots common.

$$2. \quad \Delta = 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$$

$$3. \quad (i) \quad f\left(\frac{x}{k}\right) = 0$$

$$(ii) \quad -a = 10, \beta = 16,$$

$$-\alpha = 6, b = 9$$

$$\therefore x^2 + ax + b = x^2 - 10x + 9 = 0$$

$$x = 1, 9$$

$$4. \quad \Delta < 0, a \text{ \& } ax^2 + bx + c \text{ have opposite signs}$$

$$5. \quad \Delta = 0 \Rightarrow y = f(x) \text{ touches x-axis}$$

$$6. \quad (x-2)(x-5) < 0$$

$$7. \quad x = 1 \text{ not satisfies.}$$

$$8. \quad \text{from the synopsis.}$$

$$9. \quad f(1) + f(2) = 0$$

$$\Rightarrow 5a + 3b + 2c = 0$$

$$-1 \text{ is a root} \Rightarrow 3a - 3b + 3c = 0$$

$$\Rightarrow 8a + 5c = 0$$

$$\Rightarrow \frac{c}{a} = \frac{-8}{5}$$

$$10. \quad AM \geq GM$$

$$\Rightarrow \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \left(\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c} \right)^{\frac{1}{3}} \rightarrow (1)$$

$$\frac{(a+b+c)}{3} \geq (abc)^{\frac{1}{3}} \rightarrow (2)$$

multiply (1) and (2)

11. $A : \Delta > 0$, and $f(-1) > 0$ and $f(1) > 0$

12. (i) let $f(x) = x^2$

then $g(x) = x^2 + 2x + 2$

(ii) $\frac{p}{2} = \frac{-7}{q} = \frac{3p}{6}$

13. $A = x^2 - 5x + 6 = (x-2)(x-3) = 0 \Rightarrow x = 2, 3$

$$B = |x|^2 - 7|x| = |x|(|x| - 7) = 0 \Rightarrow x = 0, \pm 7$$

$$C = \Delta = 16 - 20 < 0, \text{no real root}$$

$$D = |x|^2 - 5|x| + 6 = (|x| - 2)(|x| - 3) = 0$$

$$\Rightarrow x = \pm 2, \pm 3$$

14. for $ax^2 + bx + c = 0$, $\alpha\beta = \frac{c}{a}$.

15. If the roots of a quadratic equation $ax^2 + bx + c = 0$ are in ratio $m:n$, then

$$\frac{(m+n)^2}{b^2}$$

$$mn \quad ac$$

$$16. \quad A: \alpha + \beta = -\frac{b}{a} < 0, \alpha\beta = \frac{c}{a} > 0$$

$$B: \alpha\beta = \frac{c}{a} = 1$$

$$C: a + b + c = 0 \Rightarrow a(1)^2 + b(1) + c = 0$$

$$D: a + c = b \Rightarrow a(-1)^2 + b(-1) + c = 0$$

$$17. \quad A: (x-3)(x-1) > 0 \Rightarrow x < 1 \text{ or } x > 3$$

$$B: (x-2)(x-3) \leq 0 \Rightarrow 2 \leq x \leq 3$$

$$C: x^2 + 6x - 27 > 0 \Rightarrow (x+9)(x-3) > 0$$

$$\Rightarrow x < -9 \text{ or } x > 3$$

$$-x^2 + 3x + 4 > 0 \Rightarrow (x+1)(x-4) < 0 \Rightarrow -1 < x < 4$$

$$\therefore 3 < x < 4$$

$$D: x^2 - 3x - 4 < 0 \Rightarrow (x+1)(x-4) < 0 \Rightarrow -1 < x < 4$$

$$x^2 - 3x + 2 > 0 \Rightarrow (x-2)(x-1) > 0 \Rightarrow x < 1 \text{ or } x > 2$$

$$\therefore x \in (-1, 1) \cup (2, 4)$$

$$18. \quad \text{If } \alpha : \beta = m : n \text{ then } \frac{(m+n)^2}{mn} = \frac{b^2}{ac}$$

$$(i) \quad m : n = 1 : 1$$

(ii) $m:n = 2:1$

(iii) $m:n = 3:1$

(iv) $\alpha = \beta^n \Rightarrow (ac^2)^{\frac{1}{n+1}} + (a^2c)^{\frac{1}{n+1}} + b = 0$

19. (i) $L.H.S > 0$ but $R.H.S < 0$

(ii) $(x+2)(2x-1) \leq 0 \Rightarrow x \in \left[-2, \frac{1}{2}\right]$

(iii) $x^2 - 3x + 2 > 0 \Rightarrow (x-2)(x-1) > 0 \Rightarrow x < 1 \text{ or } x > 2$

$x^2 - 3x - 4 \leq 0 \Rightarrow (x-4)(x+1) \leq 0 \Rightarrow -1 \leq x \leq 4$

(iv) $(x+1)(x-2) < 0$ and $x-2 \neq 0$

20. I) $f(2) > 0, D \geq 0, \alpha + \beta > 4$

II) $f(2) > 0, D \geq 0, \alpha + \beta < 4$

III) $f(1).f(2) < 0, D \geq 0$

IV) $f(1) > 0, f(2) > 0, D \geq 0, 2 < \alpha + \beta < 4$

21. S.O.B.S

22. $\sin|x| = 2|\cos x|$

$\Rightarrow \sin x = 2 \cos x$ in Q_1

$\sin x = -2 \cos x$ in Q_2

108

EXERCISE (JA)

2.

Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$

(B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$

(D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

108 Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value

3. of $\frac{a_{10} - 2a_8}{2a_9}$ is

- (A) 1 (B) 2 (C) 3 (D) 4

108 A value of b for which the equations

4. $x^2 + bx - 1 = 0$

$x^2 + x + b = 0,$

have one root in common is -

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

108 Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

108 Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

- (A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

108 If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is

- (1) Less than $4ab$ (2) Greater than $-4ab$
(3) Less than $-4ab$ (4) Greater than $4ab$

108 If α and β are the roots of the equation

8. $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$

- (1) -2 (2) -1
(3) 1 (4) 2

108 Let for $a \neq a_1 \neq 0$,

9. $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$.

If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is

- (1) 6 (2) 18
(3) 3 (4) 9

- 109** Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are:

- (1) $-6, -1$ (2) $-4, -3$
 (3) $6, 1$ (4) $4, 3$

- 109** The real number k for which the equation

1. $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$

- (1) Lies between 1 and 2
 (2) Lies between 2 and 3
 (3) Lies between -1 and 0
 (4) Does not exist

- 109** If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$,

2. $a, b, c \in R$, have a common root, then $a : b : c$ is

- (1) $1 : 2 : 3$ (2) $3 : 2 : 1$
 (3) $1 : 3 : 2$ (4) $3 : 1 : 2$

109 3. Quadratic Equations

Let α and β be the roots of equation $x^2 - 6x - 2 = 0$.

If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$

is equal to

- (1) 6 (2) -6
 (3) 3 (4) -3

- 109** The sum of all real values of x satisfying the

4. equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is

- (1) -4 (2) 6
 (3) 5 (4) 3

- 109** If, for a positive integer n , the quadratic equation,

5. $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$
 has two consecutive integral solutions, then n is equal to

- (1) 9 (2) 10
 (3) 11 (4) 12

109 Let $S = \{x \in \mathbb{R} : x \geq 0 \text{ and}$

6. $2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$. Then S

- (1) Is an empty set
- (2) Contains exactly one element
- (3) Contains exactly two elements
- (4) Contains exactly four elements

109 If both the roots of the quadratic equation
7. $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval $[1, 5]$ then m lies in the interval

- (1) $(-5, -4)$
- (2) $(3, 4)$
- (3) $(4, 5)$
- (4) $(5, 6)$

109 The number of all possible positive integral values
8. of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is

- (1) 4
- (2) 5
- (3) 2
- (4) 3

109 Consider the quadratic equation $(c - 5)x^2 - 2cx +$
9. $(c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval $(0, 2)$ and its other root lies in the interval $(2, 3)$. Then the number of elements in S is

- (1) 11
- (2) 18
- (3) 12
- (4) 10

110 The value of λ such that sum of the squares of the
0. roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is

- (1) 2
- (2) 1
- (3) $\frac{15}{8}$
- (4) $\frac{4}{9}$

110 If one real root of the quadratic equation $81x^2 + kx$
1. $+ 256 = 0$ is cube of the other root, then a value of k is

- (1) -300
- (2) 144
- (3) -81
- (4) 100

110 Let α and β the roots of the quadratic equation

2. $x^2 \sin \theta - x (\sin \theta \cos \theta + 1) + \cos \theta = 0$
 $(0 < \theta < 45^\circ)$, and $\alpha < \beta$. Then

$\sum_{n=0}^{\infty} \left(\alpha^n + \frac{(-1)^n}{\beta^n} \right)$ is equal to

(1) $\frac{1}{1+\cos \theta} - \frac{1}{1-\sin \theta}$ (2) $\frac{1}{1-\cos \theta} + \frac{1}{1+\sin \theta}$

(3) $\frac{1}{1-\cos \theta} - \frac{1}{1+\sin \theta}$ (4) $\frac{1}{1+\cos \theta} + \frac{1}{1-\sin \theta}$

110 If λ be the ratio of the roots of the quadratic

3. equation in x , $3m^2x^2 + m(m-4)x + 2 = 0$, then
the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is

(1) $4 - 2\sqrt{3}$ (2) $4 - 3\sqrt{2}$

(3) $2 - \sqrt{3}$ (4) $-2 + \sqrt{2}$

110 The number of integral values of m for which the

4. quadratic expression, $(1+2m)x^2 - 2(1+3m)x + 4(1+m)$, $x \in \mathbb{R}$, is always positive, is

(1) 8 (2) 3

(3) 6 (4) 7

110 If α and β be the roots of the equation $x^2 - 2x +$

5. $2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$

is

(1) 4 (2) 5

(3) 3 (4) 2

110 The sum of the solutions of the equation

6. $|\sqrt{x}-2| + \sqrt{x}(\sqrt{x}-4) + 2 = 0, (x > 0)$ is equal to

(1) 4 (2) 10

(3) 9 (4) 12

110 If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

(1) d, e, f are in A.P.

(2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(4) d, e, f are in G.P.

110 The number of integral values of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :

(1) Infinitely many (2) 3

(3) 2 (4) 1

110 Let $p, q \in R$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then

(1) $q^2 - 4p - 16 = 0$ (2) $p^2 - 4q + 12 = 0$

(3) $p^2 - 4q - 12 = 0$ (4) $q^2 + 4p + 14 = 0$

111 If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is

(1) $8\sqrt{3}$ (2) $10\sqrt{5}$

(3) $4\sqrt{3}$ (4) $8\sqrt{5}$

111 If α and β are the roots of the equation

1. $375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$

is equal to

(1) $\frac{21}{346}$ (2) $\frac{7}{116}$

(3) $\frac{29}{358}$ (4) $\frac{1}{12}$

111 If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to

(1) 0 (2) $\alpha\gamma$

(3) $\beta\gamma$ (4) $\alpha\beta$

111 Let α and β be two real roots of the equation

3. $(k+1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, where $k(\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is

- (1) 10 (2) $10\sqrt{2}$
(3) 5 (4) $5\sqrt{2}$

111 Let α and β be the roots of the equation

4. $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

- (1) $p_3 = p_5 - p_4$
(2) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
(3) $p_5 = 11$
(4) $p_5 = p_2 \cdot p_3$

111 Let S be the set of all real roots of the equation,

5. $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S

- (1) Contains at least four elements
(2) Is a singleton
(3) Contains exactly two elements
(4) Is an empty set

111 The number of real roots of the equation,

6. $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is

- (1) 4 (2) 2
(3) 3 (4) 1

111 Let $a, b \in \mathbb{R}$, $a \neq 0$ be such that the equation,

7. $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to

- (1) 25 (2) 24
(3) 26 (4) 28

111 Let α and β be the roots of the equation,

8. $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \dots$, then

- (1) $5S_6 + 6S_5 = 2S_4$ (2) $6S_6 + 5S_5 + 2S_4 = 0$
(3) $6S_6 + 5S_5 = 2S_4$ (4) $5S_6 + 6S_5 + 2S_4 = 0$

111 Let $f(x)$ be a quadratic polynomial such that

9. $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in

- (1) $(-1, 0)$ (2) $(-3, -1)$
(3) $(0, 1)$ (4) $(1, 3)$

112 If α and β are the roots of the equation

0. $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then

$\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to

- (1) $\frac{9}{4}(9 - q^2)$ (2) $\frac{9}{4}(9 + p^2)$
(3) $\frac{9}{4}(9 + q^2)$ (4) $\frac{9}{4}(9 - p^2)$

112 The set of all real values of λ for which the

1. quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is

- (1) $(-3, -1)$ (2) $(2, 4]$
(3) $(0, 2)$ (4) $(1, 3]$

112 Let $\lambda \neq 0$ be in R . If α and β are the roots of the

2. equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then

$\frac{\beta\gamma}{\lambda}$ is equal to

- (1) 18 (2) 9
(3) 27 (4) 36

112 The product of the roots of the equation

3. $9x^2 - 18|x| + 5 = 0$, is

- (1) $\frac{25}{9}$ (2) $\frac{25}{81}$
(3) $\frac{5}{9}$ (4) $\frac{5}{27}$

112 If α and β are the roots of the equation,

4. $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to

- (1) $\frac{1}{24}$ (2) $\frac{27}{32}$
 (3) $\frac{3}{8}$ (4) $\frac{27}{16}$

112 If α and β be two roots of the equation
 5. $x^2 - 64x + 256 = 0$. Then the value of

$$\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}} \text{ is}$$

- (1) 3 (2) 2
 (3) 4 (4) 1

112 If α and β are the roots of the equation

6. $2x(2x + 1) = 1$, then β is equal to

- (1) $2\alpha^2$
 (2) $-2\alpha(\alpha + 1)$
 (3) $2\alpha(\alpha - 1)$
 (4) $2\alpha(\alpha + 1)$

112 The least positive value of 'a' for which the

7. equation, $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is _____.

112 The integer 'k', for which the inequality
 8. $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R, is :

- (1) 2 (2) 3
 (3) 4 (4) 0

112 Let α and β be the roots of $x^2 - 6x - 2 = 0$. If

9. $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is :

- (1) 2 (2) 4
 (3) 3 (4) 1

113 Let α and β be two real numbers such that

0. $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$.

Then, the value of p_n^2 is _____.

113 The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is :

1.

- (1) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$ (2) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$
 (3) $2 + \frac{2}{5}\sqrt{30}$ (4) $5 + \frac{2}{5}\sqrt{30}$

113 The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to

2.

- (1) $2 + \sqrt{3}$
 (2) $3 + 2\sqrt{3}$
 (3) $4 + \sqrt{3}$
 (4) $1.5 + \sqrt{3}$

113 The number of real roots of the equation

3. $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is

- (1) 1 (2) 2
 (3) 6 (4) 4

113 If α, β are roots of the equation

4. $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n , then the value of

$\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2} \right)$ is equal to _____.

113 The number of real solutions of the equation,

5. $x^2 - |x| - 12 = 0$ is

- (1) 4 (2) 2
 (3) 1 (4) 3

113 If $a + b + c = 1$, $ab + bc + ca = 2$ and $abc = 3$,

6. then the value of $a^4 + b^4 + c^4$ is equal to _____.

113 Let α, β be two roots of the equation

7. $x^2 + (20)^{\frac{1}{4}}x + (5)^{\frac{1}{2}} = 0$. Then $\alpha^8 + \beta^8$ is equal to

- (1) 160 (2) 10
 (3) 50 (4) 100

113 8. Let $\alpha = \max_{x \in \mathbb{R}} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}$ and

$$\beta = \min_{x \in \mathbb{R}} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}.$$

If $8x^2 + bx + c = 0$ is a quadratic equation whose

roots are $\alpha^{\frac{1}{5}}$ and $\beta^{\frac{1}{5}}$, then the value of $c - b$ is equal to:

- (1) 43 (2) 42
(3) 50 (4) 47

113 The number of real roots of the equation

9. $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to ____.

114 The sum of all integral values of $k (k \neq 0)$ for which

0. the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real

114 roots, is ____.

1. Let $\lambda \neq 0$ be in $\mathbb{R}$. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$, and α and γ are the roots of the equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to ____.

114 The set of all values of $k > -1$, for which the equation $(3x^2 + 4x + 3)^2 - (k + 1)(3x^2 + 4x + 3)(3x^2 + 4x + 2) + k(3x^2 + 4x + 2)^2 = 0$ has real roots, is

- (1) $\left(\frac{1}{2}, \frac{3}{1}\right] - \{1\}$ (2) $\left[-\frac{1}{2}, 1\right)$
(3) $[2, 3)$ (4) $\left[1, \frac{5}{2}\right]$

114 cosec 18° is a root of the equation

3.

- (1) $x^2 - 2x + 4 = 0$ (2) $x^2 + 2x - 4 = 0$
(3) $x^2 - 2x - 4 = 0$ (4) $4x^2 + 2x - 1 = 0$

114 The number of pairs (a, b) of real numbers, such that

4. whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is :

- (1) 8 (2) 4
(3) 6 (4) 2

114 Let $f(x)$ be a polynomial of degree 3 such that

5. $f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10 f(10)$ is equal to ____.

114 If α and β are the roots of the quadratic equation,

6. $x^2 + x \sin \theta - 2 \sin \theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then

$$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}) \cdot (\alpha - \beta)^{24}} \text{ is equal to}$$

(1) $\frac{2^{12}}{(\sin \theta - 8)^6}$ (2) $\frac{2^{12}}{(\sin \theta - 4)^{12}}$

(3) $\frac{2^6}{(\sin \theta + 8)^{12}}$ (4) $\frac{2^{12}}{(\sin \theta + 8)^{12}}$

114 If for some $p, q, r \in \mathbf{R}$, not all have same sign, one

7. of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x$

$-8 = 0$, then $\frac{q^2 + r^2}{p^2}$ is equal to _____.

114 If α, β are the roots of the equation

8. $x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_3 3}}\right) + 3 \left(3^{(\log_3 5)^{\frac{1}{2}}} - 5^{(\log_3 3)^{\frac{2}{3}}} - 1\right) = 0$,

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and

$\beta + \frac{1}{\alpha}$, is

(1) $3x^2 - 20x - 12 = 0$ (2) $3x^2 - 10x - 4 = 0$

(3) $3x^2 - 10x + 2 = 0$ (4) $3x^2 - 20x + 16 = 0$

114 Let a, b be the roots of the equation

9. $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1, \frac{1}{\beta^2} + 1$ be

the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are

(1) non-real complex number

(2) real and both negative

(3) real and both positive

(4) real and exactly one of them is positive

115 Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3$, $f(-$

0. $2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to

(1) -4 (2) $\frac{13}{2}$

(3) $\frac{23}{2}$ (4) 4

- 115 Let $a, b (a > b)$ be the roots of the quadratic
 1. equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in \mathbb{N}$,

then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to

_____.

- 115 If the sum of the squares of the reciprocals of the
 2. roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

- (1) 18 (2) 24
 (3) 36 (4) 96

- 115 The sum of all the real roots of the equation
 3. $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

- (1) $\log_e 3$ (2) $-\log_e 3$
 (3) $\log_e 6$ (4) $-\log_e 6$

- 115 Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15$
 4. $= 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to

- (1) 37 (2) 58
 (3) 68 (4) 92

- 115 The sum of the cubes of all the roots of the equation
 5. $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is _____.

- 115 Let p and q be two real numbers such that $p + q = 3$
 6.

and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

- 115 If the sum of all the roots of the equation $e^{2x} - 11e^x -$
 7. $45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.



- 115 Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$
 8. and α, γ be the roots of the equation

$$x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0, \lambda > 0. \text{ If}$$

$\beta + \lambda = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

- 115 The number of real solutions of the equation
 9. $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is _____.

- 116 Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1 , then the sum of the roots of $f(x) = 0$ is equal to:

- (1) $\frac{11}{3}$ (2) $\frac{7}{3}$
(3) $\frac{13}{3}$ (4) $\frac{14}{3}$

- 116 Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$ and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

- 116 Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p$, $p \neq 0$, and

$f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and $f(f(x)) = 0$ have a common real root, then $f(-3)$ is equal to _____.

- 116 The sum of all real value of x for which

3. $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ is equal to _____.

- 116 The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is

- (1) 4 (2) 5
(3) 6 (4) 8

1 □

- 116 For $x \in \mathbb{R}$, then number of real roots of the equation $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is ____.

- 116 Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is

- (A) 0 (B) 8000 (C) 8080 (D) 16000

- 116 Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$.

7. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

If $a_4 = 28$, then $p + 2q =$

- (A) 14 (B) 7 (C) 12 (D) 21

116 Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$.

8.

For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

$a_{12} =$

- (A) $2a_{11} + a_{10}$ (B) $a_{11} - a_{10}$ (C) $a_{11} + a_{10}$ (D) $a_{11} + 2a_{10}$

116 Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are

9.

the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

- (A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

117 Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

0.

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

117 EXERCISE (JA)

1.

Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

117 Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value

2.

of $\frac{a_{10} - 2a_8}{2a_9}$ is

- (A) 1 (B) 2 (C) 3 (D) 4

117 A value of b for which the equations

3.

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is -

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

- 117 4. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

(A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

- 117 5. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

(A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

- | | |
|--------|---|
| 117 6. | The least value of $f(x)$ is
A) $3/4$ B) $9/2$ C) $-9/8$ D) $3/4$ |
| 117 7. | Number of integral values of x for which $f(x) < 0$ is
A) 0 B) 1 C) 2 D) 3 |
| 117 8. | If $ax^2 + bx + c = 0$ has imaginary roots and $a - b + c > 0$, then the set of points (x, y) satisfying the equation $\left a\left(x^2 + \frac{y}{a}\right) + (b+1)x + c\right = ax^2 + bx + c + x + y $ consists of the region in the xy -plane which is
A) On or above the bisector of I and III quadrant
B) On or above the bisector of II and IV quadrant
C) On or below the bisector of I and III quadrant
D) On or below the bisector of II and IV quadrant |
| 117 9. | If α, β are the roots of the quadratic equation $x^2 - \left(3 + 2\sqrt{10\log_2 3} - 3\sqrt{\log_3 2}\right)x - 2\left(3^{\log_3 2} - 2^{\log_2 3}\right) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to
A) 3 B) 5 C) 7 D) 11 |

118
0.

The product of the roots of equation $x^{\log_{10} x} = \frac{10^4}{x^3}$, is

A) $\frac{1}{10}$

B) $\frac{1}{100}$

C) $\frac{1}{1000}$

D) $\frac{1}{10000}$

SINGLE CORRECT ANSWER TYPE118
1.

If the equation $4x^2 - 4(5x + 1) + p^2 = 0$ has one root equals to two more than the other, then the value of p is equal to

A) $\pm \frac{\sqrt{236}}{3}$

B) ± 5

C) $5 \text{ or } -1$

D) $4 \text{ or } -3$

118
2.

If the equation $(x^2 + a|x| + a + 1)(x^2 + (a + 1)|x| + a) = 0$ has no real root, then the range of values of 'a' is

A) $(0, 1)$

B) $(-\infty, 0)$

C) $(0, \infty)$

D) $(-1, 0)$

MULTIPLE CORRECT ANSWER TYPE118
3.

If the equation $ax^2 + bx + c = 0 (a > 0)$ has two real roots α and β such that $\alpha < -2$ and $\beta > 2$, then which of the following statements is/are true?

A) $a - |b| + c < 0$

B) $c < 0, b^2 - 4ac > 0$

C) $4a - 2|b| + c < 0$

D) $9a - 3|b| + c < 0$

118
4.

The possible values of 'a' for which $e^{2x} - (a - 2)e^x + a < 0$ holds for atleast one $x \in (0, \infty)$ are:

A) 6

B) 9

C) 12

D) 16

118
5.

The value(s) of 'p' for which the equation $ax^2 - px + ab = 0$ and $x^2 - ax - bx + ab = 0$ may have a common root, given a, b are non zero real numbers, is:

A) $a + b^2$

B) $a^2 + b$

C) $a(1 + b)$

D) $b(1 + a)$

INTEGER ANSWER TYPE118
6.

The value of "a" for which all roots of quadratic equation,

$$f(x) = (a - 2)x^2 + 2ax + a + 3 = 0 \text{ lies in } (-2, 1) \text{ belongs to } \left(-\infty, -\frac{1}{4}\right) \cup (m, n] \text{ then}$$

value of $n - m$ is:

3 5

$$\frac{c}{2} \text{ and } \log_c d = \frac{c}{4}. \text{ If } (a - c) = 9,$$

$0, a) \cup (b, c)$ then (abc) has the

$$3x + \sin x) = \log_{\frac{-x^2-6x}{10}}(\sin 2x)$$
$$(4-r)$$

- 119
1. For the equation $1 + \log_x \left(\frac{\tau}{10} \right) = (\log_{10}(\log_{10} p) - 1) \log_x 10$, match the value of p given in column II corresponding to the number of roots given in column I.

Column I		Column II	
a)	For no root	p	$p \in \{10^3\}$
b)	For exactly one root	q	$p \in \{10^3\}$
c)	For two distinct roots	r	$p \in (1, 10^4) - \{10^3\}$
		s	$p \in (10^4, \infty)$

- A) a-s; b-p,q; c-r B) a-p; b-s,q; c-r
 C) a-r; b-p,q; c-s D) a-s; b-p,r; c-q
17. A value of λ for which the equations $x^2 + \lambda x - 1 = 0$, $x^2 + x + \lambda = 0$ have one root in common is
 A) $-\sqrt{2}$ B) $-i\sqrt{3}$ C) $-i\sqrt{5}$ D) $\sqrt{2}$
18. Let a,b,c,d be the roots of $x^4 - x^3 - x^2 - 2 = 0$. Also consider $p(x) = x^6 - x^5 - x^3 - x^2 - x$. Then the value of $p(a) + p(b) + p(c) + p(d)$ is equal to
 A) 4 B) 12 C) 6 D) 13

19. If $x^3 - ax^2 + bx + c = 0$ has the roots $\alpha^2 + \beta^3 + \gamma^4, \beta^2 + \gamma^3 + \alpha^4$ and $\gamma^2 + \alpha^3 + \beta^4$ where α, β, γ are the roots of $x^3 - x^2 - 1 = 0$ then the value of $2a + b + c$ is equal to _____
A) 14 B) 15 C) 17 D) 16
20. If x and y satisfy the equation $xy - 2x^2 - 13x + 5y - 22 = 0$, then which of the following options is/are **CORRECT** ?
A) Number of ordered pair (x, y) is 4 where $x, y \in \mathbb{Z}$
B) Number of ordered pair (x, y) is 1 where $x, y \in \mathbb{N}$
C) if $x \geq y, x, y \in \mathbb{N}$ then number of ordered pair is zero
D) if $x \leq y, x, y \in \mathbb{N}$ then number of ordered pairs are two
21. Let α and β be the roots of $x^2 - x - 1 = 0$, with $\alpha > \beta$. For all positive integers n , define

$$a_n = \frac{\alpha^n - 3^n}{\alpha - 3}, n \geq 1$$

$$b_1 = 1 \text{ and } b_n = a_{n-1} + a_{n+1}, n \geq 2.$$
 Then which of the following options is/are **CORRECT**?
A) $a_1 + a_2 + a_3 + \dots + a_n = a_{n+2} - 1$ for all $n \geq 1$
B) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{11}{89}$ **C)** $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{12}{89}$ **D)** $b_n = \alpha^n + 3^n$ for all $n \geq 1$
22. Given that 'a' is a root of the equation $x^2 - x - 3 = 0$ then $\frac{a^3 + 1}{a^5 - a^4 - a^3 + a^2}$ equals

23. The polynomial $f(x) = x^{2022} + 17x^{2021} + 1$ has distinct zeroes $r_1, r_2, r_3, \dots, r_{2022}$. A polynomial $p(x)$ of degree 2022 has the property that $P\left(r_j + \frac{1}{r_j}\right) = 0, j = 1, 2, \dots, 2022$. The value of $\frac{259}{17} \frac{P(1)}{P(-1)}$ is equal to ?
24. The equation $x^4 + ax^3 + bx^2 + ax + 1 = 0$ has at least one real root and if the minimum value of $E(a, b) = a^2 + b^2$ can be expressed as $\frac{p}{q}$ (p and q are relatively prime), then $(p^2 + q^2)$ is
25. If $(1 - P)(1 + 3x + 9x^2 + 27x^3 + 81x^4 + 243x^5) = 1 - P^6, P \neq 1$, then find the value of $\frac{P}{x}$.
26. If the roots of $x^5 - 40x^4 + px^3 + qx^2 + rx + S = 0$ are in G.P. and sum of their reciprocals is 10, then the value of $\sqrt{|2S|}$ is _____

Sr.REVISION ASSIGNMENT

(Week-11_Day-2_18-09-2024)

MATHEMATICS

(QUADRATIC EQUATIONS)

MCQs with One Correct Answer

- 119
2. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are :
 A) non-real complex numbers
 B) real and both negative
 C) real and both positive
 D) real and exactly one of them is positive
- 119
3. If α, β are the roots of the equation $x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right) + 3\left(3^{(\log_3 5)^{1/5}} - 5^{(\log_5 3)^{1/3}} - 1\right) = 0$
 Then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$
 A) $3x^2 - 20x - 12 = 0$
 B) $3x^2 - 10x - 4 = 0$
 C) $3x^2 - 10x + 2 = 0$
 D) $3x^2 - 20x + 16 = 0$
- 119
4. The minimum value of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is
 A) 4
 B) 5
 C) 6
 D) 8

119 5.	Let α and β be the roots of the equation $x^2 + (2i - 1) = 0$. Then, the value of $ \alpha^8 + \beta^8 $ is equal to : A) 50 B) 250 C) 1250 D) 1500
119 6.	Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to : A) 1 B) α C) $1 + \alpha$ D) $1 + 2\alpha$
119 7.	Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1 , then the sum of the roots of $f(x) = 0$ is equal to A) $\frac{11}{3}$ B) $\frac{7}{3}$ C) $\frac{13}{3}$ D) $\frac{14}{3}$
119 8.	The number of distinct real roots of $x^4 - 4x + 1 = 0$ is : A) 4 B) 2 C) 1 D) 0
119 9.	If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to : A) 18 B) 24 C) 36 D) 96
120 0.	The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is A) $\log_e 3$ B) $-\log_e 3$ C) $\log_e 6$ D) $-\log_e 6$
120 1.	Let α, β be two roots of the equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to A) 10 B) 100 C) 50 D) 160
120 2.	The number of real solutions of the equation, $x^2 - x - 12 = 0$ is : A) 2 B) 3 C) 1 D) 4
120 3.	The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to A) $2 + \sqrt{3}$ B) $3 + 2\sqrt{3}$ C) $4 + 2\sqrt{3}$
120 4.	If $\alpha, \beta \in \mathbb{R}$ such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to A) -3 B) -7 C) 7 D) 3
120 5.	If $e^{(\cos^{-1} x + \cos^{-1} x + \cos^{-1} x + \dots \infty) \log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2 \sin x}{\sin x + \sqrt{3} \cos x} \left(0 < x < \frac{\pi}{2} \right)$ is A) $2\sqrt{3}$ B) $\frac{3}{2}$ C) $\sqrt{3}$ D) $\frac{1}{2}$

120 6.	Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is A) 0 B) 8000 C) 8080 D) 16000
120 7.	If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is A) 2 B) 3 C) 1 D) 4
120 8.	If α and β are the roots of the equation $2x(2x + 1) = 1$, then β is equal to A) $2\alpha(\alpha + 1)$ B) $-2\alpha(\alpha + 1)$ C) $2\alpha(\alpha - 1)$ D) $2\alpha^2$
120 9.	If α and β are the roots of the quadratic equation $x^2 + x \sin \theta - 2 \sin \theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12})(\alpha - \beta)^{24}}$ is equal to :
121 0.	Let p, q are r be real numbers ($p \neq q, r \neq 0$), such that the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the sum of squares of these roots is equal to . A) $p^2 + q^2 + r^2$ B) $p^2 + q^2$ C) $2(p^2 + q^2)$ D) $\frac{p^2 + q^2}{2}$
121 1.	The sum of all the real values of x satisfying the equation $2^{(x-1)(x^2+5x-50)} = 1$ is ; . A) 16 B) 14 C) - 4 D) - 5
121 2.	Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose θ_1 and β_1 are the roots of the equation $x^2 - 2x \sec \theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x \tan \theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$. Then $\alpha_1 + \beta_2$ equals A) $2(\sec \theta - \tan \theta)$ B) $2 \sec \theta$ C) $-2 \tan \theta$ D) 0
121 3.	If $2 + 3i$ is one of the roots of the equation $2x^3 - 9x^2 + kx - 13 = 0$, $k \in R$, then the real of this equation : A) exists and is equal to $-\frac{1}{2}$ B) exists and is equal to $\frac{1}{2}$ C) exists and is equal to 1 D) does not exist
121 4.	The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has A) one purely imaginary root B) all real roots C) two real and two purely imaginary roots D) neither real nor purely imaginary roots

121
5. If $a \in \mathbb{R}$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greater integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

- A) $(-2, -1)$ B) $(-\infty, -2) \cup (2, \infty)$ C) $(-1, 0) \cup (0, 1)$ D) $(1, 2)$

121
6. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is

- A) 1 B) 2 C) 3 D) 4

DD COLONY, HYDERABAD
(Week-11_Day-3_19-09-2024)
MATHEMATICS
(QUADRATIC EQUATIONS)

121
7. Let (x_0, y_0) be the solution of the following equations $(2x)^{\log 2} = (3y)^{\log 3}; 3^{\log x} = 2^{\log y}$. Then x_0 is

- A) $\frac{1}{6}$ B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) 6

121
8. Let p and q be real numbers such that $p \neq q$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

- A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
 B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
 D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

121
9. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

- A) $\frac{2}{9}(p - q)(2q - p)$ B) $\frac{2}{9}(q - p)(2p - q)$
 C) $\frac{2}{9}(q - 2p)(2q - p)$ D) $\frac{2}{9}(2p - q)(2q - p)$

122
0. Let a, b, c be the sides of a triangle where $a \neq b \neq c$ and $\lambda \in \mathbb{R}$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then

- A) $\lambda < \frac{4}{3}$ B) $\lambda > \frac{5}{3}$ C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

122
1. If one root is square of the other root of the equation $x^2 + px + q = 0$, then the relation between p and q is

- A) $p^3 - q(3p - 1) + q^2 = 0$ B) $p^3 - q(3p + 1) + q^2 = 0$
 C) $p^3 + q(3p - 1) + q^2 = 0$ D) $p^3 + q(3p + 1) + q^2 = 0$

122
2. For the equation $3x^2 + px + 3 = 0$, $p > 0$, if one of the root is square of the other, then p is equal to

122
3. If $b > a$, then the equation $(x - \alpha)(x - b) - 1 = 0$ has

- A) both roots in (a, b)
 B) both roots in $(-\infty, a)$
 C) both roots in $(b, +\infty)$
 D) one root in $(-\infty, \alpha)$ and the other in $(b, +\infty)$

122
4. If α and β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then

- A) $0 < \alpha < \beta$ B) $\alpha < 0 < \beta < |\alpha|$ C) $\alpha < \beta < 0$ D) $\alpha < 0 < |\alpha| < \beta$

122
5. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then

- A) $\alpha < 2$ B) $2 \leq a \leq 3$ C) $3 < a \leq 4$ D) $a > 4$

- 122
6. Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is
A) 15 B) 9 C) 7 D) 8
- 122
7. Let α, β be the roots of the equation $(x-a)(x-b) = c, c \neq 0$. Then the roots of the equation $(x-\alpha)(x-\beta) + c = 0$ are
- 122
8. Let a, b, c be real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies.
A) $\gamma = \frac{\alpha + \beta}{2}$ B) $\gamma = \alpha + \frac{\beta}{2}$ C) $\gamma = \alpha$ D) $\alpha < \gamma < \beta$
- 122
9. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has
A) no root B) one root
C) two equal roots D) infinitely many roots
- 123
0. If $(x^2 + px + 1)$ is a factor of $(ax^3 + bx + c)$, then
A) $a^2 + c^2 = -ab$ B) $a^2 - c^2 = -ab$
C) $a^2 - c^2 = ab$ D) none of these
- 123
1. Both the roots of the equation $(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$ are always
A) positive B) real C) negative D) none of these
- 123
2. If $1, m, n$ are real, $1 \neq m$, then the roots by the equation $(1-m)x^2 - 5(1+m)x - 2(1-m) = 0$ are
A) Real and equal B) Complex
C) Real and unequal D) None of these
- 123
3. The product of all positive real values of x satisfying the equation $x^{(16(\log_5 x)^3 - 68 \log_5 x)} = 5^{-16}$ is _____
- 123
4. The sum of all real values of x for which $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ is equal to
- 123
5. Let $S = \{z \in \mathbb{C} : z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} (\operatorname{Re}(z) + \operatorname{Im}(z))$ is equal to _____

Numeric/New Stem Based Questions

123
6.

Let $\alpha, \beta (\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If

$p_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____

123
7.

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is

01

123
8.

If the sum of all the roots of the equation $e^{2x} - 11e^x - 45e^{-x} + \frac{01}{2} = 0$ is $\log_e P$, then P is equal

123
9.

Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____

124
0.

The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is

124
1.

The smallest value of k, for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

Sr.REVISION ASSIGNMENT
(Week-11_Day-4_20-09-2024)
MATHEMATICS
(QUADRATIC EQUATIONS & LOGARITHMS)

124
2.**MCQs with One Correct Answer.**

The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 9k^2 - 7 > 0$ is valid for every x in R , is :

- A) 2 B) 3 C) 4 D) 0

124
3.

Let $a, b \in R, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to :

- A) 25 B) 26 C) 28 D) 24

124
4.

If $5, 5r, 5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to :

- A) $\frac{3}{4}$ B) $\frac{5}{4}$ C) $\frac{7}{4}$ D) $\frac{3}{2}$

124
5.

If $\lambda \in R$ is such that the sum of the cubes of the roots of the equation $x^2 + (2 - \lambda)x + (10 - \lambda) = 0$ is minimum, then the magnitude of the difference of the roots of this equation is

- A) 20 B) $2\sqrt{5}$ C) $2\sqrt{7}$ D) $4\sqrt{2}$

124
6.

If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1, then $|b|$ is equal to :

- A) 2 B) 3 C) $\sqrt{3}$ D) $\sqrt{2}$

124
7.

If non-zero real numbers b and c are such that $\min f(x) > \max g(x)$, where

$f(x) = x^2 = 2bx + 2c^2$ and $g(x) = -x^2 - 2x^2 + b^2 (x \in R)$; then $\left| \frac{c}{b} \right|$ lies in the interval

- A) $\left(0, \frac{1}{2}\right)$ B) $\left[\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ C) $\left[\frac{1}{\sqrt{2}}, \sqrt{2}\right]$ D) $(\sqrt{2}, \infty)$

124
8.

If equations $ax^2 + bx + c = 0 (a, b, c \in R, a \neq 0)$ and $2x^2 + 3x + 4 = 0$ have a common root, then $a:b:c$ equals :

- A) 1 : 2 : 3 B) 2 : 3 : 4 C) 4 : 3 : 2 D) 3 : 2 : 1

124
9.

The least integral value α of x such that $\frac{x-5}{x^2+5x-14} > 0$, satisfies :

- A) $\alpha^2 + 3\alpha - 4 = 0$ B) $\alpha^2 - 5\alpha + 4 = 0$ C) $\alpha^2 - 7\alpha + 6 = 0$ D) $\alpha^2 + 5\alpha - 6 = 0$

125
0.A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0$$

have one root in common is

- A) $-\sqrt{2}$ B) $-i\sqrt{3}$ C) $i\sqrt{5}$ D) $\sqrt{2}$

125
1.For all ' x ', $x^2 + 2ax + 10 - 3a > 0$, then the interval in which ' a ' lies is

- A) $a < -5$ B) $-5 < a < 2$ C) $a > 5$ D) $2 < a < 5$

Numeric/New Stem Based Questions125
2.If for some $p, q, r \in R$, not all have same sign, one of the roots of the equation

$$(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$$
 is also a root of the equation $x^2 + 2x - 8 = 0$, then

$$\frac{q^2 + r^2}{p^2}$$
 is equal to ____

Fill in the Blanks125
3.If the quadratic equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ ($a \neq b$) have a common root, then the numerical value of $a + b$ is _____**True/False**125
4.If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, where $ac \neq 0$, then $P(x)Q(x) = 0$ has at least two real roots.**MCQs with One or More than One Correct Answer.**125
5.Let S be the set of all non-zero real numbers α such that the quadratic equation $ax^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$.Which of the following interval is (are) a subset(s) of S ?

- A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ C) $\left(0, \frac{1}{\sqrt{5}}\right)$ D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

125
6.If a, b, c, d and p are distinct real numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0$$
 then a, b, c, d

- A) are in A.P. B) are in G.P. C) are in H.P. D) satisfy $ab = cd$
A) satisfy none of these

125
7.For real x the function $\frac{(x-a)(x-b)}{x}$ will assume all real values provided**Fill in the Blanks**

- 125
8. If the product of the roots of the equation $x^2 - 3kx + 2e^{2\ln k} - 1 = 0$ is 7, then the real for $k =$ _____

True/False

- 125
9. If $a < b < c < d$, then the roots of the equation $(x-a)(x-c) + 2(x-b)(x-d) = 0$ are real and distinct.

- 126
0. The equation $2x^2 + 3x + 1 = 0$ has an irrational root.

MCQs with One or More than One Correct Answer.

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$

- 126
1. $a_{12} =$
A) $a_{11} - a_{10}$ B) $a_{11} + a_{10}$ C) $2a_{11} + a_{10}$ D) $a_{11} + 2a_{10}$

- 126
2. $a_4 = 28$, then $p + 2q =$
A) 21 B) 14 C) 7 D) 12

Assertion and Reason/Statement Type Questions

- 126
3. Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$ and $\alpha, \frac{1}{\beta}$ are the roots of the equation $ax^2 + 2bx + c = 0$, where $\beta^2 \notin \{-1, 0, 1\}$
Statement-1 : $(p^2 - q)(b^2 - ac) \geq 0$ and
Statement -2: $b \neq pa$ or $c \neq qa$ Statement - 1 is True, Statement-2 is True;
Statement-2 is a correct explanation for statement-1.
A) Statement - 1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for statement-1
B) Statement - 1 is True, Statement -2 is False
C) Statement - 1 is False, Statement-2 is True.

- 126
4. Let a and b the roots of the equation $x^2 - 10cx - 11d = 0$ and those of $x^2 - 10ax - 11b = 0$ are c, d then the value of $a + b + c + d$, when $a \neq b \neq c \neq d$, is

- 126
5. If $x^2 + (a-b)x + (1-a-b) = 0$ where $a, b \in R$ then find the values of a for which equation has unequal real roots for all values of b .

- 126
6. If α, β are the roots of $ax^2 + bx + c = 0$, ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$ ($A \neq 0$) for some constant δ , then prove that $b^2 - 4ac = B^2 - 4AC$

126 The roots of the equation $\frac{1}{x-1} + \frac{1}{x-2} = 7$.

1) $3 \pm \sqrt{2}$ 2) $2 \pm \sqrt{3}$

126 If $x > 0$ and $x = \frac{1}{x} + 1$, then the value of x is

8. 1) $\frac{\sqrt{5}-1}{2}$ 2) $\frac{\sqrt{5}+1}{2}$

126 If m, n, k are rational and $m = k + \frac{n}{k}$, then the value of m is

9. 1) $k, \frac{n}{k}$ 2) $k, \frac{-n}{k}$

127 If 3 is a root of $x^2 + kx - 24 = 0$ it is also a root of

0. 1) $x^2 + 5x + k = 0$ 2) $x^2 + kx + 24 = 0$

127 The roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are

1. 1) $\frac{c(a-b)}{a(b-c)}, \frac{b(c-a)}{c(a-b)}$ 2) $\frac{b(c-a)}{c(a-b)}, 1$

127 A root of the equation $(x-1)(x-2) = \frac{30}{49}$ is

2. 1) $\frac{17}{7}$ 2) $\frac{15}{7}$

127 A root of the equation $(x-2)(x-3) = \frac{155 \times 7}{(77)^2}$ is

3. 1) $\frac{309}{77}$ 2) $\frac{155}{77}$

127 A root of the equation $\frac{a+c}{x+a} + \frac{b+c}{x+b} = \frac{2(a+b)}{x+a+b}$ is

4. 1) a 2) b

127 If $(x-a+2b)^2 + (x-2a+b)^2 = (a+b)^2$, then the value of x is

5. 1) $a-b$ 2) $2a-b$

127 If α, β are real and $\alpha^2, -\beta^2$ are the roots of the equation $x^2 + px + q = 0$, then

6. 1) a^2 2) 1

Relation between roots & coefficients :

127 If α, β are the roots of $x^2 + x + 1 = 0$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is

7. 1) -1 2) 1

127 If α, β are the roots of $x^2 - x + 3 = 0$, then $\alpha^2 + \beta^2$ is

8. 1) 10 2) 7

127 If $\alpha > \beta$ and α, β are the roots of $x^2 - 3x - 1$

9. 1) $\sqrt{13}$ 2) $3\sqrt{13}$

128 The values of P for which the difference

0. 2 are

- 1) ± 2 2) ± 4

128 If the roots of the equation $x^2 - px + q = 0$

1. 1) $4 + q$ 2) $4q + 1$

128 If $\alpha, \alpha + 4$ are the roots of $x^2 - 8x + K = 0$, then

2. 1) 9 2) 10

128 If α, β are the roots of $x^2 + bx + c = 0$ and, $c \neq 0$

3. 1) $b + q$

q)(EAMCET-2001)

128 If α, β are the roots of $x^2 + ax - b = 0$ and

4. $(\alpha - \gamma)(\alpha - \delta)(\beta - \delta)(\beta - \gamma) =$
1) $4b^2$ 2) b^2

128 If the roots of $x^2 - bx + c = 0$ are two consecutive

5. 2005)
1) 0 2) 1

128 The non trivial values of (a, b) such that

6. are
1) 2, -1 2) 1, -2

128 If the sum of the squares of the roots of $x^2 - 3x + 1 = 0$

7. 1998)
1) ± 2 2) ± 3

128 If the sum of the roots of $x^2 + bx + 1 = 0$ is

8. 1) 1, 2 2) 1, -2

128 If 8 and 2 are the roots of $x^2 + ax + \beta = 0$ and

9. of the equation $x^2 + ax + b = 0$ are
1) 1, -1 2) -9, 2

129 If $k > 0$ and the product of roots of the

0. roots =
1) 12 2) 4

129 If the roots of the quadratic equation $x^2 + px + q = 0$

1. value of $2 + q - p$ is

129 If $(1-p)$ is a root of quadratic equation $x^2 + px + q = 0$

2. 1) 0, 1 2) -1, 2

129 If α, β are the roots of $x^2 + 7x + 3 = 0$ the

3. 1) 59 2) 69

129 If α, β are the roots of $2x^2 + x + 3 = 0$ the

4. 1) $\frac{1}{2}$ 2) $-\frac{1}{2}$

129 If α, β are the roots of $ax^2 + bx + c = 0$ th

5. 1) $\frac{1}{a^2}(a^2 + b^2 + c^2 + ab + bc + ca)$
3) $\frac{1}{a^2}(a + b + c)$

129 If α, β are the roots of $x^2 + x + 3 = 0$ then $5\alpha +$

6. 1) 0 2) -1

129 If α, β are the roots of the equation $8x^2 - 3x + 1 = 0$

7. 1) $\frac{1}{6}$ 2) $\frac{3}{2}$

129 If n is a multiple of 6 and α, β are the roots

8. 1) 2 2) -2

129 If α, β are the roots of $3x^2 + 5x - 7 = 0$ the

9. 1) $-\frac{17}{21}$ 2) $\frac{67}{21}$

130 If α, β are the roots of $ax^2 + bx + c = 0$, the

0. 1) 0 2) ab

130 If w, w^2 are the roots of $x^2 + x + 1 = 0$

1. $(w\alpha + w^2\beta)(w^2\alpha + w\beta) =$
1) $p^2 - 3q$ 2) $p^2 - 2q$

130 If α, β are the roots of $x^2 + px + 1 = 0$ and

2. $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) =$
1) $q^2 - p^2$ 2) $p^2 - q^2$

130 If the difference between the roots of $x^2 + ax + b = 0$

3. is
1) $(3, \infty)$ 2) $(-\infty, -3)$

130 The value of c for which $|\alpha^2 - \beta^2| = \frac{1}{4}$ is

4. 1) 4 2) 0

130 If the sum of the roots of the equation $x^2 + px + q = 0$

5. 1) $p^2 = 6q$ 2) $2p^2 = 9q$

- 130 If the difference between the roots of
6. roots of $x^2 + px + q = 0$ then $a^2 - p^2 =$
1) $b-q$ 2) $2(p-q)$

- 130 If the roots of $ax^2 + bx + c = 0$ are $2, \frac{3}{2}$
7. 1) $\frac{a^2}{2}$ 2) $\frac{a^2}{4}$

- 130 In a triangle PQR , $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$
8. then
2005)
1) $a = b + c$ 2) $c = a + b$

- 130 If $\sin\theta, \cos\theta$ are the roots of the equati
9. 1) $2q$ 2) $2q + 1$

- 131 If $\tan^2\theta, \sin^2\theta$ are the roots of $ax^2 + bx +$
0. 1) $4ac$ 2) a^2

- 131 If $x^2 - 3x + 2$ is a factor of $f(x) = x^4 - px^2 + q$,
1. 1) $(-4, -5)$ 2) $(4, 5)$

Formation of quadratic equations & transfo

- 131 If α, β are the roots of $x^2 + bx - c = 0$, th
2. 1) $x^2 + \alpha x - \beta = 0$
3) $x^2 + (\alpha + \beta + \alpha\beta)x + \alpha\beta(\alpha + \beta) = 0$

- 131 If the arithmetic mean of the roots of q
3. their reciprocals is $\frac{8}{7}$, then the equatio
1) $5x^2 + 16x + 7 = 0$ 2) $5x^2 - 16x + 7 =$

- 131 The equation whose roots are the
4. equation $x^2 + ax - b = 0$ is
1) $x^2 - bx + a = 0$
3) $2ax^2 + (a^2 - 8b)x - 4ab = 0$

- 131 If $\alpha + \beta = 3$ and $\alpha^3 + \beta^3 = 7$, then α an
5.

- 131 If α, β are the roots of the equation $x^2 +$
6. and $(\alpha - \beta)^2$ is
1) $x^2 + 50x + 49 = 0$ 2) $x^2 - 50x + 49 =$

- 131 If $\lambda \neq \mu$ and $\lambda^2 = 5\lambda - 3, \mu^2 = 5\mu - 3$, then
7. 2002)
1) $x^2 - 5x - 3 = 0$
3) $3x^2 - 19x + 3 = 0$

131 If $a(5+q^2)+2bq+c=0$ and $a(5+r^2)+2br+c=0$ w

8. 1) $\frac{-2b}{a}$ 2) $\frac{2b}{a}$

131 If α, β are the roots of $ax^2 + bx + c = 0$, the

9. 1) $a^2x^2 + a(b-c)x + bc = 0$
3) $a^2x^2 + (b+c)x + bc = 0$

132 Let α, β be the roots of equation $x^2 + x + 1 = 0$

0. 1) $x^2 - x - 1 = 0$ 2) $x^2 - x + 1 = 0$

132 If α, β are the roots of $x^2 - px + q = 0$ then th

1. 1) $x^2 + 2qx + p^2q^2 = 0$ 2) $x^2 - 2qx + p^2 + q^2$

132 The equation whose roots are smaller by 1

2. 1) $2x^2 - 9x + 13 = 0$ 2) $2x^2 - x + 3 = 0$

132 If α, β are the roots of $x^2 + 3x + 1 = 0$, then th

3. 1) $x^2 + 7x + 10 = 0$ 2) $x^2 - 7x + 11 = 0$

132 If α, β are the roots of $ax^2 + bx + c = 0$, then

4. 1) $ax^2 + (4a-b)x + 4a - 2b + c = 0$
3) $ax^2 + (b-4a)x + 4a + 2b + c = 0$

132 The equation formed by decreasing each ro

5. 1) $a = -b$ 2) $b = -c$

132 The equation formed by decreasing each ro

6. 1) $b = 1, a + c = 0$ 2) $b = 2, a + c = 0$

132 If $x^2 + px + q = 0$ is the quadratic equation ,
7. of $x^2 - 3x + 1 = 0$ then

1) $p = 1, q = 5$ 2) $p = 1, q = -5$

132 If α, β are the roots of $x^2 + 2x - 1 = 0$, then t

8. 1) $x^2 - 6x + 1 = 0$ 2) $x^2 + 6x - 1 = 0$

132 If α, β are the roots of $x^2 + 6x + 9 = 0$, t

9. 1) $2x^2 + 3x + 18 = 0$ 2) $x^2 + 6x - 9 = 0$

133 The equation whose roots are the recipr

0. 1) $2 - 5x - 3x^2 = 0$ 2) $2x^2 - 5x - 3 = 0$

133 If α, β are the roots of $x^2 + x + 1 = 0$, then t

1. 1) $2x^2 + x + 1 = 0$ 2) $2x^2 - x + 1 = 0$

133 If ω is an imaginary cube root of unity then

2. 1) $x^2 + 5x + 7 = 0$ 2) $x^2 + 5x - 7 = 0$

133 The equation whose roots are 'K' times the

3. 1) $ax^2 + Kbx + Kc = 0$ 2) $ax^2 + K^2bx + K^2c = 0$

133 The equation whose roots are half of roots

4. 1) $8x^2 + 10x + 1 = 0$ 2) $4x^2 + 5x + 2 = 0$

133 The equation whose roots are double the roots

5. 1) $2x^2 + 12x + 6 = 0$ 2) $4x^2 + 12x + 3 = 0$

133 If α, β be the roots of the equation $ax^2 + bx + c = 0$

6. $\lambda \neq 0$, are

1) $\lambda\alpha, \lambda\beta$ 2) $\frac{\alpha}{\lambda}, \frac{\beta}{\lambda}$

133 If α, β are the roots of $x^2 + 2x + 5 = 0$, the

7. 1) $5x^2 - 9x + 4 = 0$ 2) $5x^2 - 8x + 4 = 0$

133 If α, β are the roots of $ax^2 + bx + c = 0$ then

8. 1) $c(1-\alpha)^2 + b(1-\alpha) + a = 0$
3) $a(x-1)^2 + b(1-x) + c = 0$

133 If α, β are the roots of the equation $ax^2 + bx + c = 0$

9. 1) $a+2b$ 2) $a+b+c$

134 If α, β are the roots of $ax^2 + bx + c = 0$ then the

10. 1) $acx^2 + bx + 1 = 0$ 2) $acx^2 + bx - 1 = 0$

134 Number of quadratic equations which are

1. 1) 1 2) 4

Ratio of the roots:

134 The value of a for which one root of by

2. large as the other is

1) $-\frac{2}{3}$ 2) $\frac{1}{3}$

134 If one root of $x^2 + kx + 12 = 0$ may be triple

3. 1) ± 8 2) 3

134 If the roots of $px^2 - 14x + 8 = 0$ is 6 times the

4. 1) 1 2) 2

134 If r is the ratio of the roots of $ax^2 + bx + c = 0$

5. 1) $\frac{b}{ac}$ 2) $\frac{2b}{a}$

- 134 If the ratio of the roots of $x^2 + bx + c = 0$ is $\frac{p}{q}$, then $p^2c - b^2q =$
6. 1) 0 2) 1

Roots with powers :

- 134 If one root of $x^2 + px + q = 0$ may be the square of the other, then $4pq$ is equal to
7. 1) $4pq$ 2) $3pq$

- 134 If one root of $x^2 - x - k = 0$ may be the square of the other, then k is equal to
8. 1) $2 \pm \sqrt{3}$ 2) $3 \pm \sqrt{2}$

- 134 If one root of the equation $x^2 + px + q = 0$ is the square of the other, then $q(3p+1)$ is equal to
9. 1) $q(3p+1)$ 2) $-q(3p+1)$

- 135 If α, β are the roots of the equation $x^2 + Px + Q = 0$, then $\alpha^2 + \beta^2$ is equal to
10. 1) $4, -2$ 2) $4, 2$

- 135 If one root of $x^2 + px + 1 = 0$ is the cube of the other, then p is equal to
1. 1) 0 2) 1

- 135 If one root is the n^{th} power of the other root, then a^n is equal to
2. 1) ab 2) a^n

Conjugate Roots :

- 135 If $3 + i$ is a root of $x^2 + ax + b = 0$, then $a =$
3. 1) 3 2) -3

- 135 If $3+4i$ is a root of the equation $ax^2 + bx + c = 0$, then a is equal to
4. 1) 0 2) $2a$

- 135 If one root of $x^2 - (3 + 2i)x + (1 + 3i) = 0$ is $2-i$, then the other root is
5. 1) $2-i$ 2) $1-i$

- 135 If one root of the equation $ix^2 - 2(i+1)x + (2-i) = 0$ is $2+i$, then the other root is
6. 1) $2+i$ 2) $1-2i$

- 135 If $x=1+i$ is a root of the equation $x^2 - ix - 1 = 0$, then the other root is
7. 1) 1 2) -1

- 135 If $x^2 + px + q = 0$ has roots $2i + 3, 2i - 3$ then $p^2 + q^2$ is equal to
8. 1) 36 2) 32

- 135 The quadratic equation with rational coefficients having roots $2i + 3, 2i - 3$ is
9. 1) $x^2 + 2x + 1 = 0$ 2) $x^2 - 6x - 7 = 0$

- 136 The quadratic equation whose one root is $2 + i$ and the other root is $2 - i$ is
10. 1) $x^2 + 22x + 4 = 0$ 2) $x^2 + 22x - 4 = 0$

136 If α, β are the roots of the equation $x^2 - 4ax + 2b = 0$

1.

$\alpha + \beta + \sqrt{\alpha^2 + \beta^2}$ is

1) $x^2 - 4ax + 2b = 0$ 2) $x^2 + 4ax - 2b = 0$

136 The quadratic equation whose one root is -

2. 1) $x^2 + 6x + 13 = 0$ 2) $x^2 - 6x + 13 = 0$

136 The quadratic equation whose one root is -

3. 1) $2x^2 - 3x + 2 = 0$ 2) $2x^2 + 3x - 2 = 0$

Equal Roots :

136 If the roots of $3x^2 + 4x + a = 0$, are equal, then

4.

1) 4 2) 3

136 If the roots of $x^2 - 2(7 + 3m)x + 55m + 45 = 0$ are equal, then

5.

1) $1, \frac{4}{9}$ 2) $-1, \frac{-4}{9}$

136 If $kx^2 + 1 = 3x + kx - 6x^2$ has real and equal roots, then

6.

1) -3, 5 2) 3, 5

136 If $p(q-r)x^2 + q(r-p)x + r(p-q) = 0$ has equal roots, then

7.

1) λ 2) $\frac{1}{p} - \frac{1}{r}$

136 If the roots of $(a^2 + b^2)x^2 - 2b(a + c)x + (b^2 + c^2) = 0$ are equal, then

8.

1) A.P. 2) H.P.

136 If one root of the equation $x^2 + px + 12 = 0$ is 3, then the value of 'p' is

9.

1) 3 2) 12

137 The roots of the equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are

10.

1) $a > b > c$ 2) $a = b = c$

137 If the quadratic expression $x^2 - (a-1)x + \frac{a^2}{4}$ is a perfect square, then

1.

1) 0, 6 2) 2, 4

Nature of the roots :

137 If $a = 3$, then the roots of the equation

2.

$x^2 - 2ax + 8a - 15 = 0$ are

1) rational and equal

3) irrational

137 The roots of the equation $x^2 - 2\sqrt{3}x + 3 = 0$ are

3.

1) rational and equal

3) irrational

137 The roots of $x^2 + a^2 = 8x + 6a$ are real, then

4. 1) $[2, 9]$ 2) $[-2, 8]$

137 If the equation $x^2 - x + m^2 = 0$ has no real roots, then

5. 1) $m = 0$ 2) $m < \frac{-1}{2}$

137 If $0 < a < b < c$ and the roots α, β are imaginary, then

6. 1) $|\alpha| = |\beta|$ 2) $|\alpha| < 1$

137 The roots of the equation $2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$ are

7. 1) real and equal 2) real and unequal

137 For $a, b, c \in \mathbb{Q}$ and $b + c \neq a$, the roots of the equation

8. 1) Rational and unequal 2) rational and equal

137 For $a, b, c \in \mathbb{R}$ and a, b, c are different; then the roots of the equation

9. 1) real and equal 2) real and unequal

138 If a, b, c are real then $(b-x)^2 - 4(a-x)(c-x) = 0$ has

10. 1) real and equal 2) real and distinct

138 If a, b, c are all positive and in H.P., then the roots of the equation

11. 1) real 2) imaginary

138 If the roots of $a^2x^2 + 2bx + c^2 = 0$ are imaginary, then

12. 1) complex numbers 2) real and unequal

138 The roots of $(x-a)(x-a-1) + (x-a-1)(x-a-2) + (x-a-2)(x-a-3) = 0$ are

13. 1) equal 2) imaginary

138 Let $f(x) = x^2 + ax + b$, where $a, b \in \mathbb{R}$. If $f(x) = 0$ has two distinct roots, then

14. $f(x) + f'(x) + f''(x) = 0$ are :
1) real and distinct 2) imaginary

138 The roots of $ax^2 + 2bx + c = 0$ and $bx^2 + 2cx + a = 0$ are

15. 1) $a = b, c = 0$ 2) $ac = b^2$

138 The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$ has roots in the

16. interval
1) $(0, 2\pi)$ 2) $(-\pi, 0)$

138 If $a \in \mathbb{Z}$ and the equation $(x-a)(x-10) + 1 = 0$ has integer roots, then

17. 1) 10, 8 2) 12, 10

138 If the roots of $\frac{x^2 - px + q}{ax - c} = \frac{K + i}{K - i}$ are numerical, then

18. 1) c 2) $\frac{1}{c}$

138 9. If $\frac{1}{x} + \frac{1}{x+a} = \frac{1}{m} + \frac{1}{m+a}$ has roots equal to

- 1) $a = 2m$ 2) $a^2 = 2m^2$

139 0. If the roots of $\frac{1}{x+k+1} + \frac{2}{x+k+2} = 1$ are equal to

- 1) -1 2) 0

139 1. If the roots of $\frac{a}{x+a+K} - \frac{b}{x+b+K} = 1$ are equal to

- 1) -b 2) 0

139 2. If the roots of $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ are equal to

- 1) $a^2 + b^2$ 2) $-(a^2 + b^2)$

139 3. If the roots of $ax^2 + bx + c = 0$ are equal in

- 1) $a < 0, c < 0$ 2) $a < 0, c > 0, b > 0$

139 4. If the roots of $\frac{a}{x-a} + \frac{b}{x-b} = k$ are equal to

- 1) $k = 0$ or $a + b = 1$ 2) $k = -1$ or $a + b = 1$

139 5. If the roots of $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ are equal to

- 1) $c = a + b$ 2) $c = 2(a + b)$

139 6. The least integral value of 'a' for which the

- 1) 3 2) 4

139 7. The value of 'a' for which the equation $3x^2 + ax + 1 = 0$ lies in

- 1) (1, 2) 2) $(-\infty, 1)$

139 8. If the roots of $ax^2 + bx + c = 0$ are both negative

- 1) $a < 0, c < 0$ 2) $a < 0, c > 0$

139 9. If the roots of $ax^2 + bx + c = 0$ are both positive

- 1) $a < 0; c < 0$ 2) $a < 0; c > 0$

140 0. The value of 'a' for which the quadratic equation has roots of opposite signs, lie in the interval

- 1) $1 < a < 5$ 2) $0 < a < 4$

Solving the equations (Mod, A.P, G.P, H.P) :

140 1. Two students while solving a quadratic equation got the roots 3 and 2. The other copied the correct roots are

- 1) 3, -2 2) -3, 2

140 In solving the equation $x^2 + ax + b = 0$ wrong and 2. If Again a wrong value of b was taken roots of the original equation are

- 1) $-3, 4$ 2) $3, 4$

140 If A.M of the roots of $x^2 - 2ax + b^2 = 0$ is c

3. 1) A.M 2) G.M

140 The harmonic mean of the roots of the equation

4. 1) -2 2) -4

140 If $(x + 2a)(x + 3b) = c$ has roots α, β , then

5. 1) a, b 2) $-a, -b$

140 The solution of $4^{x-1} - 3 \cdot 2^{x-1} + 2 = 0$ is

6. 1) $\{1, 2\}$ 2) $\{1, -1\}$

140 If m is a positive integer then the solutions

7. 1) $1, 2^m$ 2) $1, 3^m$

140 8. If $20^{(3-2x^2)} = (40\sqrt{5})^{(3x^2-2)}$, then $x =$

- 1) $\pm \sqrt{\frac{13}{12}}$ 2) $\pm \sqrt{\frac{12}{13}}$

140 9. $\{x \in \mathbb{R} : |x - 2| = x^2\} =$

- 1) $\{-1, 2\}$ 2) $\{1, 2\}$

141 0. The equation $|2x - x^2 - 3| = 1$ has

- 1) only real solution
3) 4 real solutions

141 1. For the equation $|x^2| + |x| - 6 = 0$, the roots

- 1) one and only one real number
3) real with sum zero

141 If $|x^2| + |x| + 12 = 0$, then $x =$

2. 1) 3 or -4 2) -3 or -4

141 3. The roots of the equation $|x^2 - x - 6| = x + 2$:

- 1) $-2, 2, 4$ 2) $-2, 3, -4$

141 4. Solutions of the equation $x|x+1| + 1 = 0$ are

- 1) $\frac{1}{2}(1 \pm \sqrt{5})$ 2) $\frac{-1}{2}(-1 \pm \sqrt{5})$

141
5. $\sqrt{\frac{x}{x-3}} + \sqrt{\frac{x-3}{x}} = \frac{5}{2}, x \neq 0, x \neq 3$ then $x =$

- 1) -1 2) $4, \frac{1}{4}$

141
6. $\sqrt{\frac{3x}{x+1}} + \sqrt{\frac{x+1}{3x}} = 2, x \neq 0, x \neq -1$ then the s

- 1) $\{1\}$ 2) $\left\{\frac{1}{2}\right\}$

141
7. If x_1, x_3 are the roots of $ax^2 - 6x + 1 = 0$ and in H.P. then $(a, b) =$

- 1) $(-5, 10)$ 2) $(3, 8)$

141
8. x_1, x_2 are the roots of the equation $x^2 - 3x + x^2 - 12x + B = 0$. If x_1, x_2, x_3, x_4 are in increa

- 1) $A = 2, B = 30$ 2) $A = 2, B = 32$

141
9. If $x_1 + x_2 = 3, x_3 + x_4 = 12$ and x_1, x_2, x_3, x_4 roots is

- 1) $x^2 - 3x + 2 = 0$ 2) $x^2 - 6x + 8 = 0$

142
0. If the sum of the roots of the quadratic equation of their reciprocals then $\frac{a}{c}, \frac{b}{a}$ and $\frac{c}{b}$ are in

- 1) A.P. 2) G.P.

Number of roots :

142
1. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has

- 1) no root 2) one roots

142
2. The number of real roots of $\frac{2x-3}{x-1} + 1 = \frac{6x-2}{x}$

- 1) 0 2) 1

142
3. If $a = 0$, then the equation $\frac{x-a-1}{x-a} = (a +$

- 1) one root 2) two roots

142
4. The number of real solutions of the equation

- 1) 2 2) 4

142
5. The number of solutions of the equation

- 1) 1 2) 2

142
6. The number of real roots of $3^{2x^2-7x+7} = 9$

- 1) 0 2) 2

- 142 The number of real roots of $\left(x + \frac{1}{x}\right)^3 + \left(x + \frac{1}{x}\right) = 0$ is
7. 1) 0 2) 2

- 142 The number of solutions of the system of equations $\frac{1}{\sqrt{2}} < a < 1$ is
8. 1) ∞

- 142 The number of real solutions of the equation $\sin x = x$ is
9. 1) 0 2) 1

- 143 If $e^{\sin x} - e^{-\sin x} - 4 = 0$ then the equation has
0. 1) one real solution
3) no real solution

- 143 The equation $x^2 + x^{-2} = \sin^2 x$, $0 < x \leq \frac{\pi}{2}$ has
1. 1) no real solution
3) two real solutions

- 143 The number of real solutions of the equation $\sin x = x$ is
2. 1) 1 2) 0

- 143 The number of solutions of the equation $|x| = \cos x$ is
3. 1) one 2) two

Common roots :

- 143 If $x^2 - 5x + 6 = 0$ and $x^2 - 7x + K = 0$ have a common root, then the value of K is
4. 1) 2 or 3 2) 5 or 6

- 143 If the two equations $x^2 - Kx - 21 = 0$ and $x^2 + 5x - 14 = 0$ have a common root, then the value of K is
5. 1) 3 2) 2

- 143 If $(2k-5)x^2 - 4x - 15 = 0$ and $(3K-8)x^2 - 5x - 21 = 0$ have a common root, then the value of K is
6. 1) 1 2) 2

- 143 If the equations $x^2 + ax + b = 0$ and $x^2 + px + q = 0$ have a common root, then the value of $\frac{b+q}{p-a}$ or $\frac{aq-bp}{p-a}$ is
7. 1) $\frac{b+q}{p-a}$ or $\frac{aq-bp}{p-a}$ 2) $\frac{b-q}{p-q}$ or $\frac{a-b}{p-q}$

- 143 If $x^2 + ax + b = 0$, $x^2 + bx + a = 0$ ($a \neq 0$) have a common root, then the value of $\frac{a}{b}$ is
8. 2002) 1) 3 2) 2

- 143 If the equations $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a common root, then the value of $\frac{a}{b}$ is
9. 1) 0 2) 1

- 144 If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root, then the value of $\frac{a}{b}$ is
0. 1) 3 2) 2

144 If the quadratic equation $ax^2 + 2cx + b = 0$

1. $a + 4b + 4c$ is equal to

- 1) -2 2) -1

144 If the equations $x^2 + ax + bc = 0$ and $x^2 + bx +$

2. the equation

- 1) $x^2 + (a+b+c)x + ab = 0$
3) $x^2 - cx + ab = 0$

144 If every pair from among the equations .

3. common root, then the sum of the three co

- 1) abc 2) $2abc$

144 If the equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$

4. equal roots then $2(b+d) =$

- 1) $a+c$ 2) $a-c$

144 If the equation $ax^2 + x + b = 0$ and $x^2 + bx +$

5. equal roots, then $2a^2 + b =$

- 1) 0 2) -1

144 If a, b, c are in G.P, then the equations a

6. $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in

- 1) A.P. 2) G.P.

144 If $x-2$ is a factor of $x^2 + ax + b$ and $x^2 + cx + d$

7.

- 1) 1 2) 2

144 8. The number of values of a for which $(a^2 - 3$

- 1) 0 2) 2

Expressions :

144 The maximum value of $c + 2bx - x^2$ is

9. 1) b^2c 2) $b^2 - c$

145 If $2x - 7 - 5x^2$ has maximum value at $x=a$,

0.

- 1) $-\frac{1}{5}$ 2) $\frac{1}{5}$

145 The minimum value of $2x^2 + x - 1$ is

1.

- 1) $\frac{1}{4}$ 2) $\frac{3}{2}$

145 The greatest value of $\frac{4}{4x^2 + 4x + 9}$ is

2.

- 1) $\frac{4}{9}$ 2) 4

- 145 The smallest value of $x^2 - 3x + 3$ in the interval $(-3, 3)$ is
3. 1) $\frac{3}{4}$ 2) 5

- 145 If a and b are the non-zero roots of the expression $x^2 + ax + b$ is
4. 1) $\frac{4}{9}$ 2) $-\frac{4}{9}$

- 145 The value of ' a ' for which the sum of the roots of the equation $x^2 + ax + 1 = 0$ assumes the least value is
5. 1) 0 2) 1

- 145 If each of the roots of $x^2 + x + a = 0$ exceeds $-\frac{1}{2}$, then the range of a is
6. 1) $a = \frac{1}{2}$ 2) $a < -2$

- 145 Out of the two roots of $x^2 + (1 - 2\lambda)x + (\lambda^2 - 1) = 0$, one root is less than 3, then the limits of λ are
7. 1) $\lambda < 2$ 2) $2 < \lambda < 5$

- 145 Out of the two roots of $x^2 - 2\lambda x + \lambda^2 - 1 = 0$, one root is greater than 3, then the limits of λ are
8. 1) $3 < \lambda < 5$ 2) $\lambda < 3$

- 145 If both roots of the quadratic equation $x^2 - 2x + 2 = 0$ lie in the interval $(0, 6)$, then the range of k is
9. 1) $(5, 6]$ 2) $(6, \infty)$

- 146 The set of all values of ' K ' for which both the roots of the equation $x^2 - 2Kx + K^2 - 1 = 0$ are less than '2' is given by
0. 1) $(2, \infty)$ 2) $\left(\frac{4}{5}, 2\right)$

- 146 If the roots of $x^2 - 2kx + k^2 - 1 = 0$ lie between -3 and 3, then the range of k is
1. 1) $(-3, 1)$ 2) $(-3, 3)$

- 146 If a, b, c are real if $ax^2 + bx + c = 0$ has two real roots, then the range of $\frac{c}{a}$ is
2. 1) $1 + \frac{c}{a} + \left|\frac{b}{a}\right| < 0$ 2) $1 - \frac{c}{a} + \left|\frac{b}{a}\right| < 0$

- 146 For the real values of x , maximum value of $\frac{x^2 - x + 1}{x^2 + x + 1}$ is
3. 1) 1 2) 2

- 146 If x is real, then the range of $\frac{x^2 - x + 1}{x^2 + x + 1}$ is
4. 1) $\left(-\infty, \frac{1}{3}\right)$ 2) $(3, \infty)$

146 If x is real then the range of $\frac{x}{x^2 + 2x + 9}$ is

5.

- 1) $(-\infty, 0] \cup (1, \infty)$ 2) $\left[\frac{1}{2}, 2\right]$

146 If x is real then the maximum value of $\frac{x}{x^2 + 2x + 9}$

6.

- 1) 2 2) 4

146 If x is real the maximum value of $\frac{3x^2 + 9}{3x^2 + 4}$

7.

- 1) $\frac{17}{7}$ 2) $\frac{1}{4}$

Inequalities :

146 The least integral value of x for which 33

8.

- 1) 1 2) 2

146 The greatest positive integral value of x

9.

- 1) 9 2) 10

147 The greatest negative integer satis

10.

- 1) -2 2) -3

147 The range of values of x which satisfy x^2

11.

- 1) (1, 2) 2) (2, 5)

147 If $x^2 - 3x + 2 > 0$, $x^2 - 3x - 4 < 0$, then x

12.

- 1) (1, 2) 2) (-1, 4)

147 The set of all solutions of the inequation

13.

- 1) $R - (-\infty, -5)$ 2) $R - (5, \infty)$

147 The set of values of x for which the ineq

14.

is

2007)

- 1) (-2, 5) 2) (2, 8)

147 If the graph of $y = ax^2 + bx + c$ lies comp

15.

- 1) $b^2 - 4ac > 0, a > 0$ 2) $b^2 - 4ac > 0, a < 0$

147 The set of values of x which satisfy $5x + 1$

16.

- 1) (2, 3) 2) $(-\infty, 1) \cup (2, 3)$

147 The values of x for which $\frac{x-1}{3x+4} < \frac{x-1}{3x-4}$

17.

- 1) $\left(-\infty, \frac{5}{4}\right)$ 2) $\left(-\frac{4}{3}, \frac{2}{3}\right)$

147 The solution set contained in R of the i

8. 1) $(1, 3)$ 2) $(0, 1)$

147 The set of values of 'x' satisfying the eq

9. 1) ϕ 2) $\left[\frac{5}{2}, \infty\right)$

148 If $\sqrt{x+5} < 1-x$, then

0. 1) $5 \leq x < -1$ 2) $-5 \leq x \leq 1$

148 If $\sqrt{2x-4} \leq 2-x$, then the range of x is

1. 1) $(-\infty, 2)$ 2) $(2, \infty)$

148 If $\sqrt{9x^2 + 6x + 1} < (2-x)$ then $x \square$

2. 1) $\left(-\frac{3}{2}, \frac{1}{4}\right)$ 2) $\left(-\frac{3}{2}, \frac{1}{4}\right]$

148 The set of values of x satisfying the in equa

3. 1) ϕ 2) $[1, 2)$

148 If $3x^2 + 4kx + 1 > 0$ for all real values of x,

4. 1) $\left(\frac{-\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$ 2) $\left(\frac{-1}{4}, \frac{1}{4}\right)$

148 The integer k for which the inequality x^2-2

5. 1) 2 2) 3

148 If $ax^2+bx+c=0$ has no real roots and $a+b+c$

6. 1) $4a+2b+c > 0$ 2) $a-b+c = 0$

148 The roots of $ax^2 + bx + c = 0$, where $\overline{a \neq 0}$

7. 1) $4a + c > 2b$ 2) $4a + c < 2b$

148 If the roots of the equation $bx^2 + cx + a = 0$ l

8. $3b^2x^2 + 6bcx + 2c^2$ is

- 1) less than $4ab$ 2) greater than -4

Resolving into factors :

148 If $x^2 + 4xy + 4y^2 + 4x + cy + 3$ can be writt

9. 1) -8 2) -2

149 If $x^2 + 4y^2 - 8x + 12 = 0$ is satisfied by real

0. 1) 2, 6 2) 2, 5

149 If $x^2+4y^2-8x+12=0$ is satisfied by real value

1. 1) 2, 6 2) 2, 5

EXC

- 149 If the roots of $ax^2 + bx + c = 0$ are $-1, -2$.
- 1) $1 : 2 : 3$ 2) $2 : 3 : 1$

- 149 Let α, β be the roots of $ax^2 + bx + c = 0$ and a, b, c are in A.P. then the respective discriminants of these equations are

- 1) $\frac{a^2}{p^2}$ 2) $\frac{a^2}{b^2}$

- 149 If α, β are the roots of $x^2 - p(x+1) - c = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is
- 1) 3 2) 2

- 149 If the roots of $ax^2 + bx + c = 0$ are $\frac{b+1}{k}, \frac{b+1}{k}$, then the value of $\frac{c}{a}$ is
- 1) 0 2) 2

- 149 If α, β are the values of t for which $f(x) = x^2 + tx + 1$ has two distinct real roots, then the range of t is
- 1) 20 2) 0

- 149 If α, β are the roots of $x^2 + x + 1 = 0$, and $\gamma = \alpha^2 + \beta^2$, then the value of γ is
- 1) 27 2) -27

- 149 If α, β are the roots of $x^2 + 2x + 5 = 0$. Then the value of $\alpha^2 + \beta^2$ is
- 1) $x^2 + 2x + 7 = 0$ 2) $x^2 - 7x + 2 = 0$

- 149 If α, β are the roots of $ax^2 + bx + c = 0$ then the value of $\frac{c}{a}$ is
- 1) $cx^2 + bx + a = 0$
2) $cx^2 - bx + a = 0$

- 150 If $\tan A, \tan B$ are the roots of $x^2 - 2x + 2 = 0$, then the value of $\tan(A+B)$ is
- 1) $\frac{4}{5}$ 2) $\frac{1}{2}$

- 150 If α is a root of the equation $4x^2 + 2x - 1 = 0$, then $\alpha^2 + 2\alpha - 1$ is
- 1) a root 2) not a root

- 150 If $a = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$, $\alpha = a + a^2 + a^4$ and $\beta = a^3 + a^5 + a^6$, then the value of $\alpha^2 + \beta^2$ is
- 1) $x^2 + x + 1 = 0$ 2) $x^2 + x + 2 = 0$

- 150 If α, β be the roots of the equation $(x - a)(x - b) = c$, then the value of $(x - c - \alpha)(x - c - \beta) = c$ is
- 1) a and $b + c$ 2) $a + c$ and b

150 If one root of the equation $ax^2 + bx + c = 0$

4. $a_1x^2 + b_1x + c_1 = 0$ then

1) $(aa_1 - cc_1)^2 = (bc_1 - b_1a)(b_1c - a_1b)$

3) $(bc_1 - b_1c)^2 = (ca_1 - a_1c)(ab_1 - a_1b)$

150 If the roots of the equation $(x-b)(x-c) + (x-a)(x-b) = 0$

5. 1) $a + b + c = 0$ 2) $a + bw + cw^2 = 0$

150 If $pr = 2(q + s)$ then among the equations :

6. 1) both have real roots
3) at least one has real roots

150 If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + e$

7. atleast

- 1) two real roots
3) two positive roots

150 If the equation $k(6x^2 + 3) + rx + 2x^2 - 1 = 0$

8. common then the value of $2r - p$ is

- 1) 0 2) 1

150 If $|x|^{x^2-2x} = 1$, then the values of x are

9. 1) $0, \pm 1, 2$ 2) $0, \pm 1$

151 The number of real roots of $\sqrt{x^2 - x - 6} + \sqrt{x^2 - 4x + 3} = 0$

0. 1) 1 2) 2

151 The equation $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0$ w

1. 1) all the roots real
3) three real roots namely $x = a, x = b, x = c$.

151 The real roots of the equations $|x^2 + 4x + 3| = 0$

2. 1) $4, -1 + \sqrt{3}$ 2) $-4, -1 - \sqrt{3}$

151 If $6^x + 6^y = 42, x + y = 3$ then

3. 1) $x = 1, y = 2$ 2) $x = -1, y = 2$

151 The roots of the equation $\log_2(x^2 - 4x + 5) = 0$

4. 1) 4, 5 2) 2, -3

151 If the equations $ax + by = 1, cx^2 + dy^2 = 1$ have

5. 1) 1 2) -1

151 The product of real roots of the equation $|x^2 + 4x + 3| = 0$

6. 1) $\frac{5}{4}$ 2) $\frac{5}{2}$

151
7. The minimum value of $\frac{(2+x)(3+x)}{1+x}$ is obt

1) $\sqrt{2} - 1$

2) $\sqrt{2} + 1$

151 If $a < b < c < d$ then $(x-a)(x-b)(x-c)(x-d)$

8. 1) (a, c)

2) (b, d)

151 If $a < b < c < d$ then the roots of $(x-a)(x-b)(x-c)(x-d)$

9. 1) real and equal

2) real and distinct

152 If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + c^2$ between b and c , is

1) no relation

2) $0 < c < \frac{b}{2}$

152
1. If $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1$ for all $x \in$

1) $(-2, 1)$

2) $\left(-2, \frac{2}{5}\right)$

152 If α and β ($\alpha < \beta$) are the roots of the equ

2. 1) $0 < \alpha < \beta$

2) $\alpha < 0 < \beta < |\alpha|$

152 Given that for all real x , the express
3.

Between which the expression $9\tan^2 x + 6\tan x + 5$

LEVEL -

EXI

- | | | | | |
|--------|--------|--------|--------|--------|
| 1) 1 | 2) 2 | 3) 3 | 4) 3 | 5) 3 |
| 11) 1 | 12) 2 | 13) 2 | 14) 3 | 15) 2 |
| 21) 1 | 22) 2 | 23) 4 | 24) 1 | 25) 3 |
| 31) 1 | 32) 1 | 33) 3 | 34) 1 | 35) 1 |
| 41) 2 | 42) 2 | 43) 2 | 44) 1 | 45) 4 |
| 51) 3 | 52) 1 | 53) 2 | 54) 4 | 55) 3 |
| 61) 4 | 62) 1 | 63) 3 | 64) 2 | 65) 4 |
| 71) 2 | 72) 1 | 73) 2 | 74) 4 | 75) 2 |
| 81) 2 | 82) 3 | 83) 3 | 84) 1 | 85) 4 |
| 91) 2 | 92) 1 | 93) 3 | 94) 2 | 95) 4 |
| 101) 1 | 102) 3 | 103) 3 | 104) 2 | 105) 1 |
| 111) 3 | 112) 1 | 113) 2 | 114) 4 | 115) 2 |
| 121) 3 | 122) 4 | 123) 2 | 124) 2 | 125) 1 |
| 131) 1 | 132) 1 | 133) 4 | 134) 2 | 135) 4 |
| 141) 3 | 142) 2 | 143) 4 | 144) 2 | 145) 3 |
| 151) 3 | 152) 2 | 153) 1 | 154) 3 | 155) 1 |
| 161) 1 | 162) 4 | 163) 1 | 164) 3 | 165) 1 |
| 171) 4 | 172) 4 | 173) 1 | 174) 1 | 175) 3 |
| 181) 2 | 182) 3 | 183) 4 | 184) 2 | 185) 3 |
| 191) 2 | 192) 1 | 193) 3 | 194) 3 | 195) 4 |
| 201) 3 | 202) 4 | 203) 1 | 204) 2 | 205) 4 |
| 211) 2 | 212) 2 | 213) 1 | 214) 3 | 215) 4 |
| 221) 2 | 222) 2 | 223) 4 | 224) 3 | 225) 1 |

EXE

- | | | | | |
|-------|-------|-------|-------|-------|
| 1) 4 | 2) 1 | 3) 3 | 4) 3 | 5) 3 |
| 11) 2 | 12) 3 | 13) 1 | 14) 2 | 15) 3 |
| 21) 2 | 22) 4 | 23) 3 | 24) 1 | 25) 2 |
| 31) 2 | 32) 4 | 33) 4 | 34) 3 | 35) 1 |

152 If α, β are the roots of $x^2 + x + 1 = 0$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is

4. 1) -1 2) 1 3) 2 4) i

152 If α, β are the roots of $x^2 - x + 3 = 0$, then $\alpha^4 + \beta^4 =$

5. 1) 10 2) 7 3) -10 4) 8

152 If $\alpha > \beta$ and α, β are the roots of $x^2 - 3x - 1 = 0$ then $\alpha^2 - \beta^2 =$

6. 1) $\sqrt{13}$ 2) $3\sqrt{13}$ 3) $3\sqrt{5}$ 4) $\sqrt{5}$

152 The values of P for which the difference between the roots of the equation $x^2 + Px + 8 = 0$ is 2 are

7. 1) ± 2 2) ± 4 3) ± 6 4) ± 8

152 If the roots of the equation $x^2 - px + q = 0$ differ by unity, then $p^2 =$

8. 1) $4 + q$ 2) $4q + 1$ 3) $4q - 1$ 4) $4 - q$

152 If $\alpha, \alpha + 4$ are the roots of $x^2 - 8x + K = 0$, then $K =$

9. 1) 9 2) 10 3) 11 4) 12

153 If α, β are the roots of $x^2 + bx + c = 0$ and, $\alpha + h, \beta + h$ are the roots of $x^2 + qx + r = 0$ then $h =$

0. 1) $b + q$ 2) $b - q$ 3) $2(b + q)$ 4) $\frac{1}{2}(b - q)$

153 If α, β are the roots of $x^2 + ax - b = 0$ and γ, δ are the roots of $x^2 + ax + b = 0$ then

1. $(\alpha - \gamma)(\alpha - \delta)(\beta - \delta)(\beta - \gamma) =$
1) $4b^2$ 2) b^2 3) $2b^2$ 4) $3b^2$

153 If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c =$

2. 1) 0 2) 1 3) 2 4) 3

153 If the sum of the squares of the roots of $x^2 + px - 3 = 0$ is 10, then $p =$

3. 1) ± 2 2) ± 3 3) 5 4) -5

153 If the sum of the roots of $x^2 + bx + 1 = 0$, is equal to the sum of their squares, then $b =$

4. 1) $1, 2$ 2) $1, -2$ 3) $-1, 2$ 4) $-1, -2$

153 If 8 and 2 are the roots of $x^2 + ax + \beta = 0$ and 3, 3 are the roots of $x^2 + \alpha x + b = 0$, then the roots of the equation $x^2 + ax + b = 0$ are

5. 1) $1, -1$ 2) $-9, 2$ 3) $-8, -2$ 4) $9, 1$

153 The Quotient obtained when $x^4 + 11x^3 - 44x^2 + 76x + 48$ is divided by $x^2 - 2x + 1$ is

6. $x^2 - 13x + 5$ 2) $x^2 + 13x - 19$ 3) $x^2 - 13x + 19$ 4) $x^2 + 13x + 25$

153 The remainder when $2x^5 - 3x^4 + 5x^3 - 3x^2 + 7x - 9$ is divisible by $x^2 - x - 3$ is :

7. 1) $33x + 4$ 2) $41x + 3$ 3) $41x + 44$ 4) $33x - 4$

153 The value of k so that $x^4 - 3x^3 + 5x^2 - 33x + k$ is divisible by $x^2 - 5x + 6$ is

8. 1) 45 2) 48 3) 51 4) 54

153 If $f(x) = 2x^4 - 13x^2 + ax + b$ is divisible by $x^2 - 3x + 2$, then $(a, b) =$

9. 1) $(-9, -2)$ 2) $(6, 4)$ 3) $(9, 2)$ 4) $(2, 9)$

154 Let $a \neq 0$ and $P(x)$ be a polynomial of degree greater than 2. If $P(x)$ leaves remainder a and $-a$ when divided respectively by $x + a$ and $x - a$ then the remainder when $P(x)$ is divided by $x^2 - a^2$ is

- 1) $2x$ 2) $-2x$ 3) x 4) $-x$

154 The equation whose roots are $1 \pm 2i, 0, 1$ is

1. 1) $x(x^3 + 3x^2 - 7x + 4) = 0$ 2) $x(x^3 - 3x^2 + 7x - 5) = 0$
3) $x(x^3 - 3x^2 - 7x + 4) = 0$ 4) $x(x^3 - 3x^2 + 7x - 4) = 0$

154 The equation whose roots are $\sqrt{2}, -\sqrt{2}, 3i, -3i$ is

2. 1) $x^4 + 7x^2 - 18 = 0$ 2) $x^4 - 7x^2 + 18 = 0$ 3) $x^4 + 7x^2 + 18 = 0$ 4) $x^4 - 7x^2 - 18 = 0$

154 The polynomial equation of the lowest degree having roots $1, \sqrt{3}i$ is

3. 1) $x^3 + x^2 + 3x + 3 = 0$ 2) $x^3 - x^2 + 3x - 3 = 0$ 3) $x^3 + x^2 - 3x - 3 = 0$ 4) $x^3 - x^2 - 3x + 3 = 0$

154 The biquadratic equation two of whose roots are $1 + i, 1 - \sqrt{2}$ is

4. 1) $x^4 - 4x^3 + 5x^2 - 2x - 2 = 0$ 2) $x^4 - 4x^3 - 5x^2 + 2x + 2 = 0$
3) $x^4 + 4x^3 - 5x^2 + 2x - 2 = 0$ 4) $x^4 + 4x^3 - 5x^2 - 2x + 2 = 0$

154 The equation of lowest degree with rational coefficients having a root $\sqrt{3} + \sqrt{2}$ is

5. 1) $x^4 + 10x^2 - 1 = 0$ 2) $x^4 - 10x^2 + 1 = 0$ 3) $x^4 + 10x^2 + 1 = 0$ 4) $x^4 - 10x^2 - 1 = 0$

154 If $\alpha, \beta, 1$ are roots of $x^3 - 2x^2 - 5x + 6 = 0$ ($\alpha > 1$) then $3\alpha + \beta =$

6. 1) 7 2) 5 3) 14 4) 10

154 If product of two roots of $4x^3 + 16x^2 + kx - 36 = 0$ is -6 then $k =$

7. 1) 9 2) -9 3) 8 4) -8

154 The condition that the product of two of the roots $x^3 + px^2 + qx + r = 0$ is -1 , is

8. 1) $r^2 + pr + q + 1 = 0$ 2) $q^2 + pq + q + 1 = 0$ 3) $p^2 + pq + p + 1 = 0$
4) $r^2 - pr - q + 1 = 0$

154 If the sum of the two roots of $x^3 + px^2 + qx + r = 0$ is zero then $pq =$

9. 1) $-r$ 2) r 3) $2r$ 4) $-2r$

155 If the sum of two of the roots of $x^3 + ax + b = 0$ is zero then $b =$

0. 1) a 2) 1 3) -1 4) 0

155 The condition that one root of $x^3 + px^2 + qx + r = 0$ is sum of the other two roots, is

1. 1) $q^3 = 4(pq - 2r)$ 2) $p^3 = 4(pq - 2r)$ 3) $r^3 = 4(pr - 2q)$ 4) $p^3 = 4(pq - r)$

155 1)

2.

11

21

If one root of $x^3 + 3x^2 + 5x + k = 0$ is sum of the other two roots then $k =$

1) $\frac{11}{4}$

2) $\frac{17}{6}$

3) $\frac{25}{9}$

4) $\frac{33}{8}$

If one root of $x^3 - 12x^2 + kx - 18 = 0$ is thrice the sum of remaining two roots then $k =$

1) 29

2) -29

3) 19

4) 15

If one of the roots of $18x^3 + 81x^2 + 121x + 60 = 0$ is equal to half the sum of the other two then one of its roots is

1) $-\frac{3}{2}$

2) $\frac{3}{2}$

3) $\frac{2}{3}$

4) $-\frac{2}{3}$

If one root of $24x^3 - 14x^2 - 63x + 45 = 0$ is double the other then the roots are

1) $-1, \frac{1}{2}, 2$

2) $2, 2, -1$

3) $\frac{3}{4}, \frac{3}{2}, -\frac{5}{3}$

4) $-\frac{3}{2}, -\frac{3}{4}, -\frac{1}{3}$

If two of the roots of $2x^3 - 3x^2 - 3x + 2 = 0$ are differ by 3 then roots are

1) $-1, \frac{1}{3}, 2$

2) $-\frac{3}{2}, -\frac{4}{3}, -\frac{5}{3}$

3) $-1, \frac{1}{2}, 2$

4) $2, 2, -1$

The difference between two roots of the equation $x^3 - 13x^2 + 15x + 189 = 0$ is 2. Then the roots of the equation are

1) $-3, 7, 9$

2) $-3, -1, -9$

3) $3, 5, 7$

4) $-3, -1, 9$

If two roots of $x^3 - 9x^2 + 14x + 24 = 0$ are in the ratio 3 : 2 then the roots are

1) $6, 4, -1$

2) $3, 2, 4$

3) $\frac{1}{2}, \frac{1}{3}, \frac{49}{6}$

4) $6, 4, 2$

If the sum of two of the roots of $x^4 - 2x^3 - 3x^2 + 10x - 10 = 0$ is zero then the roots are

1) $\pm\sqrt{5}, 1 \pm i$

2) $\pm\sqrt{5}, 1, -1$

3) $\frac{1}{2}, -\frac{1}{5}, \pm 1$

4) $\sqrt{2}, \sqrt{5}, \pm 2$

If $x^4 + 4x^3 - 2x^2 - 12x + 9 = 0$ has a pair of equal roots then the roots are

1) $-1, -1, 2, 2$

2) $1, 1, -2, -2$

3) $-1, -1, 3, 3$

4) $1, 1, -3, -3$

KEY

- 2) 2 3) 4 4) 3 5) 4 6) 2 7) 1 8) 2 9) 1 10) 2
 11) 1 12) 2 13) 1 14) 2 15) 4 16) 2 17) 4 18) 1 19) 1 20) 3
 21) 3 22) 1 23) 1 24) 1 25) 4

155
3.

The number of integers not satisfying $\lceil x-1 \rceil + \lceil 7-x \rceil - 6 \neq 0$ is ($\lceil \cdot \rceil$ denotes G.I.F)

A) 7

B) 6

C) 8

D) 2

155 4.	Let $\{x_1, x_2, \dots, x_k\}$ be the set of positive integers 'n' such that $n^2 + 256$ is a perfect square. Then A) $\sum_{i=1}^k x_i = 135$ B) $\sum_{i=1}^k x_i = 268$ C) $\sum_{i=1}^k x_i = 160$ D) $\sum_{i=1}^k x_i = 240$
155 5.	If $\alpha, \beta; \beta, \gamma; \gamma, \alpha$ are the roots of $a_i x^2 + b_i x + c_i = 0$ for $i = 1, 2, 3$ ($\alpha, \beta, \gamma > 0$) and If $\sum \alpha + \sum \alpha\beta + \alpha\beta\gamma = \prod_{i=1}^3 \left(\frac{a_i - b_i + c_i}{a_i} \right)^{1/2} + k$, then $k =$ ____ A) 2 B) -2 C) 1 D) -1
155 6.	$a, b, c > 0$ are in G.P and $(a + di)x^2 + 2(b + ei)x + (c + if) = 0$ has one real root then $\frac{a}{b}, \frac{b}{c}, \frac{c}{f}$ are in ____ ($a, b, c, d, e, f \in R, i = \sqrt{-1}$) A) A.P B) G.P C) H.P D) A.G.P
155 7.	Let α, β ($0 < \alpha < \beta$) be the roots of $ax^2 + bx + c = 0$ and γ, δ ($\gamma > \delta$) be the roots of $a^3 x^2 + abcx + c^3 = 0$ then $\frac{\gamma^2}{\delta}$ is independent of A) α, but not β B) β, but not α C) Both α and β D) neither α nor β
155 8.	The degree of the equation whose roots are integer values of 'x' for which $\log_{[x^2]}(4 - x) + \log_3 \{\sqrt{x}\}$ is real is ____ ([.] denotes G.I.F, { } denotes fraction part). A) 4 B) 1 C) 2 D) 3
155 9.	Let $f(x) = ax^2 + bx + c, a < 0, c > 0$ and $f(\alpha) = 0, \alpha > 0$, Let $A = \sum_{i=1}^n f(x_{i-1}), B = \sum_{i=1}^n f(x_i), C = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right)$ where $0 \leq x_{i-1} < x_i < \alpha$ then A) $A > B$ B) $B > A$ C) $A + B < 2C$ D) $A + B > 2C$
156 0.	The values of x satisfying $\frac{2\{x\} - 1}{x - 2\{x\}} \geq 0$ given by all the values belongs to ____ ($\{x\}$ denotes fraction part) A) $\left[n, n + \frac{1}{2}\right], n \in I^-$ B) $\left[n + \frac{1}{2}, n + 1\right], n \in I^+$ C) $(1, \infty) - I^+$ D) $\left(0, \frac{1}{2}\right]$

156 1.	The exhaustive set of solutions if $\{x-2\} - x(x-2) > 0$ is _____ ({ } denotes fraction part) A) $\left(\frac{3-2\sqrt{3}}{2}, 2\right) - \{0\}$ B) $\left(\frac{3-\sqrt{13}}{2}, 2\right) - \{0\}$ C) $\left(\frac{3-\sqrt{14}}{2}, \sqrt{3}\right) - \{0\}$ D) $\left(\frac{3-2\sqrt{3}}{2}, \sqrt{3}\right) - \{0\}$
156 2.	Let α, β be the roots of $x^2 - 2x - a^2 + 1 = 0$ and γ, δ are the roots of $x^2 - 2(a+1)x + a(a-1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$. Then the set of values of 'a' is a subset of A) $\left(-\frac{3}{4}, \frac{3}{4}\right)$ B) $(-1, 1)$ C) $\left(-\frac{1}{4}, 1\right)$ D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
156 3.	If 1, 1, α are the roots of $px^3 + q(x^2 + x) + r = 0$, then A) $pr < 0$ B) $qr < 0$ C) $pq < 0$ D) $\alpha < 0$
156 4.	If the roots of $x^2 + (a-b)x + 1 - a - b = 0, \forall b \in R$ are real and unequal then the set of values of 'a' contains. A) $(0, \infty) - \{1\}$ B) $(1, \infty)$ C) $[2, \infty)$ D) $(-\infty, 1)$
156 5.	If x, m satisfy $\log_{1/2} x^2 \geq \log_{1/2} (x+2)$ and $49x^2 - 4m^4 \leq 0$, then the exhaustive set of values of x is given by $[a, b] - \{0\}$ and exhaustive set of values of m is given by $R - (c, d)$, then A) $ab + cd = 5$ B) $a + b + c + d = 1$ C) $ab + cd = -9$ D) $2ac - bd = 0$
156 6.	The number of solutions of the equation $x^3 - [x] = 3, ([.]$ denotes G.I.F) is _____ A) 0 B) 1 C) at least 2 D) finite, but at least 3
156 7.	Let $\alpha, \beta; \alpha, \gamma; \alpha, \delta$ are the roots of $ax^2 + 2bx + c = 0, 2bx^2 + cx + a = 0, cx^2 + ax + 2b = 0$ where $a, b, c \in N$ Least possible integer value of $a + b + c$ is A) 4 B) 5 C) 3 D) 7
156 8.	Let $\alpha, \beta; \alpha, \gamma; \alpha, \delta$ are the roots of $ax^2 + 2bx + c = 0, 2bx^2 + cx + a = 0, cx^2 + ax + 2b = 0$ where $a, b, c \in N$ $\sum_{r=0}^{99} \alpha^r =$ _____ A) 0 B) 1 C) 100 D) 51
156 9.	Consider the equation $x^2 + bx + c = k$ and $b^2 - 4c > 0$. The given equation has four real roots if A) $0 < k < \frac{b^2 - 4c}{4}$ B) $k > \frac{b^2 - 4c}{4}$ C) $k > \frac{4c - b^2}{4}$ D) $k < \frac{4c - b^2}{4}$

157
0.

Consider the equation $|x^2 + bx + c| = k$ and $b^2 - 4c > 0$.

The given equation has exactly two real roots if

A) $0 < k < \frac{b^2 - 4c}{4}$ B) $k > \frac{b^2 - 4c}{4}$ C) $k > \frac{4c - b^2}{4}$ D) $k < \frac{4c - b^2}{4}$

37	A	38	A-Del	39	D	40	C-Del	41	A
42	C	43	AC	44	AD	45	B	46	BC
47	BCD	48	BC	49	BCD	50	B	51	B
52	B	53	A	54	B				

41. If α, β are the roots of $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ then $aS_{n+1} + bS_n + cS_{n-1} =$
42. The quadratic equation $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root common. The other roots of the first and second equations are integers in the ratio 4:3. Then the value of $|a - c|$ is
43. Let a, b be arbitrary real numbers. Then the smallest natural number b for which the equation $x^2 + 2(a+b)x + (a-b+8) = 0$ has unequal real roots for all $a \in R$
44. The number of integers 'n' such that the equation $nx^2 + (n+1)x + (n+2) = 0$ has rational roots only, is
45. Sum of non-real roots of $(x^2 + x - 2)(x^2 + x - 3) = 12$ is k then $|k| =$
46. The largest natural number 'a' for which the maximum value of $f(x) = a - 1 + 2x - x^2$ is smaller than the minimum value of $g(x) = x^2 - 2ax + 10 - 2a$ is
47. If $(x^2 + x + 2)^2 + (3 - \lambda)(x^2 + x + 2)(x^2 + x + 1) + (\lambda - 4)(x^2 + x + 1)^2 = 0$ has at least one real root, then complete set of value of ' λ ' contains 'n' integers where $n =$
48. The smallest value of k , for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4 is

157
1. If the equation $x^2 + 2\alpha x + \alpha^2 - 1 = 0$ and $x^2 + 2\beta x + \beta^2 - 1 = 0$ have a common root ($\alpha \neq \beta$) then the value of the expression $2\alpha^2 - 4\alpha\beta - |\alpha - \beta| + |\alpha - \beta|\beta^2$, is

157
2. If α, β are roots of the equation $2x^2 + 6x + b = 0$ (where $b < 0$), Then the least integral value of $\frac{1}{2} \left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} \right)$ is equal to

157
3. The set of all the value of parameter m, for which $x^2 + 2(m-1)x + m + 5$ is positive for $x > 1$, is $(-k, \infty)$, then find 'k' _____

157
4. Let $f(x) = kx^2 + x(3 - 4k) - 12$ and the complete set of values of k for which $f(x) < 0 \forall x \in (-3, 3)$ and $f(-4) > 0$ is $(p, q]$ then the value of $(4p + 3q)$ is equal to

- 157
5. If both roots of the equation $x^2 + 3x + a - 5 = 0$ are negative and distinct, then sum of integral values of a is λ . The value of $\lambda - 10$ is equal to
- 157
6. α and β are roots of quadratic $x^2 + (k - 5)x + 3 = 0$ and $\alpha < 1 < \beta$ and $k \in I$. If largest possible value of k is n , then $\left[\log_{(n^2+2)}(n^2 + n + 1) \right]$ is equal to
([k] denotes greatest integer function less than or equal to k.)

- 157
7. If $\sqrt[3]{x-1} + \sqrt[3]{x-3} + \sqrt[3]{x-5} = 0$, then which of the following statement is/are correct
(A) 2 values of x satisfies the equation
(B) Equation has only one odd solution
(C) Equation has only one prime solution
(D) Equation has no real solution

Ans. (B, C)

11. If both the roots of the equation $(c - 5)x^2 - 2cx + (c - 4) = 0$ are positive, one root is less than 2 and other root is lying between 2 and 3 then possible value of c is/are :
(A) 14
(B) 16
(C) 20
(D) 22

Ans. (A, B, C, D)

- 157
8. Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, then
(A) $P(x) = 0$ has imaginary roots
(B) $P(x) = 0$ has roots of opposite sign
(C) $P(1) = 4$
(D) $P(1) = 6$

Ans. (AC)

- 157
9. If $px^2 + qx + r = 0$ is a quadratic equation having roots α and β where $p, q, r \in \mathbb{R}$ such that $p + q + r < 0$, $p - q + r < 0$ and $r > 0$, then
(A) $pr < 0$
(B) $3[\alpha] + 3[\beta] = -3$
(C) $9p + 3q + r < 0$
(D) $9p - 3q + r > 0$

Ans. (A, B, C)

158
0. The common root between the equations obtained by $r = 1$ and $r = 3$ is

(A) $-\sqrt{\frac{3}{2}}$ or $\sqrt{\frac{3}{2}}$

(B) $-\sqrt{\frac{2}{3}}$ or $\sqrt{\frac{2}{3}}$

(C) $-\sqrt{6}$ or $\sqrt{6}$

(D) 1

Ans. (A)

158
1. The number of the triplets (p_1, p_2, p_3) for which such equations exist is

(A) 2

(B) 3

(C) 6C_2

(D) infinite

Ans. (A)

158
2. If the equation $x^3 + bx^2 + cx + 1 = 0$, ($b < c$), has only one real root α . Then the value of $2 \tan^{-1}(\operatorname{cosec} \alpha) + \tan^{-1}(2 \sin \alpha \sec^2 \alpha)$, is :

(A) $-\pi$

(B) $-\frac{\pi}{2}$

(C) $\frac{\pi}{2}$

(D) π

Ans. (A)

158
3. Sum of all the coefficients of $P(x)$ is

(A) 20

(B) 19

(C) 17

(D) 15

Ans. (B)

158
4. If the range of $P(x)$ is $[m, \infty)$ then the value of 'm' is

(A) -12

(B) -15

(C) -17

(D) -5

Ans. (C)

158
5. The value of $P(45)$ is

(A) 2007

(B) 2008

(C) 2009

(D) 2483

Ans. (D)

158 Matrix match type

6. **Column-I** **Column-II**

(A) The roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are real and in G.P. If the sum of their reciprocals is 10, then δ can be (P) 39

(B) Let $f(x) = (x^2 + 2x + 3)^2 + 2(x^2 + 2x + 3) + 3$, if m is the minimum value of $f(x)$ then value of $3m - 1$ is (Q) 32

(C) If 17 arithmetic means are inserted between $\frac{7}{2}$ and $-\frac{83}{2}$, then the absolute value of 17th AM is (R) -39

(D) In the equation $x^2 - \frac{10}{9}x + c = 0$, one root is the square of the other root. If $c > 0$ is the rational number (m/n) in the simplest form, then the value of $(m + n)$ is (S) 35

Ans. [(A) $\rightarrow$ (Q); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (P); (D) $\rightarrow$ (S)]

158 Matrix match type

7. **Column-I** **Column-II**

(A) If the equation $|x^2 + 4x + 3| + 2m - mx = 0$ has exactly three solutions, then the absolute value of $(m + \sqrt{60})$ is (P) 6

(B) $S = 3 - \frac{1}{2} - \frac{1}{6} - \frac{1}{12} - \frac{1}{20} - \frac{1}{30} - \frac{1}{42} - \frac{1}{56}$, then the value of $8S$ is (Q) 4

(C) $E = \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \frac{1}{63} + \frac{1}{99} + \frac{1}{143}$, then the value of $13E$ is (R) 8

(D) The quadratic equation $x^2 + (2K + 1)x + (K^2 + 2) = 0$ has two real roots such that one root is twice the other root. (S) 17

Ans. [(A) $\rightarrow$ (R); (B) $\rightarrow$ (S); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)]

22. Match the column

Column-I **Column-II**

(A) Let α and β be the solutions of the quadratic equation $x^2 - 1154x + 1 = 0$, then the value of $\sqrt[4]{\alpha} + \sqrt[4]{\beta}$ is equal to. (P) 3

(B) Let a, b and c be distinct non zero real numbers such that $\frac{1-a^3}{a} = \frac{1-b^3}{b} = \frac{1-c^3}{c}$. The value of $(a^3 + b^3 + c^3)$, is (Q) 4

(C) Polynomial $P(x)$ contains only terms of odd degree. When $P(x)$ is divided by $(x - 3)$, the remainder is 6. If $P(x)$ is divided by $(x^2 - 9)$, then the remainder is $g(x)$. Then the value of $g(2)$ is. (R) 5

(D) If $a, b \in \mathbb{R}$ such that $a + b = 1$ and $(1 - 2ab)(a^3 + b^3) = 12$. The value of $(a^2 + b^2)$ is equal to. (S) 6

Ans. [(A) $\rightarrow$ (S); (B) $\rightarrow$ (P); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (P)]

- 23.** Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $Q(x) = x^4 - x^3 - x^2 - 1$. If x_1, x_2, x_3 and x_4 are the roots of equation $Q(x) = 0$. Then the value of $90 \sum P(x_i)$, $i = 1, 2, 3, 4$ is

Ans. (540)

- 24.** If the equation $2x^2 + bx + b = 0$ and $x^2 + 2x + 2b = 0$; $b \neq 0$; $b \in \mathbb{C}$ ($\mathbb{C}$, is set of complex numbers). have a common root then the value of $\left(\left|\frac{b-8}{b}\right|\right)^4$.

Ans. [256]

- 25.** Let a, b, c are the roots of the equation $x^3 - 2x^2 + 3x - 4 = 0$, then the value of $\left(a + \frac{1}{a}\right)\left(b + \frac{1}{b}\right)\left(c + \frac{1}{c}\right)$ is

Ans. (2)

- 26.** Let $f(x) = x^3 - 3x^2 + [a]$. If $a \in [\alpha, \beta]$ be the complete range of values of 'a' for which $f(x) = 0$ has three real and distinct solutions, then $(\beta - \alpha)$ is equal to (where $[.]$ denotes greatest integer function) :

Ans. (3)

- 158** For what value of 'a', the equation $(a^2 - a - 2)x^2 + (a^2 - 4)x + (a^2 - 3a + 2) = 0$, will have more than two solutions ? Does there exist a real value of 'x' for which the above equation will be an identity in 'a' ?

- 158** If α and β are the roots of the equation $2x^2 + 3x + 4 = 0$, then find the values of

- 9.** (i) $\alpha^2 + \beta^2$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

- 159** If α and β are the roots of the equation $ax^2 + bx + c = 0$, then find the equation whose roots are given by

- 0.** (i) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$ (ii) $\alpha^2 + 2, \beta^2 + 2$

- 159** If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3, \beta^2 = 5\beta - 3$, then find the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

1.

- 159** In copying a quadratic equation of the form $x^2 + px + q = 0$, the coefficient of x was wrongly written as -10 in place of -11 and the roots were found to be 4 and 6. Find the roots of the correct equation.

- 159** **3.** (i) Find the value of the expression $2x^3 + 2x^2 - 7x + 72$ when $x = \frac{3+5\sqrt{-1}}{2}$.

- (ii) Find the value of the expression $2x^3 + 2x^2 - 7x + 72$ when $x = \frac{-1+\sqrt{15}}{2}$

- (iii) Solve the following equation $2^{2x} + 2^{x+2} - 32 = 0$

- 159** Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β

- 159** If α, β are roots of $x^2 - px + q = 0$ and $\alpha - 2, \beta + 2$ are roots of $x^2 - px + r = 0$, then prove that $16q + (r + 4 - q)^2 = 4p^2$.

- 159** If one root of the equation $ax^2 + bx + c = 0$ is equal to n^{th} power of the other root, then show that

- 6.** $(ac^n)^{1/(n+1)} + (a^n c)^{1/(n+1)} + b = 0$.

159 If the sum of the roots of quadratic equation $(a + 1)x^2 + (2a + 3)x + (3a + 4) = 0$ is -1 , then find the product of the roots.

159 Find the least prime integral value of '2a' such that the roots α, β of the equation $2x^2 + 6x + a = 0$ satisfy the inequality $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < 2$.

159 If α and β be two real roots of the equation $x^3 + px^2 + qx + r = 0$ ($r \neq 0$) satisfying the relation $\alpha\beta + 1 = 0$, then prove that $r^2 + pr + q + 1 = 0$.

Quadratic Equation



160 If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then find the value of

0. $\left(\alpha - \frac{1}{\beta\gamma}\right)\left(\beta - \frac{1}{\gamma\alpha}\right)\left(\gamma - \frac{1}{\alpha\beta}\right)$.

160 (i) Solve the equation $24x^3 - 14x^2 - 63x + \lambda = 0$, one root being double of another. Hence find the value(s) of λ .

1. (ii) Solve the equation $18x^3 + 81x^2 + \lambda x + 60 = 0$, one root being half the sum of the other two. Hence find the value of λ .

160 If α, β, γ are roots of equation $x^3 - 6x^2 + 10x - 3 = 0$, then find cubic equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$.

160 If α, β and γ are roots of $2x^3 + x^2 - 7 = 0$, then find the value of $\sum_{\alpha, \beta, \gamma} \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$.

160 Find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$ if two of its roots are equal.

160 If $2 + i\sqrt{3}$ is a root of the equation $x^2 + px + q = 0$ (where $p, q \in \mathbb{R}$ and $i^2 = -1$), then find the ordered pair (p, q) .

160 If the roots of the equation $x^2 - 2cx + ab = 0$ are real and unequal, then prove that the roots of $x^2 - 2(a + b)x + a^2 + b^2 + 2c^2 = 0$ will be imaginary.

160 For what values of k the expression $kx^2 + (k + 1)x + 2$ will be a perfect square of a linear polynomial.

160 Show that if roots of equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + c^2 - ab = 0$ are equal, then either $b = 0$ or $a^3 + b^3 + c^3 = 3abc$.

160 If $a, b, c \in \mathbb{R}$, then prove that the roots of the equation $\frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c} = 0$ are always real and cannot have roots if $a = b = c$.

161 If the roots of the equation $\frac{1}{(x+p)} + \frac{1}{(x+q)} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then show that $p + q = 2r$ and that the product of the roots is equal to $(-1/2)(p^2 + q^2)$.

161 (i) If $-2 + i\beta$ is a root of $x^3 + 63x + \lambda = 0$ (where $\beta \in \mathbb{R} - \{0\}$, $\lambda \in \mathbb{R}$ and $i^2 = -1$), then find roots of equation.

1. (ii) If $\frac{-1}{2} + i\beta$ is a root of $2x^3 + bx^2 + 3x + 1 = 0$ (where $b, \beta \in \mathbb{R} - \{0\}$ and $i^2 = -1$), then find the value(s) of b .

161 Solve the equation $x^4 + 4x^3 + 5x^2 + 2x - 2 = 0$, one root being $-1 + \sqrt{-1}$.

161 Draw graph of $y = 12x^3 - 4x^2 - 3x + 1$. Hence find number of positive zeroes.
3.

161 Draw the graph of the following expressions :

4. (i) $y = x^2 + 4x + 3$ (ii) $y = 9x^2 + 6x + 1$ (iii) $y = -2x^2 + x - 1$

Quadratic Equation



161 Find the range of following quadratic expressions :

5. (i) $f(x) = -x^2 + 2x + 3 \quad \forall x \in \mathbb{R}$
 (ii) $f(x) = x^2 - 2x + 3 \quad \forall x \in [0, 3]$
 (iii) $f(x) = x^2 - 4x + 6 \quad \forall x \in (0, 1]$

161 . If x be real, then find the range of the following rational expressions :

6. (i) $y = \frac{x^2 + x + 1}{x^2 + 1}$ (ii) $y = \frac{x^2 - 2x + 9}{x^2 - 2x - 9}$

161 Find the range of values of k , such that $f(x) = \frac{kx^2 + 2(k+1)x + (9k+4)}{x^2 - 8x + 17}$ is always negative.
7.

161 $x^2 + (a-b)x + (1-a-b) = 0$, $a, b \in \mathbb{R}$. Find the condition on 'a' for which

8. (i) Both roots of the equation are real and unequal $\forall b \in \mathbb{R}$.
 (ii) Roots are imaginary $\forall b \in \mathbb{R}$

161 If both roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, then show that $a > 11/9$.

9.

162 Find all the values of 'K' for which one root of the equation $x^2 - (K+1)x + K^2 + K - 8 = 0$, exceeds 2 & the other root is smaller than 2.

162 Find all the real values of 'a', so that the roots of the equation

1. $(a^2 - a + 2)x^2 + 2(a-3)x + 9(a^4 - 16) = 0$ are of opposite sign.

162 Find all the values of 'a', so that exactly one root of the equation $x^2 - 2ax + a^2 - 1 = 0$, lies between the numbers 2 and 4, and no root of the equation is either equal to 2 or equal to 4.

162 If α & β are the two distinct roots of $x^2 + 2(K-3)x + 9 = 0$, then find the values of K such that $\alpha, \beta \in (-6, 1)$.

162 If one of the roots of the equation $ax^2 + bx + c = 0$ be reciprocal of one of the roots of

4. $a_1x^2 + b_1x + c_1 = 0$, then prove that $(a_1 - c_1c_1)^2 = (b_1c_1 - a_1b_1)(b_1c_1 - a_1b_1)$.

162 Find the value of 'a' so that $x^2 - 11x + a = 0$ and $x^2 - 14x + 2a = 0$ have a common root.

5.

162 If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root and a, b, c are non-zero real numbers, then

6. find the value of $\frac{a^3 + b^3 + c^3}{abc}$.

162 If $x^2 + px + q = 0$ and $x^2 + qx + p = 0$, ($p \neq q$) have a common root, show that $1 + p + q = 0$; show that

7. their other roots are the roots of the equation $x^2 + x + pq = 0$.

162 Draw the graphs of following :

8. (i) $y = 2x^3 + 9x^2 - 24x + 15$ (ii) $y = -3x^4 + 4x^3 + 12x^2 - 2$

162 Find values of 'k' if equation $x^3 - 3x^2 + 2 = k$ has

9. (i) 3 real roots (ii) 1 real root

163 The roots of the equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are

0. (A) $\frac{c-a}{b-c}, 1$ (B) $\frac{a-b}{b-c}, 1$ (C) $\frac{b-c}{a-b}, 1$ (D) $\frac{c-a}{a-b}, 1$

163 If α, β are the roots of quadratic equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$,

1. then $(\alpha - \gamma) \cdot (\alpha - \delta)$ is equal to :

- (A) $q + r$ (B) $q - r$ (C) $-(q + r)$ (D) $-(p + q + r)$

163 Two real numbers α & β are such that $\alpha + \beta = 3$, $\alpha - \beta = 4$, then α & β are the roots of the quadratic equation:

- (A) $4x^2 - 12x - 7 = 0$ (B) $4x^2 - 12x + 7 = 0$ (C) $4x^2 - 12x + 25 = 0$ (D) none of these

163 For the equation $3x^2 + px + 3 = 0$, $p > 0$ if one of the roots is square of the other, then p is equal to:

3. (A) $1/3$ (B) 1 (C) 3 (D) $2/3$

163 Consider the following statements :

4. S_1 : If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then value of $b^2 - 4c$ is equal to 1.

S_2 : If α, β are roots of $x^2 - x + 3 = 0$ then value of $\alpha^4 + \beta^4$ is equal 7.

S_3 : If α, β, γ are the roots of $x^3 - 7x^2 + 16x - 12 = 0$ then value of $\alpha^2 + \beta^2 + \gamma^2$ is equal to 17.

State, in order, whether S_1, S_2, S_3 are true or false

- (A) TTT (B) FTF (C) TFT (D) FTT

163 If two roots of the equation $x^3 - px^2 + qx - r = 0$, ($r \neq 0$) are equal in magnitude but opposite in sign, then:

5. (A) $pr = q$ (B) $qr = p$ (C) $pq = r$ (D) None of these

163 If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:

6. (A) zero (B) -1 (C) -7 (D) 1

163 Let α, β, γ be the roots of $(x - a)(x - b)(x - c) = d$, $d \neq 0$, then the roots of the equation

7. $(x - \alpha)(x - \beta)(x - \gamma) + d = 0$ are :

- (A) $a + 1, b + 1, c + 1$ (B) a, b, c (C) $a - 1, b - 1, c - 1$ (D) $\frac{a}{b}, \frac{b}{c}, \frac{c}{a}$

163 If α, β, γ are the roots of the equation $x^3 + ax + b = 0$ then value of $\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2}$ is equal to :

8.

- (A) $\frac{3b}{2a}$ (B) $\frac{-3b}{2a}$ (C) $3b$ (D) $2b$

163 If two of the roots of equation $x^4 - 2x^3 + ax^2 + 8x + b = 0$ are equal in magnitude but opposite in sign,

9. then value of $4a + b$ is equal to :

- (A) 16 (B) 8 (C) -16 (D) -8

Quadratic Equation



164 If one roots of equation $x^2 - \sqrt{3}x + \lambda = 0$, $\lambda \in \mathbb{R}$ is $\sqrt{3} + 2$ then other root is

0. (A) $\sqrt{3} - 2$ (B) -2 (C) $2 - \sqrt{3}$ (D) 2

164 If roots of equation $2x^2 + bx + c = 0$; $b, c \in \mathbb{R}$, are real & distinct then the roots of equation

1. $2cx^2 + (b - 4c)x + 2c - b + 1 = 0$ are

- (A) imaginary (B) equal (C) real and distinct (D) can't say

164 Let one root of the equation $x^2 + \ell x + m = 0$ is square of other root. If $m \in \mathbb{R}$ then

2. (A) $\ell \in \left(-\infty, \frac{1}{4}\right] \cup \{1\}$ (B) $\ell \in (-\infty, 0]$ (C) $\ell \in \left(-\infty, \frac{1}{9}\right]$ (D) $\ell \in \left(\frac{1}{4}, 1\right]$

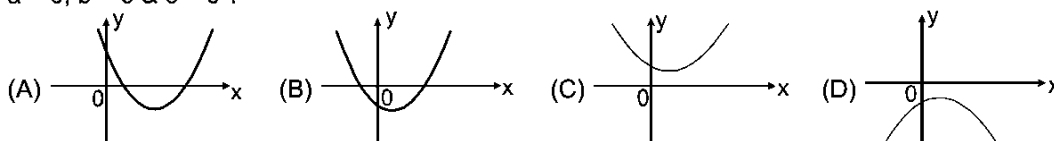
- 164 If a, b, c are integers and $b^2 = 4(ac + 5d^2)$, $d \in \mathbb{N}$, then roots of the quadratic equation $ax^2 + bx + c = 0$ are
3. (A) Irrational (B) Rational & different (C) Complex conjugate (D) Rational & equal

- 164 Let a and b be real numbers such that $4a + 2b + c = 0$ and $ab > 0$. Then the equation $ax^2 + bx + c = 0$ has
4. (A) real roots (B) imaginary roots (C) exactly one root (D) none of these

- 164 Consider the equation $x^2 + 2x - n = 0$, where $n \in \mathbb{N}$ and $n \in [5, 100]$. Total number of different values of 'n' so that the given equation has integral roots, is
5. (A) 4 (B) 6 (C) 8 (D) 3

- 164 If α & β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then
6. (A) $0 < \alpha < \beta$ (B) $\alpha < 0 < \beta^2 < \alpha^2$ (C) $\alpha < \beta < 0$ (D) $\alpha < 0 < \alpha^2 < \beta^2$

- 164 Which of the following graph represents the expression $f(x) = ax^2 + bx + c$ ($a \neq 0$) when $a > 0, b < 0$ & $c < 0$?
- 7.



- 164 The expression $y = ax^2 + bx + c$ has always the same sign as of 'a' if :
8. (A) $4ac < b^2$ (B) $4ac > b^2$ (C) $4ac = b^2$ (D) $ac < b^2$

- 164 The entire graph of the expression $y = x^2 + kx - x + 9$ is strictly above the x-axis if and only if
9. (A) $k < 7$ (B) $-5 < k < 7$ (C) $k > -5$ (D) none of these

- 165 If $a, b \in \mathbb{R}$, $a \neq 0$ and the quadratic equation $ax^2 - bx + 1 = 0$ has imaginary roots then $a + b + 1$ is:
0. (A) positive (B) negative (C) zero (D) depends on the sign of b

- 165 If a and b are the non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is
1. (A) $\frac{3}{2}$ (B) $\frac{9}{4}$ (C) $-\frac{9}{4}$ (D) 1

- 165 If $y = -2x^2 - 6x + 9$, then
2. (A) maximum value of y is -11 and it occurs at $x = 2$
 (B) minimum value of y is -11 and it occurs at $x = 2$
 (C) maximum value of y is 13.5 and it occurs at $x = -1.5$
 (D) minimum value of y is 13.5 and it occurs at $x = -1.5$

- 165 If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ are such that $\min f(x) > \max g(x)$, then the relation between b and c , is
3. (A) no relation (B) $0 < c < b/2$ (C) $c^2 < 2b$ (D) $c^2 > 2b^2$

Quadratic Equation



- 165 If $b > a$, then the equation $(x - a)(x - b) - 1 = 0$, has:

4. (A) both roots in $[a, b]$ (B) both roots in $(-\infty, a)$
 (C) both roots in $[b, \infty)$ (D) one root in $(-\infty, a)$ & other in (b, ∞)

- 165 If α, β are the roots of the quadratic equation $x^2 - 2p(x - 4) - 15 = 0$, then the set of values of 'p' for which one root is less than 1 & the other root is greater than 2 is:
5. (A) $(7/3, \infty)$ (B) $(-\infty, 7/3)$ (C) $x \in \mathbb{R}$ (D) none of these

- 165 If α, β be the roots of $4x^2 - 16x + \lambda = 0$, where $\lambda \in \mathbb{R}$, such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral solutions of λ is
6. (A) 5 (B) 6 (C) 2 (D) 3

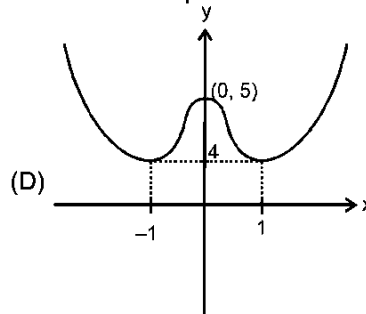
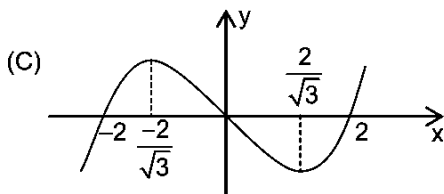
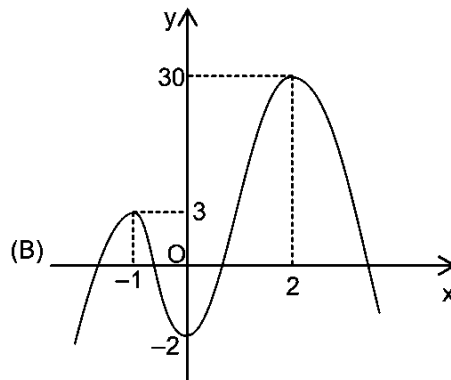
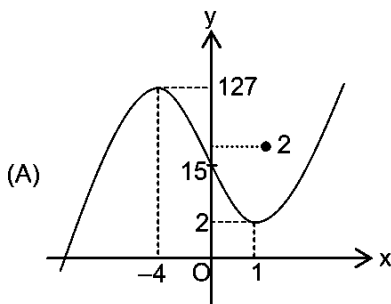
- 165 Set of real values of k if the equation $x^2 - (k-1)x + k^2 = 0$ has atleast one root in $(1, 2)$ is
7. (A) $(2, 4)$ (B) $[-1, 1/3]$ (C) $\{3\}$ (D) ϕ

- 165 If the equations $k(6x^2 + 3) + rx + 2x^2 - 1 = 0$ and $6k(2x^2 + 1) + px + 4x^2 - 2 = 0$ have both roots common, then the value of $(2r - p)$ is
 8. (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) none of these

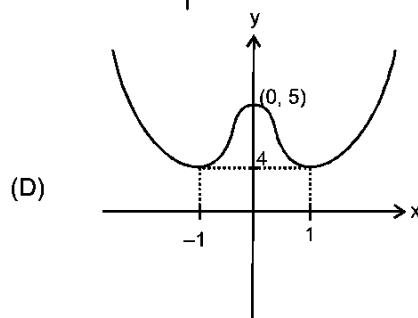
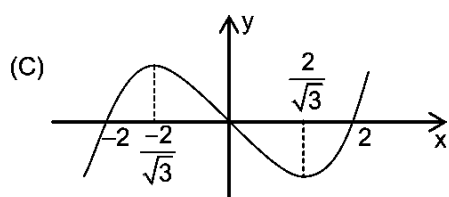
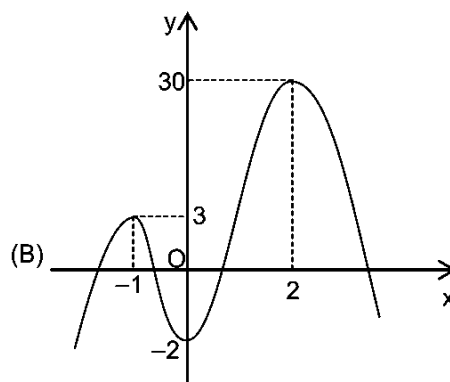
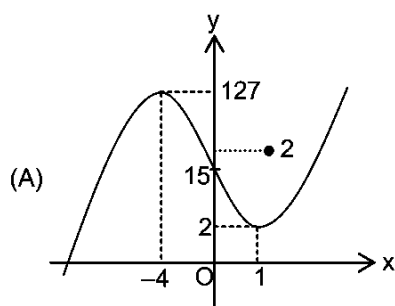
- 165 If $3x^2 - 17x + 10 = 0$ and $x^2 - 5x + \lambda = 0$ has a common root, then sum of all possible real values of λ is
 9. (A) 0 (B) $-\frac{29}{9}$ (C) $\frac{26}{9}$ (D) $\frac{29}{3}$

- 166 If a, b, p, q are non-zero real numbers, then two equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have :
 0. (A) no common root (B) one common root if $2a^2 + b^2 = p^2 + q^2$
 (C) two common roots if $3pq = 2ab$ (D) two common roots if $3qb = 2ap$

- 166 The graphs of $y = \frac{x^3 - 4x}{4}$ is
 1.



- 166 The graphs of $y = x^4 - 2x^2 + 5$ is
2.



PART - III : MATCH THE COLUMN

166 Column - I

3. (A) If $\alpha, \alpha + 4$ are two roots of $x^2 - 8x + k = 0$, then possible value of k is
- (B) If α, β are roots of $x^2 + 2x - 4 = 0$ and $\frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $x^2 + qx + r = 0$ then value of $\frac{-3}{q+r}$ is
- (C) If α, β are roots of $ax^2 + c = 0, ac \neq 0$, then $\alpha^3 + \beta^3$ is equal to
- (D) If roots of $x^2 - kx + 36 = 0$ are Integers then number of values of $k =$

Column - II

- (p) 4
- (q) 0
- (r) 12
- (s) 10

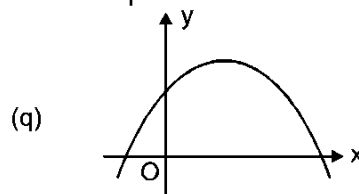
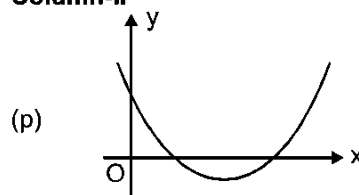
- 166 If graph of the expression $f(x) = ax^2 + bx + c$ ($a \neq 0$) are given in column-II, then Match the items in column-I with in column-II (where $D = b^2 - 4ac$)

Column-I

(A) $\frac{abc}{D} > 0$

(B) $\frac{abc}{D} < 0$

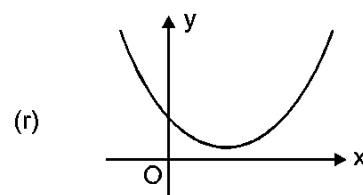
Column-II



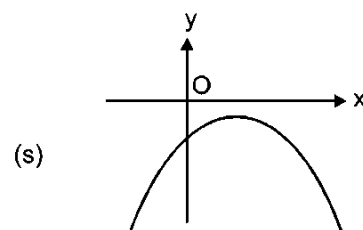
Quadratic Equation



(C) $abc > 0$



(D) $abc < 0$



- 166 Let $y = Q(x) = ax^2 + bx + c$ be a quadratic expression. Match the inequalities in **Column-I** with possible graphs in **Column-II**.

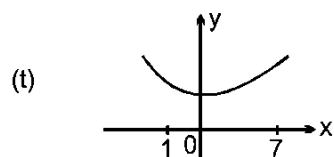
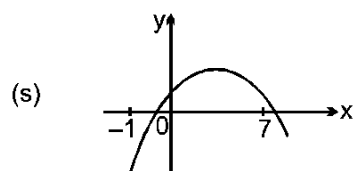
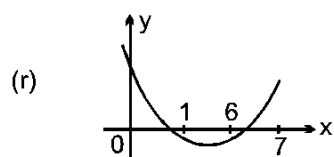
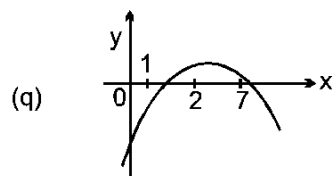
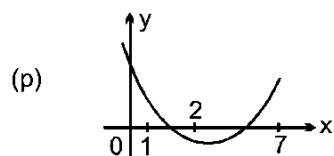
Column-I

(A) $Q(x) > 0, \forall x \in (2, 7)$

(B) $Q(x) > 0, \forall x \in (-\infty, 1)$

(C) $Q(x) < 0, \forall x \in (1, 6)$

(D) $Q(x) < 0, \forall x \in (-\infty, -1)$

Column-II

- 166 Let $a > 0, b > 0$ & $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$

6. (A) are real & negative (B) have negative real parts
(C) are rational numbers (D) have positive real parts

- 166
7. If the roots of the equation $x^2 + 2ax + b = 0$ are real and distinct and they differ by at most $2m$, then b lies in the interval
 (A) $(a^2 - m^2, a^2)$ (B) $[a^2 - m^2, a^2)$ (C) $(a^2, a^2 + m^2)$ (D) none of these
3. The set of possible values of λ for which $x^2 - (\lambda^2 - 5\lambda + 5)x + (2\lambda^2 - 3\lambda - 4) = 0$ has roots, whose sum and product are both less than 1, is
 (A) $\left(-1, \frac{5}{2}\right)$ (B) $(1, 4)$ (C) $\left[1, \frac{5}{2}\right]$ (D) $\left(1, \frac{5}{2}\right)$
4. If $p, q, r, s \in \mathbb{R}$, then equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ has
 (A) 6 real roots (B) atleast two real roots
 (C) 2 real and 4 imaginary roots (D) 4 real and 2 imaginary roots
5. If coefficients of biquadratic equation are all distinct and belong to the set $\{-9, -5, 3, 4, 7\}$, then equation has
 (A) atleast two real roots
 (B) four real roots, two are conjugate surds and other two are also conjugate surds
 (C) four imaginary roots
 (D) None of these
6. Let $p, q, r, s \in \mathbb{R}$, $x^2 + px + q = 0$, $x^2 + rx + s = 0$ such that $2(q + s) = pr$ then
 (A) atleast one of the equation have real roots.
 (B) either both equations have imaginary roots or both equations have real roots.
 (C) one of equations have real roots and other equation have imaginary roots
 (D) atleast one of the equations have imaginary roots.
7. The equation, $\pi^x = -2x^2 + 6x - 9$ has:
 (A) no solution (B) one solution (C) two solutions (D) infinite solutions
8. If $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1$ for all $x \in \mathbb{R}$, then λ belongs to the interval
 (A) $(-2, 1)$ (B) $\left[-2, \frac{2}{5}\right)$ (C) $\left(\frac{2}{5}, 1\right)$ (D) none of these
9. Let conditions C_1 and C_2 be defined as follows : $C_1 : b^2 - 4ac \geq 0$, $C_2 : a, -b, c$ are of same sign. The roots of $ax^2 + bx + c = 0$ are real and positive, if
 (A) both C_1 and C_2 are satisfied (B) only C_2 is satisfied

(C) only C_1 is satisfied

(D) none of these

10. If 'x' is real, then $\frac{x^2 - x + c}{x^2 + x + 2c}$ can take all real values if :

(A) $c \in [0, 6]$ (B) $c \in [-6, 0]$ (C) $c \in (-\infty, -6) \cup (0, \infty)$ (D) $c \in (-6, 0)$

Quadratic Equation



11. If both roots of the quadratic equation $(2 - x)(x + 1) = p$ are distinct & positive, then complete set of values of p is:

(A) $(2, \infty)$ (B) $(2, 9/4)$ (C) $(-\infty, -2)$ (D) $(-\infty, \infty)$

12. If two roots of the equation $(a - 1)(x^2 + x + 1)^2 - (a + 1)(x^4 + x^2 + 1) = 0$ are real and distinct, then 'a' lies in the interval

(A) $(-2, 2)$ (B) $(-\infty, -2) \cup (2, \infty)$ (C) $(2, \infty)$ (D) $(-\infty, -2)$

13. The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 respectively, then the ordered pair (x_1, x_2) is:

(A) $(-5, -7)$ (B) $(1, -1)$ (C) $(-1, 1)$ (D) $(5, 7)$

14. If a, b, c are real and $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in the interval:

(A) $\left[\frac{1}{2}, 2\right]$ (B) $[0, 2]$ (C) $\left[-\frac{1}{2}, 1\right]$ (D) $\left[-1, \frac{1}{2}\right]$

PART - II : NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. Find sum of square of real roots of equation $x(x + 1)(x + 2)(x + 3) = 120$

2. Find product of all real values of x satisfying $(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$

166 If a, b are the roots of $x^2 + px + 1 = 0$ and c, d are the roots of $x^2 + qx + 1 = 0$. Then find the value of $(a - c)(b - c)(a + d)(b + d)/(q^2 - p^2)$.

166 α, β are roots of the equation $\lambda(x^2 - x) + x + 5 = 0$. If λ_1 and λ_2 are the two values of λ for which the

roots α, β are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 4$, then the value of $\left(\frac{\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}}{15}\right)$ is

167 Let one root of equation $(\ell - m)x^2 + \ell x + 1 = 0$ be double of the other. If ℓ be real and $m \leq k$ then find the least value of k.

167 Let α, β be the roots of the equation $x^2 + ax + b = 0$ and γ, δ be the roots of $x^2 - ax + b - 2 = 0$. If $\alpha\beta\gamma\delta = 24$ and $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{11}{5}$, then find the value of a.

167 If $a > b > 0$ and $a^3 + b^3 + 27ab = 729$ then the quadratic equation $ax^2 + bx - 9 = 0$ has roots α, β ($\alpha < \beta$). Find the value of $4\beta - a\alpha$.

- 167 Let α and β be roots of $x^2 - 6(5t^2 - 3t + 7)x - 2 = 0$ with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then find the
 3. minimum value of $\frac{a_{100} - 2a_{98}}{a_{99}}$ (where $t \in \mathbb{R}$)

Quadratic Equation



- 167 If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - Kx^3 + Kx^2 + Lx + M = 0$, where K, L & M are real numbers,
 4. then the minimum value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is $-n$. Find the value of n .

- 167 Consider $y = \frac{2x}{1+x^2}$, where x is real, then the range of expression $y^2 + y - 2$ is $[a, b]$. Find the value of
 5. $(b - a)$.

- 167 If the roots of the equation $3x^3 + Px^2 + Qx - 37 = 0$ are each one more than the roots of the equation
 6. $x^3 - Ax^2 + Bx - C = 0$, where A, B, C, P & Q are constants, then the value of $A + B + C$ is equal to :

- 167 If one root of the equation $t^2 - (12x)t - (f(x) + 64x) = 0$ is twice of other, then find the maximum value of
 7. the function $f(x)$, where $x \in \mathbb{R}$.

- 167 The values of k , for which the equation $x^2 + 2(k - 1)x + k + 5 = 0$ possess atleast one positive root, are
 8. $(-\infty, -b]$. Find value of b .

- 167 Find the least value of 'a' for which atleast one of the roots of the equation $x^2 - (a - 3)x + a = 0$ is
 9. greater than 2.

- 168 If the quadratic equations $3x^2 + ax + 1 = 0$ & $2x^2 + bx + 1 = 0$ have a common root, then the value of the
 0. expression $5ab - 2a^2 - 3b^2$ is

- 168 The equations $3x^2 - 7ax + b = 0$, $x^3 - px^2 + qx = 0$, where $a, b, p, q \in \mathbb{R} - \{0\}$ have one common root &
 1. the second equation has two equal roots. Find value of $\frac{3q + b}{aq}$.

- 168 If $x - y$ and $y - 2x$ are two factors of the expression $x^3 - 3x^2y + \lambda xy^2 + \mu y^3$, then $\lambda^2 + \mu^2$ is
 2. _____

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

- 168 Possible values of 'p' for which the equation $(p^2 - 3p + 2)x^2 - (p^2 - 5p + 4)x + p - p^2 = 0$ does not
 3. possess more than two roots is/are
 (A) 0 (B) 1 (C) 2 (D) 4

- 168 If a, b are non-zero real numbers and α, β the roots of $x^2 + ax + b = 0$, then
 4. (A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$
 (B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$
 (C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$
 (D) $(\alpha - 1), (\beta - 1)$ are the roots of the equation $x^2 + x(a + 2) + 1 + a + b = 0$

- 168 If α, β are the roots of $ax^2 + bx + c = 0$ ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of,
 5. $Ax^2 + Bx + C = 0$ ($A \neq 0$) for some constant δ , then

- (A) $\delta = \frac{1}{2} \left(\frac{B}{A} - \frac{b}{a} \right)$ (B) $\delta = \frac{1}{2} \left(\frac{b}{a} - \frac{B}{A} \right)$
 (C) $\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$ (D) $\frac{b^2 + 4ac}{a^2} = \frac{B^2 + 4AC}{A^2}$

168 If one root of the equation $4x^2 + 2x - 1 = 0$ is ' α ', then

6. (A) α can be equal to $\frac{-1+\sqrt{5}}{4}$ (B) α can be equal to $\frac{1+\sqrt{5}}{4}$
 (C) other root is $4\alpha^3 - 3\alpha$. (D) other root is $4\alpha^3 + 3\alpha$

Quadratic Equation



168 If α, β are roots of $x^2 + 3x + 1 = 0$, then

7. (A) $(7 - \alpha)(7 - \beta) = 0$ (B) $(2 - \alpha)(2 - \beta) = 11$
 (C) $\frac{\alpha^2}{3\alpha + 1} + \frac{\beta^2}{3\beta + 1} = -2$ (D) $\left(\frac{\alpha}{1+\beta}\right)^2 + \left(\frac{\beta}{\alpha+1}\right)^2 = 18$

168 If both roots of $x^2 - 32x + c = 0$ are prime numbers then possible values of c are

8. (A) 60 (B) 87 (C) 247 (D) 231

168 Let $f(x) = x^2 - a(x + 1) - b = 0$, $a, b \in \mathbb{R} - \{0\}$, $a + b \neq 0$. If α and β are roots of equation $f(x) = 0$, then

9. the value of $\frac{1}{\alpha^2 - a\alpha} + \frac{1}{\beta^2 - a\beta} - \frac{2}{a+b}$ is equal to
 (A) 0 (B) $f(a) + a + b$ (C) $f(b) + a + b$ (D) $f\left(\frac{a}{2}\right) + \frac{a^2}{4} + a + b$

169 If $f(x)$ is a polynomial of degree three with leading coefficient 1 such that $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, then

0. (A) $f(4) = 22$ (B) $f\left(\frac{6}{5}\right) = \left(\frac{6}{5}\right)^3$
 (C) $f(x) = x^3$ holds for exactly two values of x . (D) $f(x) = 0$ has a root in interval $(0, 1)$.

169 Let $P(x) = x^{32} - x^{25} + x^{18} - x^{11} + x^4 - x^3 + 1$. Which of the following are **CORRECT** ?

1. (A) Number of real roots of $P(x) = 0$ are zero.
 (B) Number of imaginary roots of $P(x) = 0$ are 32.
 (C) Number of negative roots of $P(x) = 0$ are zero.
 (D) Number of imaginary roots of $P(x) + P(-x) = 0$ are 32.

169 If α, β are the real and distinct roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the

2. equation $x^2 - 4qx + 2q^2 - r = 0$ has always (given $\alpha \neq -\beta$)
 (A) two real roots (B) two negative roots
 (C) two positive roots (D) one positive root and one negative root

169 $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is

3. (a, b, c, d $\in \mathbb{R}$)
 (A) $-d/a$ (B) d/a (C) $(b - a)/a$ (D) $(a - b)/a$

169 If $-5 + i\beta, -5 + i\gamma$ (where $\beta^2 \neq \gamma^2$; $\beta, \gamma \in \mathbb{R}$ and $i^2 = -1$) are roots of $x^3 + 15x^2 + cx + 860 = 0$, $c \in \mathbb{R}$, then

4. (A) $c = 222$
 (B) all the three roots are imaginary
 (C) two roots are imaginary but not complex conjugate of each other.
 (D) $-5 + 7i\sqrt{3}, -5 - 7i\sqrt{3}$ are imaginary roots.

169 Let $f(x) = ax^2 + bx + c > 0, \forall x \in \mathbb{R}$ or $f(x) < 0, \forall x \in \mathbb{R}$. Which of the following is/are **CORRECT** ?

5. (A) If $a + b + c > 0$ then $f(x) > 0, \forall x \in \mathbb{R}$ (B) If $a + c < b$ then $f(x) < 0, \forall x \in \mathbb{R}$
 (C) If $a + 4c > 2b$ then $f(x) < 0, \forall x \in \mathbb{R}$ (D) $ac > 0$.

- 169 Let $x_1 < \alpha < \beta < \gamma < x_4$, $x_1 < x_2 < x_3$. If $f(x)$ is a cubic polynomial with real coefficients such that
6. $(f(\alpha))^2 + (f(\beta))^2 + (f(\gamma))^2 = 0$, $f(x_1)f(x_2) < 0$, $f(x_2)f(x_3) < 0$ and $f(x_1)f(x_3) > 0$ then which of the following are **CORRECT** ?
- (A) $\alpha \in (x_1, x_2)$, $\beta \in (x_2, x_3)$ and $\gamma \in (x_3, x_4)$ (B) $\alpha \in (x_1, x_3)$, $\beta, \gamma \in (x_3, x_4)$
- (C) $\alpha, \beta \in (x_1, x_2)$ and $\gamma \in (x_4, \infty)$ (D) $\alpha \in (x_1, x_3)$, $\beta \in (x_2, x_3)$ and $\gamma \in (x_2, x_4)$

Quadratic Equation



- 169 If $f(x)$ is cubic polynomial with real coefficients, $\alpha < \beta < \gamma$ and $x_1 < x_2$ be such that $f(\alpha) = f(\beta) = f(\gamma) = f'(x_1) = f'(x_2) = 0$ then possible graph of $y = f(x)$ is (assuming y-axis vertical)



- 169 Let $f(x) = \frac{3}{x-2} + \frac{4}{x-3} + \frac{5}{x-4}$, then $f(x) = 0$ has

8. (A) exactly one real root in $(2, 3)$ (B) exactly one real root in $(3, 4)$
- (C) 3 different roots (D) atleast one negative root

- 169 If the quadratic equations $ax^2 + bx + c = 0$ ($a, b, c \in \mathbb{R}$, $a \neq 0$) and $x^2 + 4x + 5 = 0$ have a common root,
9. then a, b, c must satisfy the relations:

- (A) $a > b > c$ (B) $a < b < c$
- (C) $a = k; b = 4k; c = 5k$ ($k \in \mathbb{R}$, $k \neq 0$) (D) $b^2 - 4ac$ is negative.

- 170 If the quadratic equations $x^2 + abx + c = 0$ and $x^2 + acx + b = 0$ have a common root, then the equation containing their other roots is/are :

- (A) $x^2 + a(b+c)x - a^2bc = 0$ (B) $x^2 - a(b+c)x + a^2bc = 0$
- (C) $a(b+c)x^2 - (b+c)x + abc = 0$ (D) $a(b+c)x^2 + (b+c)x - abc = 0$

- 170 Consider the following statements.

1. **S₁** : The equation $2x^2 + 3x + 1 = 0$ has irrational roots.
- S₂** : If $a < b < c < d$, then the roots of the equation $(x-a)(x-c) + 2(x-b)(x-d) = 0$ are real and distinct.
- S₃** : If $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root and $a, b, c \in \mathbb{N}$, then the minimum value of $(a+b+c)$ is 10.
- S₄** : The value of the biquadratic expression $x^4 - 8x^3 + 18x^2 - 8x + 2$, when $x = 2 + \sqrt{3}$, is 1

Which of the following are **CORRECT** ?

- (A) **S₂** and **S₄** are true. (B) **S₁** and **S₃** are false.
- (C) **S₁** and **S₂** are true. (D) **S₃** and **S₄** are false.

- 170 If the equations $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ & $x^2 + (a+b)x + 36 = 0$ have a common positive
2. root, then which of the following are true ?

- (A) $ab = 56$ (B) common positive root is 3
- (C) sum of uncommon roots is 21. (D) $a + b = 15$.

- 170 If $x^2 + \lambda x + 1 = 0$, $\lambda \in (-2, 2)$ and $4x^3 + 3x + 2c = 0$ have common root then $c + \lambda$ can be

3.

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) $\frac{3}{2}$

170 Let quadratic equation $p(x) = 0$ (where $p(x) = x^2 + bx + c$) and equation $p(p(p(x))) = 0$ has a common root, then which of the following statement is/are correct.

4. (A) If $b, c \in \mathbb{R}$, then $b^2 - 4c \geq 0$
 (B) If $P(0) = 1$, then $p(1) = 0$
 (C) equations $p(p(p(x))) = 0$ and $p(p(p(p(p(x)))) = 0$ has at least two common root.
 (D) zero is root of equation $p(p(p(p(p(p(x)))))) = 0$

170 The least value of expression $x^2 + 2xy + 2y^2 + 4y + 7$ is :

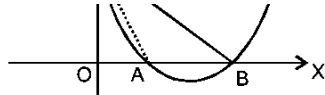
5. (A) 1 (B) 2 (C) 3 (D) 4

170 Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. If x, y satisfy equation $P(x).Q(y) = 28$, then the value of $11y - 26x$ is -

6. (A) 6 (B) 36 (C) 8 (D) 42

170

7.



Roots of $y = 0$ are

- (A) $\{2, 1\}$ (B) $\{4, 2\}$ (C) $\{1, 1/2\}$ (D) $\{8, 4\}$

170 The equation whose roots are $(\alpha + \beta)$ & $(\alpha - \beta)$, where α, β ($\alpha > \beta$) are roots obtained in previous question, is

8. (A) $x^2 - 4x + 3 = 0$ (B) $x^2 - 8x + 12 = 0$ (C) $4x^2 - 8x + 3 = 0$ (D) $x^2 - 16x + 48 = 0$

170 If the equation has four real and distinct roots, then λ lies in the interval

9. (A) $(-\infty, -6) \cup (6, \infty)$ (B) $(0, \infty)$ (C) $(6, \infty)$ (D) $(-\infty, -6)$

171 If the equation has no real root, then λ lies in the interval

0. (A) $(-\infty, 0)$ (B) $(-\infty, 6)$ (C) $(6, \infty)$ (D) $(0, \infty)$

171 If the equation has only two real roots, then set of values of λ is

1. (A) $(-\infty, -6)$ (B) $(-6, 6)$ (C) $\{6\}$ (D) ϕ

171 Roots of equation $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ are

2. (A) $2 \pm \sqrt{3}, 3 \pm \sqrt{2}$ (B) $2 \pm \sqrt{3}, 3 \pm 2\sqrt{2}$
 (C) $3 \pm \sqrt{2}, 3 \pm 2\sqrt{2}$ (D) $8 \pm \sqrt{3}, 3 \pm \sqrt{2}$

171 Roots of equation $x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$ are

3. (A) $1, \frac{3 \pm \sqrt{5}}{2}, \frac{1 \pm i\sqrt{3}}{2}$ (B) $1, \frac{5 \pm \sqrt{3}}{2}, \frac{3 \pm i}{2}$
 (C) $1, \frac{3 \pm \sqrt{5}}{2}, \frac{3 \pm i}{2}$ (D) $1, \frac{5 \pm \sqrt{3}}{2}, \frac{1 \pm i\sqrt{3}}{2}$

171 Roots of equation $x^6 - 4x^4 + 4x^2 - 1 = 0$ are

4. (A) $\pm 1, \frac{1 \pm i\sqrt{5}}{2}, \frac{-1 \pm \sqrt{5}}{2}$ (B) $\pm 1, \frac{1 \pm \sqrt{5}}{2}, \frac{-1 \pm i\sqrt{5}}{2}$
 (C) $\pm 1, \frac{1 \pm \sqrt{5}}{2}, \frac{-1 \pm \sqrt{5}}{2}$ (D) $\pm 1, \frac{-1 \pm \sqrt{5}}{2}, \frac{-1 \pm i\sqrt{5}}{2}$

171 **Solution.** $x^2 - (\alpha + \beta)x + \alpha\beta + 1 = 0$: $D = (\alpha + \beta)^2 - 4\alpha\beta - 4 = (\beta - \alpha)^2 - 4 < 0$. No real roots.

5. (Graphically: the minimum of $(x - \alpha)(x - \beta)$ is $-\frac{(\beta - \alpha)^2}{4} > -1$.)

171

6.

Single Correct • Assignment #A, Q.17

Let $f(x) = x^2 + ax + b$ with $f(2) + f(3) < 2$. Then $f(x) = 1$ has

- (A) both roots real and distinct (B) both roots real and equal
(C) non-real roots (D) roots whose nature depends on a, b

Answer: (A)

Solution. Let $g(x) = f(x) - 1$ (upward parabola). $g(2) + g(3) = f(2) + f(3) - 2 < 0$, so at least one of $g(2), g(3)$ is negative. An upward parabola that takes a negative value has two real distinct roots.

171

7.

Single Correct • Assignment #A, Q.18

If $a + b + c = 0$, the roots of $3ax^2 + 5bx + 7c = 0$ are

- (A) positive (B) negative (C) real and distinct (D) imaginary

Answer: (C)

Solution. $D = 25b^2 - 84ac = 25(a + c)^2 - 84ac = 25a^2 - 34ac + 25c^2 = 25\left(a - \frac{17}{25}c\right)^2 + \frac{336}{25}c^2 > 0$ (unless $a = c = 0$, which kills the equation). Real and distinct.

171

8.

Single Correct • Assignment #A, Q.29 & #B, Q.5

The quadratic in x : $(k^2 - 2k + 2)x^2 - 2\sqrt{k^4 + 4}x + (k^2 + 2k + 2) = 0$, $k \in \mathbb{R}$, has

- (A) real and distinct roots (B) equal roots
(C) imaginary roots (D) roots of opposite sign

171 Solution. Let $K = a^2 + b^2 + 1$. Expanding: $x^2 + 2(K - a - b)x + (2ab - K^2) = 0$.

9. $\frac{D}{4} = (K - a - b)^2 + K^2 - 2ab \geq (K - a - b)^2 + K^2 - (a^2 + b^2) > 0$

since $K > a^2 + b^2$. Real and different.

Key Note: Assignment #A's key gives (B); Assignment #B (Q.36, same question) gives (C). The discriminant is strictly positive, so (B) is correct.

172

0.

Single Correct • Assignment #A, Q.36 & #B, Q.32

If $p, q, r, s \in \mathbb{R}$, then $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ must have

- (A) 6 real roots (B) at least two real roots
(C) 2 real and 4 imaginary roots (D) 4 real and 2 imaginary roots

Answer: (B)

Solution. Discriminants: $D_1 = p^2 - 12q$, $D_2 = r^2 + 4q$, $D_3 = s^2 + 8q$. If $q \geq 0$, $D_2 \geq 0$; if $q < 0$, $D_1 > 0$ (and D_3 may be anything). Either way at least one quadratic has real roots — at least two real roots.

172

1.

Single Correct • Assignment #B, Q.1

If the roots of $\ell x^2 + 2mx + n = 0$ are real and distinct, then the roots of $(\ell + n)(\ell x^2 + 2mx + n) = 2(\ell n - m^2)(x^2 + 1)$ are

- (A) real and equal (B) real and different
(C) imaginary (D) may be real or imaginary

Answer: (C)

Solution. Let $D = m^2 - \ell n > 0$. The new equation is $Ax^2 + 2m(\ell + n)x + C = 0$ with $A = \ell(\ell + n) + 2D$, $C = n(\ell + n) + 2D$.

$$\begin{aligned}\frac{\Delta}{4} &= m^2(\ell + n)^2 - AC = (D + \ell n)(\ell + n)^2 - [\ell n(\ell + n)^2 + 2D(\ell + n)^2 + 4D^2] \\ &= -D(\ell + n)^2 - 4D^2 < 0.\end{aligned}$$

Imaginary roots.

172

2.

Multiple Correct • Assignment #B, Q.55

If a, b, c are non-zero reals and $(ax^2 + bx + c)(ax^2 - bx - c) = 0$, which can be true?

- (A) four real roots (B) two real and two imaginary roots
(C) three real and one imaginary root (D) four imaginary roots

Solution. Discriminants $b^2 - 4ac$ and $b^2 + 4ac$ add to $2b^2 > 0$, so at least one factor has real roots: (D) impossible. Non-real roots come in conjugate pairs: (C) impossible. (A) happens when $|4ac| \leq b^2$, (B) otherwise.

172

3.

Multiple Correct • Assignment #B, Q.60

If $x^2 + (2a + 4)x + (6a^2 + 4ab + b^2 + 8) = 0$ has real roots, then

- (A) $a^b \cdot b^a = 1$ (B) product of roots is -4
(C) product of roots is 16 (D) $(\sin a)^a + \left(\sec \frac{b}{2}\right)^{b/2} = 1$

Answer: (A), (C), (D)

Solution. $\frac{D}{4} = (a + 2)^2 - (6a^2 + 4ab + b^2 + 8) = -(b + 2a)^2 - (a - 2)^2 \geq 0 \Rightarrow a = 2, b = -4$.
(A) $2^{-4} \cdot (-4)^2 = 1 \checkmark$. Product = $24 - 32 + 16 + 8 = 16$ (C) $\checkmark$. (D) $(\sin 2)^2 + (\sec(-2))^{-2} = \sin^2 2 + \cos^2 2 = 1 \checkmark$.

172

4.

Multiple Correct • Assignment #B, Q.62

If the roots of $(4^a + 4^b + 4^c)x^2 - 2(2^{a+b} + 2^{b+c} + 2^{c+d})x + (4^b + 4^c + 4^d) = 0$ are real, which is/are always correct?

(A) $a + b = c + d$

(B) $b = \frac{a+c}{2}$

(C) $a + d = b + c$

(D) $c = \frac{b+d}{2}$

Answer: (B), (C), (D)

Solution. The LHS is $(2^a x - 2^b)^2 + (2^b x - 2^c)^2 + (2^c x - 2^d)^2$. A real root makes all three brackets zero: $x = 2^{b-a} = 2^{c-b} = 2^{d-c}$, so a, b, c, d are in AP. Then (B), (C), (D) hold; (A) does not in general.

172 **Solution.** $D_P = b^2 - 4ac$ and $D_Q = c^2 - 4ab$ have opposite signs (Q.77). With $a > 0$, the one with negative discriminant is positive for all x , and the other has real roots and is not. Exactly one.

172

6.

Subjective • Assignment #A, Q.53

If $x^2 + (a - b)x + (1 - a - b) = 0$ ($a, b \in \mathbb{R}$), find a for which the equation has unequal real roots for all b .

Answer: $a > 1$

Solution. Need $D = (a - b)^2 - 4(1 - a - b) > 0$ for every b : $b^2 + (4 - 2a)b + (a^2 + 4a - 4) > 0 \forall b$. So its discriminant < 0 : $(4 - 2a)^2 - 4(a^2 + 4a - 4) = 32 - 32a < 0 \Rightarrow a > 1$.

Topic 2 | Sign of a Quadratic Expression

172

7.

Single Correct • Assignment #A, Q.13 & #B, Q.7

If $a + b + c > \frac{9c}{4}$ and $ax^2 + 2bx - 5c = 0$ has non-real roots, then

(A) $a > 0, c > 0$

(B) $a > 0, c < 0$

(C) $a < 0, c < 0$

(D) $a < 0, c > 0$

Answer: (B)

Solution. Let $f(x) = ax^2 + 2bx - 5c$. Non-real roots: $4b^2 + 20ac < 0 \Rightarrow ac < -\frac{b^2}{5} \leq 0$, so a and c have opposite signs; also $f(x)$ has the sign of a for every x . The condition $4(a + b + c) > 9c$ reads $4a + 4b - 5c > 0$, i.e. $f(2) > 0$. Hence $a > 0$ and therefore $c < 0$.

Key Note: Assignment #A's key gives (A); Assignment #B (Q.7, same question) gives (B). The derivation above confirms (B).

172

8.

Single Correct • Assignment #B, Q.22

If $4(a - 1)x^2 + 4(a - 1)x + (119 + 5a - 4a^2) > 0$ for all $x \in \mathbb{R}$, the number of integral values of a is

(A) 4

(B) 5

(C) 9

(D) 10

- 172 **Solution.** With $t = \operatorname{cosec}^2 x \in [1, \infty)$: $(t + m)^2 > 3$ for all $t \geq 1$. If $m < -1$, $t = -m$ is allowed and gives $0 > 3$ — fails. If $m \geq -1$, $t + m \geq 1 + m \geq 0$ is smallest at $t = 1$: need $1 + m > \sqrt{3} \Rightarrow m > \sqrt{3} - 1$.

173
0.

Numerical • Assignment #B, Q.132

The number of integral $a \in (8, \infty)$ for which $\log_{10}(4x^2 - ax + 3) \geq 0$ for all $x \in [1, 3]$ is

Answer: 0

Solution. Need $4x^2 - ax + 3 \geq 1$, i.e. $a \leq 4x + \frac{2}{x}$ on $[1, 3]$. $4x + \frac{2}{x}$ is increasing for $x \geq 1$ (derivative $4 - \frac{2}{x^2} > 0$), minimum 6 at $x = 1$. So $a \leq 6$, incompatible with $a > 8$. Zero values.

173
1.

Subjective • Assignment #B, Q.119

Let $f(x) = x^2 - 2ax - a^2 - \frac{3}{4}$. Find all a with $|f(x)| \leq 1$ for all $x \in [0, 1]$.

Answer: $a \in \left[-\frac{1}{2}, \frac{\sqrt{2}}{4}\right]$

Solution. The maximum of f on $[0, 1]$ is at an end-point and the minimum at the vertex $x = a$ if $a \in [0, 1]$.

- $f(0) = -a^2 - \frac{3}{4} \geq -1 \Rightarrow |a| \leq \frac{1}{2}$ (and $f(0) \leq 1$ automatically).
- $f(1) = \frac{1}{4} - 2a - a^2 \leq 1 \iff (a + \frac{1}{2})(a + \frac{3}{2}) \geq 0$, true for $a \geq -\frac{1}{2}$; $\geq -1 \iff a \in [-\frac{5}{2}, \frac{1}{2}]$.
- For $a \in [0, \frac{1}{2}]$: $f(a) = -2a^2 - \frac{3}{4} \geq -1 \Rightarrow a \leq \frac{\sqrt{2}}{4}$.

Together: $a \in \left[-\frac{1}{2}, \frac{\sqrt{2}}{4}\right]$.

Topic 3 | Relations between Roots & Coefficients

173

Single Correct • Assignment #A, Q.27 & #B, Q.2

2. If $\alpha, \beta \in \mathbb{R}$ are the roots of $x^2 + 4x + b = 0$ and $|\alpha - \beta| \leq 2$, the complete range of b is

(A) (3, 4) (B) (4, 5) (C) [3, 4] (D) (3, 5)

- 173 **Solution.** α, β are roots of $t^2 - kt + k = 0$; real iff $k^2 - 4k \geq 0 \iff k \leq 0$ or $k \geq 4$. Smallest positive: 4.

173
4.

Single Correct • Assignment #A, Q.5 & #B, Q.39

α, β are roots of $x^2 - 2ax + 7 = 0$ and γ, δ of $x^2 - 2bx + 1 = 0$. If $\alpha\delta = \beta\gamma$, then $\frac{a^2}{b^2} =$

(A) 5 (B) 6 (C) 7 (D) 8

Answer: (C)

Solution. $\alpha\delta = \beta\gamma \Rightarrow \frac{\alpha}{\beta} = \frac{\gamma}{\delta} = k$. For any pair, $\frac{(\text{sum})^2}{\text{product}} = \frac{(k+1)^2}{k}$ depends only on the ratio k :
 $\frac{(2a)^2}{7} = \frac{(2b)^2}{1} \Rightarrow \frac{a^2}{b^2} = 7$.

173

5.

Single Correct • Assignment #A, Q.6 & #B, Q.40

α, β are roots of $x^2 - x + b = 0$ and $S_k = \alpha^k + \beta^k$. The value(s) of b for which S_2, S_3, S_4 are in AP is

- (A) 0 (B) 1 (C) $\frac{1}{5}$ (D) no real b

Answer: (A)

Solution. Newton's recurrence: $S_{k+2} = S_{k+1} - bS_k$ with $S_0 = 2, S_1 = 1$. $S_2 = 1 - 2b, S_3 = 1 - 3b, S_4 = S_3 - bS_2 = 1 - 4b + 2b^2$. AP: $2S_3 = S_2 + S_4 \Rightarrow 2 - 6b = 2 - 6b + 2b^2 \Rightarrow b = 0$.

173

6.

Single Correct • Assignment #A, Q.8 & #B, Q.23

f is quadratic with $f(1) = 6, f(-1) = 30, f(3) > 0$, and one root of $f(x) = 0$ is twice the other. Then $f(3) =$

- (A) 54 (B) 78 (C) 96 (D) 110

Answer: (D)

Solution. $f = ax^2 + bx + c: a + b + c = 6, a - b + c = 30 \Rightarrow b = -12, a + c = 18$. Roots $r, 2r: 3r = \frac{12}{a}, 2r^2 = \frac{c}{a} \Rightarrow c = \frac{32}{a}$. Then $a + \frac{32}{a} = 18 \Rightarrow a = 16$ or 2 . $a = 16, c = 2: f(3) = 144 - 36 + 2 = 110 > 0 \checkmark$. $a = 2, c = 16: f(3) = -2 < 0 \times$.

173

7.

Single Correct • Assignment #A, Q.39 & #B, Q.28

If the sum of the roots of $ax^2 + bx + c = 0$ equals the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2} =$

- (A) 2 (B) -2 (C) 1 (D) -1

Solution. $\alpha + \beta = \frac{a^2 + \beta^2}{(\alpha\beta)^2} \Rightarrow -\frac{b}{a} = \frac{b^2 - 2ac}{c^2} \Rightarrow -bc^2 = ab^2 - 2a^2c$. Divide by $a^2c: -\frac{bc}{a^2} = \frac{b^2}{ac} - 2 \Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$.

173

8.

Single Correct • Assignment #A, Q.41 & #B, Q.6

If $x^2 - px + p = 0$ has positive integral roots α, β , then $\alpha^{2016} - \beta^{2016} =$

- (A) 4 (B) 2016 (C) $p^2 - 4$ (D) 0

Answer: (D)

Solution. $\alpha + \beta = \alpha\beta = p \Rightarrow \alpha\beta - \alpha - \beta + 1 = 1 \Rightarrow (\alpha - 1)(\beta - 1) = 1$. Positive integers force $\alpha = \beta = 2$. So the difference is 0.

173

9.

Single Correct • Assignment #A, Q.43 & #B, Q.9

If α, β are roots of $ax^2 + bx + c = 0$, the roots of $a(2x + 1)^2 - b(2x + 1)(3 - x) + c(3 - x)^2 = 0$ are

- (A) $\frac{2\alpha+1}{\alpha-3}, \frac{2\beta+1}{\beta-3}$ (B) $\frac{3\alpha+1}{\alpha-2}, \frac{3\beta+1}{\beta-2}$ (C) $\frac{2\alpha-1}{\alpha-2}, \frac{2\beta-1}{\beta-2}$ (D) none

Answer: (B)

Solution. Divide by $(3 - x)^2$ and let $t = \frac{2x+1}{3-x}$: $at^2 - bt + c = 0$, whose roots are $-\alpha, -\beta$. $\frac{2x+1}{3-x} = -\alpha \Rightarrow 2x + 1 = -3\alpha + \alpha x \Rightarrow x = \frac{3\alpha+1}{\alpha-2}$; similarly for β .

174

0.

Single Correct • Assignment #B, Q.35

α, β are roots of $x^2 + px + q = 0$ ($p \neq 0$) and also of $x^{2n} + p^n x^n + q^n = 0$. If $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are roots of $x^n + 1 + (x+1)^n = 0$, then $n \in \mathbb{Z}$ is

(A) an odd prime

(B) an odd integer

(C) an even integer

(D) none of these

Answer: (C)

Solution. From the second equation, α^n, β^n are roots of $y^2 + p^n y + q^n = 0$, so $\alpha^n + \beta^n = -p^n$. Putting $x = \frac{\alpha}{\beta}$ in the third and multiplying by β^n : $\alpha^n + \beta^n + (\alpha + \beta)^n = 0 \Rightarrow -p^n + (-p)^n = 0 \Rightarrow (-1)^n = 1$. So n is even.

174 Solution. In an AP, $\alpha + \delta = \beta + \gamma$, i.e. $-k_1 = -k_2$ (A). The total is $-2k_1$, so (C) fails (unless $k_1 = 0$).

1. (B) Write $a \mp 3d, a \mp d$: $\alpha\delta = a^2 - 9d^2 = -5, \beta\gamma = a^2 - d^2 = 3 \Rightarrow d^2 = 1, a^2 = 4$. Sum = $4a = \pm 8$.

174

2.

Multiple Correct • Assignment #B, Q.63

If α, β are roots of $x^2 - 6x - 4 = 0$, then $\frac{\alpha^{17} + \beta^{17} - 4(\alpha^{15} + \beta^{15})}{(\alpha^8 + \beta^8)^2 - 2^{17}}$ is a

(A) real number

(B) composite number

(C) prime number

(D) rational number

Answer: (A), (B), (D)

Solution. $\alpha^2 - 4 = 6\alpha \Rightarrow \alpha^{17} - 4\alpha^{15} = \alpha^{15}(\alpha^2 - 4) = 6\alpha^{16}$. Numerator = $6(\alpha^{16} + \beta^{16})$. Denominator = $\alpha^{16} + \beta^{16} + 2(\alpha\beta)^8 - 2^{17} = \alpha^{16} + \beta^{16} + 2 \cdot 4^8 - 2^{17} = \alpha^{16} + \beta^{16}$. Value = 6: real, composite, rational.

174 Solution. Each root r satisfies $r - 3 = \frac{1}{r} \iff r^2 - 3r - 1 = 0$. So the equation is $x^2 - 3x - 1 = 0$:

3. $b = -3, c = -1$. $b^2 + 6c + bc = 9 - 6 + 3 = 6$.

174

4.

Numerical • Assignment #B, Q.102

The roots of $ax^2 - 45x + 100 = 0$ ($a \in \mathbb{N}$) are two consecutive natural numbers. The greater root is

Answer: 5

Solution. Roots $n, n+1$: $\frac{n(n+1)}{2n+1} = \frac{100}{45} = \frac{20}{9} \Rightarrow 9n^2 - 31n - 20 = 0 \Rightarrow n = 4$. Greater root 5 (with $a = 5$).

174

5.

Numerical • Assignment #B, Q.126

α, β are roots of $x^2 - bx - b = 0$ ($b > 0$), and $|\alpha|, |\beta|$ are roots of $x^2 + px + q = 0$. If the minimum of $p^2 - 8q$ is k , find $|k|$.

Answer: 4

Solution. $\alpha\beta = -b < 0$: opposite signs, so $|\alpha| + |\beta| = |\alpha - \beta| = \sqrt{b^2 + 4b}$ and $|\alpha||\beta| = b$. $p^2 - 8q = (b^2 + 4b) - 8b = b^2 - 4b = (b - 2)^2 - 4 \geq -4$. $k = -4$, $|k| = 4$.

174

6.

Numerical • Assignment #B, Q.134

If α, β are roots of $x^2 + 2x + 2 = 0$, find $\alpha^3 + 2\beta$.

Answer: 0

Solution. $\alpha^2 = -2\alpha - 2 \Rightarrow \alpha^3 = -2\alpha^2 - 2\alpha = 2\alpha + 4$. So $\alpha^3 + 2\beta = 2(\alpha + \beta) + 4 = -4 + 4 = 0$.

174

7.

Subjective • Assignment #A, Q.54

α, β are roots of $ax^2 + bx + c = 0$ ($a \neq 0$). Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β .

174 Solution. Let $p(x) = x^3 + P_0x^2 + P_1x + P_2 = (x - \alpha)(x - \beta)(x - \gamma)$. Then

8.

$$p(1) = (1 - \alpha)(1 - \beta)(1 - \gamma), \quad p(-1) = -(1 + \alpha)(1 + \beta)(1 + \gamma).$$

So the product $= p(1) \cdot [-p(-1)] = (1 + P_0 + P_1 + P_2)(1 - P_0 + P_1 - P_2) = [(1 + P_1) + (P_0 + P_2)][(1 + P_1) - (P_0 + P_2)] = (1 + P_1)^2 - (P_0 + P_2)^2$.

174

9.

Single Correct • Assignment #A, Q.3

$x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If their third roots are x_1 and x_2 respectively, then $(x_1, x_2) =$

(A) $(-5, -7)$ (B) $(1, -1)$ (C) $(-1, 1)$ (D) $(5, 7)$

Answer: (A)

Solution. Common roots satisfy the difference: $2x^2 + (r - q) = 0$, whose roots are $\pm t$ — so the two common roots add to 0. Sum of roots: first cubic $0 + x_1 = -5$, second $0 + x_2 = -7$. Hence $(x_1, x_2) = (-5, -7)$.

175

0.

Single Correct • Assignment #A, Q.4 & #B, Q.38

If all roots of $x^4 - 8x^3 + ax^2 - bx + 16 = 0$ are positive reals, then $a + b =$

(A) 32

(B) 24

(C) 56

(D) 8

Answer: (C)

Solution. Sum of roots $= 8$, product $= 16$. $AM = \frac{8}{4} = 2$ and $GM = 16^{1/4} = 2$. $AM = GM$ forces all four roots equal to 2: $(x - 2)^4 = x^4 - 8x^3 + 24x^2 - 32x + 16$, so $a = 24$, $b = 32$, $a + b = 56$.

175

1.

Single Correct • Assignment #A, Q.12 & #B, Q.21

If α, β, γ are roots of $111x^3 - 11x - 1 = 0$, then $(\alpha\beta)^{-2} + (\beta\gamma)^{-2} + (\gamma\alpha)^{-2} =$

(A) 2332

(B) 1331

(C) 1210

(D) 2442

175 Solution. Let the roots of f be r_i , with $r_1 r_2 r_3 = -1$. $P(x) = k \prod (x - r_i^2)$; $P(0) = -k(r_1 r_2 r_3)^2 = -k = -1 \Rightarrow k = 1$. $P(9) = \prod (3 - r_i)(3 + r_i) = f(3) \cdot [-f(-3)] = 31 \cdot 29 = 899$.

175

3.

Single Correct • Assignment #B, Q.29

If $\alpha = \log_3 4$ is a root of $x^3 + ax^2 + bx + 1 = 0$ ($a, b \in \mathbb{R}$), the equation which always has the root $\beta = \log_2 48$ is

(A) $x^3 + 2(b-6)x^2 + 4(a-4b+12)x + 8(4b-2a-7) = 0$ (B) $x^3 + bx^2 + ax + 1 = 0$

(C) $(65 + 16a + 4b)x^3 + 2(48 + 8a + b)x^2 + 4(12 + a)x + 8 = 0$ (D) $(65 + 16a + 4b)x^3 + (48 + 8a + b)x^2 + (12 + a)x + 8 = 0$

Answer: (A)

Solution. $\beta = \log_2 16 + \log_2 3 = 4 + \frac{1}{\log_3 2} = 4 + \frac{2}{\log_3 4} = 4 + \frac{2}{\alpha} \Rightarrow \alpha = \frac{2}{\beta-4}$. Substitute in $\alpha^3 + a\alpha^2 + b\alpha + 1 = 0$ and multiply by $(\beta-4)^3$: $8 + 4a(\beta-4) + 2b(\beta-4)^2 + (\beta-4)^3 = 0$. Expanding: $\beta^3 + 2(b-6)\beta^2 + 4(a-4b+12)\beta + 8(4b-2a-7) = 0$.

175

4.

Single Correct • Assignment #B, Q.31

If $x^3 + 2x^2 - 4x + 5 = 0$ has roots α, β, γ , then $\frac{(\alpha^3 + 5)(\beta^3 + 5)(\gamma^3 + 5)}{8\alpha\beta\gamma} =$

(A) 104

(B) 8

(C) 13

(D) -104

Answer: (C)

Solution. From the equation, $\alpha^3 + 5 = -2\alpha^2 + 4\alpha = -2\alpha(\alpha - 2)$. So $\prod(\alpha^3 + 5) = -8\alpha\beta\gamma \prod(\alpha - 2)$ and the ratio is $-\prod(\alpha - 2) = \prod(2 - \alpha) = p(2) = 8 + 8 - 8 + 5 = 13$.

175

5.

Single Correct • Assignment #B, Q.37

If α, β, γ are roots of $ax^3 + bx^2 + cx + d = 0$, then $a\left(\frac{\alpha^3}{\beta} + \frac{\beta^3}{\gamma} + \frac{\gamma^3}{\alpha}\right) + b\left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha}\right) + c\left(\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha}\right) =$

(A) a (B) b (C) c (D) d

175 Solution. Let $P = \alpha\beta\gamma$. $\alpha + \frac{1}{\beta\gamma} = \frac{P+1}{\beta\gamma}$, so $\beta\gamma = 2(P+1)$, $\alpha\gamma = -10(P+1)$, $\alpha\beta = 18(P+1)$. Then

6.

$\alpha = \frac{P}{\beta\gamma} = \frac{k}{2}, \beta = -\frac{k}{10}, \gamma = \frac{k}{18}$ with $k = \frac{P}{P+1} \cdot \frac{\beta-\alpha}{\alpha-\gamma} = \frac{-\frac{1}{10}-\frac{1}{2}}{\frac{1}{2}-\frac{1}{18}} = \frac{-\frac{3}{5}}{\frac{4}{9}} = -\frac{27}{20}$.

175 Single Correct • Assignment #B, Q.43

7. $\alpha, \beta, \gamma, \delta$ are the roots of $x^4 - 3x + 1 = 0$. Then $\alpha^3 + \beta^3 + \gamma^3 + \delta^3 =$

(A) 3

(B) 12

(C) 9

(D) -9

Answer: (C)**Solution.** $\alpha^4 = 3\alpha - 1 \Rightarrow \alpha^3 = 3 - \frac{1}{\alpha}$. So $\sum \alpha^3 = 12 - \sum \frac{1}{\alpha}$. $\sum \frac{1}{\alpha} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{3}{1} = 3$. Hence $\sum \alpha^3 = 9$.

175 Multiple Correct • Assignment #A, Q.33 & #B, Q.47

8.

If α, β, γ are the roots of $x^3 - 3x + 1 = 0$, then(A) $\prod(\alpha + 1) = -3$ (B) $\sum(\alpha + 1) = 0$ (C) $\sum(\alpha + 1)(\beta + 1) = -3$ (D) $\sum \alpha^2 = 6$ **Answer: (A), (D)****Solution.** $\sum \alpha = 0, \sum \alpha\beta = -3, \alpha\beta\gamma = -1, p(x) = x^3 - 3x + 1$. (A) $\prod(-1 - \alpha) = p(-1) = 3 \Rightarrow \prod(1 + \alpha) = -3 \checkmark$. (B) $\sum(\alpha + 1) = 3 \times$. (C) $\sum \alpha\beta + 2\sum \alpha + 3 = 0 \times$. (D) $\sum \alpha^2 = 0 - 2(-3) = 6 \checkmark$.175 **Solution.** For $y^3 - 5y^2 + 6y - 3 = 0$: $e_1 = 5, e_2 = 6, e_3 = 3$. $p_2 = 25 - 12 = 13$; $p_3 = e_1p_2 - e_2p_1 +$
9. $3e_3 = 65 - 30 + 9 = 44$.

176 Numerical • Assignment #B, Q.117

0.

If $a + b + c = 3, a^2 + b^2 + c^2 = 29, abc = 11$, find $a^5 + b^5 + c^5$.176 **Solution.** $\alpha^3 + \alpha^2 + 3 = -\alpha^2 + \alpha = -\alpha(\alpha - 1)$. Ratio = $(-1)^3 \prod(\alpha - 1) = \prod(1 - \alpha) = p(1) =$
1. $1 + 2 - 1 + 3 = 5$.**Topic 5 | Constructing Polynomials & Minimal Equations**

176 Single Correct • Assignment #A, Q.20

2. $p(x) = 0$ is a polynomial equation of least degree with rational coefficients having $\sqrt[3]{7} + \sqrt[3]{49}$ as a root. The product of all roots of $p(x) = 0$ is

(A) 7

(B) 49

(C) 56

(D) 63

Answer: (C)**Solution.** Let $u = \sqrt[3]{7}$ and $t = u + u^2 = u(1 + u)$. Then $t^3 = u^3(1 + u)^3 = 7(1 + 3u + 3u^2 + u^3) = 7(8 + 3t) = 56 + 21t$. So $t^3 - 21t - 56 = 0$ (irreducible: no rational root among divisors of 56). Product of roots = 56.

- 176 Solution.** $(u+v)^3 = u^3 + v^3 + 3uv(u+v)$ with $u^3 + v^3 = 90$ and $uv = \sqrt[3]{45^2 - 2 \cdot 29^2} = \sqrt[3]{343} = 7$:
 3. $\alpha^3 = 90 + 21\alpha$, so the value is 0.

176
4.

Numerical • Assignment #B, Q.114

$P(x)$ is monic of degree 4 with $P(1) = 3$, $P(3) = 11$, $P(5) = 27$. Find $P(-2) + 7P(6)$.

Answer: 1112

Solution. $x^2 + 2$ takes the values 3, 11, 27 at 1, 3, 5, so $P(x) = (x-1)(x-3)(x-5)(x-r) + x^2 + 2$.
 $P(-2) = (-3)(-5)(-7)(-2-r) + 6 = 210 + 105r + 6$; $7P(6) = 7[15(6-r) + 38] = 896 - 105r$.
 Sum = 1112 (r cancels).

176
5.

Subjective • Assignment #A, Q.50

$f(x) = x^2 + ax + b$ has a zero $\frac{4+3\sqrt{3}}{2+\sqrt{3}}$, with a, b integers. Also $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ and
 $g\left(\frac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$ with c, d integers. Find a, b, c, d .

Answer: $a = 2$, $b = -11$, $c = 4$, $d = -1$

Solution. Rationalise: $t = \frac{(4+3\sqrt{3})(2-\sqrt{3})}{4-3} = 8 - 4\sqrt{3} + 6\sqrt{3} - 9 = -1 + 2\sqrt{3}$. Its conjugate $-1 - 2\sqrt{3}$ is the other zero: sum -2 , product $1 - 12 = -11$, so $f = x^2 + 2x - 11$: $a = 2$, $b = -11$. Divide:
 $g(x) = (x^2 + 2x - 11)(x^2 + 1) + (2x + 1)$. Since $f(t) = 0$, $g(t) = 2t + 1 = 4\sqrt{3} - 1$: $c = 4$, $d = -1$.

176
6.

Subjective • Assignment #B, Q.120

Find the monic polynomial of eighth degree with rational coefficients having one root

$$\sqrt{8 + \sqrt{6} + 4\sqrt{3} + 3\sqrt{2}}.$$

Answer: $x^8 - 32x^6 + 240x^4 - 320x^2 - 368$ (no printed key)

Solution. Let $y = x^2$ and $w = y - 8 = 3\sqrt{2} + (4\sqrt{3} + \sqrt{6})$. Remove $\sqrt{3}$: $(w - 3\sqrt{2})^2 = (4\sqrt{3} + \sqrt{6})^2 = 54 + 24\sqrt{2} \Rightarrow w^2 - 36 = \sqrt{2}(6w + 24)$. Remove $\sqrt{2}$: $(w^2 - 36)^2 = 2(6w + 24)^2 \Rightarrow w^4 - 144w^2 - 576w + 144 = 0$. With $w = y - 8$ this becomes $y^4 - 32y^3 + 240y^2 - 320y - 368 = 0$; putting $y = x^2$:

$$x^8 - 32x^6 + 240x^4 - 320x^2 - 368 = 0.$$

(Numerical check: at $x^2 = 8 + \sqrt{6} + 4\sqrt{3} + 3\sqrt{2}$ the value is 0.)

- 176 Solution.** $x^3 - 2x^2 + 2x - 1 = (x-1)(x^2 - x + 1)$: roots 1, $-\omega$, $-\omega^2$ (the last two non-real). A real quadratic sharing two roots with the cubic cannot share 1 and one non-real root (non-real roots come in conjugate pairs), so it shares the pair of $x^2 - x + 1$: $ax^2 + bx + a = a(x^2 - x + 1) \Rightarrow b = -a \Rightarrow a + b = 0$.

176

8.

Single Correct • Assignment #A, Q.10 & #B, Q.19

The number of values of α for which $y = \frac{\alpha x^2 + 7x - 2}{\alpha + 7x - 2x^2}$ has at least one common linear factor in numerator and denominator is

- (A) 0 (B) 1 (C) 2 (D) 3

Answer: (D)

Solution. A common linear factor means a common root r : $\alpha r^2 + 7r - 2 = 0$ and $-2r^2 + 7r + \alpha = 0$. Subtracting: $(\alpha + 2)r^2 - (\alpha + 2) = 0 \Rightarrow (\alpha + 2)(r^2 - 1) = 0$.

- $\alpha = -2$: numerator and denominator are identical — common factors. ✓
- $r = 1$: $\alpha + 5 = 0 \Rightarrow \alpha = -5$ (check: $-2 + 7 - 5 = 0$). ✓
- $r = -1$: $\alpha - 9 = 0 \Rightarrow \alpha = 9$ (check: $-2 - 7 + 9 = 0$). ✓

Three values: $-2, -5, 9$.

176

9.

Single Correct • Assignment #B, Q.11

If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root, then $\left(\frac{a^3 + b^3 + c^3}{abc}\right)^2 =$

- (A) 1 (B) 3 (C) 9 (D) 16

Answer: (C)

Solution. For a common root α , cross-multiplication gives $\frac{a^2}{ab-c^2} = \frac{\alpha}{bc-a^2} = \frac{1}{ca-b^2}$, so $(bc - a^2)^2 = (ab - c^2)(ca - b^2)$. Expanding: $a^4 - 3a^2bc + ab^3 + ac^3 = 0 \Rightarrow a(a^3 + b^3 + c^3 - 3abc) = 0$. With $a \neq 0$: $\frac{a^3 + b^3 + c^3}{abc} = 3$, square = 9.

177

0.

Multiple Correct • Assignment #B, Q.45

If $(a + 2)x^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root, where $a, b, c \in \mathbb{N}$, then

- (A) $b^2 - 4ac < 0$ (B) minimum value of $a + b + c$ is 16
(C) $b^2 < 4ac + 8c$ (D) minimum value of $a + b + c = 7$

177

1.

Multiple Correct • Assignment #B, Q.51

If $x^3 + ax^2 + 11x + 6$ and $x^3 + bx^2 + 14x + 8$ have a common factor $x^2 + px + q$, then

- (A) $a + p = b + q$ (B) $ap < bq$ (C) pq divides ab (D) $p + q$ divides $a + b$

Answer: (A), (C)

Solution. The common factor divides $4(\text{first}) - 3(\text{second}) = x(x^2 + (4a - 3b)x + 2)$ and (not having root 0) equals $x^2 + (4a - 3b)x + 2$: $q = 2$. It also divides $(\text{first}) - (\text{second}) = (a - b)x^2 - 3x - 2$, so this is a multiple: $\frac{-2}{a-b} = q = 2 \Rightarrow b = a + 1$, and $p = \frac{-3}{a-b} = 3$. Then $4a - 3b = 3 \Rightarrow a = 6, b = 7$. Check: $(x + 1)(x + 2)(x + 3)$ and $(x + 1)(x + 2)(x + 4)$ ✓. (A) $9 = 9$ ✓; (B) $18 < 14$ ✗; (C) $6 \mid 42$ ✓; (D) $5 \nmid 13$ ✗.

177

2.

Multiple Correct • Assignment #B, Q.58

If $bx^2 + acx + b^2c = 0$ and $cx^2 + abx + b^2c = 0$ have a common root (a, b, c non-zero, distinct), then

(A) $\frac{a}{c} + \frac{b}{a} + \frac{a}{b} = 0$

(B) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 0$

(C) $\frac{1}{bc^2} + \frac{1}{a^2c} + \frac{1}{cb^2} = 0$

(D) $\frac{1}{ac^2} + \frac{1}{ba^2} + \frac{1}{cb^2} = 0$

Answer: (A), (C)

Solution. Subtract: $(b - c)x^2 - a(b - c)x = 0 \Rightarrow x = 0$ or $x = a$. $x = 0$ would need $b^2c = 0$; so the common root is a : $ba^2 + ca^2 + b^2c = 0$. (A) $\times abc$: $a^2b + b^2c + a^2c = 0 \checkmark$. (C) $\times a^2b^2c^2$: $a^2b + b^2c + a^2c = 0 \checkmark$. (B), (D) give different relations.

177

3.

Multiple Correct • Assignment #B, Q.70

If $x^2 + 3x + 5$ is the GCD of $x^3 + ax^2 + bx + 1$ and $2x^3 + 7x^2 + 13x + 5$, then $[a + b]$ is greater than or equal to

(A) 7

(B) 8

(C) 9

(D) 10

177 Solution. A quadratic with three distinct roots is identically zero: $2\sin\theta + 1 = 0$, $\sqrt{3}\tan\theta - 1 = 0$,
4. $\cot\theta - \sqrt{3} = 0$. $\sin\theta = -\frac{1}{2}$ and $\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta$ in the third quadrant: $\theta = \frac{7\pi}{6} + 2k\pi$. In $[-\pi, 2\pi]$:
 $\frac{7\pi}{6}, -\frac{5\pi}{6}$. Two values.

177

5.

Numerical • Assignment #B, Q.107

$x^2 + px + 12 = 0$, $x^2 - qx + 15 = 0$ and $x^2 + (p - q)x + 36 = 0$ have a common negative root. Find $|p + q|$.

Answer: 1

Solution. Adding the first two: $2r^2 + (p - q)r + 27 = 0$; the third: $r^2 + (p - q)r + 36 = 0$. Subtract: $r^2 = 9 \Rightarrow r = -3$. Then $9 - 3p + 12 = 0 \Rightarrow p = 7$; $9 + 3q + 15 = 0 \Rightarrow q = -8$. $|p + q| = 1$.

177

6.

Numerical • Assignment #B, Q.109

If $x^2 - 4x + 1 = 0$ and $x^2 - (4\sin\theta)x - 7 = 0$ have one root common, the number of $\theta \in (0, 2\pi)$ is

177 Solution. $2 \times (\text{second}) - (\text{first})$: $(4 - b)x + 3b = 0 \Rightarrow x = \frac{3b}{b-4}$ ($b = 4$ gives $-12 = 0$, impossible).
7. Substituting in the second: $b(2b^2 - b + 8) = 0$. The quadratic factor has $D = 1 - 64 < 0$, so $b = 0$ only. One value.

177

8.

Numerical • Assignment #B, Q.125

The sum of all real m for which $(x^2 - 2mx - 4(m^2 + 1))(x^2 - 4x - 2m(m^2 + 1)) = 0$ has exactly three different real roots is

Answer: 3

Solution. Let $k = m^2 + 1$. The first factor always has two distinct real roots. *Double root of the second* ($4 + 2mk = 0 \Rightarrow m = -1$): root 2, but the first factor $x^2 + 2x - 8$ also has root 2 — only 2 distinct. **X** *Exactly one shared root:* subtracting, $(4 - 2m)(x - k) = 0$. $m = 2$ makes the factors identical **X**; otherwise the shared root is $x = k$, and $k^2 - 2mk - 4k = 0 \Rightarrow m^2 - 2m - 3 = 0 \Rightarrow m = 3$ (or -1 , rejected). $m = 3$: $(x - 10)(x + 4)(x - 10)(x + 6)$ — roots 10, -4 , -6 ✓. Sum = 3.

177

9.

Numerical • Assignment #B, Q.133

Each pair of $x^2 + ax + b = 0$, $x^2 + cx + d = 0$, $x^2 + ex + f = 0$ has exactly one root in common, and no root is common to all three. Find $\frac{(a + c + e)^2}{ac + ce + ea - b - d - f}$.

Answer: 4

Solution. Roots (p, q) , (q, r) , (r, p) . Then $a + c + e = -2(p + q + r)$ and $ac + ce + ea = p^2 + q^2 + r^2 + 3(pq + qr + rp)$, $b + d + f = pq + qr + rp$, so the denominator is $(p + q + r)^2$. Ratio = 4.

178

0.

Subjective (Proof) • Assignment #A, Q.51

If $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ ($a \neq 0$, $b \neq c$) have a common root, prove that their other roots satisfy $x^2 + ax + bc = 0$.

Answer: Proved below

Solution. Subtracting: $(b - c)x - a(b - c) = 0 \Rightarrow$ common root $x = a$. Substituting: $a^2 + ab + ca = 0 \Rightarrow a + b + c = 0$. Other roots (product of roots / a): first $\frac{ca}{a} = c$, second $\frac{ab}{a} = b$. Equation with roots b, c : $x^2 - (b + c)x + bc = 0$, i.e. $x^2 + ax + bc = 0$ (using $b + c = -a$). ■

178 Solution. Factorise: $f(x) = (ax + 2)(x - 2)$, roots 2 and $-\frac{2}{a}$. If $a \leq 0$, $f < 0$ on an unbounded set 1. (infinitely many integers). So $a > 0$ and $f < 0$ exactly on $(-\frac{2}{a}, 2)$. Exactly three integers there means they are 1, 0, -1 : need $-2 \leq -\frac{2}{a} < -1 \Rightarrow 1 < \frac{2}{a} \leq 2 \Rightarrow a \in [1, 2)$.

178

2.

Single Correct • Assignment #A, Q.15 & #B, Q.10

The set of a for which $x^2 - (a + 2)x - (a + 3) < 0$ holds for at least one positive x is

(A) $[-3, \infty)$ (B) $(-3, \infty)$ (C) $(-\infty, -3)$ (D) $(-\infty, 3]$

Answer: (B)

Solution. $x = -1$ is a root ($1 + a + 2 - a - 3 = 0$), so the expression is $(x + 1)(x - (a + 3))$, negative exactly between -1 and $a + 3$. A positive x exists in that interval iff $a + 3 > 0$, i.e. $a > -3$.

178

3.

Single Correct • Assignment #A, Q.16 & #B, Q.12

The set of m for which every solution of $x^2 - 3x + 2 \leq 0$ also satisfies $2x^2 - (m+1)x - (2m+3) < 0$ is

- (A) $(\frac{3}{4}, \infty)$ (B) $(-\frac{2}{3}, \infty)$ (C) $(-\frac{2}{3}, \frac{3}{4})$ (D) $(-\infty, \infty)$

Answer: (A)

Solution. The first inequality gives $x \in [1, 2]$. For the upward parabola $g(x) = 2x^2 - (m+1)x - (2m+3)$ to be negative on all of $[1, 2]$ it suffices and is necessary that $g(1) < 0$ and $g(2) < 0$: $g(1) = -3m - 2 < 0 \Rightarrow m > -\frac{2}{3}$; $g(2) = 3 - 4m < 0 \Rightarrow m > \frac{3}{4}$. Hence $m > \frac{3}{4}$.

178

4.

Single Correct • Assignment #A, Q.22

If $ax^2 + bx + c = 0$ has two positive real roots, then $ax^2 + (b+6a)x + (c+3b) = 0$ has

- (A) no solution (B) at least one positive solution
(C) at least one negative solution (D) none of the above

178 Solution. $cx^2 + bx + a = 0$ has the reciprocal roots. $r_1 > 1 \Rightarrow \frac{1}{r_1} \in (0, 1)$; $r_2 < -3 \Rightarrow \frac{1}{r_2} \in (-\frac{1}{3}, 0)$.

5. So $\alpha < 1$ and $\beta > -\frac{1}{3}$.

178

6.

Single Correct • Assignment #B, Q.33

If $a^2 \sin^2 x - 2 \sin x - (2a+1) = 0$ ($a > 0$) has exactly one root in $[0, \frac{\pi}{2}]$, the range of a is

- (A) $[2, \infty)$ (B) $[1, \infty)$ (C) $[3, \infty)$ (D) none

Answer: (C)

Solution. $t = \sin x \in [0, 1]$ corresponds one-to-one with $x \in [0, \frac{\pi}{2}]$. $g(t) = a^2 t^2 - 2t - (2a+1)$ has $g(0) < 0$, so exactly one positive root. It lies in $[0, 1]$ iff $g(1) \geq 0$: $a^2 - 2a - 3 \geq 0 \Rightarrow (a-3)(a+1) \geq 0 \Rightarrow a \geq 3$.

178

7.

Multiple Correct • Assignment #A, Q.30 & #B, Q.68

For $x^2 + (2-m)x - 4 = 0$, identify the correct statement(s):

- (A) if $m < 5$, exactly one root is smaller than -1 (B) if $m < 5$, both roots are smaller than -1
(C) if $m > 5$, both roots are greater than -1 (D) roots are distinct reals for each $m \in \mathbb{R}$

Answer: (A), (C), (D)

Solution. $D = (2-m)^2 + 16 > 0$ always (D). Let $f(x) = x^2 + (2-m)x - 4$; $f(-1) = m - 5$. $m < 5$: $f(-1) < 0$, so -1 lies strictly between the roots — exactly one root is less than -1 (A), not both (B false). $m > 5$: $f(-1) > 0$, so -1 is outside [roots]. The roots have product $-4 < 0$, so one is positive (> -1); hence both are > -1 (C).

178

8.

Multiple Correct • Assignment #B, Q.54

If $2y^3 + \ell y^2 + my + 4 = 0$ with $\ell + 15 < 2m$ and $2\ell + m + 10 < 0$, then at least one root must lie in

- (A) $(-1, 0)$ (B) $(-2, -1)$ (C) $(0, 2)$ (D) $(1, 2)$

178 **Solution.** $x + 3 = \alpha$ or β : $x = \alpha - 3 \in (-2, 0)$ and $x = \beta - 3 > 0$. Different signs.
9.

179
0.

Numerical • Assignment #B, Q.129

Both roots of $ax^2 + bx + c = 0$ ($a, b, c \neq 0$) lie in $(2, 3)$; both roots of $a(x+1)^2 + b(x^2-1) + c(x-1)^2 = 0$ are always greater than k . Maximum integral k is

Answer: 2

Solution. Divide by $(x-1)^2$: $at^2 + bt + c = 0$ with $t = \frac{x+1}{x-1}$, so $t \in (2, 3)$. Then $x = 1 + \frac{2}{t-1}$ with $t-1 \in (1, 2)$ gives $x \in (2, 3)$. Both roots exceed 2; largest integer $k = 2$.

179
1.

Subjective • Assignment #A, Q.49

Find K so that $x^2 + 2(K-1)x + K+5 = 0$ has at least one positive root.

Answer: $K \leq -1$

Solution. Real roots: $(K-1)^2 - (K+5) \geq 0 \iff K^2 - 3K - 4 \geq 0 \iff K \leq -1$ or $K \geq 4$.

- $K \geq 4$: sum $= -2(K-1) < 0$, product $K+5 > 0$ — both negative. ✗
- $-5 < K \leq -1$: sum > 0 , product > 0 — both positive. ✓
- $K \leq -5$: product ≤ 0 — one root positive (at $K = -5$: roots 0 and 12). ✓

Answer: $K \leq -1$.

179
2.

Subjective • Assignment #A, Q.56

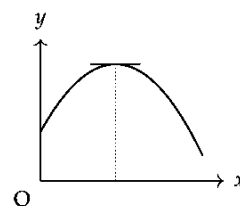
α, β are roots of $x^2 - 2x - a^2 + 1 = 0$ and γ, δ of $x^2 - 2(a+1)x + a(a-1) = 0$ with $\alpha, \beta \in (\gamma, \delta)$. Find a .

179

Multiple Correct • Assignment #A, Q.32 & #B, Q.46

3. The graph of $f(x) = px^2 - qx - r$ is shown (downward parabola, vertex to the right of the y -axis, positive y -intercept). Then

- (A) $r(p-q-r) < 0$ (B) $q < 0$
(C) $\sqrt{q^2 + 4pr} > |q|$ (D) $p < 0$



Answer: (A), (B), (C), (D)

Solution. Opens downward: $p < 0$ (D). y -intercept $f(0) = -r > 0 \Rightarrow r < 0$. Vertex at $x = \frac{q}{2p} > 0$ with $p < 0 \Rightarrow q < 0$ (B). (C) $q^2 + 4pr > q^2$ since $pr > 0$. ✓ (A) $p - q - r = f(1)$. Since $f(0) > 0$ and the vertex is at a positive abscissa, $f > 0$ on $[0, 2x_v]$; in the figure $x = 1$ lies in this positive part, so $f(1) > 0$ and $r f(1) < 0$. ✓

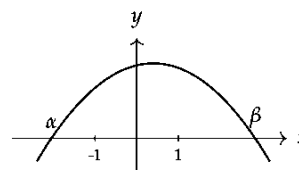
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4.

Multiple Correct • Assignment #A, Q.35 & #B, Q.49

The graph of $f(x) = ax^2 + bx + c$ is shown: a downward parabola cutting the x -axis at $\alpha < -1$ and $\beta > 1$, with vertex to the right of the y -axis. Which are correct?

- (A) $\frac{c}{a} < -1$ (B) $|\beta - \alpha| > 2$ (C) $f(x) > 0 \forall x \in (0, \beta)$ (D) $abc < 0$



Answer: (A), (B), (C), (D)

Solution. $a < 0$; $f(0) = c > 0$; vertex $-\frac{b}{2a} > 0 \Rightarrow b > 0$. So $abc < 0$ (D). (A) $\frac{c}{a} = \alpha\beta$ with $\alpha < -1$, $\beta > 1 \Rightarrow \alpha\beta < -1$. (B) $\beta - \alpha > 1 + 1 = 2$. (C) $(0, \beta)$ lies between the roots, where a downward parabola is positive.

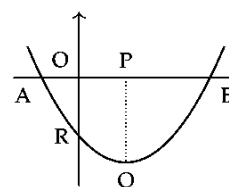
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5.

Multiple Correct • Assignment #B, Q.59

The graph of $y = ax^2 + bx + c$ is shown; $PQ = 9$ and $OR = 5$ (Q is the vertex, P its foot on the x -axis, R the y -intercept), and area of $\triangle OBQ = \frac{45}{4}$. Then

- (A) $AB = 3$ (B) $AB = 4$ (C) $\left|\frac{b}{a}\right| > 1$ (D) $\left|\frac{b}{a}\right| < 1$



179 Solution. Axis $x = -\frac{b}{2a} = -\frac{1}{2} \Rightarrow b = a$. Sum of roots $= -\frac{b}{a} = -1$.

6.

179

7.

Subjective • Assignment #A, Q.48

For what values of p does the vertex of $y = x^2 + 2px + 13$ lie at a distance 5 from the origin?

Answer: $p \in \{-4, -3, 3, 4\}$

Solution. Vertex $(-p, 13 - p^2)$. Distance²: $p^2 + (13 - p^2)^2 = 25$. With $u = p^2$: $u^2 - 25u + 144 = 0 \Rightarrow u = 9, 16$. So $p = \pm 3, \pm 4$.

Topic 9 | Range of Rational Expressions & Extreme Values

179

8.

Single Correct • Assignment #A, Q.19

For real x , $\frac{(x-a)(x-b)}{x-c}$ assumes all real values provided

- (A) $a > b > c$ (B) $a < b < c$ (C) $a > c > b$ (D) none of these

Answer: (C)

Solution. $y(x-c) = (x-a)(x-b) \Rightarrow x^2 - (a+b+y)x + (ab+cy) = 0$. Every y must give real x :

$$(a+b+y)^2 - 4(ab+cy) = y^2 + 2(a+b-2c)y + (a-b)^2 \geq 0 \quad \forall y.$$

Its discriminant ≤ 0 : $(a+b-2c)^2 - (a-b)^2 \leq 0 \iff 4(b-c)(a-c) \leq 0 \iff c$ lies between a and b (strictly, else the function degenerates to a line with a hole). Only (C) has c between a and b .

179

9.

Single Correct • Assignment #B, Q.3

Let $f(x) = \frac{x^2 - 5x + 6}{x^2 - 3x + 2}$. The range of f is

(A) $(-\infty, 1) \cup (1, \infty)$

(B) $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$

(C) $(-\infty, \infty)$

(D) $(-\infty, -1) \cup (-1, \infty)$

Solution. $f = \frac{(x-2)(x-3)}{(x-1)(x-2)} = \frac{x-3}{x-1} = 1 - \frac{2}{x-1}$, $x \neq 1, 2$. $1 - \frac{2}{x-1}$ never equals 1; and the hole at $x = 2$ removes the value $\frac{2-3}{2-1} = -1$. Range = $\mathbb{R} - \{1, -1\}$.

180

0.

Single Correct • Assignment #A, Q.21

The least value of $x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$ is

(A) 0

(B) 1

(C) no least value

(D) none of these

Answer: (B)

Solution. Complete squares: $(x-1)^2 + (2y-3)^2 + 3(z-1)^2 + 14 - 1 - 9 - 3 = (x-1)^2 + (2y-3)^2 + 3(z-1)^2 + 1$. Least value 1 at $x = 1, y = \frac{3}{2}, z = 1$.

180

1.

Single Correct • Assignment #B, Q.30

If $3a + 4b = 5$ ($a, b \in \mathbb{R}$), the minimum of $a^2 + b^2$ is

(A) 1

(B) 4

(C) 5

(D) 10

Answer: (A)

Solution. $a^2 + b^2$ is the squared distance from the origin to a point on the line $3a + 4b = 5$; the minimum is $\left(\frac{5}{\sqrt{9+16}}\right)^2 = 1$ (at $a = \frac{3}{5}, b = \frac{4}{5}$).

180

2.

Multiple Correct • Assignment #B, Q.50

If $x^2 + y^2 = 4$ and $E = x^2 + 2xy - y^2$, then

(A) maximum of E is $4\sqrt{2}$

(B) minimum of E is $-4\sqrt{2}$

(C) $x \in [-2, 2]$

(D) $y \in [-2, 2]$

Answer: (A), (B), (C), (D)

Solution. $x = 2\cos\theta, y = 2\sin\theta$: $E = 4(\cos^2\theta - \sin^2\theta) + 8\sin\theta\cos\theta = 4(\cos 2\theta + \sin 2\theta) \in [-4\sqrt{2}, 4\sqrt{2}]$. Clearly $x, y \in [-2, 2]$.

180

3.

Multiple Correct • Assignment #B, Q.65

Let $E = (x+y)(y+z)$ with $x, y, z > 0$ and $xyz(x+y+z) = 1$. If λ is the minimum of E , then λ is

(A) an integer

(B) even

(C) prime

(D) algebraic

180 4. Solution. $y + z = 4 - x, y^2 + z^2 = 6 - x^2 \geq \frac{(4-x)^2}{2} \Rightarrow 3x^2 - 8x + 4 \leq 0 \Rightarrow x \in [\frac{2}{3}, 2]$. By symmetry the same holds for y and z .

180

5.

Matrix Match • Assignment #A, Q.57 & #B, Q.97

$$\text{Let } f(x) = \frac{x^2 - 11x + 18}{x^2 - 7x + 12}.$$

- (A) Complete solution of $0 < f(x) < 1$ (P) $(3, 4)$
 (B) Complete solution of $f(x) < 1$ (Q) $(\frac{3}{2}, 2) \cup (9, \infty)$
 (C) Complete solution of $f(x) < 0$ (R) $(2, 3) \cup (4, 9)$
 (D) Complete solution of $f(x) > 2$ (S) $(-\infty, 2) \cup (3, 4) \cup (9, \infty)$
 (T) $(\frac{3}{2}, 3) \cup (4, \infty)$

Answer: (A)→(Q); (B)→(T); (C)→(R); (D)→(P)

Solution. $f = \frac{(x-2)(x-9)}{(x-3)(x-4)}$. Signs on $(-\infty, 2), (2, 3), (3, 4), (4, 9), (9, \infty)$: $+, -, +, -, +$. (C) $f < 0$: $(2, 3) \cup (4, 9)$. (B) $f - 1 = \frac{-4x+6}{(x-3)(x-4)} < 0 \iff \frac{4x-6}{(x-3)(x-4)} > 0 \iff x \in (\frac{3}{2}, 3) \cup (4, \infty)$. (A) intersect (B) with $f > 0$: $(\frac{3}{2}, 2) \cup (9, \infty)$. (D) $f - 2 = \frac{-x^2+3x-6}{(x-3)(x-4)}$; the numerator is always negative, so $f > 2 \iff (x-3)(x-4) < 0 \iff x \in (3, 4)$.

180

6.

Subjective • Assignment #A, Q.52

If the range of $f(x) = \frac{x^2 + ax + b}{x^2 + 2x + 3}$ is $[-5, 4]$ with $a, b \in \mathbb{N}$, find $a^2 + b^2$.

Answer: 277

Solution. $(y-1)x^2 + (2y-a)x + (3y-b) = 0$ must have real x exactly for $y \in [-5, 4]$: $(2y-a)^2 - 4(y-1)(3y-b) \geq 0 \iff 8y^2 - (4b-4a+12)y - (a^2-4b) \leq 0$. Its roots are $-5, 4$: sum $\frac{4b-4a+12}{8} = -1 \Rightarrow b-a = -5$; product $-\frac{a^2-4b}{8} = -20 \Rightarrow a^2-4b = 160$. $a^2-4a+20 = 160 \Rightarrow a = 14$ (natural), $b = 9$. $a^2 + b^2 = 196 + 81 = 277$.

- 180 **Solution.** Roots r, s : $r+s = \frac{4}{\lambda}$, $rs = \frac{9}{\lambda}$, so $\frac{rs}{r+s} = \frac{9}{4} \Rightarrow 4rs - 9r - 9s = 0 \Rightarrow (4r-9)(4s-9) = 81$.
 7. Each factor is $\equiv 3 \pmod{4}$ (and > -9 since $r, s \geq 1$). Positive factor pairs of 81: $(1, 81), (3, 27), (9, 9)$; only $3 \equiv 3$ and $27 \equiv 3 \pmod{4}$ work: $4r-9 = 3, 4s-9 = 27 \Rightarrow r = 3, s = 9$. (Negative pairs give a non-positive root.) Sum = 12 (with $\lambda = \frac{1}{3}$).

180

8.

Single Correct • Assignment #A, Q.24

For integer n from 1 to 100, the number of n for which $x^2 + x - n = 0$ has integral roots is

- (A) 6 (B) 4 (C) 9 (D) none of these

Answer: (C)

Solution. Integral roots need $D = 1 + 4n$ to be an odd perfect square: $1 + 4n = (2k+1)^2 \Rightarrow n = k(k+1)$ (then roots $k, -k-1$ are integers). $k(k+1) \leq 100$: $k = 1, \dots, 9$ ($9 \cdot 10 = 90, 10 \cdot 11 = 110$). Nine values.

180

9.

Single Correct • Assignment #A, Q.25

The number of positive integers n for which $n^2 + 96$ is a perfect square is

- (A) 4 (B) 8 (C) 12 (D) infinite

Answer: (A)

Solution. $m^2 - n^2 = 96 \Rightarrow (m - n)(m + n) = 96$ with both factors of the same parity, hence both even, $m - n < m + n$: $(2, 48), (4, 24), (6, 16), (8, 12)$ give $n = 23, 10, 5, 2$. Four values.

181

0.

Single Correct • Assignment #A, Q.26

Which statement is true?

- (A) The only integral x for which $x^2 + 19x + 92$ is a perfect square is -8 . (B) ... is -11 .
(C) The number of integral x for which $x^2 + 19x + 92$ is a perfect square is two. (D) ... is infinite.

Answer: (C)

Solution. $x^2 + 19x + 92 = k^2 \Rightarrow (2x + 19)^2 + 7 = (2k)^2 \Rightarrow (2k - 2x - 19)(2k + 2x + 19) = 7$. With $k \geq 0$: factors 1, 7 give $2x + 19 = \pm 3 \Rightarrow x = -8$ or $x = -11$ (both give $k^2 = 4$). Exactly two.

181 Solution. $kx^2 + (2k - 1)x + (k - 2) = 0$: $D = (2k - 1)^2 - 4k(k - 2) = 4k + 1$, which must be a perfect square. $k = 5 \rightarrow 21, k = -3 \rightarrow -11, k = -6 \rightarrow -23, k = 12 \rightarrow 49 = 7^2 \checkmark$.

181

2.

Multiple Correct • Assignment #A, Q.31 & #B, Q.66

Integral values of x for which $x^2 + 17x + 71$ is a perfect square of an integer are

- (A) -10 (B) -7 (C) 10 (D) 7

Answer: (A), (B)

Solution. $x^2 + 17x + 71 = k^2 \Rightarrow (2x + 17)^2 - 5 = (2k)^2 \Rightarrow (2x + 17 - 2k)(2x + 17 + 2k) = 5$. So $2x + 17 = \pm 3 \Rightarrow x = -7$ or -10 (value 1 in both cases).

181

3.

Multiple Correct • Assignment #B, Q.52

If $x^2 - \frac{(a^2 + 11)x}{9} + \frac{15}{4}(a + \beta) + 16 = 0$ has two integral roots α, β , then

- (A) $\alpha + \beta = 20$ (B) $|\alpha - \beta| = 120$ (C) $|\alpha - \beta| = 6$ (D) $\alpha + \beta = 128$

Answer: (A), (C)

Solution. Product: $\alpha\beta = \frac{15}{4}(a + \beta) + 16 \Rightarrow (4\alpha - 15)(4\beta - 15) = 481 = 13 \cdot 37$. Each factor $\equiv 1 \pmod{4}$, which allows only 1, 13, 37, 481: $\{\alpha, \beta\} = \{4, 124\}$ or $\{7, 13\}$. Sum condition $\alpha + \beta = \frac{a^2 + 11}{9}$: for $\{7, 13\}$, $\alpha = 13$ gives $\frac{180}{9} = 20 \checkmark$; $\{4, 124\}$ fails. So $\alpha = 13, \beta = 7$: sum 20, difference 6.

181

4.

Multiple Correct • Assignment #B, Q.53

If $ax^2 + bx + c$ ($a > 1$; a, b, c integers) equals a prime p for two distinct integral values of x , then $ax^2 + bx + c$ cannot equal $2p$ for

- (A) $x = 0$ (B) $x = -1$ (C) $x = 1$ (D) $x = 3$

- 181 Solution.** Sum of roots $= a + 3$ must be an integer, so $a \in \mathbb{Z}$. For an integral root, $(x - a)(x - 3) = -1$
 5. forces $x - 3 = \pm 1$, $x - a = \mp 1$: $x = 4 \Rightarrow a = 5$; $x = 2 \Rightarrow a = 1$. (Then $(x - 4)^2 = 0$ and $(x - 2)^2 = 0$ — both roots integral.) Two values, sum 6, product 5.

181

6.

Multiple Correct • Assignment #B, Q.72

a, b integers, $b \neq -1$. If $x^2 + ax + b + 1 = 0$ has a positive integral root, which is/are true?

- (A) the other root is also a positive integer (B) the other root is an integer
 (C) $a^2 + b^2$ is a prime (D) $a^2 + b^2$ has a factor other than 1 and itself

Answer: (B), (D)

Solution. Roots r (positive integer) and $s = -a - r$ (an integer) (B). $rs = b + 1$, $r + s = -a$: $a^2 + b^2 = (r + s)^2 + (rs - 1)^2 = (r^2 + 1)(s^2 + 1)$. $s = 0$ would give $b = -1$ (excluded), so both factors are ≥ 2 : composite (D). (A) fails, e.g. $r = 1, s = -2$.

181

7.

Numerical • Assignment #B, Q.131

$f(x)$ is a cubic with integer coefficients, $f(0) = 3$, $f(1) = 11$, and $f(x) = 0$ has exactly 2 integer roots. The number of such polynomials is

Answer: 0

Solution. Integer roots divide 3 and 1 is not a root, so they come from $\{-1, 3, -3\}$. Write $f = (x - r)(x - s)(ax + b)$ with $a, b \in \mathbb{Z}$. $f(0) = rsb = 3$ and $f(1) = (1 - r)(1 - s)(a + b) = 11$: $\{-1, 3\}$: $b = -1$, $-4(a - 1) = 11$ ✗; $\{-1, -3\}$: $b = 1$, $8(a + 1) = 11$ ✗; $\{3, -3\}$: $-9b = 3$ ✗. (Repeated roots fail similarly.) None.

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181

Single Correct • Assignment #A, Q.23

8. The set of real p for which $|2x + 3| + |2x - 3| = px + 6$ has more than two solutions is
 (A) $[0, 4)$ (B) $(-4, 4)$ (C) $\mathbb{R} - \{4, -4, 0\}$ (D) $\{0\}$

181 Solution. Triangle inequality with signs chosen so that x cancels: $|2ax - 3| + |ax + 1| + |5 - ax| \geq$

9. $|(2ax - 3) - (ax + 1) + (5 - ax)| = 1 > \frac{1}{2}$. No solution.

182

0.

Single Correct • Assignment #A, Q.40 & #B, Q.26

The number of integers x satisfying $(x^2 - x - 1)^{x+2} = 1$ is

- (A) 2 (B) 3 (C) 4 (D) 5

Answer: (C)

Solution.

- Base 1: $x^2 - x - 2 = 0 \Rightarrow x = 2, -1$.
- Exponent 0, base $\neq 0$: $x = -2$ (base 5).
- Base -1 , even exponent: $x^2 - x = 0 \Rightarrow x = 0$ (exponent 2 ✓), $x = 1$ (exponent 3 ✗).

$x \in \{2, -1, -2, 0\}$: four.

182

1.

Single Correct • Assignment #A, Q.42 & #B, Q.24

If $\frac{9 - |2x + 3|}{2 - |x|} < 0$, the number of integral x is

- (A) 3 (B) 4 (C) 5 (D) more than 5

Answer: (A)

Solution. Numerator $> 0 \iff |2x + 3| < 9 \iff x \in (-6, 3)$. Denominator $> 0 \iff |x| < 2$. Opposite signs: (num > 0 , den < 0): $x \in (-6, -2) \cup (2, 3)$; (num < 0 , den > 0): impossible since $(-2, 2) \subset (-6, 3)$. Integers: $-5, -4, -3$. Three.

182

2.

Single Correct • Assignment #A, Q.46 & #B, Q.15

The number of real x satisfying $|x - \frac{1}{x}| + |x + \frac{1}{x}| = 2$ is

- (A) 1 (B) 2 (C) 3 (D) 4

182 Solution. Let $A = px^2 - qx + r$ and $B = x^2 - 5x + 6$. The LHS is $|A + B|$. And $|A + B| = ||A| - |B|| \iff AB \leq 0$. A has no real roots so keeps one sign; $A(2) = 4p - 2q + r > 0$ by the given condition, so $A > 0$ for all x . Hence we need $B \leq 0$: $x \in [2, 3]$.

182

4.

Single Correct • Assignment #B, Q.25

If $a_n = \sin^n \theta + \cos^n \theta$ and $4a_{n+2} - 2\sqrt{6}a_{n+1} + a_n = 0 \forall n$, then $\sin \theta + \cos \theta =$

- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{\sqrt{2}}$

Answer: (C)

Solution. A sequence $u^n + v^n$ satisfies $4a_{n+2} - 2\sqrt{6}a_{n+1} + a_n = 0$ for all n when u, v are the roots of $4t^2 - 2\sqrt{6}t + 1 = 0$. So $\sin \theta, \cos \theta$ are those roots and $\sin \theta + \cos \theta = \frac{2\sqrt{6}}{4} = \sqrt{\frac{3}{2}}$. (Consistency: product $\frac{1}{4}$, and $\sin^2 + \cos^2 = \frac{3}{2} - \frac{1}{2} = 1$ ✓.)

182

5.

Multiple Correct • Assignment #B, Q.44

For the equation $\log_{x^2+6x+8}(\log_{2x^2+2x+3}(x^2-2x)) = 0$:

(A) number of solutions is two

(B) sum of solutions is -4

(C) number of solutions is one

(D) sum of solutions is -1

Answer: (C), (D)

Solution. $\log_{2x^2+2x+3}(x^2-2x) = 1 \Rightarrow x^2-2x = 2x^2+2x+3 \Rightarrow x^2+4x+3 = 0 \Rightarrow x = -1, -3$.
Base check: $x = -1$: $x^2+6x+8 = 3 \checkmark$, $2x^2+2x+3 = 3 \checkmark$. $x = -3$: $x^2+6x+8 = -1 \times$. One solution, $x = -1$ (sum -1).

182

6.

Multiple Correct • Assignment #B, Q.61

For $x^4 + (x+2)^4 = 16(x+1)^4$:

(A) sum of real roots = -2 (B) sum of imaginary roots = -2 (C) product of real roots = $\frac{8}{7}$ (D) product of imaginary roots = $\frac{8}{7}$

182 Solution. Divide by x^2 ; with $t = x + \frac{1}{x}$: $t^2 + 2at - 1 = 0$. A value $t < -2$ gives two distinct negative roots. $x(t = -2)$ gives only $x = -1$. The negative root $t = -a - \sqrt{a^2+1} < -2 \iff \sqrt{a^2+1} > 2-a \iff a > \frac{3}{4}$. All options qualify.

182

8.

Multiple Correct • Assignment #B, Q.71

If $4x^4 + 2x^3 - 2x^2 + 3ax - a^2 = 0$ has four real roots, then a can be

(A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{16}$ (D) $\frac{1}{8}$

Answer: (A), (B), (C), (D)

Solution. Treat as a quadratic in a : $a^2 - 3xa - (4x^4 + 2x^3 - 2x^2) = 0$, discriminant $16x^4 + 8x^3 + x^2 = x^2(4x+1)^2$, so $a = 2x^2 + 2x$ or $a = x - 2x^2$. Hence $(2x^2 + 2x - a)(2x^2 - x + a) = 0$. Four real roots need $4 + 8a \geq 0$ and $1 - 8a \geq 0$: $a \in [-\frac{1}{2}, \frac{1}{8}]$. All options lie here.

182

9.

Multiple Correct • Assignment #B, Q.73

$x^4 - (3m+2)x^2 + m^2 = 0$ has four real roots in AP. Possible m :

(A) $-\frac{6}{19}$ (B) $\frac{6}{19}$ (C) 6 (D) -6

Answer: (A), (C)

Solution. Roots $\pm p, \pm q$ ($0 < p < q$) in AP: $-q, -p, p, q$ with $2p = q - p \Rightarrow q = 3p$. $p^2 + q^2 = 10p^2 = 3m+2$ and $p^2q^2 = 9p^4 = m^2 \Rightarrow m = \pm 3p^2$. $m = 3p^2$: $10p^2 = 9p^2 + 2 \Rightarrow p^2 = 2, m = 6$. $m = -3p^2$: $19p^2 = 2, m = -\frac{6}{19}$.

183 Solution. Exactly two real x : one t with $|t| > 2$ and the other with $|t| < 2$ (same sign, product 9). For $k > 0$ (roots negative): $h(-2) = 26 - 2k < 0 \Rightarrow k > 13$. For $k < 0$: $k < -13$. (13, 16) fits.

183

1.

Matrix Match • Assignment #A, Q.58 & #B, Q.98

- (A) If $|x+2| + |x-3| = 5$, the sum of integral x satisfying it lies in (P) $(-3, -1)$
 (B) If $\sqrt{x+\sqrt{x}} - \sqrt{x-\sqrt{x}} = \frac{3}{2}\sqrt{\frac{x}{x+\sqrt{x}}}$, the value(s) of x may lie in (Q) $[3, 5]$
 (C) If $a, b, c > 0$ with $2a + 3b + 4c = 9$, the maximum of $a^4b^2c^3$ lies in (R) $(1, 3)$
 (D) If 6 lies between the roots of $x^2 + 2(m-3)x + 9 = 0$, m may belong to (S) $(1, 2]$
 (T) $[5, 10]$

Answer: (A)→(Q); (B)→(R, S); (C)→(Q); (D)→(P)

Solution. (A) $|x+2| + |x-3| \geq 5$ with equality iff $x \in [-2, 3]$. Integers $-2, \dots, 3$ sum to $3 \in [3, 5]$.

(B) Put $x = t^2$ ($t \geq 1$). Multiplying by $\sqrt{t^2+t}$: $t^2+t - \sqrt{t^4-t^2} = \frac{3}{2}t \Rightarrow \sqrt{t^2-1} = t - \frac{1}{2} \Rightarrow t = \frac{5}{4}$, $x = \frac{25}{16} \approx 1.56 \in (1, 3)$ and $(1, 2]$.

(C) AM-GM on nine parts: $4 \times \frac{a}{2}$, $2 \times \frac{3b}{2}$, $3 \times \frac{4c}{3}$ (sum 9). Max when each part = 1: $a = 2, b = \frac{2}{3}, c = \frac{3}{4}$, giving $16 \cdot \frac{4}{9} \cdot \frac{27}{64} = 3 \in [3, 5]$.

(D) $f(6) < 0$: $36 + 12(m-3) + 9 < 0 \Rightarrow m < -\frac{3}{4}$; $(-3, -1)$ is contained in this.

183

2.

Matrix Match • Assignment #A, Q.59 & #B, Q.99

Match the inequations (Column I) with sets of x satisfying them (Column II).

- (A) $|x+1| + |x-2| + |x-3| \geq 6$ (P) $x < 0$
 (B) $||x| - 1| < 1 - x$ (Q) $x = 3$
 (C) $|x-4| < \sqrt{3-x}$ (R) $x \leq 3$
 (D) $\sqrt{x^3-4x^2+4x-3} + \sqrt{3-x} + x^2 \geq 9$ (S) $x \geq 4$
 (T) $x \in \phi$

Solution. (A) Piecewise: $x \geq 3$: $3x-4 \geq 6 \Rightarrow x \geq \frac{10}{3}$; $[2, 3)$: $x+2 \geq 6$ fails; $[-1, 2)$: $6-x \geq 6 \Rightarrow x \leq 0$; $x < -1$: $4-3x \geq 6$ always. Solution $x \leq 0$ or $x \geq \frac{10}{3}$, which contains P and S.

(B) Needs $1-x > 0$. For $0 \leq x < 1$: $1-x < 1-x$ fails. For $x < 0$: $|x+1| < 1-x \iff (x+1)^2 < (1-x)^2 \iff x < 0 \checkmark$. So $x < 0$.

(C) Domain $x \leq 3$; with $u = 3-x \geq 0$: $1+u < \sqrt{u}$ is impossible. ϕ .

(D) Domain: $x \leq 3$ and $(x-3)(x^2-x+1) \geq 0 \Rightarrow x \geq 3$. So $x = 3$: $0+0+9 \geq 9 \checkmark$.

183

3.

Matrix Match • Assignment #A, Q.60 & #B, Q.100

- (A) $x^2 + y^2 + 2z^2 = 4x - 2z + 2yz - 5$ (x, y, z real); possible value of (P) 1
 $x - y - z$
 (B) $(1 + \frac{1}{2})(1 + \frac{1}{2^2})(1 + \frac{1}{2^4})(1 + \frac{1}{2^8}) \dots \infty$ (Q) 2
 (C) If α is a root of $x^4 + x^2 - 1 = 0$, then $(\alpha^6 + 2\alpha^4)^7$ (R) -3
 (D) Sum of all real roots of $(x-1)^2 - 2|x-1| - 3 = 0$ (S) -1
 (T) 4

Answer: (A)→(T); (B)→(Q); (C)→(P); (D)→(Q)

Solution. (A) $(x-2)^2 + (y-z)^2 + (z+1)^2 = 0 \Rightarrow x = 2, z = -1, y = -1; x - y - z = 4$.

(B) Multiply by $(1 - \frac{1}{2})$: the product telescopes to $\frac{1}{1-\frac{1}{2}} = 2$.

(C) $\alpha^4 = 1 - \alpha^2 \Rightarrow \alpha^6 + 2\alpha^4 = \alpha^2(1 - \alpha^2) + 2\alpha^4 = \alpha^2 + \alpha^4 = 1$. Value 1.

(D) $t = |x-1|$: $t^2 - 2t - 3 = 0 \Rightarrow t = 3 \Rightarrow x = 4, -2$; sum 2.

183

4.

Numerical • Assignment #B, Q.105

Number of solutions of $(\sqrt{3} + \sqrt{2})^{x^2-15x} + (\sqrt{3} - \sqrt{2})^{x^2-15x} = 2\sqrt{3}$.

Answer: 4

Solution. $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 1$. With $u = (\sqrt{3} + \sqrt{2})^t$, $t = x^2 - 15x$: $u + \frac{1}{u} = 2\sqrt{3} \Rightarrow u = \sqrt{3} \pm \sqrt{2} \Rightarrow t = \pm 1$. $x^2 - 15x - 1 = 0$ and $x^2 - 15x + 1 = 0$ each have two real roots (discriminants 229, 221). Four solutions.

183

5.

Numerical • Assignment #B, Q.106

Number of rational x satisfying $x^{(\log_5 x)^2 + 5 \log_5 x + 2} = 5^8$.

183

6.

Solution. Regroup: $2^{x-1}(4x^2 - 1) = 2^{|x-3|+2}(4x^2 - 1) \Rightarrow (4x^2 - 1)(2^{x-1} - 2^{|x-3|+2}) = 0$. So $x = \pm \frac{1}{2}$, or $x - 1 = |x - 3| + 2 \iff |x - 3| = x - 3 \iff x \geq 3$. No negative integer. Answer 0.

183

7.

Numerical • Assignment #B, Q.116

Find the sum of all real solutions of $(x + 1)(x^2 + 1)(x^3 + 1) = 30x^3$.

Answer: 3 (no printed key)

Solution. $x = 0$ is not a root. Divide by x^3 and let $t = x + \frac{1}{x}$: $\frac{(x+1)^2}{x} \cdot \frac{x^2+1}{x} \cdot \frac{x^2-x+1}{x} = (t+2)t(t-1) = 30 \Rightarrow t^3 + t^2 - 2t - 30 = 0 \Rightarrow (t-3)(t^2 + 4t + 10) = 0$. Only $t = 3$ is real, giving $x^2 - 3x + 1 = 0$ with two real roots of sum 3.

183

8.

Numerical • Assignment #B, Q.121

Find the number of real solutions of $2x^4 - 3x^2 - 2x \sin x + 3 = 0$.

Answer: 0

Solution. LHS = $2(x^2 - 1)^2 + (x - \sin x)^2 + \cos^2 x$ (expand to check). Zero would need $x^2 = 1$, $x = \sin x$ and $\cos x = 0$ simultaneously — impossible. No real solution.

183

9.

Numerical • Assignment #B, Q.124

Number of real solution(s) of $3^{3x} - 3(3^{2x}) - 49(3^x) - 45 = 0$.

Answer: 1

Solution. $t = 3^x > 0$: $t^3 - 3t^2 - 49t - 45 = (t+1)(t-9)(t+5) = 0$. Only $t = 9$ is positive: $x = 2$.

184

0.

Numerical • Assignment #B, Q.127

For $(2x + 3)(3x + 4)(5x + 6)(30x + 31) = 10$, find the greatest integer less than the product of the real roots.

184 Solution. $t = e^x > 0$: $t^2 + (1 - 2k)t + k^2 - 1 = 0$, $D = 5 - 4k$. $k = 1$: $t^2 - t = 0 \Rightarrow t = 1$, giving $x = 0$
1. — has a solution. $k = 2$: $D < 0$ — no solution. Least $k = 2$.

184

2.

Numerical • Assignment #B, Q.136

The complete range of k for which $x^4 - 2kx^2 + x + k^2 - k = 0$ has all real solutions is $\left[\frac{m}{n}, \infty\right)$ with m, n coprime. Find $m + n$.

Answer: 7

Solution. Quadratic in k : $k^2 - (2x^2 + 1)k + (x^4 + x) = 0$, discriminant $(2x - 1)^2$, so $k = x^2 + x$ or $k = x^2 - x + 1$: $(x^2 + x - k)(x^2 - x + 1 - k) = 0$. All real: $1 + 4k \geq 0$ and $4k - 3 \geq 0 \Rightarrow k \geq \frac{3}{4}$. $m + n = 7$.

184

3.

Subjective • Assignment #A, Q.55

For $a \leq 0$, determine all real roots of $x^2 - 2a|x - a| - 3a^2 = 0$.

Answer: $x = (1 - \sqrt{2})a$ or $x = (\sqrt{6} - 1)a$

Solution. ($a = 0$ gives $x = 0$, consistent with both formulas.) Let $a < 0$.

$x \geq a$: $x^2 - 2ax - a^2 = 0 \Rightarrow x = a(1 \pm \sqrt{2})$. $a(1 + \sqrt{2}) < a$ is rejected; $a(1 - \sqrt{2}) > 0 \geq a \checkmark$.

$x < a$: $x^2 + 2ax - 5a^2 = 0 \Rightarrow x = -a \pm \sqrt{6}|a| = -a \mp \sqrt{6}a$. $x = -a(1 + \sqrt{6}) > 0$ is rejected; $x = a(\sqrt{6} - 1) < a \checkmark$ (since $\sqrt{6} - 1 > 1$ and $a < 0$).

184

4.

Subjective • Assignment #B, Q.110

Solve $(x \geq y \geq z \geq 2)$: $x^3 - 6z^2 + 12z - 8 = 0$, $y^3 - 6x^2 + 12x - 8 = 0$, $z^3 - 6y^2 + 12y - 8 = 0$.

Answer: $(x, y, z) = (2, 2, 2)$

Solution. $6z^2 - 12z + 8 = z^3 - (z - 2)^3$, so the first equation reads $x^3 = z^3 - (z - 2)^3 \leq z^3$ (as $z \geq 2$), i.e. $x \leq z$. With $x \geq y \geq z$ this forces $x = y = z$ and $(z - 2)^3 = 0$: $x = y = z = 2$ (which satisfies all three).

184

5.

Subjective • Assignment #B, Q.111

Solve for $x > 0$: $\sqrt{x + 2\sqrt{x + \cdots + 2\sqrt{x + 2\sqrt{3x}}}} = x$.

184 Solution. Subtract and divide by $x - y \neq 0$: $-(x + y) = x^2 + xy + y^2 - 3(x + y) + 2 \Rightarrow x^2 + xy + y^2 - 2x - 2y + 2 = 0$. As a quadratic in x : $\Delta = (y - 2)^2 - 4(y^2 - 2y + 2) = -3y^2 + 4y - 4 < 0$ for all y (its discriminant $16 - 48 < 0$). No real solution.

Topic 12 | Miscellaneous

184

Multiple Correct • Assignment #A, Q.34 & #B, Q.48

7. The values of k for which $2x + ky = k$, $3x - y = 10$ has solution(s) with $x > 0$, $y > 0$ are
 (A) $k < 0$ (B) $k < -\frac{2}{3}$ (C) $k > 0$ (D) $k > \frac{20}{3}$

Answer: (B), (D)

Solution. $y = 3x - 10 \Rightarrow 2x + 3kx - 10k = k \Rightarrow x = \frac{11k}{3k+2}$, $y = \frac{3k-20}{3k+2}$. If $3k+2 > 0$: need $k > 0$ and $k > \frac{20}{3} \Rightarrow k > \frac{20}{3}$. If $3k+2 < 0$: need $k < 0$ and $3k < 20 \Rightarrow k < -\frac{2}{3}$. Solution set: $k < -\frac{2}{3}$ or $k > \frac{20}{3}$; options (B) and (D) are exactly these parts.

184

8.

Multiple Correct • Assignment #B, Q.64

P_1, P_2 are the real values of P for which $3x^2 + 2Pxy + 2y^2 + 4x - 4y + 1$ can be resolved into linear factors. Then

- (A) $P_1 + P_2 = 8$ (B) $P_1 + P_2 = -8$
 (C) P_1, P_2 both positive (D) P_1, P_2 both negative

Answer: (B), (D)

Solution. For $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$: condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ with $a = 3, b = 2, c = 1, h = P, g = 2, f = -2$: $6 - 8P - 12 - 8 - P^2 = 0 \Rightarrow P^2 + 8P + 14 = 0 \Rightarrow P = -4 \pm \sqrt{2}$ (both satisfy $h^2 \geq ab = 6$). Sum -8 , both negative.

184

9.

Multiple Correct • Assignment #B, Q.67

$P(x) = x^2 + bx + c$ ($b, c \in \mathbb{R}$) with $P(P(1)) = P(P(2)) = 0$ and $P(1) \neq P(2)$. Then

- (A) $2b + 1 = 0$ (B) $2c + 3 = 0$ (C) $2b + 3 = 0$ (D) $2c + 1 = 0$

- 185 **Solution.** With $g(x) = x^2 + 7x + 5$, the product is $g(\alpha)g(\beta)$. Since $\alpha^2 = -2a\alpha - 3$: $g(\alpha) = (7 - 2a)\alpha + 2$. Let $u = 7 - 2a$: $g(\alpha)g(\beta) = 3u^2 + 2u(\alpha + \beta) + 4 = 3u^2 + 2u(u - 7) + 4 = 5u^2 - 14u + 4$ (using $\alpha + \beta = -2a = u - 7$). Minimum at $u = \frac{7}{5} \Rightarrow a = \frac{14}{5} = 2.8$, $[a] = 2$.

185

1.

Numerical • Assignment #B, Q.113

Evaluate $\frac{(7^4 + 64)(15^4 + 64)(23^4 + 64)(31^4 + 64)(39^4 + 64)}{(3^4 + 64)(11^4 + 64)(19^4 + 64)(27^4 + 64)(35^4 + 64)}$.

Answer: 337

Solution. Sophie Germain: $n^4 + 64 = n^4 + 4 \cdot 2^4 = ((n+2)^2 + 4)((n-2)^2 + 4)$. Write $g(m) = m^2 + 4$. Numerator = $g(9)g(5) \cdot g(17)g(13) \cdot g(25)g(21) \cdot g(33)g(29) \cdot g(41)g(37)$; denominator = $g(5)g(1) \cdot g(13)g(9) \cdot g(21)g(17) \cdot g(29)g(25) \cdot g(37)g(33)$. Everything cancels except $\frac{g(41)}{g(1)} = \frac{1685}{5} = 337$.

185

2.

Numerical • Assignment #B, Q.115

If $x + \sqrt[4]{5 - x^4} = 2$, find $x\sqrt[4]{5 - x^4}$ ($x \in \mathbb{R}$).

Answer: $\frac{8 - \sqrt{42}}{2}$

Solution. Let $y = \sqrt[4]{5 - x^4} \geq 0$, $p = xy$: $x + y = 2$, $x^4 + y^4 = 5$. $x^4 + y^4 = (x^2 + y^2)^2 - 2p^2 = (4 - 2p)^2 - 2p^2 = 16 - 16p + 2p^2 = 5 \Rightarrow 2p^2 - 16p + 11 = 0 \Rightarrow p = \frac{8 \pm \sqrt{42}}{2}$. Real x, y need $4 \geq 4p$, so $p \leq 1$: $p = \frac{8 - \sqrt{42}}{2} \approx 0.76$ (and then $x, y > 0$ ✓).

185

3.

Numerical • Assignment #B, Q.128

If $x + \frac{1}{x} = 3$ and $x^5 + \frac{1}{x^5} = a$, find the sum of the digits of a .

Answer: 6

Solution. $x^2 + \frac{1}{x^2} = 7$, $x^3 + \frac{1}{x^3} = 27 - 9 = 18$. $x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x}) = 126 - 3 = 123$. Digit sum 6.

185 Solution. Key identity: $P(y) - 1 = y(y - 1)$. Let $a_1 = P(\alpha)$, $a_2 = P(a_1)$, $a_3 = P(a_2)$, $a_4 = P(a_3) = 0$.

4. $a_4 - 1 = a_3(a_3 - 1) = a_3a_2(a_2 - 1) = a_3a_2a_1(a_1 - 1)$. Since $a_4 = 0$, the required product equals -1 ; modulus 1.

185

5.

Subjective (Proof) • Assignment #B, Q.118

Prove that the sum of cubes of the legs of a right-angled triangle is less than the cube of the hypotenuse.

Answer: Proved below

Solution. Legs a, b , hypotenuse c : $a^2 + b^2 = c^2$ with $a < c$, $b < c$. $a^3 + b^3 = a \cdot a^2 + b \cdot b^2 < c a^2 + c b^2 = c(a^2 + b^2) = c^3$. ■

185

6.

If $kx^2 - 4x + 3k + 1 > 0$ for atleast one $x > 0$, then if $k \in S$, then S contains :

- A) $(1, \infty)$ B) $(0, \infty)$ C) $(-1, \infty)$ D) $\left(-\frac{1}{4}, \infty\right)$

185

7.

If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral values(s) of 'a' is/are :

- A) -1 B) -2 C) 1 D) 2

185

8.

If $f(x)$ can take both positive and negative values then t must lie in the interval

- A) $\left(\frac{-1}{4}, \frac{1}{4}\right)$ B) $\left(-\infty, \frac{-1}{4}\right) \cup \left(\frac{1}{4}, \infty\right)$ C) $\left(\frac{-1}{4}, \frac{1}{4}\right) - \{0\}$ D) $(-4, 4)$

185 9.	If $f(x)$ is non-negative $\forall x \geq 0$ then t lies in the interval. A) $\left[\frac{1}{5}, \frac{1}{4}\right]$ B) $\left[\frac{1}{4}, \infty\right)$ C) $\left[\frac{-1}{4}, \frac{1}{4}\right]$ D) $\left[\frac{1}{5}, \infty\right)$
186 0.	Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If $\alpha_1, \alpha_2, \alpha_3$ and α_4 are all the roots of the equation $f(x) = 0$, then $\frac{ \alpha_1 ^2 + \alpha_2 ^2 + \alpha_3 ^2 + \alpha_4 ^2}{4}$ is equal to $\left[a + ib = \sqrt{a^2 + b^2} \text{ where } a, b \in R \text{ \& } \sqrt{-1} = i\right]$
186 1.	Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - x^3 - x^2 - 1 = 0$, then $P(\alpha) + P(\beta) + P(\gamma) + P(\delta) =$.
186 2.	Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - x^3 - x^2 - 1 = 0$, then $P(\alpha) + P(\beta) + P(\gamma) + P(\delta) =$
186 3.	The number of ordered pairs (a, b), where a, b are integers satisfying the inequality $\min(x^2 + (a-b)x + (1-a-b)) > \max(-x^2 + (a+b)x - (1+a+b)) \forall x \in R$, is:
186 4.	If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral value(s) of 'a' is/are: A) -1 B) -2 C) 1 D) 2
186 5.	If the minimum value of the quadratic expression $y = ax^2 + bx + c$ is negative attained at negative value of x, then: A) $a > 0$ B) $b > 0$ C) $c > 0$ D) $D > 0$ (where D is discriminant)
186 6.	If $2x^2 + 5x + 7 = 0$ and $ax^2 + bx + c = 0$ have at least one root common such that $a, b, c \in \{1, 2, \dots, 100\}$, then the difference between the maximum and minimum values of $a + b + c$ is: A) 196 B) 284 C) 182 D) 126
186 7.	If $(a+1)(b+1)(c+1)(d+1)=3$, $(a+2)(b+2)(c+2)(d+2)=7$, $(a+3)(b+3)(c+3)(d+3)=13$, $(a+4)(b+4)(c+4)(d+4)=21$. Then the value of $(a+5)(b+5)(c+5)(d+5)$ is A) More than 55 B) More than 45 C) Less than 65 D) An even number
186 8.	If the equation $x^3 - 6x^2 + 9x + \lambda = 0$ has exactly one root in $(1, 3)$ then the number of integral values of $[\lambda + 1]$ (where $[.]$ denotes the greatest integer function) is _____

186 9.	If $A = (49 + 20\sqrt{6})^{1/4}$, $B = 8\sqrt{3} + \frac{8\sqrt{6}}{\sqrt{3}} + \frac{16}{\sqrt{3}} + \frac{16\sqrt{2}}{3} + \dots$ and $\alpha - \beta = (6\sqrt{6})^k$, where α, β are the roots of the quadratic equation $A(\sqrt{3} - \sqrt{2})x^2 + \frac{B}{(\sqrt{3} + \sqrt{2})}x + C = 0$ and $k = \log_6 10 - 2\log_6 \sqrt{5} + \log_6 \sqrt{(\log_6 18 + \log_6 72)}$, then the value of C is
187 0.	Let $p(x) = 0$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $p(x) = 0$ will be
187 1.	The set of real parameter 'b' for which the equation $x^4 - 4bx^2 + x - 2b + 4b^2 = 0$ has all real solutions is given by $\left[\frac{p}{q}, \infty\right)$ where p and q are natural numbers having their highest common factor as 2, then the value of $(p + q)$ will be $\begin{vmatrix} a & b-y & c-z \end{vmatrix}$
187 2.	If $a, b, c \in R - \{0\}$, $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root, then the value of $\frac{a^2}{3bc} \cdot \frac{b^2}{3ac} \cdot \frac{c^2}{3ab}$ will be
187 3.	If α, β^2 are the roots of $5x^2 - 20x + 3k = 0$ and β, α^2 are the roots of $kx^2 - 50x + 45 = 0$, where $\alpha, \beta \in R$ and $k \neq 0$, then which of the following is/are Correct? A) $\alpha\beta = 3$ B) $k = 5$ C) $\alpha^3 - 10\alpha + 3\beta = 0$ D) $\beta^3 - 4\beta + \alpha = 0$

187

4.

Column-I	Column-II
(I) The number of real solution(s) of the equation $x^2 - x - 2 = 0$ is/are	(P) 0
(II) If one of the roots is the square of the other for the equation $3x^2 + px + 3 = 0, p > 0$ then $ p =$	(Q) 1
(III) The number of real value(s) of k for which the system of equations $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k - 1$ have infinitely many solutions is/are	(R) 2
(IV) Number of real root(s) of $x^2 - \frac{2}{x-2} = 4 - \frac{2}{x-2}$ is/are	(S) 3
	(T) 6

- A) (I)-(S); (II)-(R); (III)-(Q); (IV)-(R) B) (I)-(S); (II)-(P); (III)-(Q); (IV)-(Q)
 C) (I)-(R); (II)-(S); (III)-(Q); (IV)-(Q) D) (I)-(R); (II)-(S,T); (III)-(Q); (IV)-(R)

187

5.

Matching

Column - I	Column - II
A) Number of distinct real roots of $x^4 + 4x^3 + 12x^2 + 4x + 1 = 0$ is	P) 4
B) Number of distinct real roots of $x^3 - x^2 + x^2 - x + 15 = 0$ is	Q) 3
C) Number of distinct real roots of the equation $x^7 + 20x^6 - x^5 - 576 = 0$ is	R) 0
D) If α is greatest positive root of $\sqrt{6}x^4 - \left(\sqrt{3} + \frac{3}{2}\sqrt{2}\right)x^2 + \frac{3}{2} = 0$ then value of $[4\alpha^4]$ is ([.] denotes GIF)	S) 2

- A) A - S, B - R, C - Q, D - Q B) A - R, B - R, C - Q, D - Q
 C) A - R, B - S, C - P, D - Q D) A - P, B - R, C - Q, D - S

187

6.

The number of pairs of natural numbers (x, y) which satisfy the equation $2x(xy - 2y - 3) = (x + y)(3x + y)$ is _____

187

7.

For the equation $x^4(x-5) + 9x^2(x-1) = 1 - 5x$ which of the following is/are correct

- A) It has 2 imaginary roots B) It has 2 irrational roots
 C) It has 2 rational roots D) It has 3 real roots

187
8.

Let $f(x) = \left(\frac{a^4 - 9}{a^2 + 3}\right)x^3 - 3x + \tan 3$, then the roots of the equation

$$\frac{1}{x - f(1)} - \frac{2}{x - f(2)} - \frac{3}{x - f(3)} = 0 \text{ necessarily possesses}$$

- A) Atleast one root between $f(1)$ and $f(2)$ when $|a| < 3$
 B) Atleast one root between $f(2)$ and $f(3)$ when $|a| < 3$
 C) Atleast one root between $f(2)$ and $f(3)$ when $|a| > 3$
 D) Atleast one root between $f(2)$ and $f(3)$ for all real values of a

187
9.

If the quadratic equation $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ has two integral roots, Then sum of all possible values of θ in interval $(0, 2\pi)$ is $k\pi$. The value of k is _____

- A) 2 B) 3 C) 1 D) 4

188
0.

point C_{10} . If $b = 5$ and $c = 3$, then

The absolute value of sum of abscissas of points $B_1, B_2, B_3, \dots, B_{10}$ is _____

188
1.

The value of the sum $\sum_{n=1}^{10} \cot(B_n C_n P)$ is _____

188
2.

The equation $x^2 - a = \sqrt{x + a}$ has real or imaginary roots depending on values of a . List-I represents the nature of root and List-II represents the corresponding exhaustive values of a . match the following **Column-I** with **Column-II**

Column-I		Column-II	
A)	No real root	P)	$(0, 1) \cup \left\{-\frac{1}{4}\right\}$
B)	One real root	Q)	$\left(-\infty, -\frac{1}{4}\right)$
C)	Exactly two real roots	R)	$\left[-\frac{1}{4}, 0\right] \cup [1, \infty)$
D)	Atleast two real roots	S)	$\left(0, \frac{3}{4}\right) \cup \left\{-\frac{1}{4}\right\}$
		T)	$\left[-\frac{1}{4}, 0\right] \cup \left[\frac{3}{4}, \infty\right)$

The Correct option is:

A) $A \rightarrow Q, B \rightarrow P, C \rightarrow R, D \rightarrow R$

B) $A \rightarrow Q, B \rightarrow P, C \rightarrow T, D \rightarrow S$

C) $A \rightarrow P, B \rightarrow S, C \rightarrow T, D \rightarrow T$

D) $A \rightarrow Q, B \rightarrow S, C \rightarrow T, D \rightarrow T$

- 188
3. common negative integral root then
The value of $(a + b)$ is _____

- 188
4. Then the absolute value of common root is _____

Question Stem for Question Nos. 47 and 48

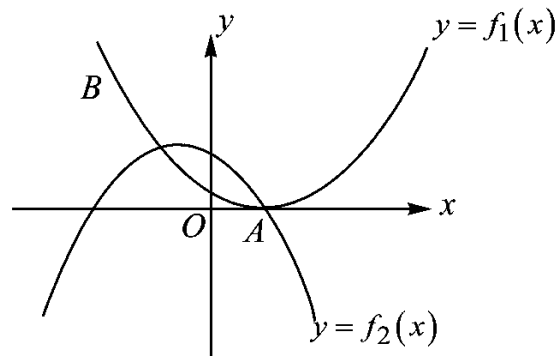
Question Stem

If $\alpha \neq 1$ is an n^{th} roots of unity and $1 + 5\alpha + 9\alpha^2 + \dots + (4n-3)\alpha^{n-1} = \frac{an}{\alpha-1}$

$$1 + 5\cos\frac{2\pi}{n} + 9\cos\frac{4\pi}{n} + \dots + (4n-3)\cos\left(\frac{2(n-1)\pi}{n}\right) = b \cdot n$$

$$\text{and } 5\sin\frac{2\pi}{n} + 9\sin\frac{4\pi}{n} + \dots + (4n-3)\sin\left(\frac{2(n-1)\pi}{n}\right) = c\left(n\cot\frac{\pi}{n}\right) \text{ for } n \geq 2$$

- 188
5. Graphs of two quadratic expressions $y = f_1(x)$ and $y = f_2(x)$ are shown in figure.
Which of the following statement(s) is/are true?



- A) Sum of roots of $f_1(x) = 0$ is greater than sum of roots of $f_2(x) = 0$
 B) Sum of roots of $f_1(x) = 0$ is less than sum of roots of $f_2(x) = 0$
 C) Both roots of $f_1(x) = 0$ and $f_2(x) = 0$ are common
 D) The equation $f_1(x) \cdot f_2(x) = 0$ has 2 distinct roots

- 188
6. Given 2 quadratic equations $x^2 - ax + b = 0$ and $x^2 - px + q = 0$. If

$(a^2 - 4b)(p^2 - 4q) \geq 0$, then the number of real root(s) of

$(x^2 - ax + b)(x^2 - px + q) = 0$ can be

- A) 0 B) 2 C) 4 D) 1

188
7.

If λ is the number of integral root(s) of $ax^2 + bx + c = 0$, given that a, b, c are odd integers which are not less than 2024, then sum of possible value(s) of λ will be ____

188
8.

The number of triples (x, a, b) where x is a real number and a, b belong to the set $\{1, 2, 3, 4, 5, 6, 7\}$ such that $x^2 - a\{x\} + b = 0$, (where $\{ \}$ denotes fractional part function) is R , then sum of the digits in R is

188
9.

Let $f(x) = a(x-2)(x-b)$, where $a, b \in R$ and $a \neq 0$. Also

$$f(f(x)) = a^3 \left(x^2 - (2+b)x + 2b - \frac{2}{a} \right) \left(x^2 - (2+b)x + 2b - \frac{b}{a} \right), a \neq 0, \text{ has exactly one}$$

real root as 5. The slope of the straight line passing through $O(0,0)$ and tangent to $y = f(x)$, can be

- A) -4 B) 2 C) $\frac{4}{9}$ D) $\frac{2}{9}$

189
0.

The leading coefficient of $f(x)$, is

- A) $1/6$ B) $1/9$ C) $1/12$ D) $1/18$

189
1.

The degree of $f(x)$, is

- A) 2 B) 3 C) 4 D) 5

189
2.

The condition that the equation has one purely imaginary root is

- A) $(c\bar{a} - a\bar{c})^2 = -(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$ B) $(c\bar{a} + a\bar{c})^2 = (b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$
C) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b)$ D) $(c\bar{a} + a\bar{c})^2 = -(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$

189
3.

If equation has two purely imaginary roots, then which of the following must **NOT** be true.

- A) $a\bar{b}$ is purely imaginary B) $b\bar{c}$ is purely imaginary
C) $c\bar{a}$ is purely real D) $a\bar{c}$ is purely imaginary

189 4.	Let α and β be the roots of $x^2 - 8x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{4a_9}$ is 1) 2 2) 1 3) 4 4) 3
189 5.	The integral value of a, for which the quadratic expression $ax^2 + (a-2)x - 2$ is negative to exactly two integral value of x, is 1) 1 2) 2 3) 5 4) 8 30. $\sqrt{2}$
189 6.	Let $a, b, c, d \in \{1, 2, 3, 4, 5\}$ such that $y = 2x^2 + ax + b$ and $y = 2x^2 + cx + d$ have no point in common then maximum possible value of $(a-c)^2 + (b-d)$ is 1) 4 2) 5 3) 6 4) 9
189 7.	equal to _____. If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1 , then b^4 is equal to:
189 8.	The value of 'a' for which the quadratic expression $ax^2 + 2a-3 x - 6$ is positive for exactly three integral values of x A) $\left[-\frac{3}{5}, -\frac{1}{2}\right]$ B) $\left[-\frac{3}{5}, -\frac{1}{2}\right]$ C) $\left[-\frac{3}{5}, -\frac{1}{6}\right]$ D) $\left[-\frac{3}{5}, -\frac{1}{6}\right]$
189 9.	Let a, b, c, d be positive integers such that $a \geq b \geq c \geq d$, then which of the following can be correct about the roots of the equation $x^4 - ax^3 - bx^2 - cx - d = 0$ A) All roots can be integer. B) Atmost two roots can be integer. C) Atmost one root can be integer. D) No roots can be integer.
190 0.	If α is a real root of the equation $x^5 - x^3 + x - 2 = 0$, then find the value of $\left[\alpha^6\right]$. (For any real number a, $[a]$ represent the greatest integer not exceeding a) A) $\left[\alpha^6\right] = 2$ B) $\left[\alpha^6\right] = 3$ C) $\left[\alpha^6\right] = 4$ D) $\left[\alpha^6\right] = 5$
190 1.	Number of integral solutions (x, y) is/are $(x, y \in I)$

190
2.

The value of $\left(\frac{x_1 + x_2 + \dots + x_n + y_1 + y_2 + \dots + y_n}{n}\right) \times \left(-\frac{12}{5}\right)$ is

Question Stem for Question Nos. 45 and 46

Question Stem

For the equation $(2x^2 - 1)^2 + 2ax(2x^2 - 1) + x^2(4a^2 - 1) = 0$

190
3.

The number of negative integral values of a for which above equation has all roots positive

190
4.

Find the largest integral value of a for which above equation has two positive and two negative roots

Question Stem for Question Nos. 47 and 48

Question Stem

Consider a quadratic equation $ax^2 + bx + c = 0$ and quadratic equation does not have any integral roots.

190
5.

Given that $5a + 2b + c = 0$, then exactly one of the root of the above quadratic equation will lie in the interval $(n, 2n + 1)$, $n \in I$. Find the value of n ?

190
6.

Given that $a + b = 0$ and $4a + 6b + 9c = 0$ then both the roots would lie in the interval $(n, n + 1)$, $n \in I$. Find the value of n ?

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- **Full Marks** : +4 If only (all) the correct option(s) is (are) chosen;
- **Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen,
- **Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
- **Partial Marks** : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- **Zero Marks**: 0 If unanswered;
- **Negative Marks**: -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then
 Choosing ONLY (A), (B) and (D) will get +4 marks;
 Choosing ONLY (A), will get +1 mark;
 Choosing ONLY (B), will get +1 mark;
 Choosing ONLY (D), will get +1 mark;
 Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 Choosing any other option(s) will get -2 marks.

- 190
7. Let t be a real number satisfying $2t^3 - 9t^2 + \lambda = 0$ and $\lambda \in R$; then which of the following options (s) are correct ?
- A)** If $t = x + \frac{1}{x}$ in above equation then for exactly one value of λ the equation will have exactly 3 real and distinct solution of x .
- B)** If $t = x + \frac{4}{x}$ in above equation then for exactly one value of λ the equation will have exactly 3 real and distinct solution of x .
- C)** If $t = x + \frac{1}{x}$ in above equation then for exactly 6 integral values of λ the equation will have exactly 4 real and distinct solution of x
- D)** If $t = x + \frac{4}{x}$ in above equation then maximum positive integral value of λ for which the equation will have exactly 2 real , distinct and positive solution of x , is 12

- 190
8. If $f(x) = a_{100}x^{100}$,
 $g(x) = a_{100}x^{100} + a_{99}x^{99}$,
 $h(x) = a_{49}x^{49} + a_{48}x^{48} + a_{47}x^{47} + \dots + a_2x^2 + a_1x + a_0$, then
 ($a_0, a_1, a_2, \dots, a_{100}$ or non zero real numbers)
- A)** $f(x) + h(x) = 0$ has atleast 50 non real roots
- B)** $f(x) - h(x) = 0$ has atleast 50 non real roots
- C)** $g(x) + h(x) = 0$ has atleast 50 non real roots if $a_{99}.a_{49} > 0$
- D)** $g(x) + h(x) = 0$ has atleast 50 non real roots if $a_{99}.a_{49} < 0$

- 190
9. Let x, y, z be non zero real numbers that satisfy the system of equations:
- $$(x^2 + xy + y^2)(y^2 + yz + z^2)(z^2 + zx + x^2) = xyz, \text{ and}$$
- $$(x^4 + x^2y^2 + y^4)(y^4 + y^2z^2 + z^4)(z^4 + z^2x^2 + x^4) = x^3y^3z^3$$
- A)** Number of triplets (x, y, z) that satisfy above equations is 3
- B)** Sum of all possible values of x that satisfy above equation is less than 2
- C)** Sum of all possible values of y that satisfy above equation is less than 1
- D)** Number of triplets (x, y, z) that satisfy above equations is less than 4

- 191
0. For every real number a the equation $8x^4 - 16x^3 + 16x^2 - 8x + a = 0$,
A) has atleast one non-real root **B)** has all real roots
C) sum of all non real roots can be 1 **D)** sum of all non real roots can be 2

- 191
1. Let $P(x) = x^2 + \frac{1}{2}x + b$ and $Q(x) = x^2 + cx + d$ be two polynomials with real coefficients such that $P(x)Q(x) = Q(P(x))$ for all real x , then which of the following is/are the correct statement(s)
A) sum of real roots of $P(Q(x)) = 0$ is $-\frac{1}{2}$
B) sum of real roots of $P(Q(x)) = 0$ is 0
C) Product of non real roots of $P(Q(x)) = 0$ is 1
D) sum of non real roots of $P(Q(x)) = 0$ is $-\frac{1}{2}$

- 191
2. Let p, q, r be positive real numbers, not all equal, such that some two of the equations $px^2 + 2qx + r = 0$, $qx^2 + 2rx + p = 0$, $rx^2 + 2px + q = 0$, have a common root, say α , then which of the following is/are the correct statement(s)
A) α is real and negative.
B) The third equation has non-real roots.
C) α is real and positive.
D) The third equation has real roots.

SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

- 191
3. If the range of value of a for which the roots of the equation $x^2 - 2x - a^2 + 1 = 0$ lie between the roots of the equation $x^2 - 2(a+1)x + a(a-1) = 0$ is (p, q) then the value of $10\left(q + \frac{1}{p^2}\right)$ is

- 191
4. Let O be the vertex and Q be any point on the parabola, $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio 1:3, then locus of P is:
1) $y^2 = 2x$ **2)** $y^2 = x$ **3)** $x^2 = y$ **4)** $x^2 = 2y$

- 191
5. Let a, b, c, d be distinct real numbers such that a, b are roots of $x^2 - 5cx - 6d = 0$ and c, d are roots of $x^2 - 5ax - 6b = 0$ then $b+d=$ _____ ($a > 0, c > 0$)
1) 120 2) 144 3) 164 4) 180
- 191
6. The equation $x^2 - 4x + [x] + 3 = x[x]$ where $[.]$ denotes greatest integer function has
1) exactly 2 solutions in R 2) no solution
3) unique solution in $(-\infty, 1)$ 4) unique solution in R
- 191
7. If the equation $x^2 + (a - 2)x + 1 = 3|x|$ has exactly 3 distinct real solutions in 'x' then sum of possible value a is
1) 2 2) 4 3) 6 4) 8
- 191
8. The number of integral values of k for which one root of $2x^2 - 8x + k = 0$ lies in the interval (1,2) and it's other roots lies in the interval (2,3) is
1) 1 2) 3 3) 0 4) 2
- 191
9. Let f(x) be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1, then the sum of the roots of $f(x) = 0$ is equal to
1) $\frac{11}{3}$ 2) $\frac{7}{3}$ 3) $\frac{13}{3}$ 4) $\frac{14}{3}$
- 192
0. Find the number of integral values of 'k' for which the equation $x^2 - 4|x| + 3 - |k - 1| = 0$ has
four distinct real roots.
- 192
1. One root of the equation $x^2 - (k + 1)x + k^2 + k - 8 = 0$ exceeds 2 and other root is less than 2 then number of non-positive integral values of 'k' are_____